Nano Photonics and Plasmonics Research

Overview Lecture

Opto-electronic Devices and Circuits

Syllabus

Nano Photonics and Plasmonics Research

Instructor Information	Mohammad A. Alsunaidi	Office M4-232	Phone ---	Office Hours Mon-Thu: 11:30-12:30	Email mohammad.alsunaidi@marmara	
Course Information	Text *Optoelectronics*, S. O. Kasap	Grading	Class Work Assignments + Exp (4x5)% + Project 10%		Midterm Exam 30%	Final Exam 40%

Week	Events	Subjects	Tuesdays	Wednesdays
1 Oct 2-6	---	Nature of Light and Material Properties	Course introduction and scope.	Light as an EM Wave: Maxwell's and Wave Equations. Types of waves.
2 Oct 9-13	---		Light Propagation equation.	Light Propagation in different media.
3 Oct 16-20	Course drop/add Week		Light Reflection and Refraction.	Light Reflection and Refraction. Anti-reflection Surfaces.
4 Oct 23-27	Cumhuriyet Bayramı Oct 29		Light-matter interaction	Light as an energy quanta: The photoelectric effect and the Photon.
5 Oct 30-Nov 3	---	The PN Junction	Semiconductor Fundamentals and the PN Junction. PN junction energy bands.	Photon absorption and generation. Direct and Indirect semiconductors.
6 Nov 6-10	Assignment 1 Due Nov 10	Photodetectors	Photodetectors types and basic structures.	Characteristics and modes of operation. PN, PiN, APD and Schottky PD.
7 Nov 13-17	---		Photodiode circuit configurations and modes of operation.	Phototransistor circuit configurations and modes of operation.
Nov 20-26	**Mid-term Exams period**			
9 Nov 27-Dec 1	Experiment 1 Due Nov 17	Solar Cells	Basic and improved solar cells structures and materials. Thin film cells. Design rules.	Photo-voltaic (PV) circuit analysis.
10 Dec 4-8	---	Light Emitting Diodes (LEDs)	LEDs basic design and operation.	LED characterization: optical power, spectral distribution and modulation BW.
11 Dec 11-15	Assignment 2 Due Dec 15		LED characterization: optical power, spectral distribution and modulation BW.	Improved LED structures and materials.
12 Dec 18-22	Midterm Make-up (Excused, Dec 21-23)		UV LEDs. Material systems, basic designs and challenges.	White LEDs. Techniques, efficiencies and challenges.
13 Dec 25-29	Experiment 2 Due Dec 29		LED circuits. Basic amplifier configurations.	LED circuits. Modules, and panels.
14 Jan 1-5	New Year 2024 Monday	Opto-couplers and Opto-isolators	Opto-couplers basic design and applications.	Opto-couplers circuits.
15 Jan 8-12	Project submission Jan 12	Special topics	Students' Projects	Students' Projects
Jan 15-28	**Final Exams period**		**Jan 29-Feb 4 (Re-sit Final Exams period)**	

Light as an Electromagnetic Wave

The following subjects will be covered

- Maxwell's Equations: Review
- The EM Wave Equation and Solutions.
- The Refractive Index Relation.
- Light Wave Power Density.

Maxwell's Equations

Review of the fundamental laws of electricity and magnetism.

Fields	$\vec{E} = \frac{\vec{F}_e}{q}$	$(\vec{v} \times \vec{B}) = \frac{\vec{F}_m}{q}$
Flux densities	$\vec{D} = \varepsilon \vec{E}$	$\vec{B} = \mu \vec{H}$
Gauss's Law	$\vec{\nabla} \cdot \vec{D} = \rho_v$	$\oint \vec{D} \cdot \overrightarrow{ds} = Q_{enc}$
Gauss's Law	$\nabla \cdot \vec{B} = 0$	$\oint \vec{B} \cdot \overrightarrow{ds} = 0$
Ampere's Law	$\vec{\nabla} \times \vec{H} = \vec{J}$	$\oint \vec{H} \cdot \overrightarrow{dl} = I_{enc}$
Faraday's Law	$\mathcal{E}mf = -\frac{\partial \varphi}{\partial t}$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Charge conservation
The continuity equation

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} = 0$$

$$\vec{J} = \sigma \vec{E}$$

Basic Quantities

(V/m) $\vec{E}, \vec{D}$ (C/m^2)

(A/m) $\vec{H}, \vec{B}$ (Wb/m^2)

(A/m^2) $\vec{J}, \rho_v$ (C/m^3)

Material Properties

ε, μ, σ

(F/m) (H/m) (S/m)

Maxwell's Equations

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \qquad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{D} = \rho_v \qquad \nabla \cdot \vec{B} = 0$$

$$\vec{D} = \varepsilon \vec{E} \qquad \vec{B} = \mu \vec{H} \qquad \vec{J} = \sigma \vec{E}$$

Maxwell's Equations

Constitutive Relations

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\mu \frac{\partial (\vec{\nabla} \times \vec{H})}{\partial t}$$

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E} = -\mu \frac{\partial}{\partial t}(\vec{J} + \frac{\partial \vec{D}}{\partial t})$$

$$\vec{\nabla}^2 \vec{E} - \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E}$$

$$\vec{\nabla}^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla}^2 \vec{H} = \mu\sigma \frac{\partial \vec{H}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

The EM Wave Equation

The EM Wave Equation

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\sigma\frac{\partial\vec{H}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Write the wave equation in this form

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\vec{E} - \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} = 0 \qquad \nabla^2 U - \frac{1}{v^2}\frac{d^2U}{dt^2} = 0$$

Wave speed $v = \dfrac{1}{\sqrt{\mu\varepsilon}}$ For free space $v = \dfrac{1}{\sqrt{\mu_o\varepsilon_o}} = 3.0\times10^8$ (m/s)

1) Electromagnetic waves exist.

2) The wave is composed of two interacting quantities E and H fields.

3) EM waves travel with the speed of light.

4) Light is an EM waves.

TH Plane-wave Solution in Lossless Media

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Time-harmonic waves are generated by time-harmonic sources.

$\vec{E}(r,t) = \vec{E}(r)\,e^{j\omega t}$ r here represents the space variable. $e^{j\omega t}$ Sinusoidal, single frequency

$$\vec{\nabla}^2\vec{E}(r) + k^2\vec{E}(r) = 0$$
$$\vec{\nabla}^2\vec{H}(r) + k^2\vec{H}(r) = 0$$

Helmholtz wave equation

$$k = \omega\sqrt{\mu\varepsilon} = \frac{\omega}{v}$$

Wave number (rad/m) (Real number)

Refractive index

$$n = \sqrt{\mu_r\varepsilon_r} \qquad v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n} \qquad k = \frac{n\omega}{c}$$

The Uniform Plane-wave Solution

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

TH Wave equation $\nabla^2\vec{E}(x,y,z) + k^2\vec{E}(x,y,z) = 0$ ➡ $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right)\vec{E}(x,y,z) = 0$

Consider a one-dimensional case (z) with one field component only.

$$\frac{\partial^2 E_x}{\partial z^2} + k^2 E_x = 0$$

$$E_x(z) = A\,e^{-jkz} + B\,e^{jkz}$$

Plane wave Solution

A and B are constants.

Put back the time-harmonic factor

$$E_x(z,t) = A\,e^{j(\omega t - kz)} + B\,e^{j(\omega t + kz)}$$

Wave travelling along the $+ve$ z direction

Wave travelling along the $-ve$ z direction

Take the $+z$ wave

$$E_x(z,t) = E_o\,e^{j(\omega t - kz)}$$

$$E_x(z,t) = E_o\cos(\omega t - kz) + j\,E_o\sin(\omega t - kz)$$

Solution Solution

Take the sine wave solution

$$E_x(z,t) = E_o\sin(\omega t - kz)$$

$$\vec{E}(z,t) = E_o\sin(\omega t - kz)\;\vec{a}_x$$

Same definitions for E_o, ω, k

Time-harmonic (Sinusoidal) Light Wave

Electric field Part of the wave

Amplitude

Phase

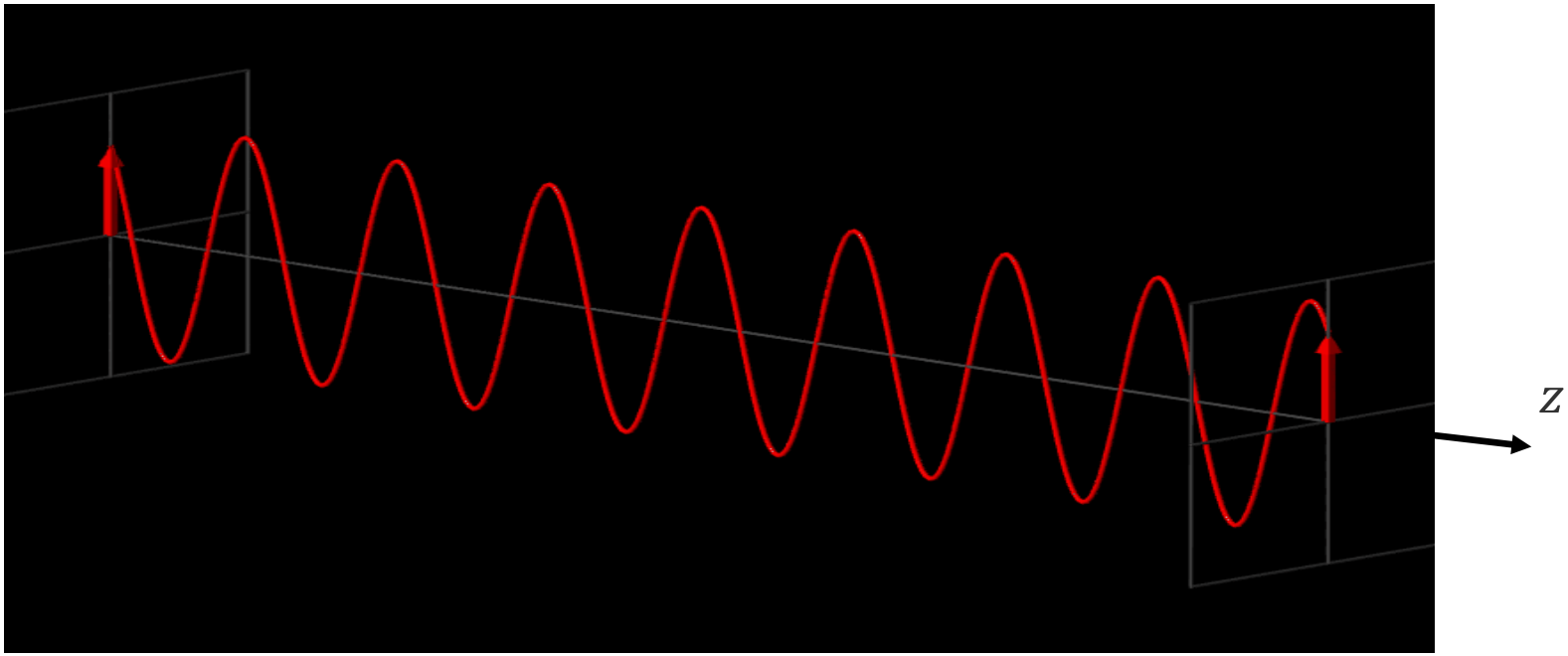
$$\vec{E}(z,t) = E_o \sin(\omega t \pm kz)\ \vec{a}_x$$

Field orientation/polarization

$\omega = 2\pi f$ Frequency

$k = \frac{\omega}{v}$ Wave number

Direction of wave propagation



Light as an Electromagnetic Wave

The following subjects will be covered

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- The Refractive Index Relation.
- Light Wave Power Density.

The Uniform Plane-wave Solution

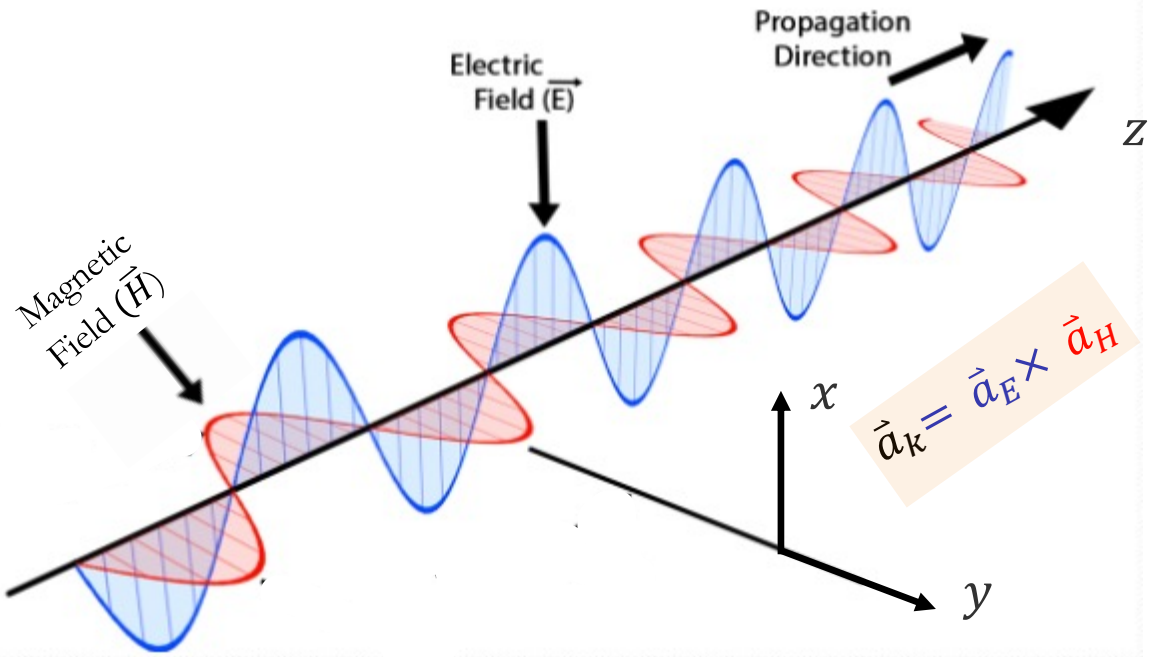
$$\vec{E}(z,t) = E_o \sin(\omega t - kz)\ \vec{a}_x$$

What about $\vec{H}$?

The EM fields are related using Maxwell's equations.

$$\vec{E}(z) = E_o\, e^{-jkz}\ \vec{a}_x$$

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$



For the 1D TH case, we have

$$\vec{H} = \frac{1}{-j\omega\mu}\ \vec{\nabla} \times \vec{E}\ = \frac{1}{-j\omega\mu}\ (-jk) E_o\, e^{-jkz}\ \vec{a}_y\ = \frac{k}{\omega\mu}\ E_o\, e^{-jkz}\ \vec{a}_y \qquad k^2 = \omega^2 \mu\varepsilon$$

$$\vec{H} = \sqrt{\frac{\varepsilon}{\mu}}\, E_o\, e^{-jkz}\, \vec{a}_y$$

$\sqrt{\frac{\mu}{\varepsilon}}$ has units of impedance (Ω)

$$\eta = \frac{E_o}{H_o} = \sqrt{\frac{\mu}{\varepsilon}}$$ Plane wave impedance

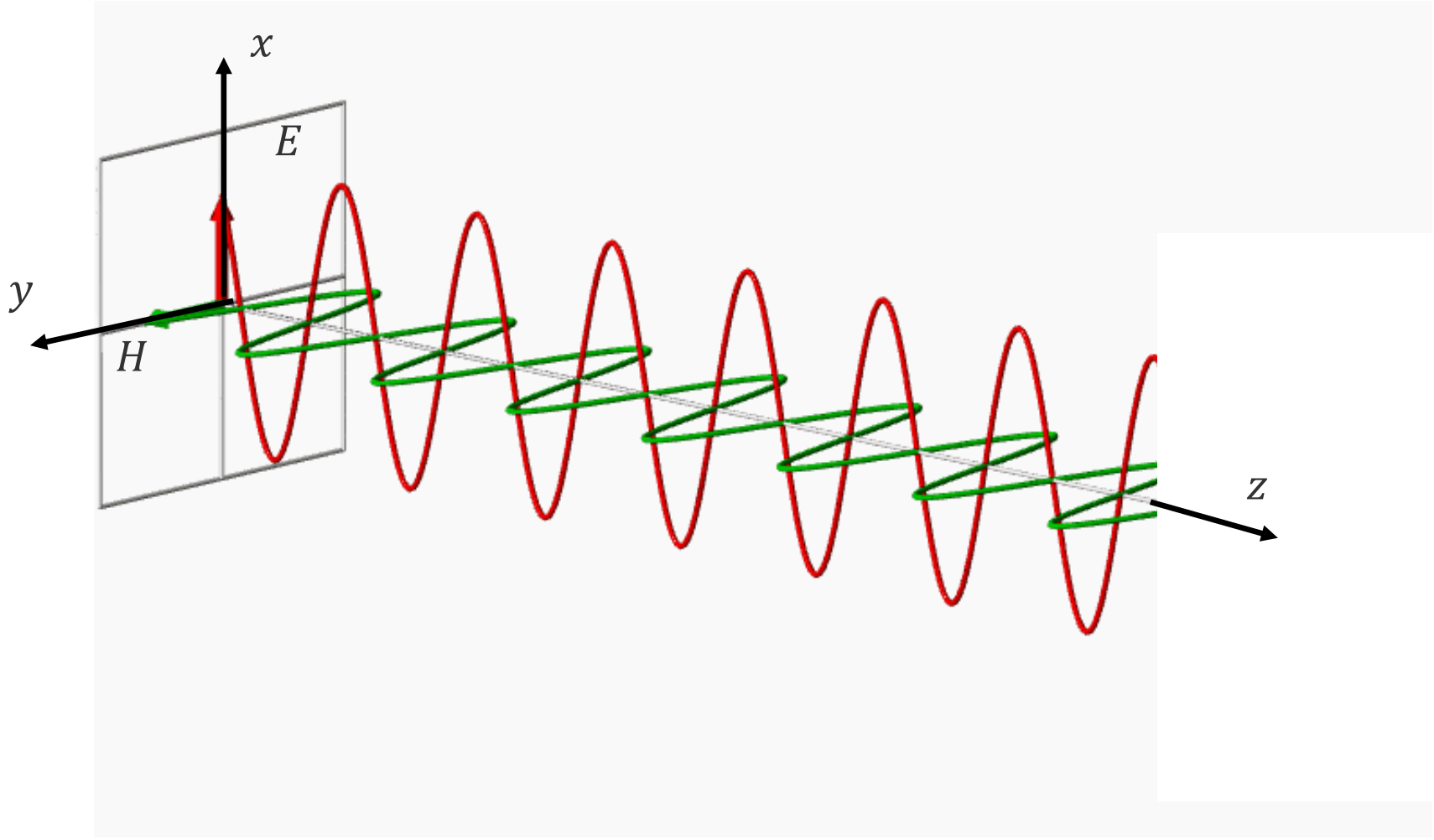
$$H_o = \frac{E_o}{\eta}$$

$$\vec{H} = H_o\, e^{-jkz}\ \vec{a}_y$$

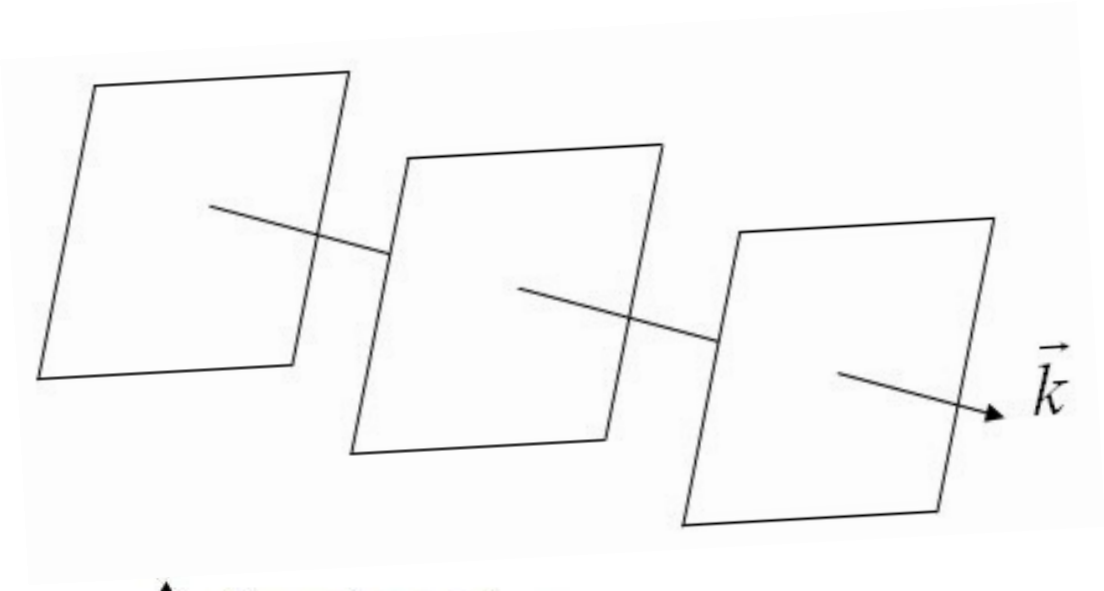
For free-space $\eta_o = \sqrt{\frac{\mu_o}{\varepsilon_o}} = 377\ \Omega$

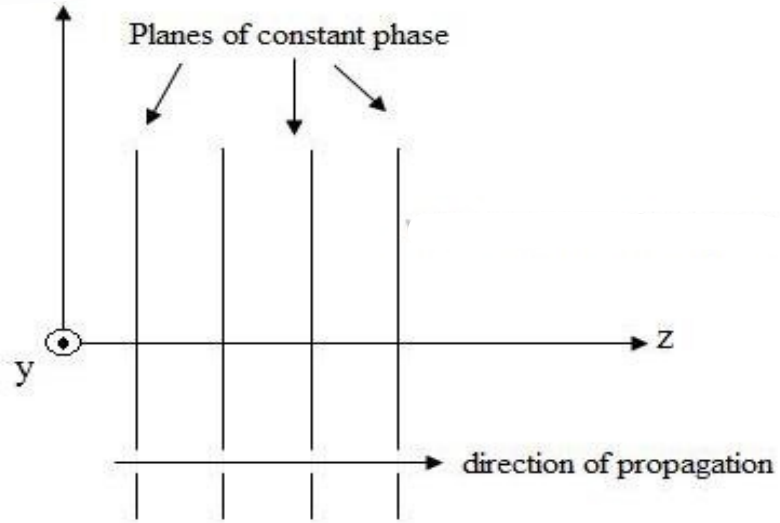
Other media $\eta = \frac{\eta_o}{n}$

The Uniform Plane-wave Solution



The Uniform Plane-wave Solution





Example

Consider an EM wave travelling in free space (n=1). The wave is polarized in the y-direction and is given by

$$\vec{E}(x,t) = 5\cos(2.7\times10^{15}t + kx)\ \vec{a}_y \quad \text{(V/m)}$$

The direction of propagation, $-ve\ x$ direction.

Frequency $\omega = 2.7\times10^{15}$ rad/s.

Phase constant (wavenumber) $k = \frac{\omega}{v} = \frac{2.7\times10^{15}}{3\times10^{8}} = 0.9\times10^{7}$ rad/m.

How many meters to complete a full 2π phase?

Wavelength $\lambda = \frac{v}{f} = 680$ nm (red light).

Write the expression for the magnetic field. $H_o = \frac{E_o}{\eta_o} = \frac{5}{377}$ $\quad \vec{a}_k = \vec{a}_E \times \vec{a}_H$

$$\vec{H}(x,t) = \frac{5}{377}\cos(2.7\times10^{15}t + kx)\ (-\vec{a}_z) \quad \text{(A/m)} \qquad -\vec{a}_x = \vec{a}_y \times \vec{a}_H$$

TH Plane-wave Solution in Lossy Media

Complex Refractive index

$$n = \sqrt{\mu_r \varepsilon_r} = n' - jn''$$

Complex propagation constant

$$k = \frac{n\omega}{c}$$

$$k = \frac{\omega}{c} n'' + j \frac{\omega}{c} n'$$

$$k = \alpha + jk$$

Field attenuation

Phase constant

$$\alpha = \frac{\omega}{c} n'' \qquad k = \frac{\omega}{c} n'$$

Refractive index

$$(n')^2 = \frac{\varepsilon_r'}{2} + \frac{1}{2}\sqrt{(\varepsilon_r')^2 + (\varepsilon_r'')^2}$$

Extinction coefficient

$$n'' = -\frac{\varepsilon_r''}{2n'}$$

Other relations

$$\varepsilon_r' = (n')^2 + (n'')^2$$

$$\varepsilon_r'' = -2n'\, n''$$

Wave speed

$$v = \frac{c}{n'}$$

Wave impedance

$$\eta = \frac{\eta_o}{n} = \frac{\eta_o}{n' - jn''}$$

The Plane-wave Solution in Lossy Media

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

Consider a one-dimensional case (z) with one component only.

$$\frac{\partial^2 E_x}{\partial z^2} - k^2 E_x = 0 \qquad k \text{ is complex}$$

$$E_x(z) = A\,e^{-kz} + B\,e^{kz}$$

Wave travelling along the $+ve\ z$ direction

Wave travelling along the $-ve\ z$ direction

Complex wave impedance

$$\eta = \frac{E_o}{H_o} = |\eta|\,\angle\theta_\eta$$

$$\vec{a}_k = \vec{a}_E \times \vec{a}_H$$

Take the $+z$ wave

$$E_x(z) = E_o\,e^{-(\alpha+jk)z}$$

$$E_x(z) = E_o\,e^{-\alpha z}\,e^{-jkz}$$

Attenuation

Phase

Take the sine wave solution

$$E_x(z,t) = E_o \sin(\omega t - kz)\,e^{-\alpha z}$$

$$\vec{E}(z,t) = E_o \sin(\omega t - kz)\,e^{-\alpha z}\,\vec{a}_x$$

$$\vec{H}(z,t) = H_o \sin(\omega t - kz)\,e^{-\alpha z}\,\vec{a}_y$$

Example

Ultraviolet sun light propagation in seawater. At UV wavelength (400 nm), the refractive index is $n = 1.34451 - j2.11\times10^{-10}$. If the magnetic field of this wave at some location inside seawater is given below, find the propagation parameters and the electric field expression.

$$\vec{H}(z,t) = 10\, e^{-\alpha z} \cos(\omega t - kz)\ \vec{a}_y \quad \text{(mA/m)}$$

The direction of propagation, $+ve\ z$ direction $\vec{a}_k = \vec{a}_z$.

$n' = 1.34451$

$$k_o = \frac{\omega}{c} = \frac{2\pi}{\lambda_o} = \frac{2\pi}{400\ nm} = 15.7\times10^{6} \quad \text{rad/m}$$

$$k = k_o n' = (15.7\times10^{6})(1.34451) = 21.12\times10^{6} \quad \text{rad/m}$$

$n'' = 2.11\times10^{-10}$

$$\alpha = k_o n'' = (15.7\times10^{6})(2.11\times10^{-10}) = 33.1\times10^{-4} \quad \text{1/m}$$

$$\eta = \frac{\eta_o}{n} \approx \frac{377}{1.34451} = 280.4 \quad \text{Ohms}$$

$$\vec{E}(z,t) = 2.804\ e^{-\alpha z} \cos(\omega t - kz)\ \vec{a}_x \quad \text{V/m}$$

Power Density in a Plane Wave

Consider the following Plane wave represented by E and H.

$$\vec{E} = E_o \cos(\omega t - kz)\, e^{-\alpha z}\ \vec{a}_x$$

$$\vec{H} = H_o \cos(\omega t - kz - \theta_\eta)\, e^{-\alpha z}\ \vec{a}_y$$

$$\vec{S} = \vec{E} \times \vec{H}$$

Poynting vector (Wave power density W/m^2)

$$\vec{S} = E_o H_o \cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

$$\vec{S} = \frac{{E_o}^2}{|\eta|}\cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

The time-average power density is given by

$$\vec{S}_{ave} = \frac{1}{T}\int_0^T \vec{S}\ dt\ \vec{a}_z$$

(W/m^2)

For a plane wave,

$$\vec{S}_{ave} = \frac{1}{2}\, Re\left[\vec{E} \times \vec{H}^*\right] \Rightarrow \vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|}\cos\theta_\eta\ e^{-2\alpha z}\ \vec{a}_z$$

Total average power within some area

$$P_{ave} = \int_A \vec{S} \cdot \overrightarrow{dA} \quad \text{(W)}$$

Example

Consider an EM wave given by $\vec{E} = 4\ \sin(8.8\pi\times10^{14}t - kx)\ \vec{a}_z$ (V/m) travelling in a non-magnetic medium of n =3.8. Find k, η and $\vec{S}_{ave}$.

Lossless medium $\alpha = 0$

Frequency $\omega = 8.8\pi\times10^{14}$ rad/s

Refractive index $n = 3.8$ (real)

$$k = \frac{\omega}{c}n = \frac{8.8\pi\times10^{14}}{3\times10^{8}}\ (3.8) = 3.5\times10^{7}\ \text{rad/m}$$

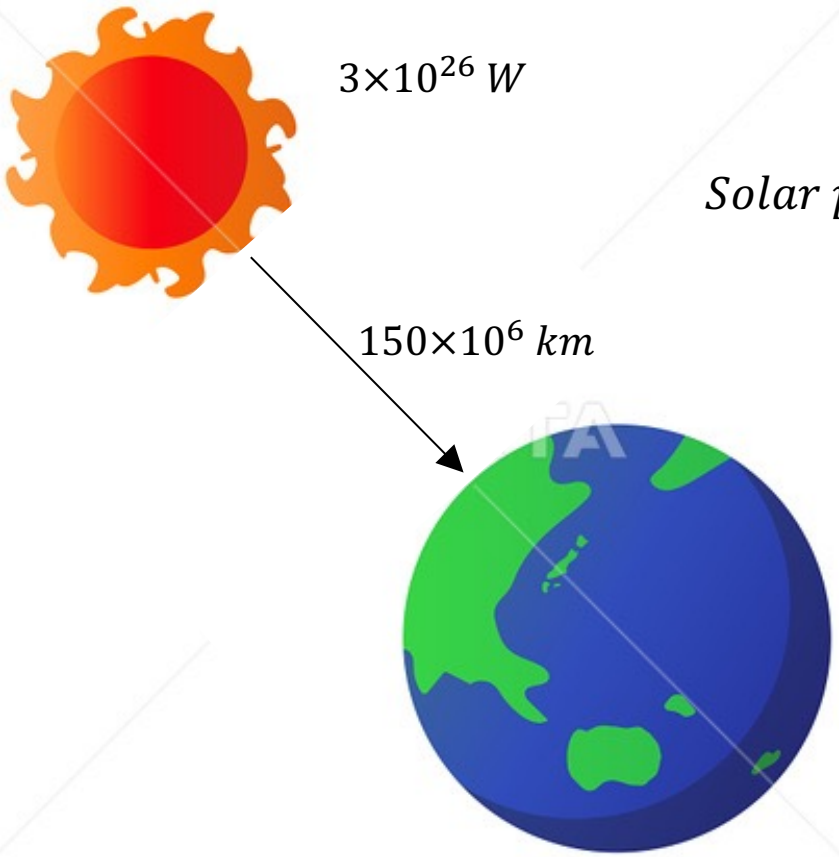
$$v = \frac{c}{n} = \frac{3\times10^{8}}{3.8} = 0.79\times10^{8}\ \text{m/s}$$

$$\eta = \frac{\eta_o}{n} = \frac{377}{3.8} = 98.6\ \Omega$$

$$\vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|}\cos\theta_\eta\ e^{-2\alpha z}\ \vec{a}_x$$

$$\vec{S}_{ave} = \frac{4^2}{2(98.6)}\ \vec{a}_x = 81.1\ \vec{a}_x\ \text{mW/m}^2$$

Example



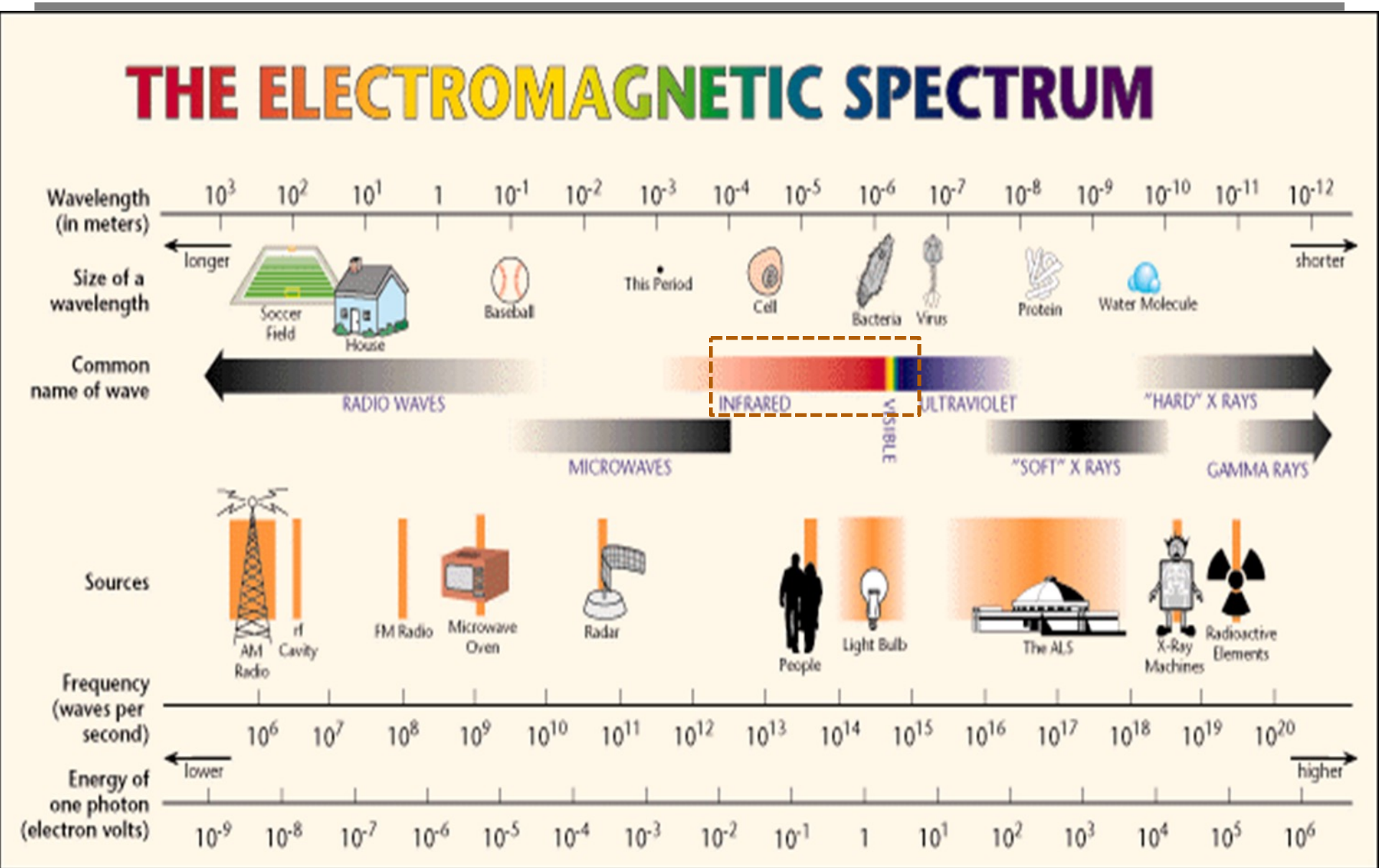
$$Total\ solar\ power = 3\times10^{26}\ W$$

$$Solar\ power\ density\ at\ Earch\ surface = \frac{3\times10^{26}}{4\pi r^2}$$

$$S_{ave} = \frac{{E_o}^2}{2|\eta|} \approx 1\ kW/m^2$$

$$E_o = 890\ V/m$$

Optical Spectrum



Light as an Electromagnetic Wave

The following subjects will be covered

- More Examples on Light Wave Propagation in Open Space.

Example

The visible light complex refractive index curve of GaAs is shown in the figure. For red light (680 nm), find (a) the value of the complex refractive index, (b) the speed of light and (c) the attenuation coefficient (d) How far would a red light beam travel inside a GaAs material before it reduces to 1% of its original power density value?

$$n = 3.1826 - j0.18$$

Speed of light

$$v = \frac{c}{n'} = 0.943\times10^{8} \quad (\mathrm{m/s})$$

The attenuation coefficient is

$$\alpha = \frac{\omega}{c} n'' = 1.5\times10^{6} \quad (1/\mathrm{m})$$

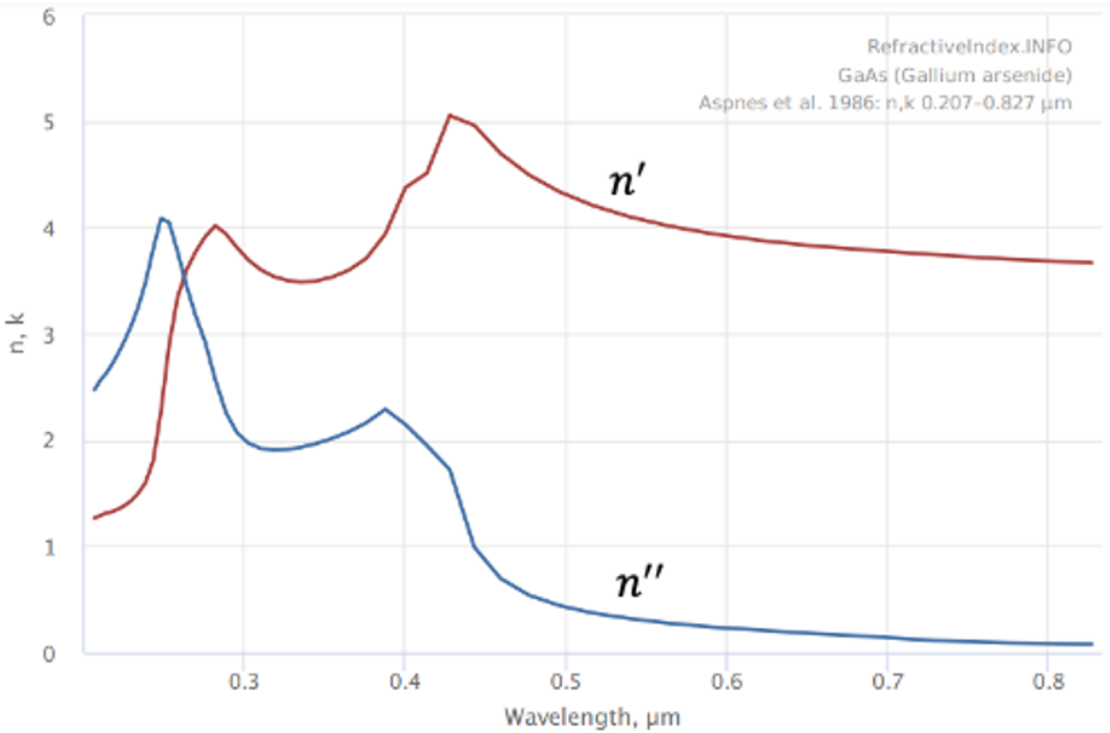
The absorption coefficient is

$$\alpha = 2\frac{\omega}{c} n'' = 3\times10^{6} \quad (1/\mathrm{m})$$

Find the distance using

$$0.01 = e^{-3\times10^{6} z}$$

$$z = 1.54\ \mu m$$



https://refractiveindex.info/

Examples

1) Discuss the wave direction. $\vec{E} = E_o \vec{a}_y \, e^{j\omega t} \, e^{-j(x+2z)/\sqrt{2}}$ (V/m)

Transverse Electric (TE)

$k_x = 1/\sqrt{2}$ $k_z = 2/\sqrt{2}$ $\vec{k} = \frac{1}{\sqrt{2}} \vec{a}_x + \frac{2}{\sqrt{2}} \vec{a}_z$ (rad/m)

2) The complex electric field intensity of a time-harmonic wave travelling in a source-free free space is given by (distance in meters)

$$\vec{E} = (80 \, \vec{a}_x - 60 \, \vec{a}_z) \, e^{-j(6x+8z)} \text{ (V/m)}$$

What is the wavelength in the wave propagation direction?

$k_x = 6$ $k_z = 8$ $k = \sqrt{6^2 + 8^2} = 10$ (rad/m) $\lambda = \frac{2\pi}{k} = 0.628$ (m)

What is the angle the wave propagation direction makes with z?

$\tan\theta = 6/8$ $\theta = 36.9°$

Transverse Magnetic (TM)

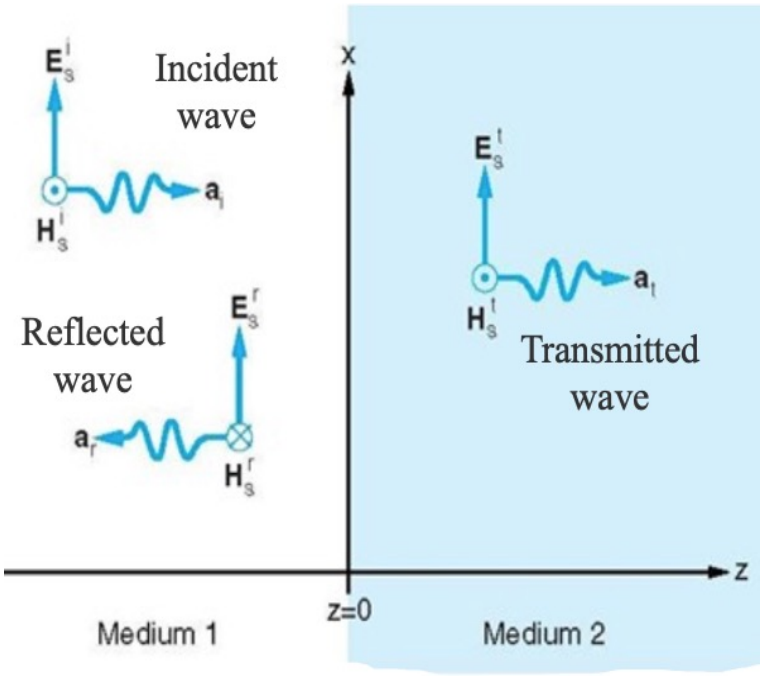
Light as an Electromagnetic Wave

Part II: Light Behavior and Properties

The following subjects will be covered

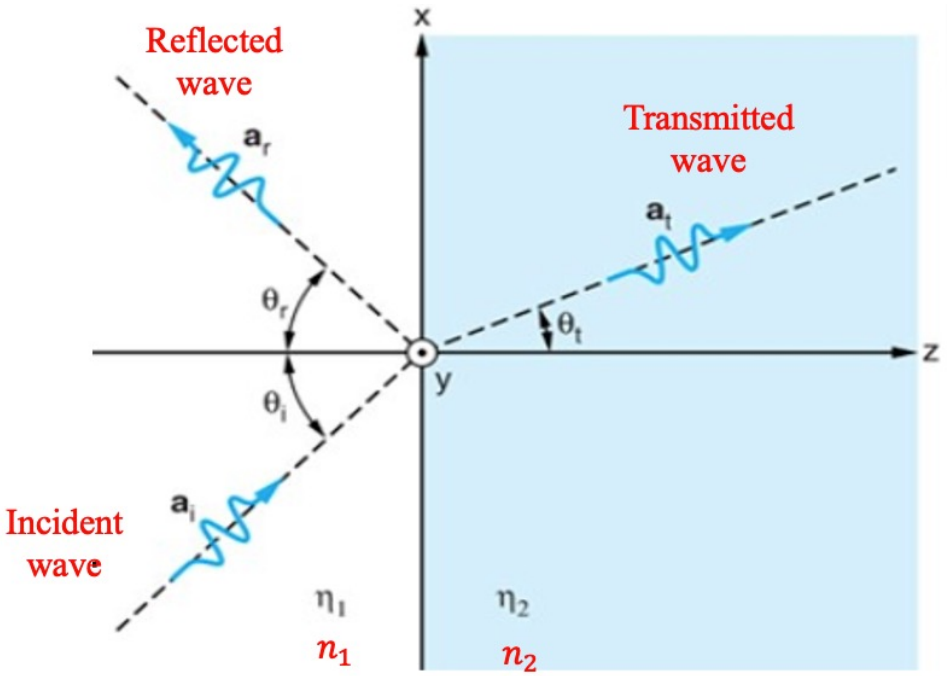
- Reflection, Refraction and Transmission of Light.
- Light Polarization.
- Light-matter Interaction.
- Scattering of Light.

Light Reflection and Transmission



n_1, k_1, α_1 n_2, k_2, α_2

Normal Incidence



Oblique Incidence

Single Interface: Normal Incidence

Incident Wave

$$\vec{E}_i(z) = E_o\, e^{-jk_1 z}\, \vec{a}_x$$

$$\vec{H}_i(z) = \frac{E_o}{\eta_1} e^{-jk_1 z}\, \vec{a}_y$$

Reflected Wave

$$\vec{E}_r(z) = \Gamma\, E_o\, e^{+jk_1 z}\, \vec{a}_x$$

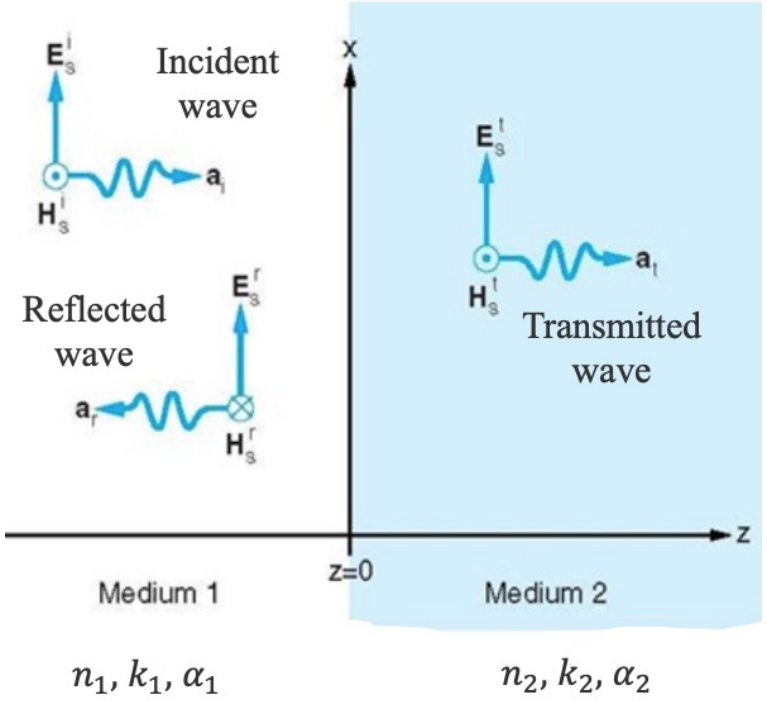
$$\vec{H}_r(z) = \frac{\Gamma E_o}{\eta_1}\, e^{+jk_1 z}\ \ (-\vec{a}_y)$$

Transmitted Wave

$$\vec{E}_t(z) = \tau E_o\, e^{-jk_2 z}\ \ \vec{a}_x$$

$$\vec{H}_t(z) = \frac{\tau\, E_o}{\eta_2}\, e^{-jk_2 z}\ \ \overrightarrow{a_y}$$

$$\eta_1 = \frac{\eta_o}{n_1} \qquad \eta_2 = \frac{\eta_o}{n_2} \qquad \eta_o = \sqrt{\mu_o/\varepsilon_o}$$



n_1, k_1, α_1 $\qquad$ n_2, k_2, α_2

Apply boundary conditions at interface $z = 0$

$$E_{tangential\ (1)} = E_{tangential\ (2)}$$

$$H_{tangential\ (1)} = H_{tangential\ (2)}$$

Single Interface: Normal Incidence

Apply boundary conditions

$$E_{1_tan} = E_o(1+\Gamma)$$

$$E_{2_tan} = E_o\tau$$

$$H_{1_tan} = \frac{E_o}{\eta_1}(1-\Gamma)$$

$$H_{2_tan} = \frac{\tau E_o}{\eta_2}$$

Field Coefficients

Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

$$\Gamma = \tau - 1$$

Transmission coefficient

$$\tau = \frac{2n_1}{n_1 + n_2}$$

Power Coefficients

Reflectance

$$R = |\Gamma|^2$$

$$R + T = 1$$

Transmittance

$$T = \frac{n_2}{n_1}|\tau|^2$$

Power Conservation

1) Γ and τ are in general complex.

2) $0 \le |\Gamma| \le 1$

$$\vec{S}^i_{ave} = \frac{n_1|E_o|^2}{2\eta_o}\,\vec{a}_k$$

$$\vec{S}^t_{ave} = n_1 T\frac{|E_o|^2}{2\eta_o}\,\vec{a}_k$$

$$\vec{S}^r_{ave} = n_1 R\frac{|E_o|^2}{2\eta_o}\,(-\vec{a}_k)$$

Single Interface: Normal Incidence

Reflectance (Power)

$$R = |\Gamma|^2 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2$$

Transmittance (Power)

$$T = 1 - R = \frac{4\, n_1 n_2}{(n_1 + n_2)^2}$$

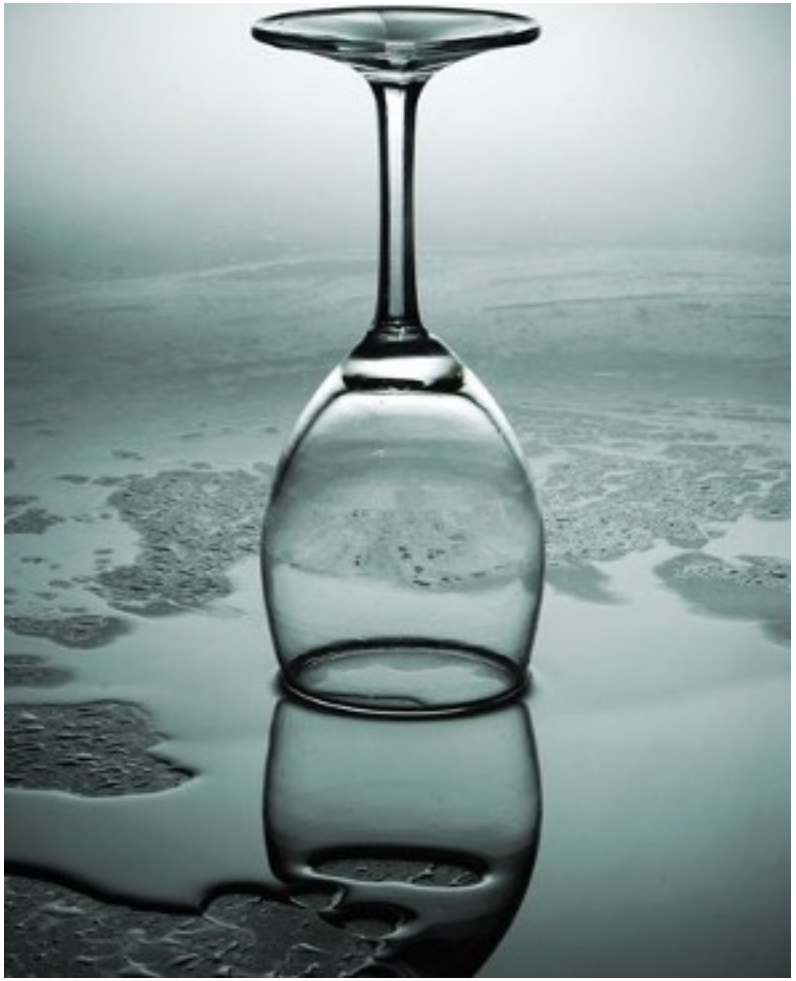
Example: Glass Reflection

Glass refractive index, $n \approx 1.52$

$$\Gamma = \frac{1.0-1.52}{1.0+1.52} = -0.2065$$

$$R = (-0.2065)^2 = 0.0426$$

$$T = 1 - 0.0426 = 0.957$$



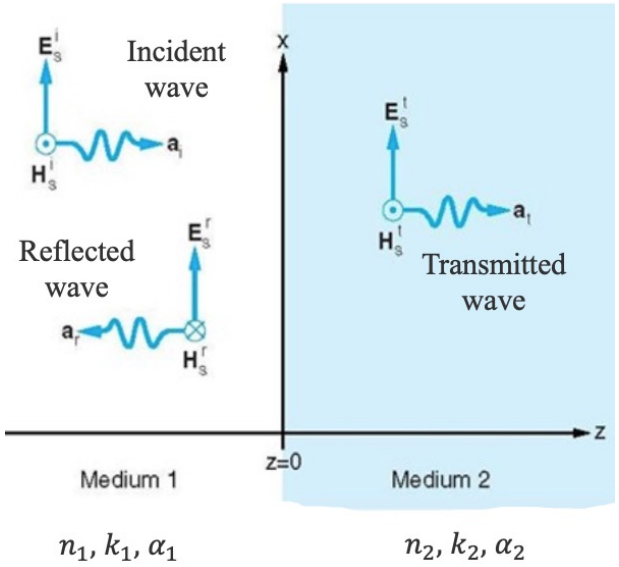
96% of light power goes through!

Remarks

1) Medium 1 perfect dielectric (n_1=real) and medium 2 perfect conductor ($n_2 \to \infty$)

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2} = -1$$ Total reflection

$$\tau = 0$$ Zero transmission



2) Medium 1 is a denser dielectric than medium 2

$$n_1 > n_2 \qquad \Gamma > 0$$

3) Medium 1 is a lighter dielectric than medium 2

$$n_1 < n_2 \qquad \Gamma < 0$$

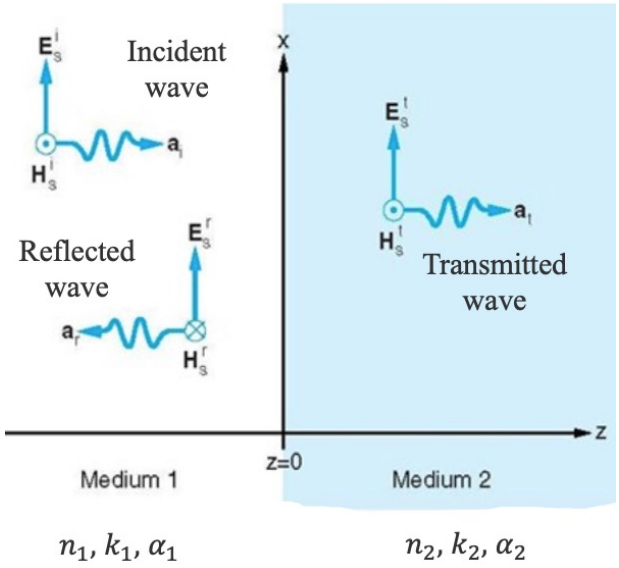
Lossy Media

Electric Fields

$$\overrightarrow{E_i}(z) = E_o\, e^{-\alpha_1 z}\, e^{-jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_r}(z) = \Gamma\, E_o\, e^{+\alpha_1 z}\, e^{+jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_t}(z) = \tau E_o\, e^{-\alpha_2 z}\, e^{-jk_2 z}\, \vec{a}_x$$



Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

Transmission coefficient

$$\tau = \frac{2n_1}{n_1 + n_2}$$

(n_1 and n_2 complex quantities)

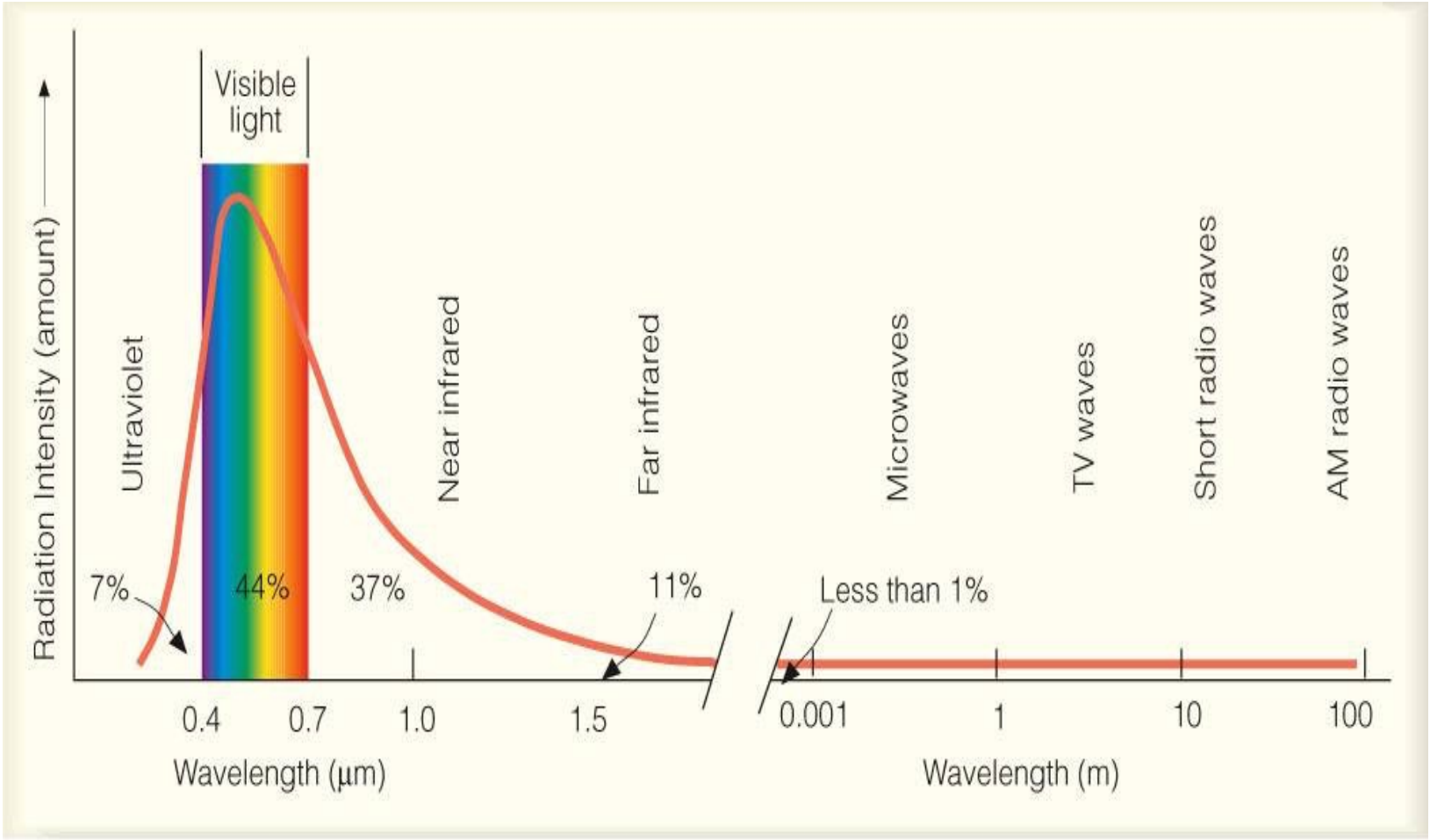
$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2}\, e^{-2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, \vec{a}_z$$

$$\vec{S}^r_{ave} = |\Gamma|^2\, \frac{|E_o|^2}{2}\, e^{+2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, (-\vec{a}_z)$$

$$\vec{S}^t_{ave} = |\tau|^2\, \frac{|E_o|^2}{2}\, e^{-2\alpha_2 z}\, Re\left(\frac{\eta_1}{\eta_2}\right)\, \vec{a}_z$$

Example: Solar Cell Design Considerations

It is required to select between Si and Ge for the design of a solar cell. The selection is based on the amount of reflection at normal incidence.



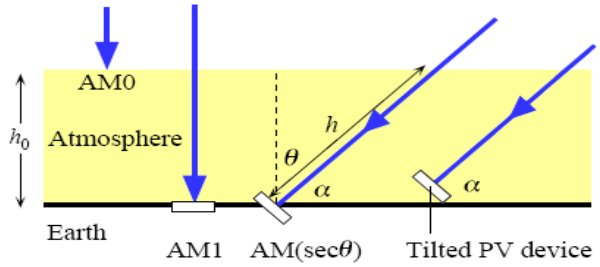
© 2007 Thomson Higher Education

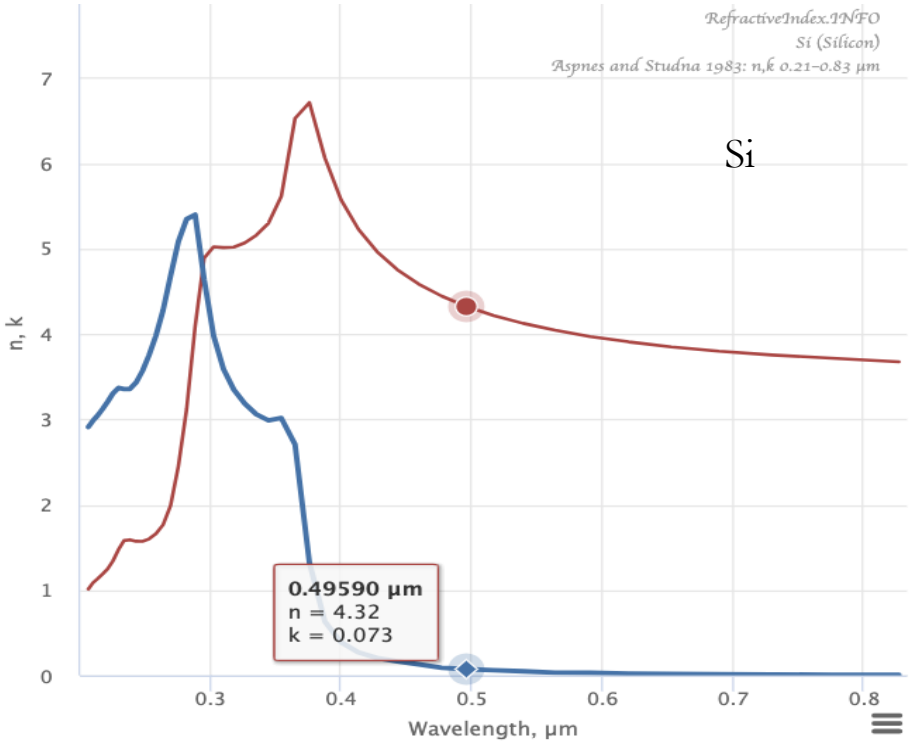
Example: Solar Cell Design Considerations

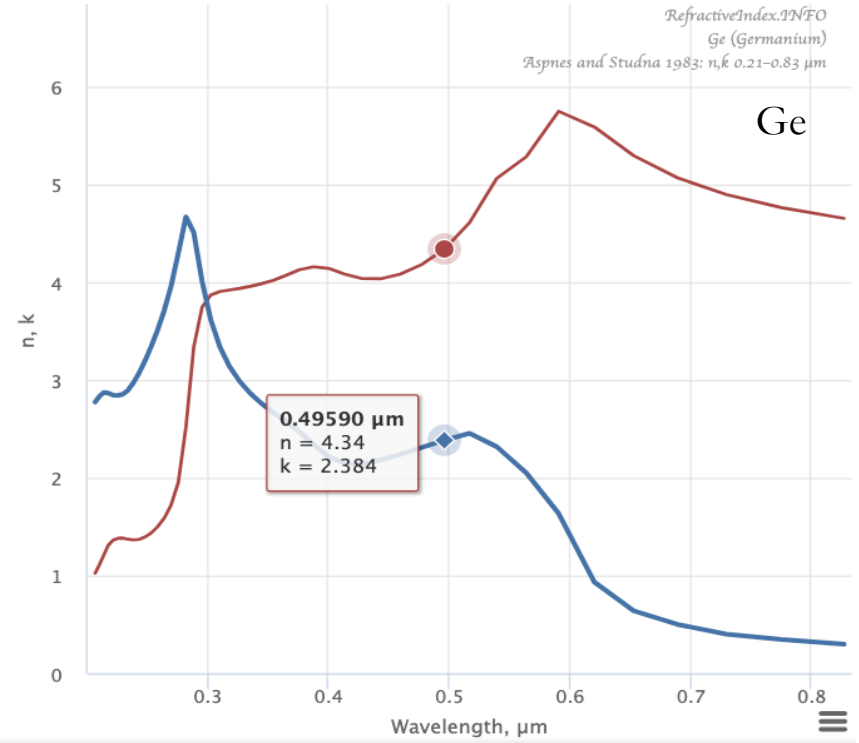
Refractive index curves for Si and Ge. Note, in the figures n and k are n' and n''.

Si (0.5 μm): $n = 4.32 - j0.073$.

Ge (0.5 μm): $n = 4.34 - j2.384$.





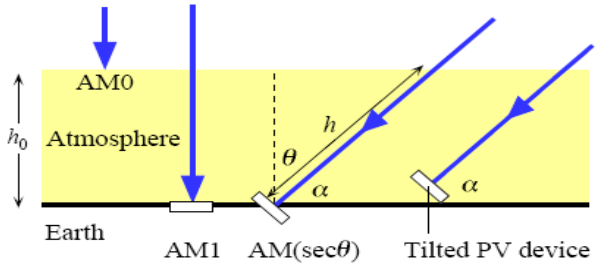


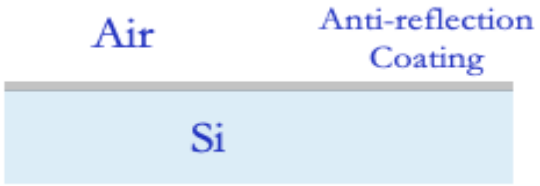
Example: Solar Cell Design Considerations

Refractive index curves for Si and Ge. Note, in the figures n and k are n' and n''.

Si (0.5 m): $n = 4.32 - j0.073$.

Ge (0.5 m): $n = 4.34 - j2.384$.



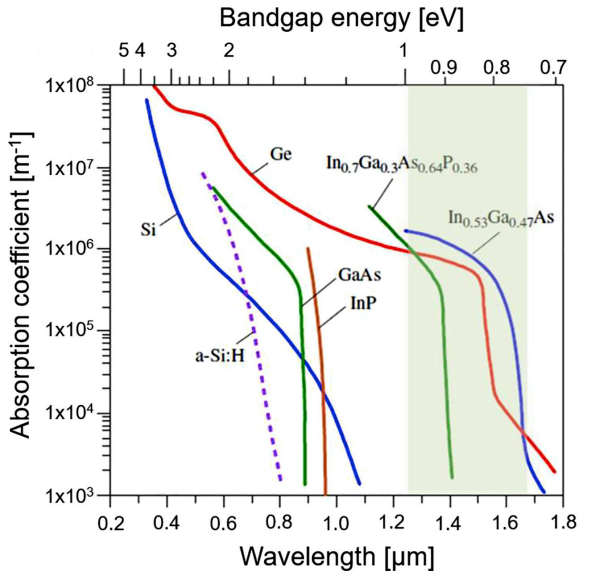


$$\Gamma_{Si} = \frac{1.0 - (4.32 - j0.073)}{1.0 + (4.32 - j0.073)} = 0.6184 \angle 170.9°$$

$$R_{Si} = 0.382$$

$$\Gamma_{Ge} = \frac{1.0 - (4.34 - j2.384)}{1.0 + (4.34 - j2.384)} = 0.7017 \angle 120.4°$$

$$R_{Ge} = 0.492$$



Light as an Electromagnetic Wave

Part II: Light Behavior and Properties

The following subjects will be covered

- Reflection and Transmission of Light.
- Light Polarization.
- Light-matter Interaction.
- Scattering of Light.

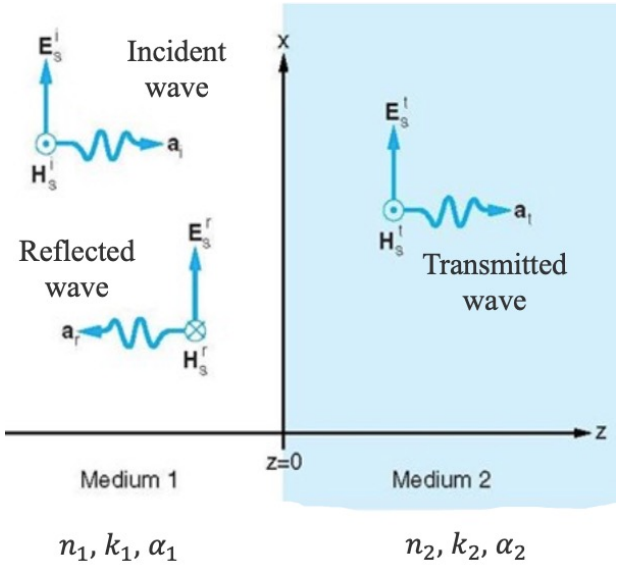
General Media

Electric Fields

$$\overrightarrow{E_i}(z) = E_o\, e^{-\alpha_1 z}\, e^{-jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_r}(z) = \Gamma\, E_o\, e^{+\alpha_1 z}\, e^{+jk_1 z}\, \vec{a}_x$$

$$\overrightarrow{E_t}(z) = \tau E_o\, e^{-\alpha_2 z}\, e^{-jk_2 z}\, \vec{a}_x$$



Reflection coefficient

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2}$$

Transmission coefficient

$$\tau = \frac{2n_1}{n_1 + n_2}$$

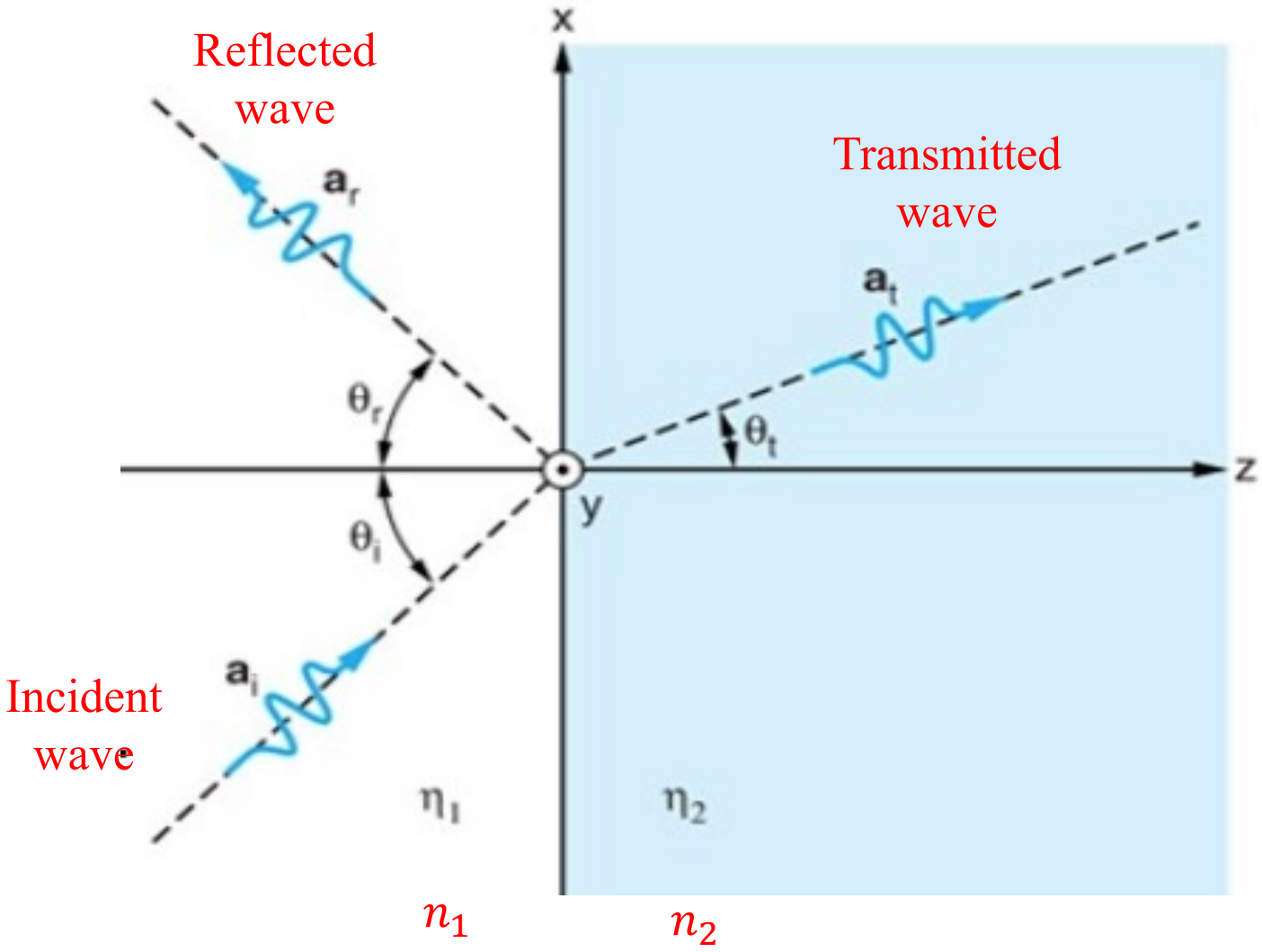
(n_1 and n_2 complex quantities)

$$\vec{S}^i_{ave} = \frac{|E_o|^2}{2}\, e^{-2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, \vec{a}_z$$

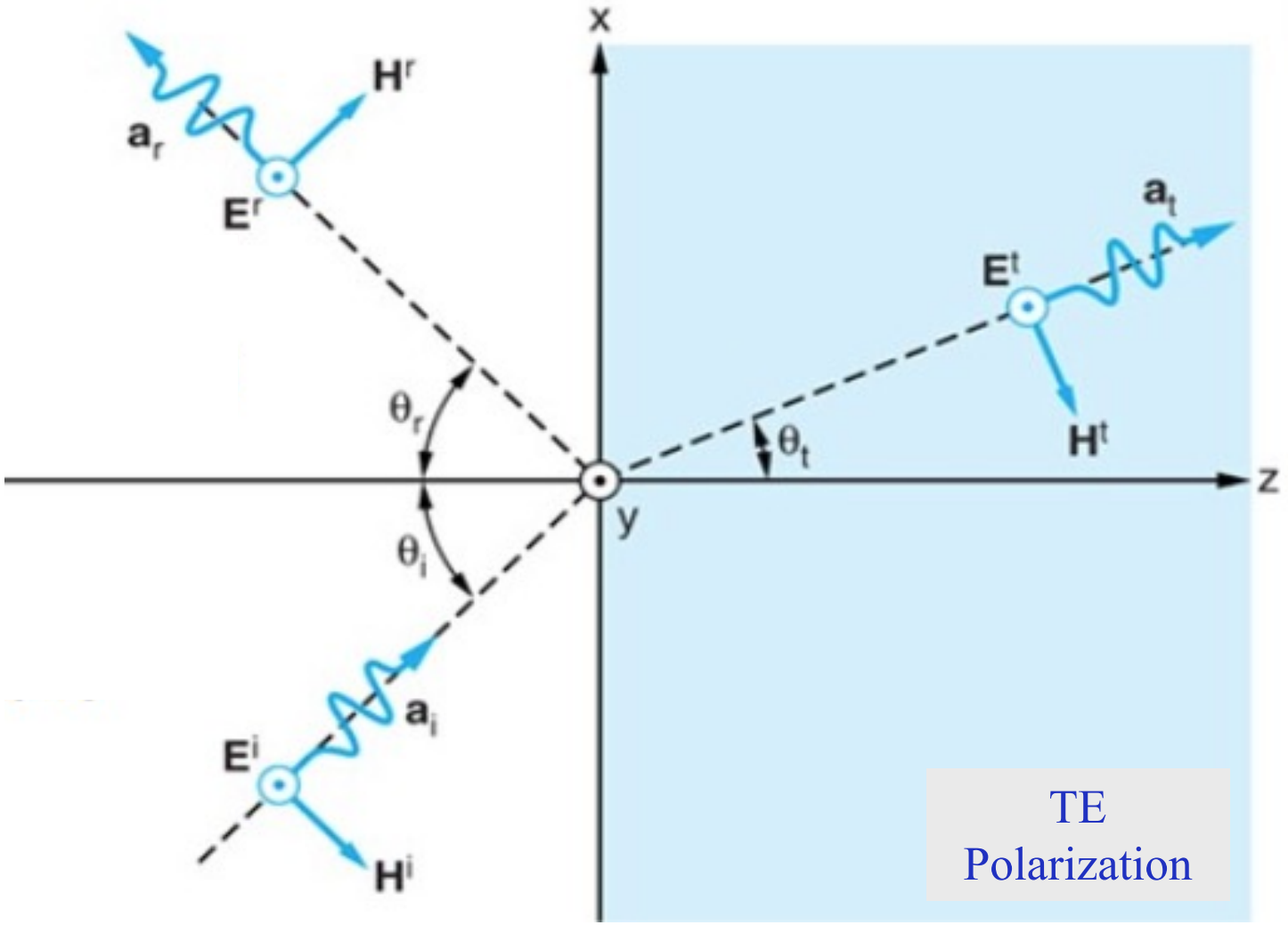
$$\vec{S}^r_{ave} = |\Gamma|^2\, \frac{|E_o|^2}{2}\, e^{+2\alpha_1 z}\, Re\left(\frac{1}{\eta_1}\right)\, (-\vec{a}_z)$$

$$\vec{S}^t_{ave} = |\tau|^2\, \frac{|E_o|^2}{2}\, e^{-2\alpha_2 z}\, Re\left(\frac{\eta_1}{\eta_2}\right)\, \vec{a}_z$$

Light Reflection and Transmission

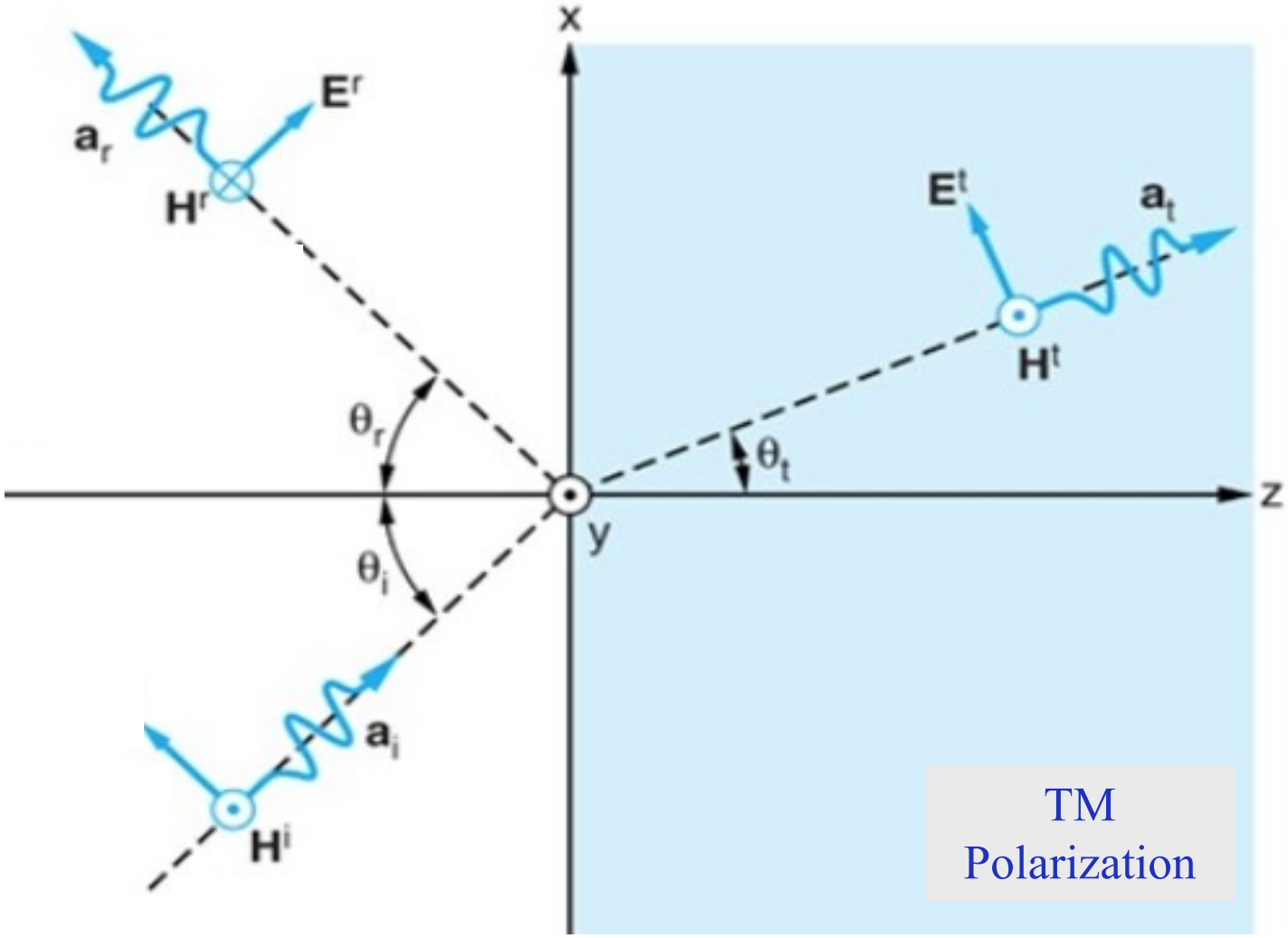


Single Interface: Oblique Incidence

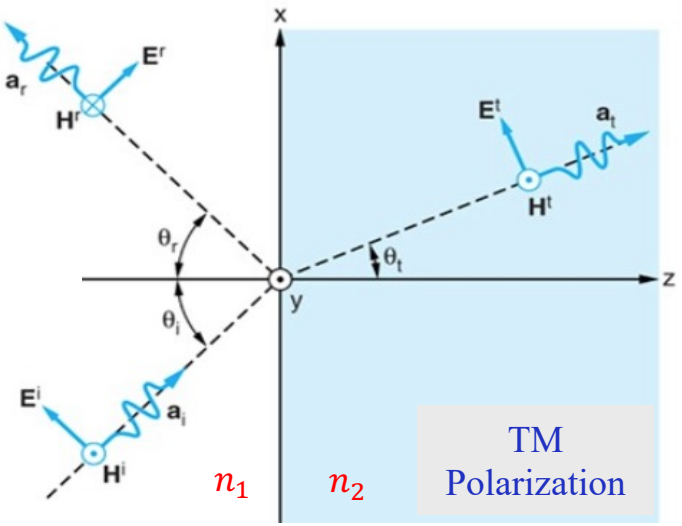


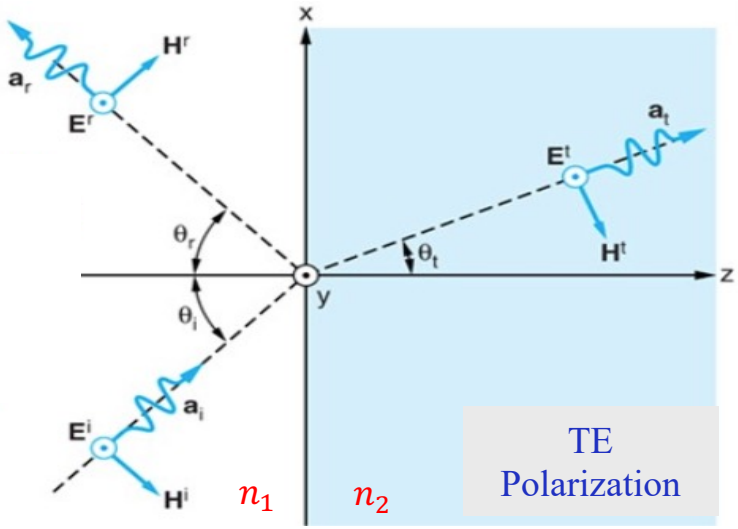
n_1 n_2

Single Interface: Oblique Incidence



Single Interface: Oblique Incidence





$$\Gamma_{TM} = \frac{n_1\,cos\theta_t - n_2\,cos\theta_i}{n_1\,cos\theta_t + n_2\,cos\theta_i}$$

$$\Gamma_{TM} = \frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)}$$

$$\tau_{TM} = \frac{2\,n_1\,cos\theta_i}{n_1\,cos\theta_t + n_2\,cos\theta_i}$$

$$\tau_{TM} = (1 + \Gamma_{TM})\frac{cos\theta_i}{cos\theta_t}$$

Snell's Law

$$sin\theta_t = \frac{n_1}{n_2}\,sin\theta_i$$

$$R = |\Gamma|^2$$

$$T = 1 - R$$

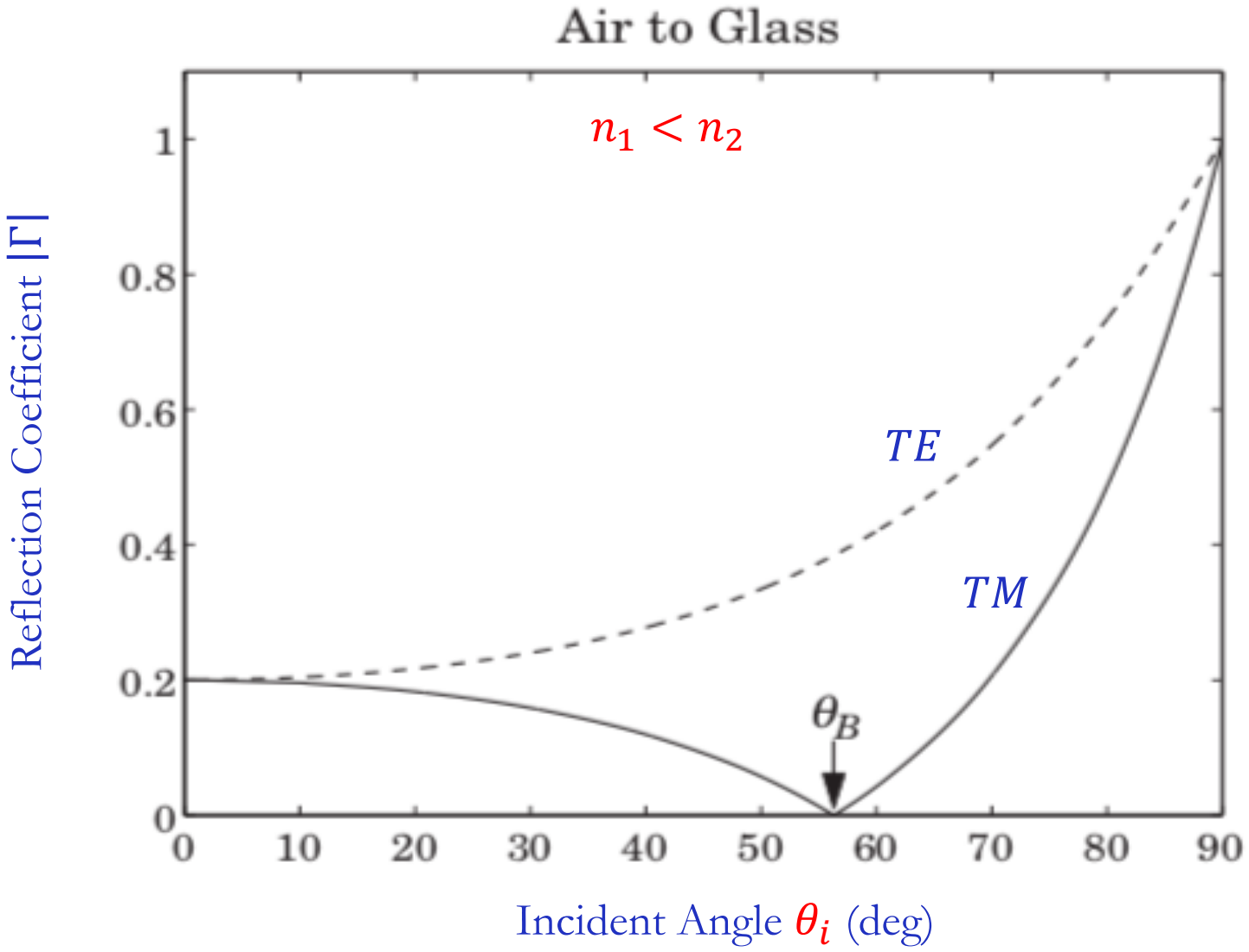
$$\Gamma_{TE} = \frac{n_1\,cos\theta_i - n_2\,cos\theta_t}{n_1\,cos\theta_i + n_2\,cos\theta_t}$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)}$$

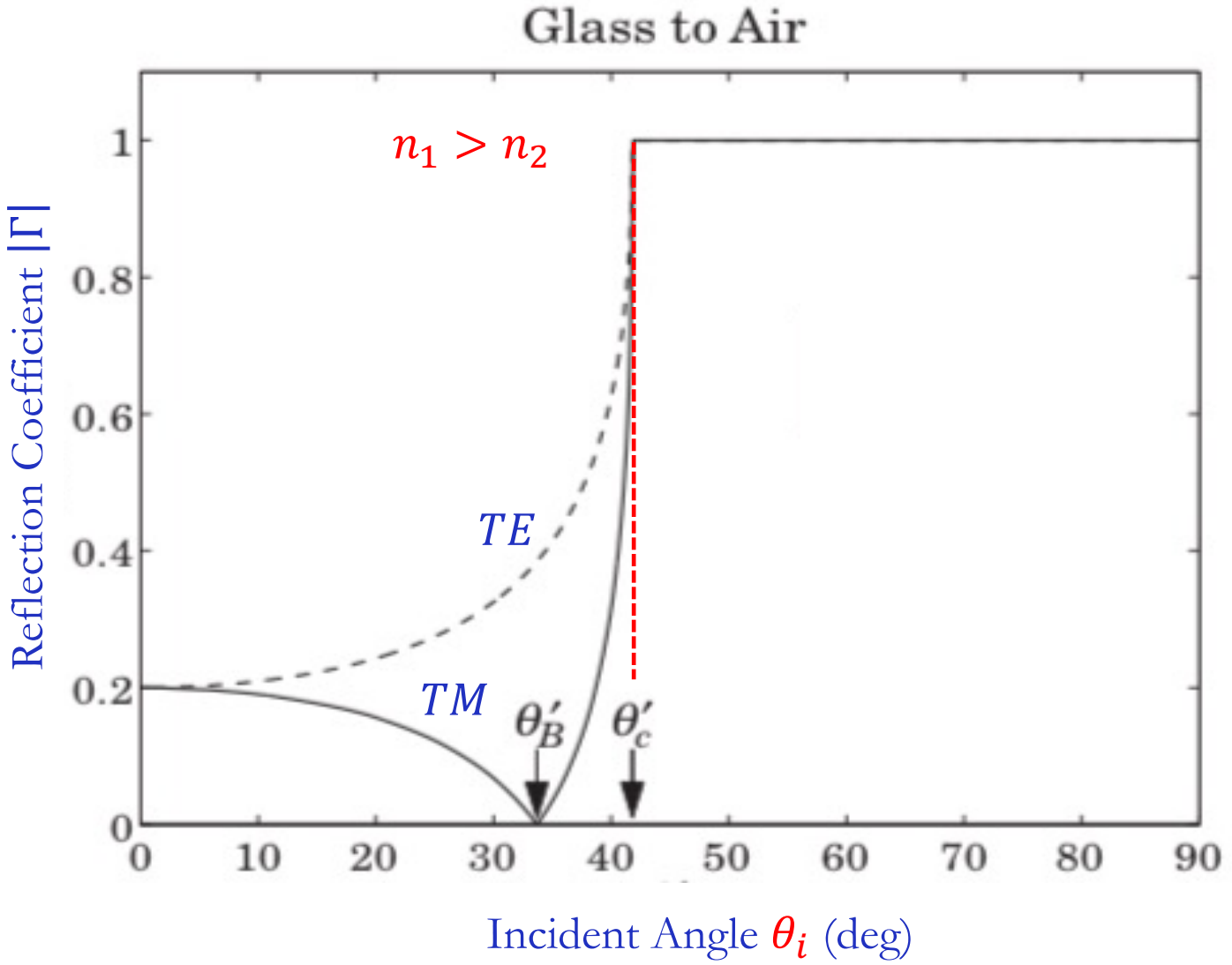
$$\tau_{TE} = \frac{2\,n_1\,cos\theta_i}{n_1\,cos\theta_i + n_2\,cos\theta_t}$$

$$\tau_{TE} = 1 + \Gamma_{TE}$$

Single Interface: Oblique Incidence



Single Interface: Oblique Incidence



Single Interface: Oblique Incidence

1) Total Transmission. $|\Gamma| = 0$, Only TM polarization.

Incident Angle $\theta_i = \tan^{-1}(n_2/n_1)$ Brewster Angle θ_B

In this case $\theta_i + \theta_t = 90^o$

Air-Glass: $\theta_B = \tan^{-1}(1.52/1)$

$\theta_B = 56.7^o$

2) Total Reflection $|\Gamma| = 1$, Both TE and TM polarizations, for $n_1 > n_2$ only.

Incident Angle $\theta_i = \sin^{-1}(n_2/n_1)$ Critical Angle θ_c

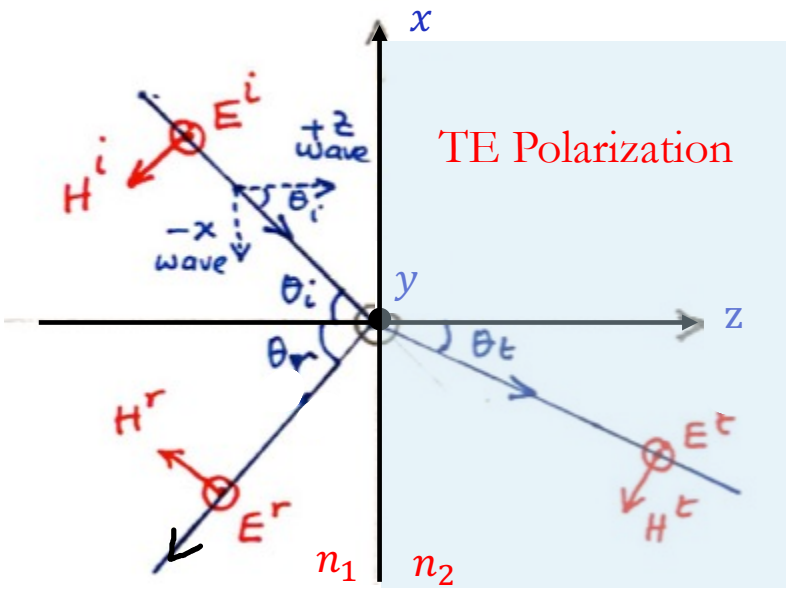
Happens for any angle $\theta_i \geq \theta_c$

Glass-Air: $\theta_c = \sin^{-1}(1/1.52)$

$\theta_c = 41.1^o$

Examples

A single-frequency plane wave is incident obliquely from air onto a planar interface with a non-magnetic medium having a refractive index of $\sqrt{2}$. The angle of incidence is 54.7°. Find the reflection and the transmission coefficients. Also find reflectance and transmittance.



$$\theta_i = 54.7°$$

From Snell's law, $n_1 \sin\theta_i = n_2 \sin\theta_t$

$$\sin\theta_t = \frac{n_2}{n_1}\sin\theta_i$$

$$\theta_t = 35.3°$$

$$\Gamma_{TE} = \frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} = -0.333$$

$$R = (-0.333)^2 = 0.111$$

$$\tau_{TE} = 1 + \Gamma_{TE} = 0.667$$

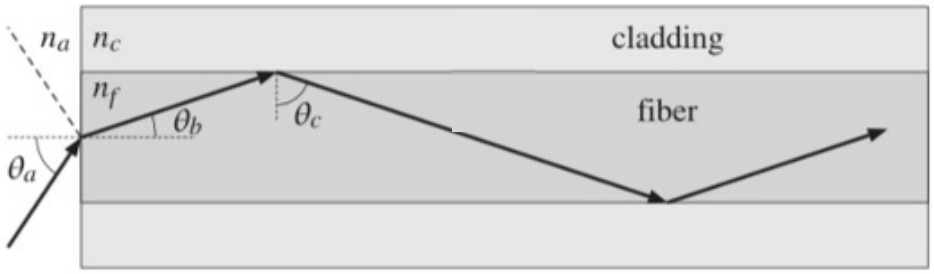
$$T = 1 - R = 0.889$$

$$\vec{E} = \left[\frac{1}{\sqrt{2}}\vec{a}_y\right] e^{-j\beta(z-\sqrt{2}x)/\sqrt{3}} \quad V/m$$

Examples

The Optical Fiber.

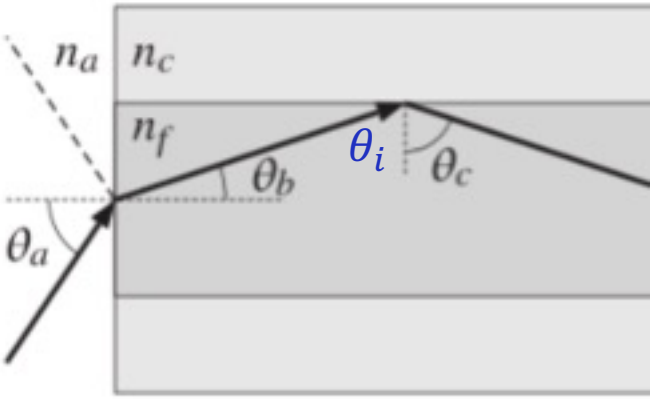
The acceptance angle θ_a should correspond to the critical angle inside the fiber at the cladding-fiber interface. $\theta_i = \theta_c = 90 - \theta_b$



Snell's Law $n_a \sin\theta_a = n_f \sin\theta_b$

$$\sin\theta_a = \frac{n_f}{n_a}\sin\theta_b = \frac{n_f}{n_a}\sin(\theta_c + 90)$$

$$\sin\theta_a = \frac{n_f}{n_a}\cos\theta_c = \frac{n_f}{n_a}\sqrt{1-(\sin\theta_c)^2}$$



Critical angle condition

$$\sin\theta_c = \frac{n_c}{n_f} \Rightarrow \sin\theta_a = \frac{n_f - n_c}{n_a}$$

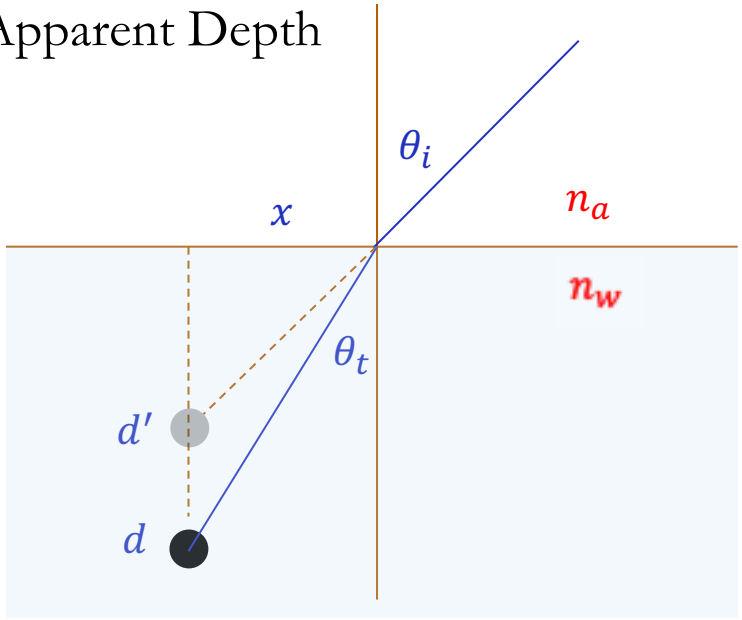
$n_a = 1, n_f = 1.49, n_c = 1.48$

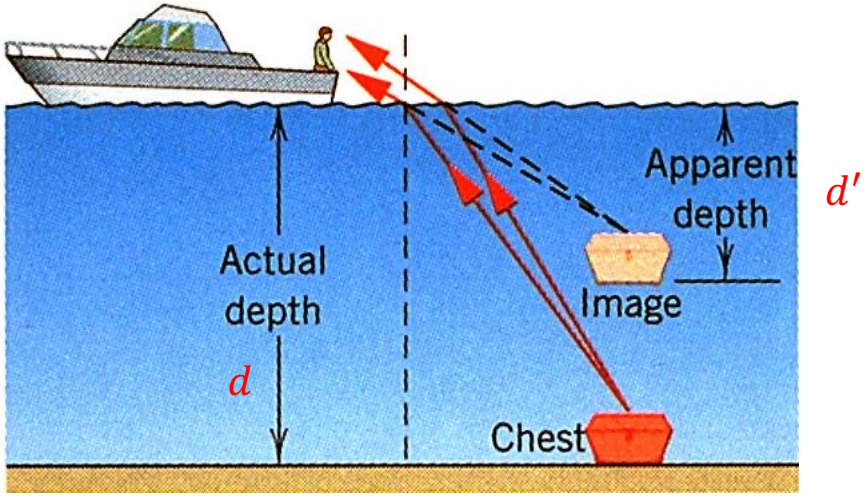
$\theta_c = 83.4°$

$\theta_a = 9.9°$

Examples

Apparent Depth





$$d = x \cot\theta_t$$

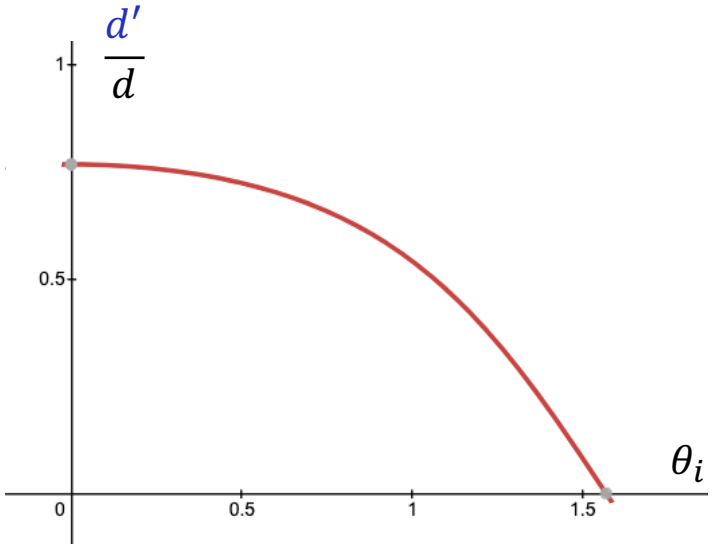
$$d' = x \cot\theta_i$$

$$n_a \sin\theta_i = n_w \sin\theta_t$$

$$\frac{d'}{d} = \frac{\cot\theta_i}{\cot\theta_t} = \frac{\cos\theta_i}{\cos\theta_t} \cdot \frac{\sin\theta_t}{\sin\theta_i}$$

$$\frac{d'}{d} = \frac{n_a}{n_w} \cdot \frac{\cos\theta_i}{\cos\theta_t}$$

$$\frac{d'}{d} = \frac{\cos\theta_i}{\sqrt{n_w^2 - (\sin\theta_i)^2}}$$



Anti-reflection Coating

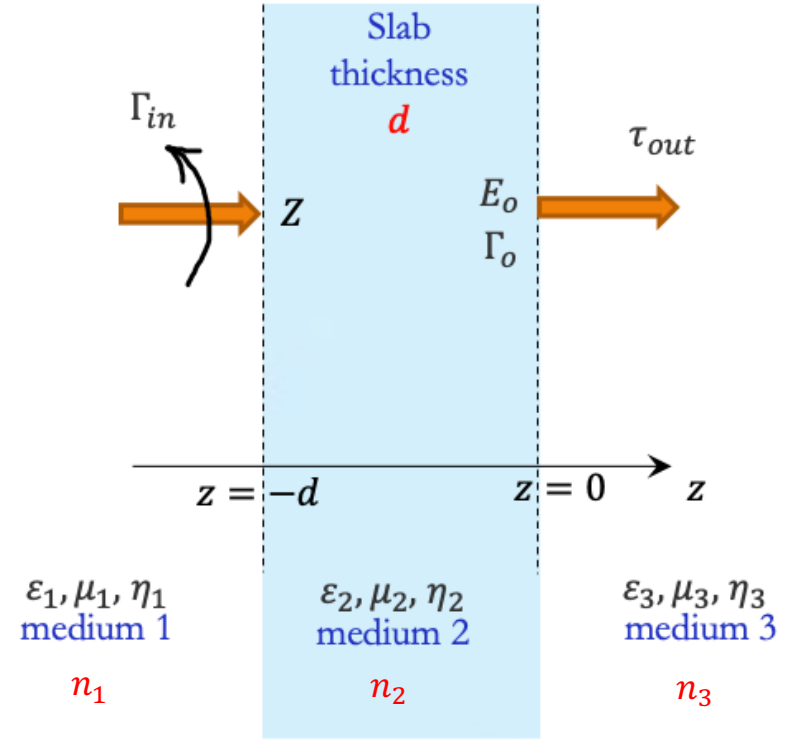
Let's consider normal incidence only. All media are lossless.

$$\Gamma_{in} = \frac{\Gamma_{12} + \Gamma_{23}\, e^{-j2k_2 d}}{1 + \Gamma_{12}\Gamma_{23}\, e^{-j2k_2 d}}$$

$$\Gamma_{12} = \frac{n_1 - n_2}{n_1 + n_2} \qquad \Gamma_{23} = \frac{n_2 - n_3}{n_2 + n_3}$$

For Γ_{12} and Γ_{23} samll

$$\Gamma_{in} \approx \Gamma_{12} + \Gamma_{23}\, e^{-j2k_2 d}$$



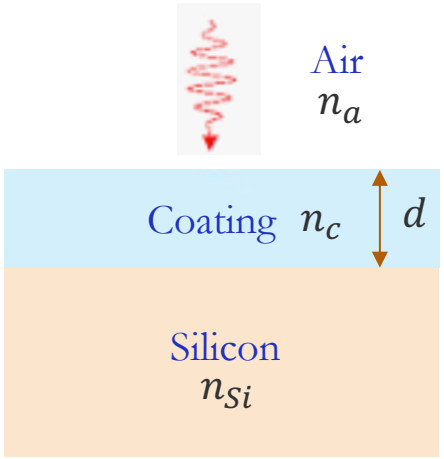
Similarly, $$\tau_{out} = \frac{\tau_{12}\tau_{23}\, e^{-jk_2 d}}{1 + \Gamma_{12}\Gamma_{23}\, e^{-j2k_2 d}}$$

$$\tau_{12} = 1 + \Gamma_{12}$$

$$\tau_{23} = 1 + \Gamma_{23}$$

Examples

1) A coating is needed on a silicon solar cell ($n_{Si} = 4.4 - 3.8$) such that incident light at mid-visible wavelengths (550 nm) from air is not reflected. Find the required coating thickness and material.



$\Gamma_{in} = 0$ ➡ $\left|\Gamma_{a-c} + \Gamma_{c-Si}\, e^{-j2k_c d}\right| = 0$

$$\Gamma_{a-c} = \frac{n_a - n_c}{n_a + n_c} \qquad \Gamma_{c-Si} = \frac{n_c - n_{Si}}{n_c + n_{Si}}$$

Look at $2k_c d = 2\frac{2\pi}{\lambda_c} d = \pi \frac{4}{\lambda_c} d$

If we take $d = \frac{\lambda_c}{4}$ Quarter-wavelength Transformer then $2k_c d = \pi$

➡ $e^{-j2k_c d} = e^{-j\pi} = -1$ ➡ $\left|\frac{n_a - n_c}{n_a + n_c} - \frac{n_c - n_{Si}}{n_c + n_{Si}}\right| = 0$

$$\left|\frac{(n_a - n_c)(n_c + n_{Si}) - (n_c - n_{Si})(n_a + n_c)}{(n_a + n_c)(n_c + n_{Si})}\right| = \left|\frac{2(n_a n_{Si} - n_c^2)}{(n_a + n_c)(n_c + n_{Si})}\right| = 0 \;\Rightarrow\; 2|(n_a n_{Si} - n_c^2)| = 0$$

If we take $n_c = \sqrt{n_a n_{Si}}$ ➡ $\Gamma_{in} = 0$

Coating material $n_c \approx 2$

$d = \frac{\lambda_c}{4} \approx 140\ nm$ Anti-reflection Coating thickness

Examples

2) Design a ¼-wavelength antireflection coating for a glass substrate ($n_g = 1.5$) . The coating is supposed to work for mid-visible range ($\lambda \approx 550\ nm$) .

Using the quarter wavelength condition $n_c = \sqrt{n_a n_g} = \sqrt{1.5} = 1.22$

$$d = \frac{\lambda_c}{4} = 112.7\ nm$$



The solution is not *that* easy!

Practically, the closest materials to the requirement are Cryolite ($n = 1.35$) and Magnesium Fluoride ($n = 1.38$, more durable for this task and more abundant).

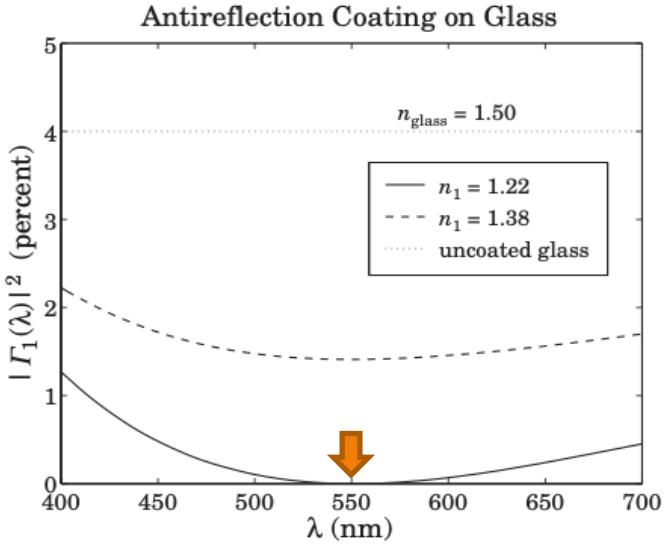
Choose Magnesium Fluoride,

$$\Gamma_{in} = \frac{\sqrt{n_a n_g} - n_c}{\sqrt{n_a n_g} + n_c} = -0.118$$

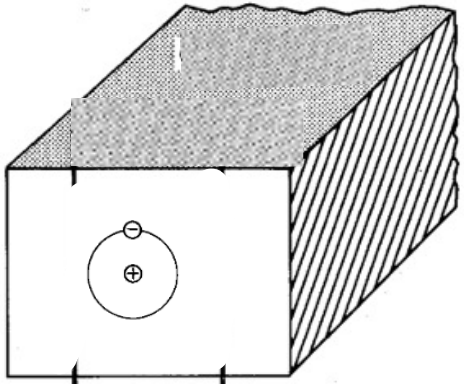
$$R = |\Gamma_{in}|^2 = 1.4\% \qquad d = \frac{\lambda_c}{4} = 99.6\ nm$$

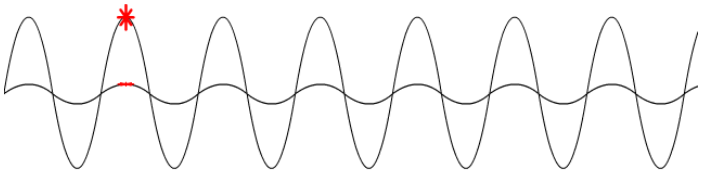
Note that without coating, (see figure)

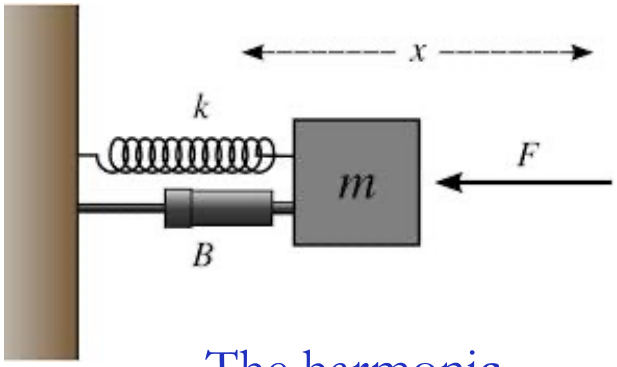
$$R = 4\%$$



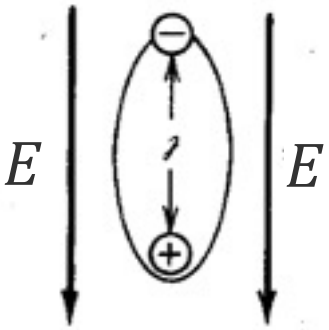
Light-matter Interaction



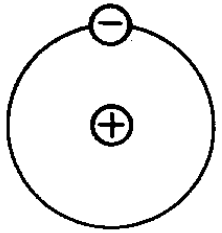




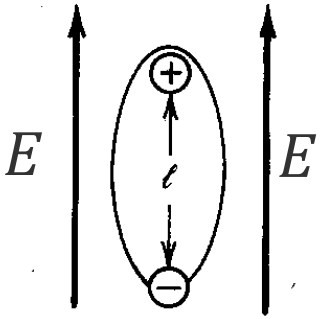
The harmonic oscillator



Atom polarized

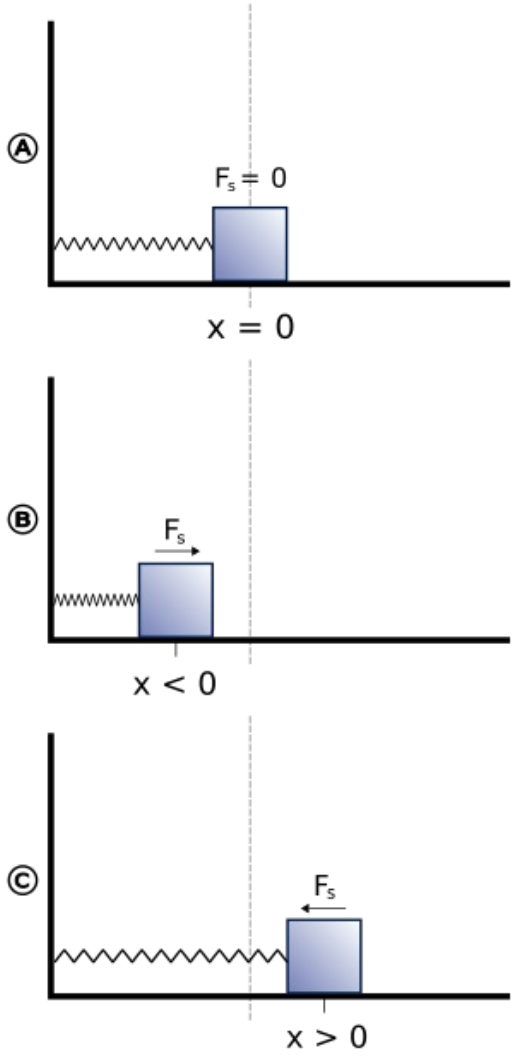


No Wave



Atom polarized

Light-matter Interaction



$$m\frac{\partial^2\vec{x}}{\partial t^2} + m\Gamma\frac{\partial\vec{x}}{\partial t} + m\omega_0^2\vec{x} = -q\vec{E}$$

- $m\frac{\partial^2\vec{x}}{\partial t^2}$: acceleration force
- $m \Rightarrow m_e$: mass of an electron
- $m\Gamma\frac{\partial\vec{x}}{\partial t}$: frictional force
- Γ: damping rate (loss/sec)
- $m\omega_0^2\vec{x}$: restoring force
- $\omega_0 = \sqrt{\frac{K}{m}}$: natural frequency
- $-q\vec{E}$: electric force

Fourier transform

$$m(-j\omega)^2\vec{x}(\omega) + m\Gamma(-j\omega)\vec{x}(\omega) + m\omega_0^2\vec{x}(\omega) = -q\vec{E}(\omega)$$

Simplify

$$\left(-m\omega^2 - j\omega m\Gamma + m\omega_0^2\right)\vec{x}(\omega) = -q\vec{E}(\omega)$$

Light-matter Interaction

$$\varepsilon_r(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - j\omega\Gamma} \qquad \omega_p^2 = \frac{Nq^2}{\varepsilon_0 m_e}$$

$$\varepsilon_r(\omega) = \varepsilon_r'(\omega) - j\varepsilon_r''(\omega)$$

$$\varepsilon_r'(\omega) = 1 + \omega_p^2 \frac{\omega_0^2 - \omega^2}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

$$\varepsilon_r''(\omega) = \omega_p^2 \frac{\omega\Gamma}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

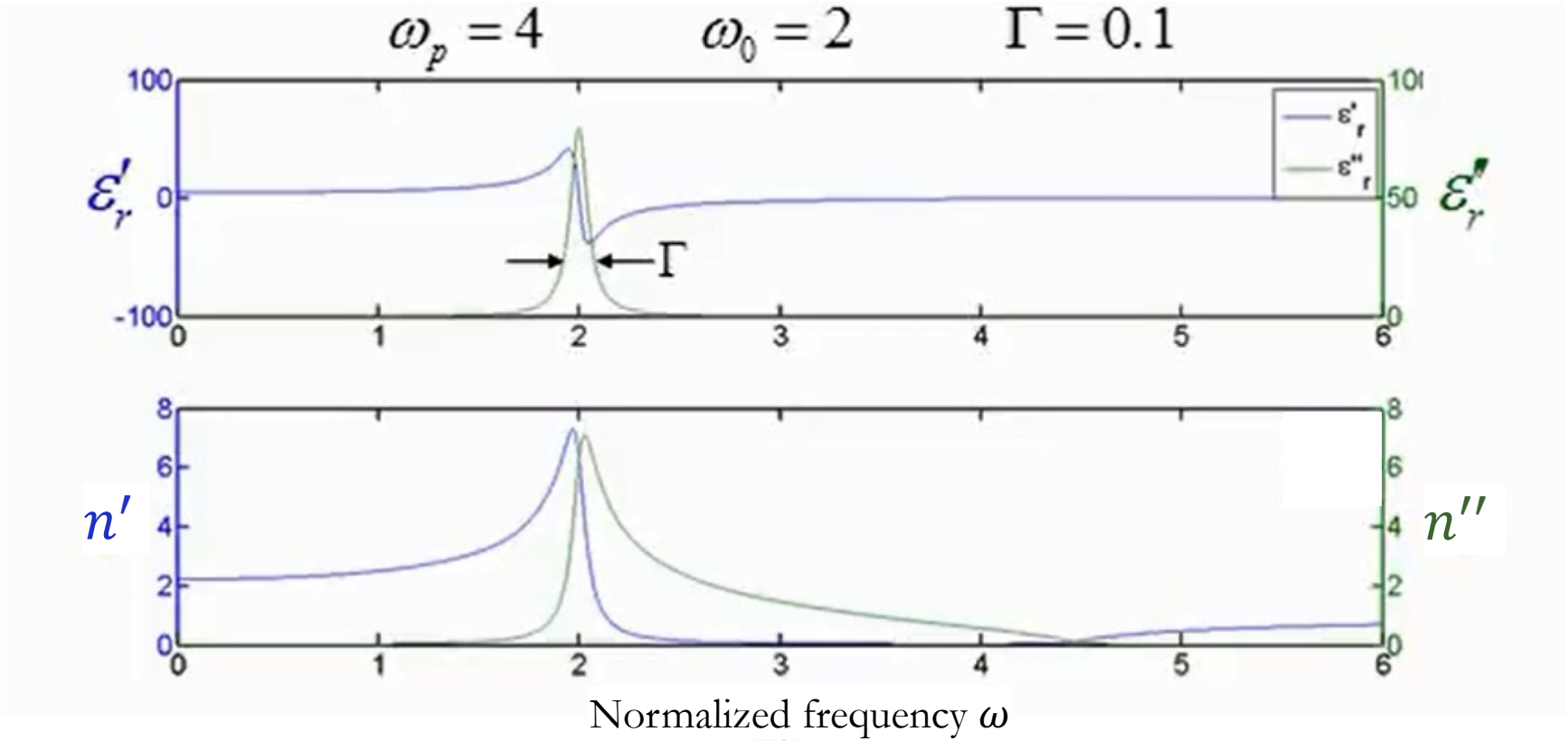
$$n = \sqrt{\mu_r \varepsilon_r} = n' - jn''$$ Complex Refractive index

$$(n')^2 = \frac{\varepsilon_r'}{2} + \frac{1}{2}\sqrt{(\varepsilon_r')^2 + (\varepsilon_r'')^2}$$

$$n'' = -\frac{\varepsilon_r''}{2n'}$$ Extinction coefficient

$$\alpha = 2\frac{\omega}{c}n''$$ Absorption Coefficient

Light-matter Interaction: Lorentz Model

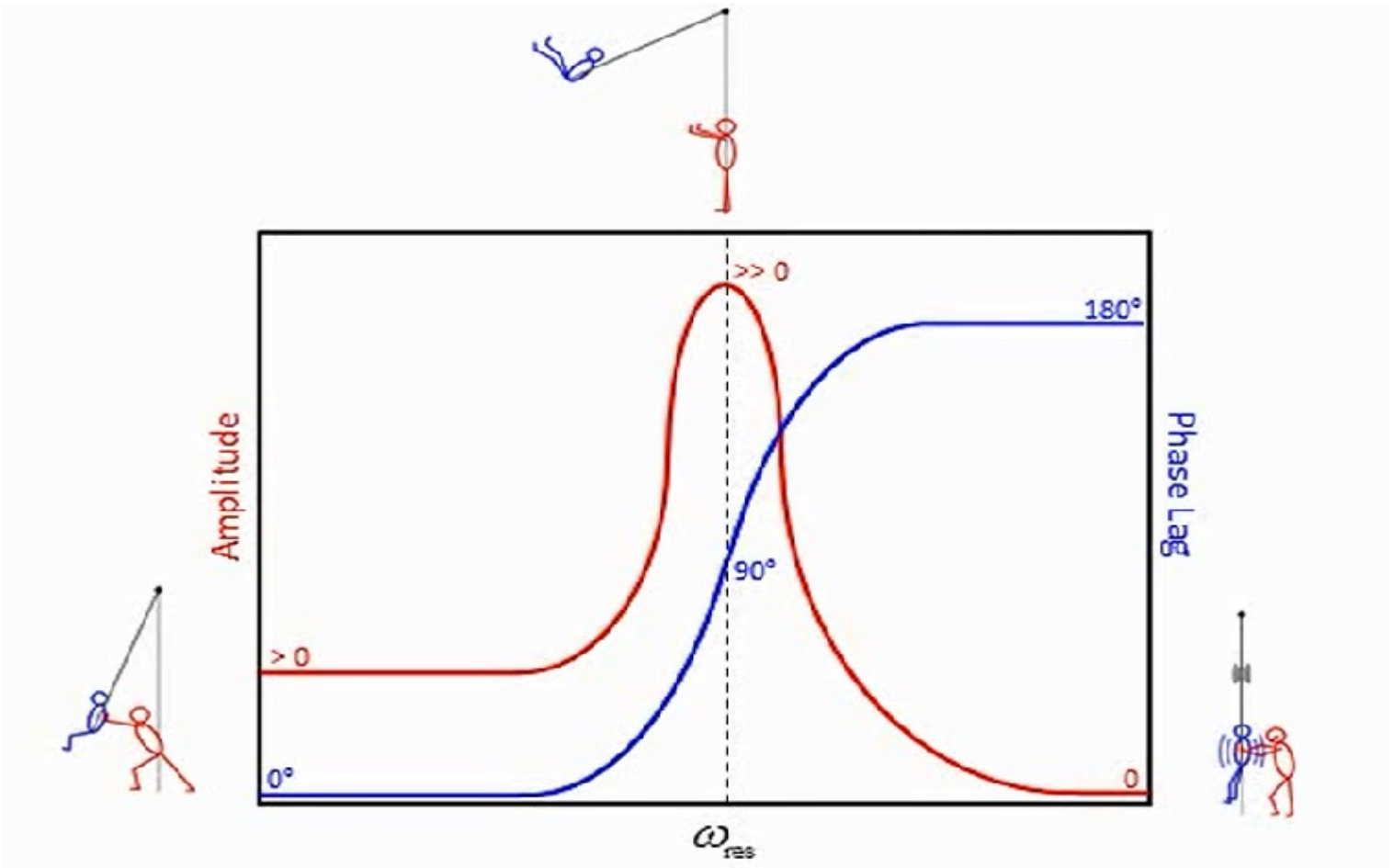


Loss is very high near resonance

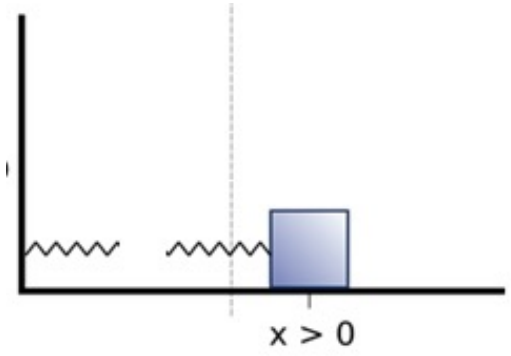
Damping rate determines width of resonance

Dielectric constant has a DC offset below resonance

Light-matter Interaction: Resonance



Light-matter Interaction: Metals



$$m\frac{\partial^2\vec{x}}{\partial t^2}+m\Gamma\frac{\partial\vec{x}}{\partial t}+m\omega_0^2\vec{x}=-q\vec{E}$$

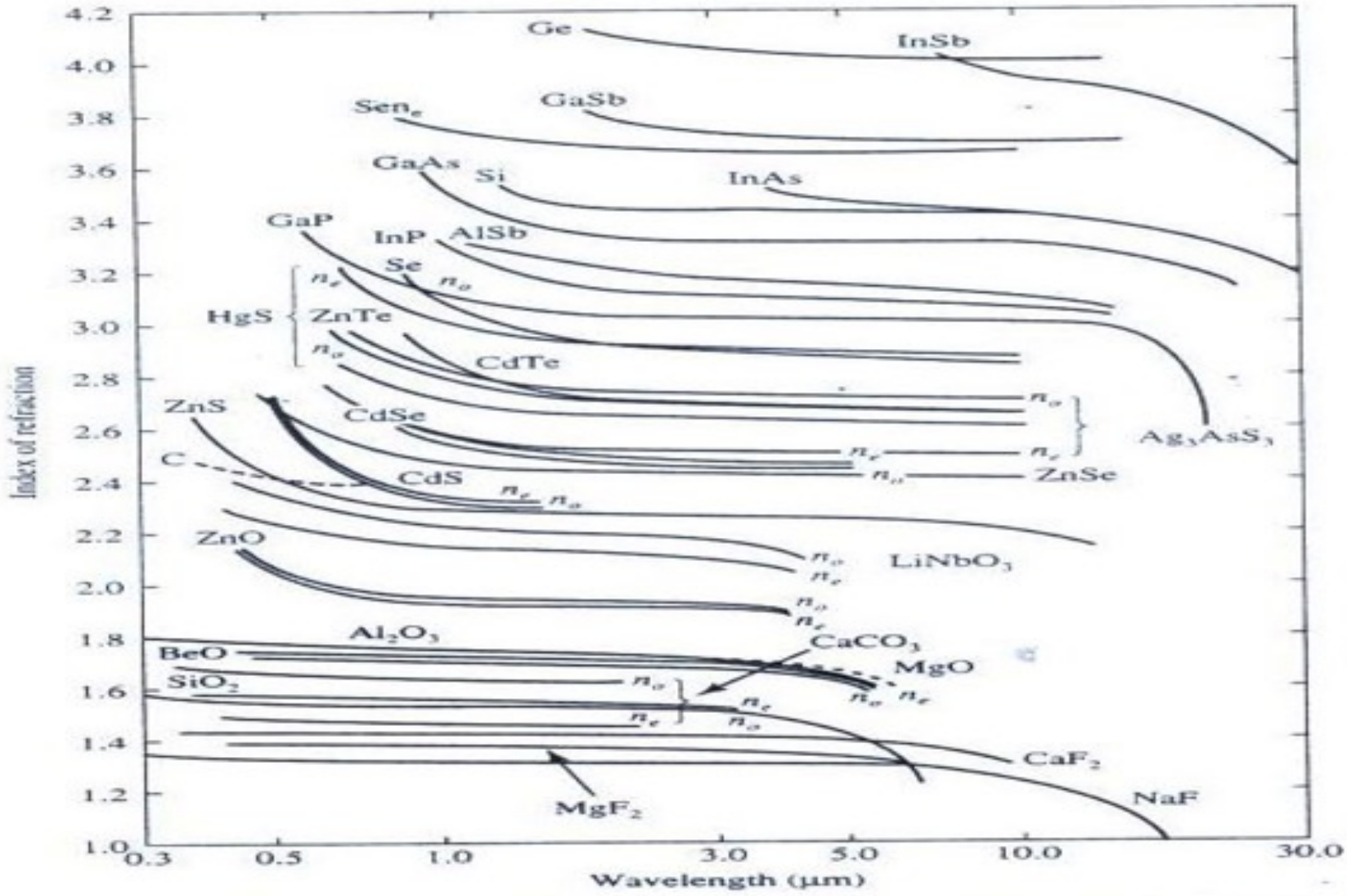
- $m\frac{\partial^2\vec{x}}{\partial t^2}$: acceleration force; $m \Rightarrow m_e$ mass of an electron
- $m\Gamma\frac{\partial\vec{x}}{\partial t}$: frictional force; Γ damping rate (loss/sec)
- $m\omega_0^2\vec{x}$: restoring force; $\omega_0=\sqrt{\frac{K}{m}}$ natural frequency
- $-q\vec{E}$: electric force

$$\varepsilon_r(\omega)=1-\frac{\omega_p^2}{\omega^2-j\omega\Gamma}$$

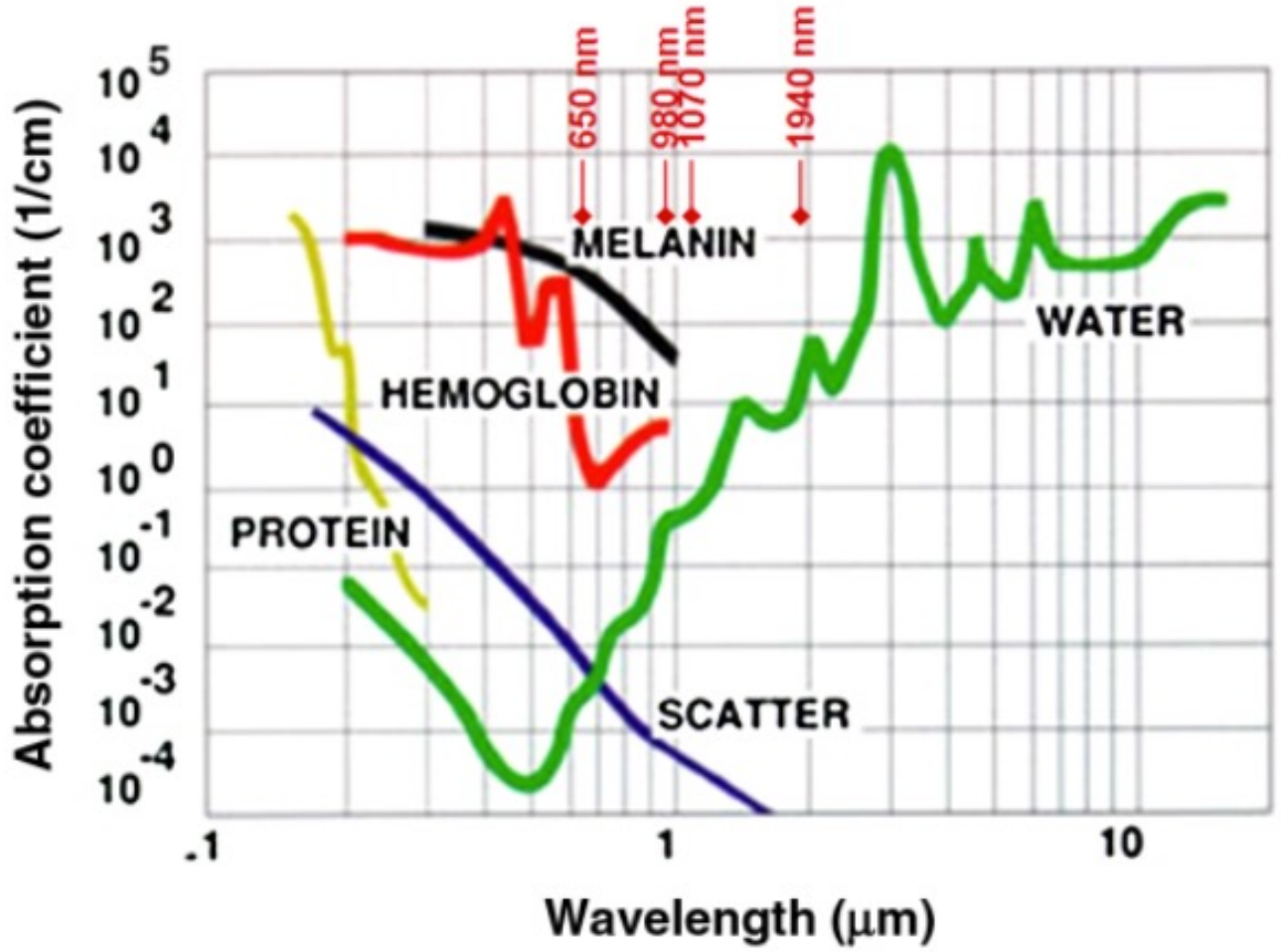
$$\varepsilon_r=\left(1-\frac{\omega_p^2\tau^2}{1+\omega^2\tau^2}\right)+j\left(\frac{\omega_p^2\tau/\omega}{1+\omega^2\tau^2}\right)$$

Drude Model

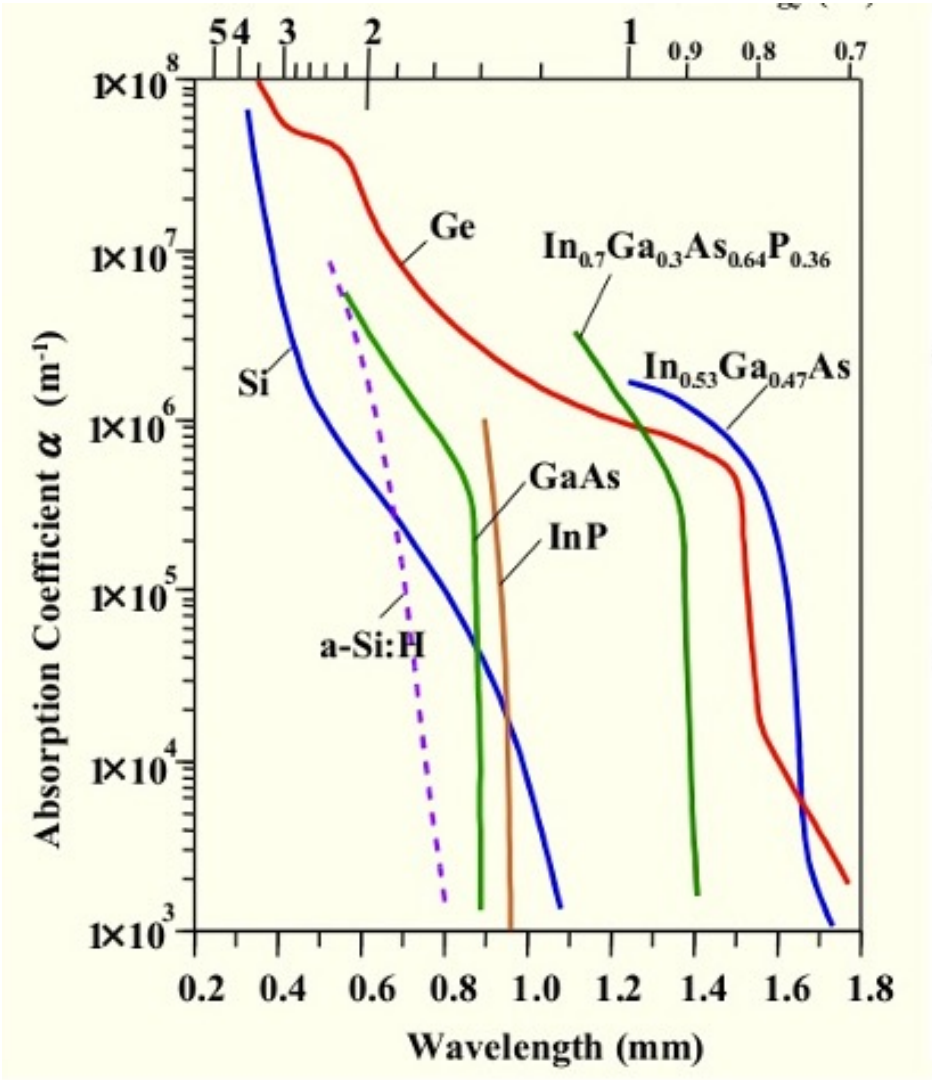
Light-matter Interaction: Examples
Refractive Index

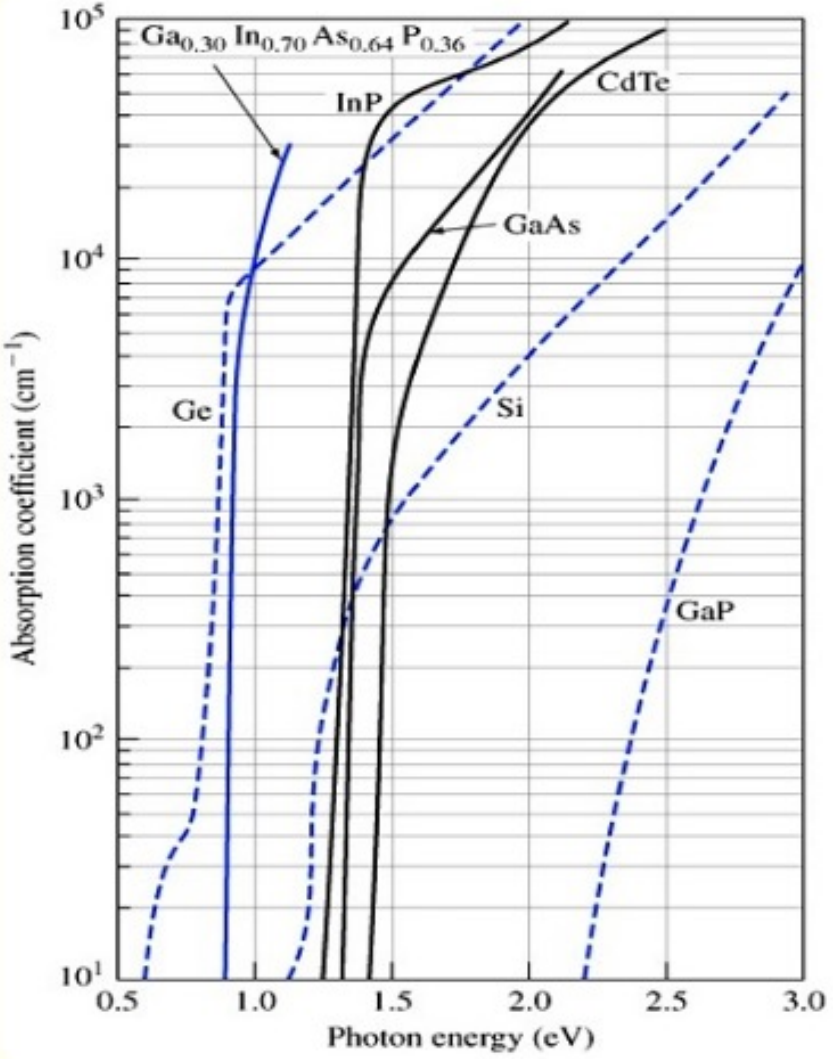


Light-matter Interaction: Examples
Absorption Coefficient



Light-matter Interaction: Examples
Absorption Coefficient





Light as Particles (Photons)

The following subjects will be covered

- The Birth of Quantum Mechanics
- The Photoelectric Effect
- Light Wave-Particle Duality.
- Light as Photons.
- Examples

The Birth of Quantum Mechanics

The process of generating light from heat was very well "known" BUT
the actual physics was not "understood"
The relation between Temperature and Light intensity and wavelength was not developed.



Max Planck
1858-1947

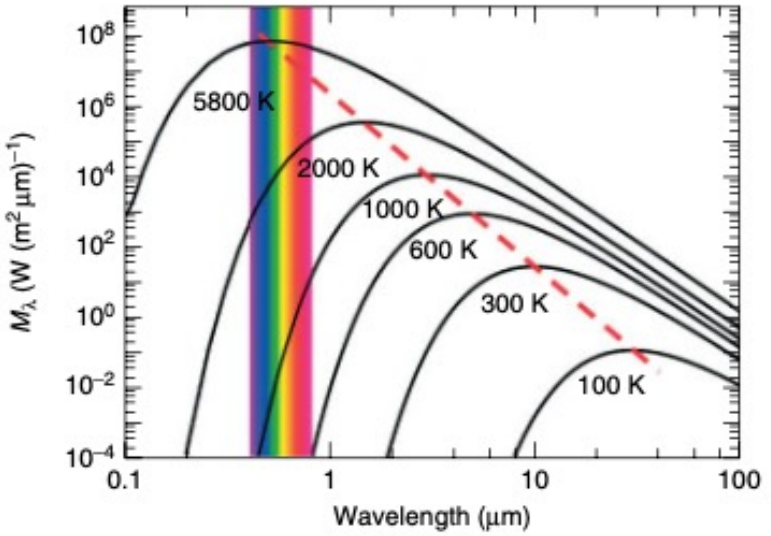
Planck began studying physics in 1874 after being advised against this by a physics professor on the grounds that "*in this field, almost everything is already discovered, and all that remains is to fill a few holes.*"

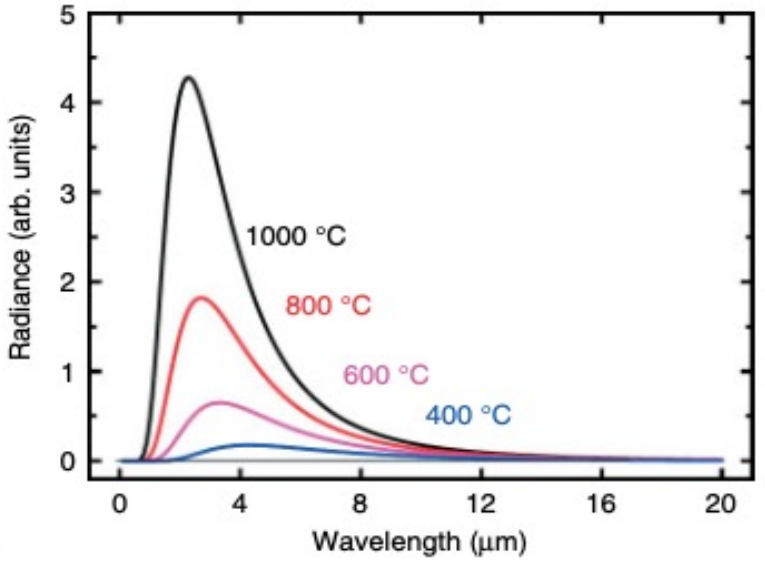
His doctoral thesis concerned thermodynamics.

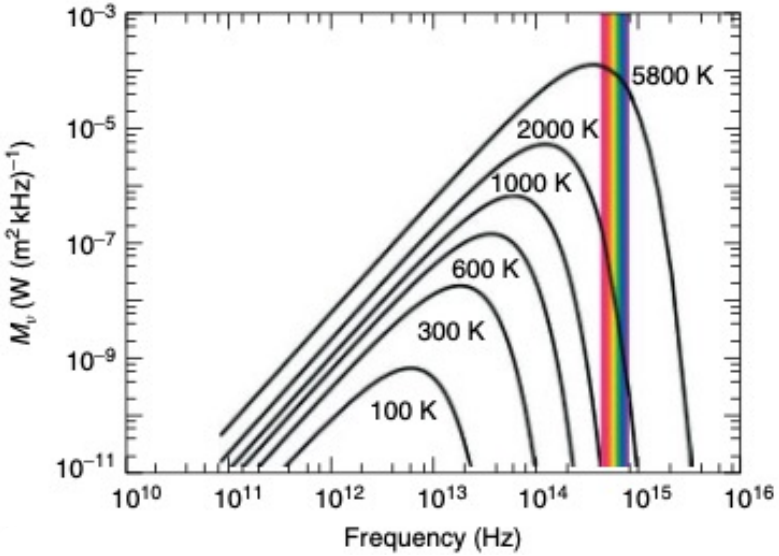
In 1894 Planck was commissioned to research how to obtain the most output from a light bulb for a given input of electrical power.

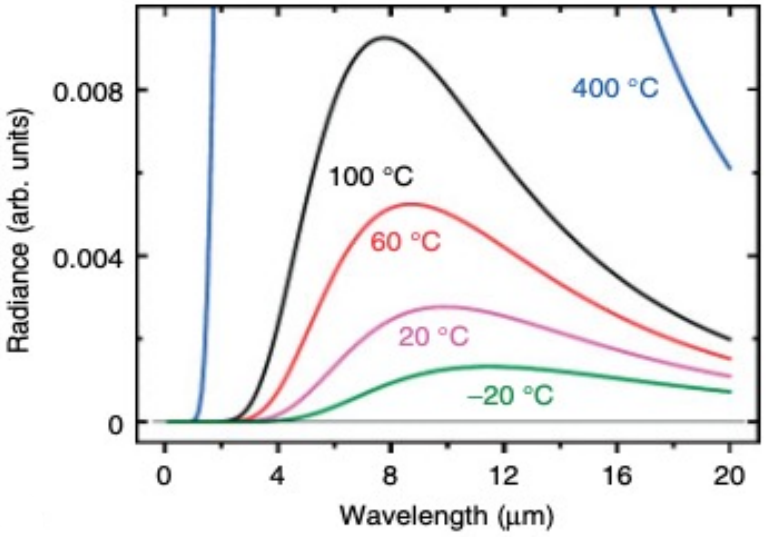


Thermal Radiation

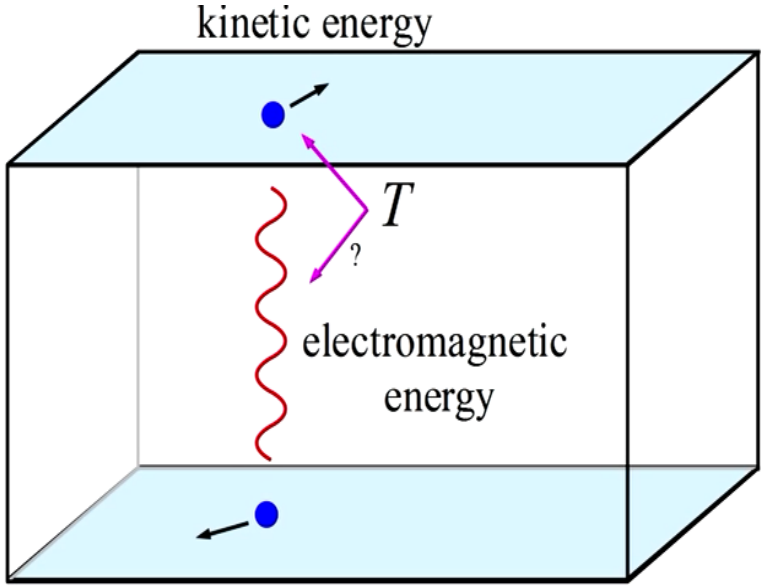








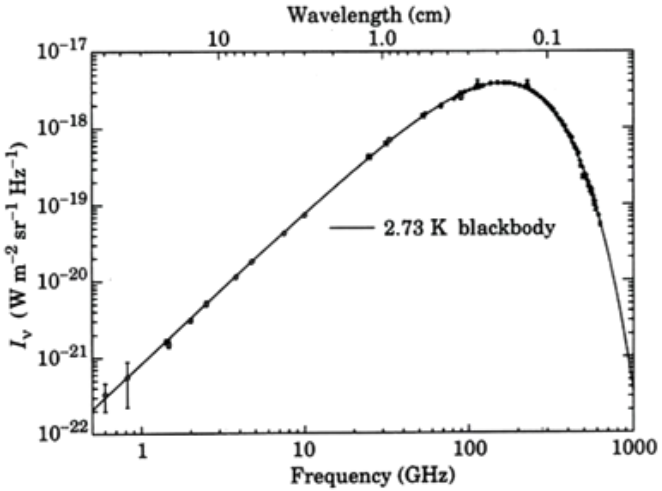
The Birth of Quantum Mechanics



Rayleigh-Jeans law

$$u(\nu) = A\,\nu^2 kT$$

(straight-line on a log-log plot)



Each mode sho Classical physics $u(\nu) = A\,\nu^2 kT$ nergy kT

In thermodynan more modes at $\langle E \rangle_{\text{mode}} = kT$ diation to
increase with te high frequencies

The Birth of Quantum Mechanics

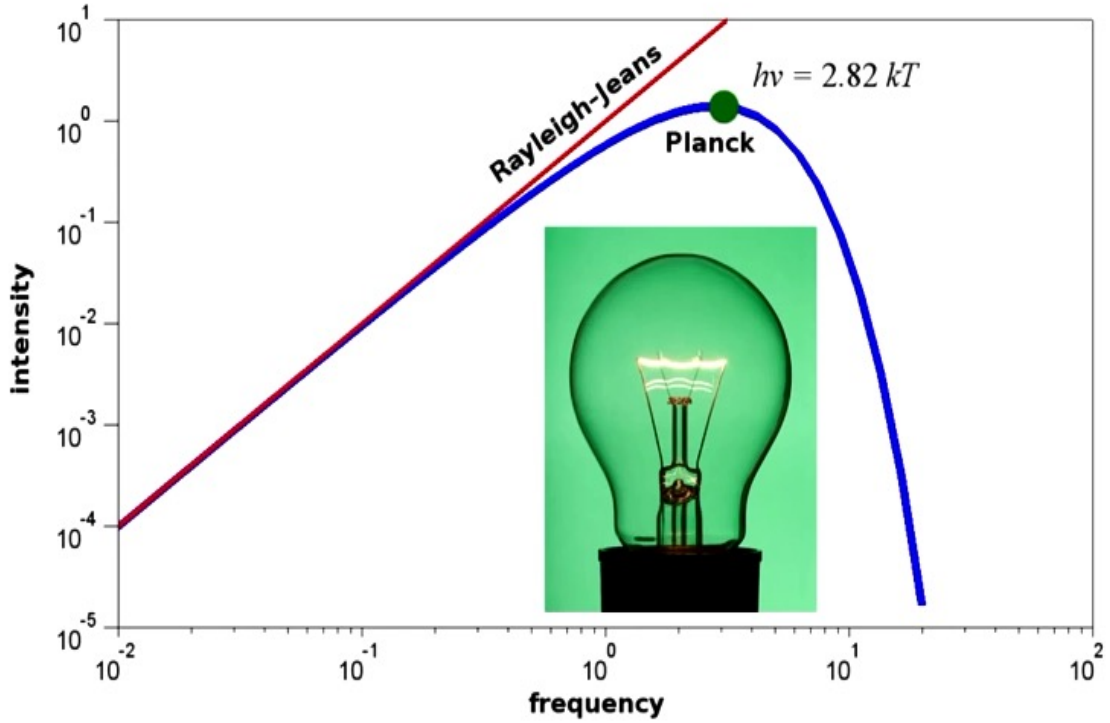
for finite h

$$u(\nu) = A\,\nu^2 \frac{h\nu}{e^{h\nu/kT} - 1}$$

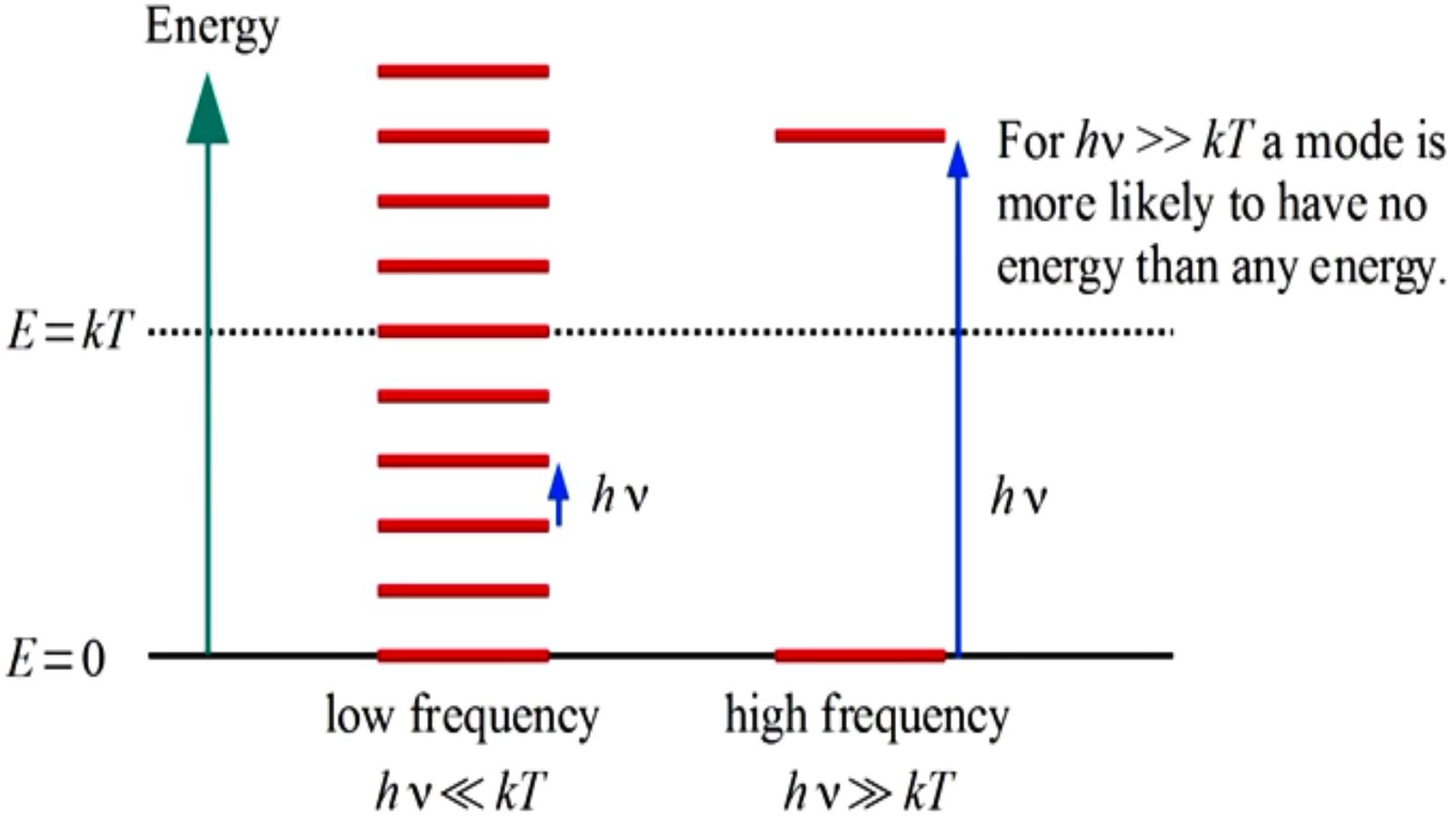
Planck's law (1900)

$$E = n h\nu$$

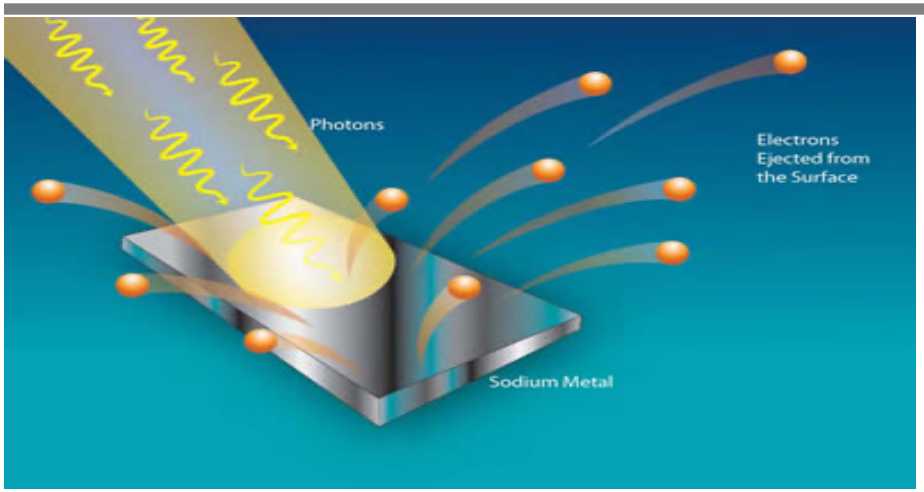
$$h = 6.626 \cdot 10^{-34}\,\text{Js}$$

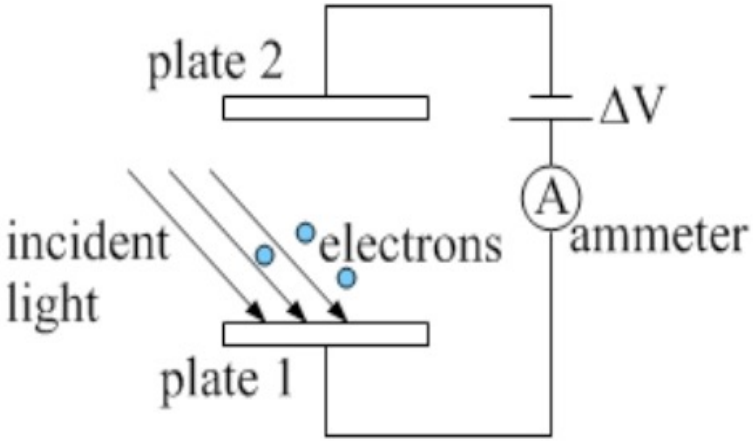


The Birth of Quantum Mechanics



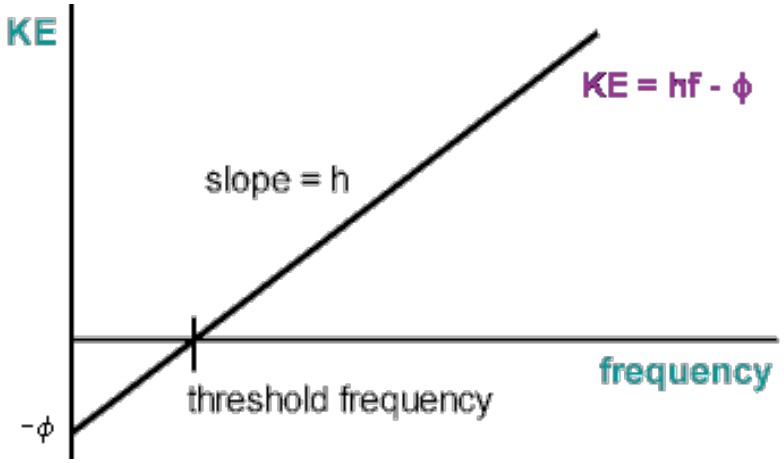
The Photo-electric Effect





Observations

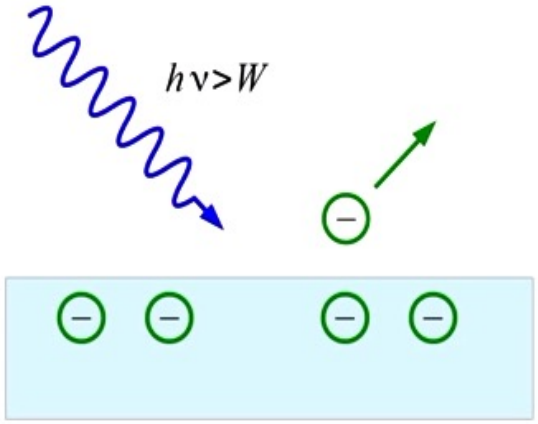
- No current detected for $\omega < \omega_{th}$ even with high intensity.
- Current detected when $\omega > \omega_{th}$ and current increases with light intensity.
- Electron kinetic energy is linear with ω.

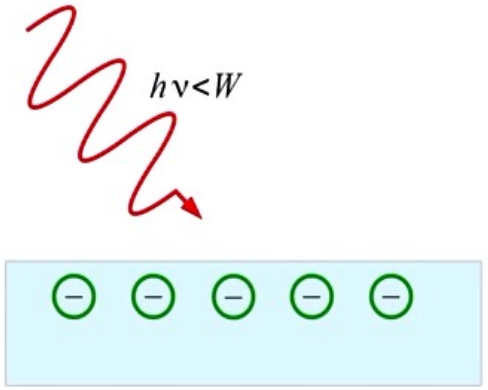


The Photo-electric Effect

Einstein 1905: If W is the energy needed to remove an electron, only photons with $h\nu > W$ will eject electrons. The number of ejected electrons is proportional to the number of photons. The kinetic energy is $h\nu - W$.







The Light Quanta: The Photon

Based on these observations, the photon concept emerges

What is a photon?

A light wave with frequency ω can be thought of as composed of a stream of photons (energy particles) that have the same wave speed (c) and travel in the same wave direction.

Photon Energy $E = hf = \hbar\omega = hc/\lambda$

Photon Momentum $p = \frac{E}{v} = \frac{\hbar\omega}{v} = \frac{h}{\lambda}$

Photon mass = 0

Photon spin = 1

Photon Stream

Average Photon Flux density in a light wave $\varphi = \frac{I}{\hbar\omega}$ $\frac{photons}{s \cdot m^2}$

Average Photon Flux $\int \varphi \, dA$ $\frac{photons}{s}$

Star light 10^6

Moon light 10^8

Sun light 10^{14}

Laser light 10^{22}

$\frac{photons}{s \cdot cm^2}$

Plank's constant
$h = 6.626\times10^{-34}\ Js$

Reduced Plank's constant ($h/2\pi$)
$\hbar = 1.055\times10^{-34}\ Js$

The Light Quanta: The Photon

Example. What is the energy in a radiation mode at $\lambda = 1\ \mu m$, in free space?

$\lambda = 1\ \mu m$ → $\omega = 2\pi\ c/\lambda$

Infrared

$\omega = 2\pi \times 3 \times 10^{14}$ rad/s

Photon Energy $E = \hbar\ \omega = 1.99 \times 10^{-19}\ J$

$$E = \frac{1.99 \times 10^{-19}}{q} = 1.24\ eV$$

Example. What is the photon flux in a 0.2-μm light, if the total power over some area is 1 nW?

$\lambda = 0.2\ \mu m$ → $\hbar\ \omega = 10 \times 10^{-19}\ J$

UV-C

$$\text{Photon flux} = \frac{1 \times 10^{-9}\ J/s}{10 \times 10^{-19}\ J} = 1 \times 10^{9} \quad \frac{photons}{s}$$

Example

Ultraviolet light with a wavelength of 240 nm illuminates plate 1 in the figure. Consequently, electrons are emitted and driven across the gap to plate 2 and hence current is read on the ammeter. The battery voltage is gradually increased until the current drops to zero. The battery voltage at this point is 1.2 V.

Find,

1) The energy of the illuminating light.

$$E = 1.24/0.24 = 5.17\ eV$$

2) Maximum kinetic energy of the electron beam.

$$KE = 1.2\ eV$$

3) The work function of the metal.

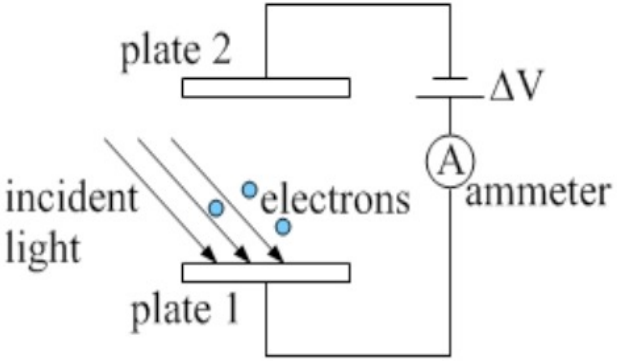
$$W = 5.17 - 1.2 = 3.93\ eV$$

4) The metal used for plate 1.

$$Al$$

5) The maximum wavelength that causes current.

$$\lambda_{max} = ch/\omega = 330\ nm$$



Metal	Work Function, ϕ, (in eV)
Na	2.28
Al	4.08
Cu	4.70
Zn	4.31
Ag	4.73
Pt	6.35
Pb	4.14
Fe	4.50

Semiconductors Fundamentals

The following subjects will be covered

- Semiconductors in General.
- The PN Junction.
- The Energy Band Diagram.
- Carrier Statistics in Semiconductors.
- PN Junction Band Diagram.
- Photon Generation and Absorption.
- Direct and Indirect Gap Materials.
- Examples.

Semiconductors

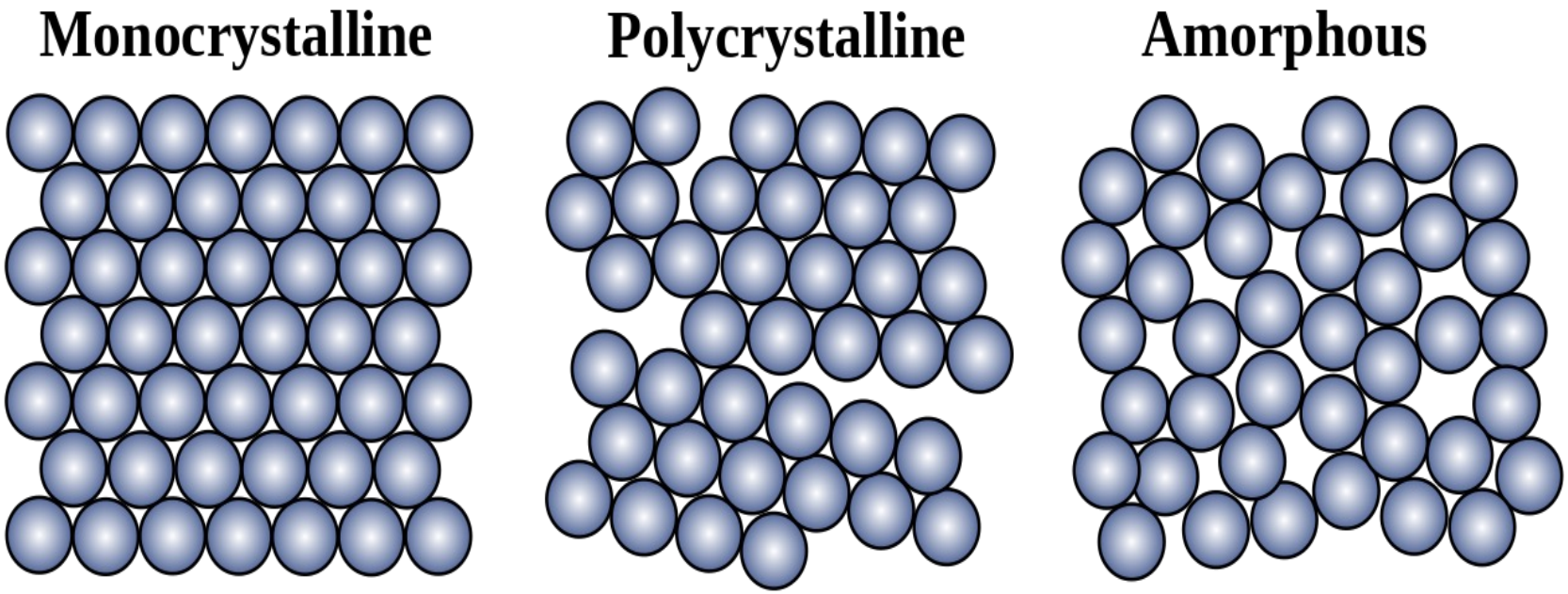
Elemental: Si, Ge.

Compounds: GaAs, InP.

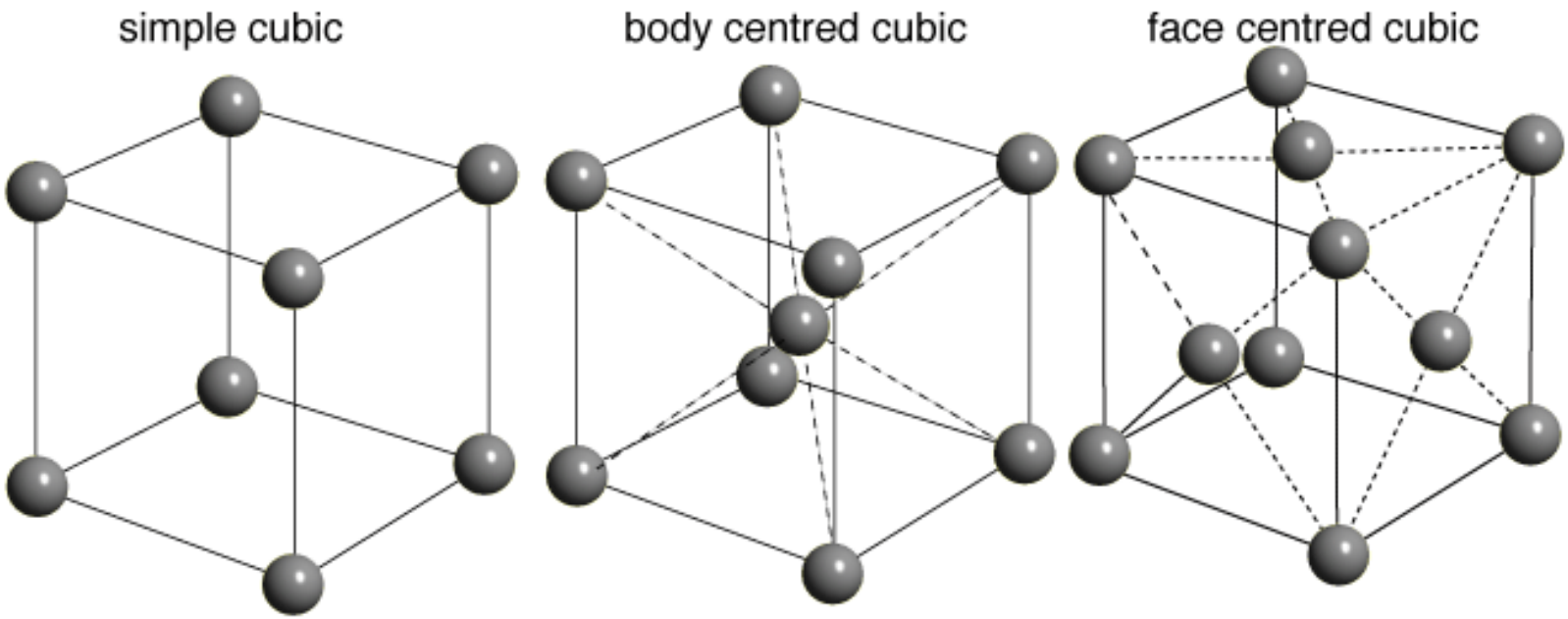
Alloys: $Al_{1-x}Ga_xAs$.

							VIIIA
		IIIA	IVA	VA	VIA	VIIA	2 He 4.003
		5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.183
IB	IIB	13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.064	17 Cl 35.453	18 Ar 39.948
29 Cu 63.54	30 Zn 65.37	31 Ga 69.72	32 Ge 72.59	33 As 74.922	34 Se 78.96	35 Br 79.909	36 Kr 83.80
47 Ag 107.870	48 Cd 112.40	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 127.60	53 I 126.904	54 Xe 131.30
79 Au 196.967	80 Hg 200.59	81 Tl 204.37	82 Pb 207.19	83 Bi 208.980	84 Po (210)	85 At (210)	86 Rn (222)

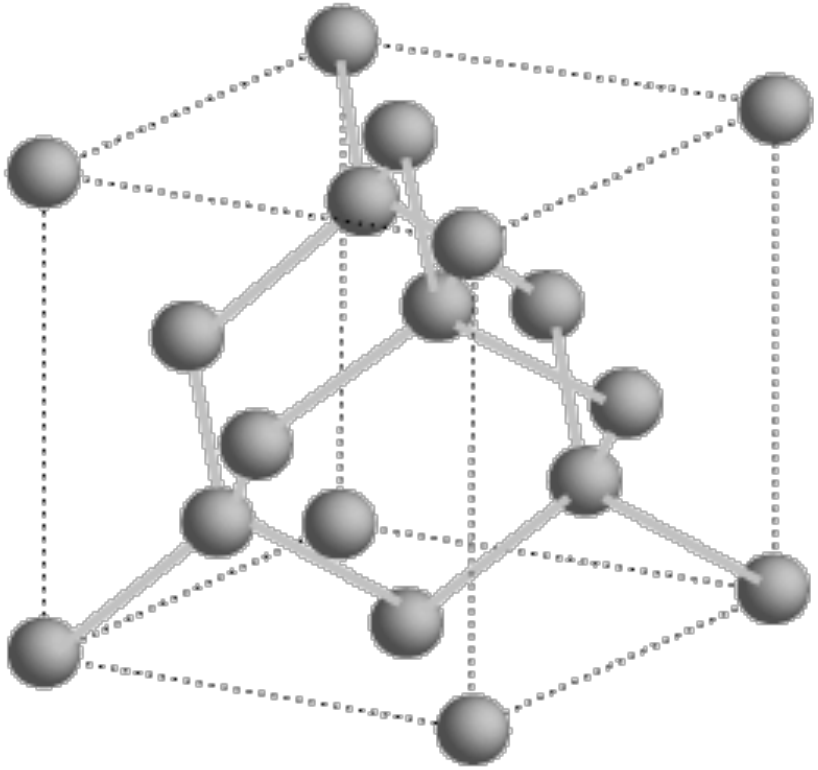
Structure



Crystal Structure: Cubic

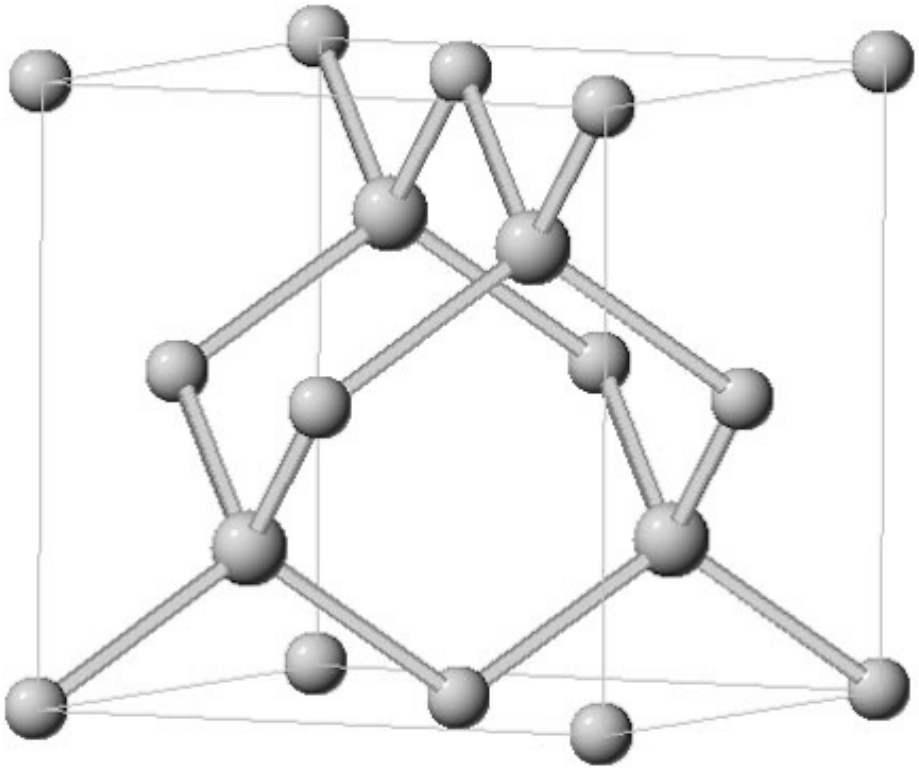


Crystal Structure: Diamond





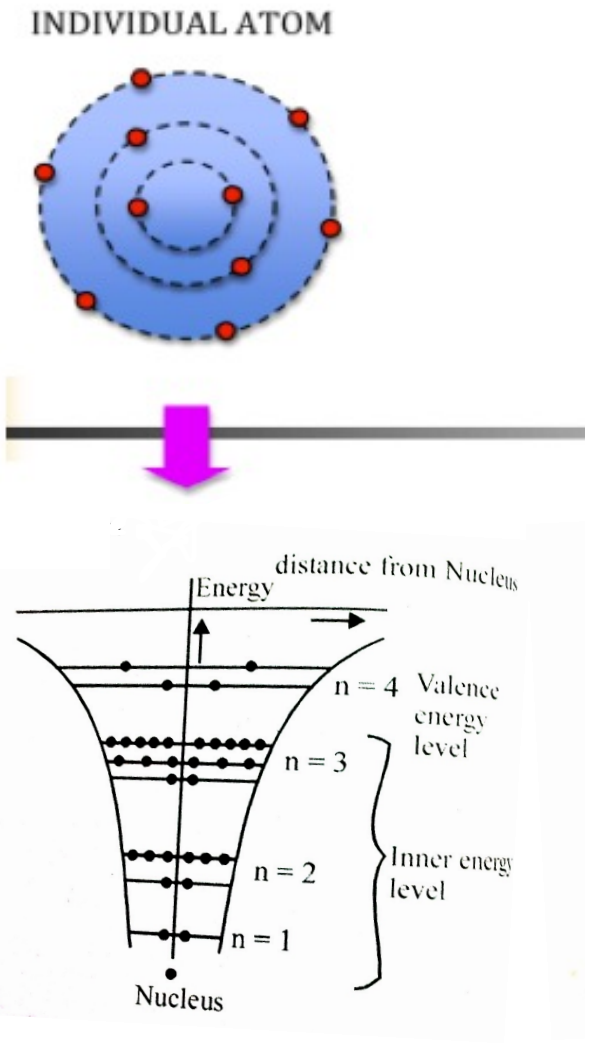
Crystal Structure: Zinc-blende



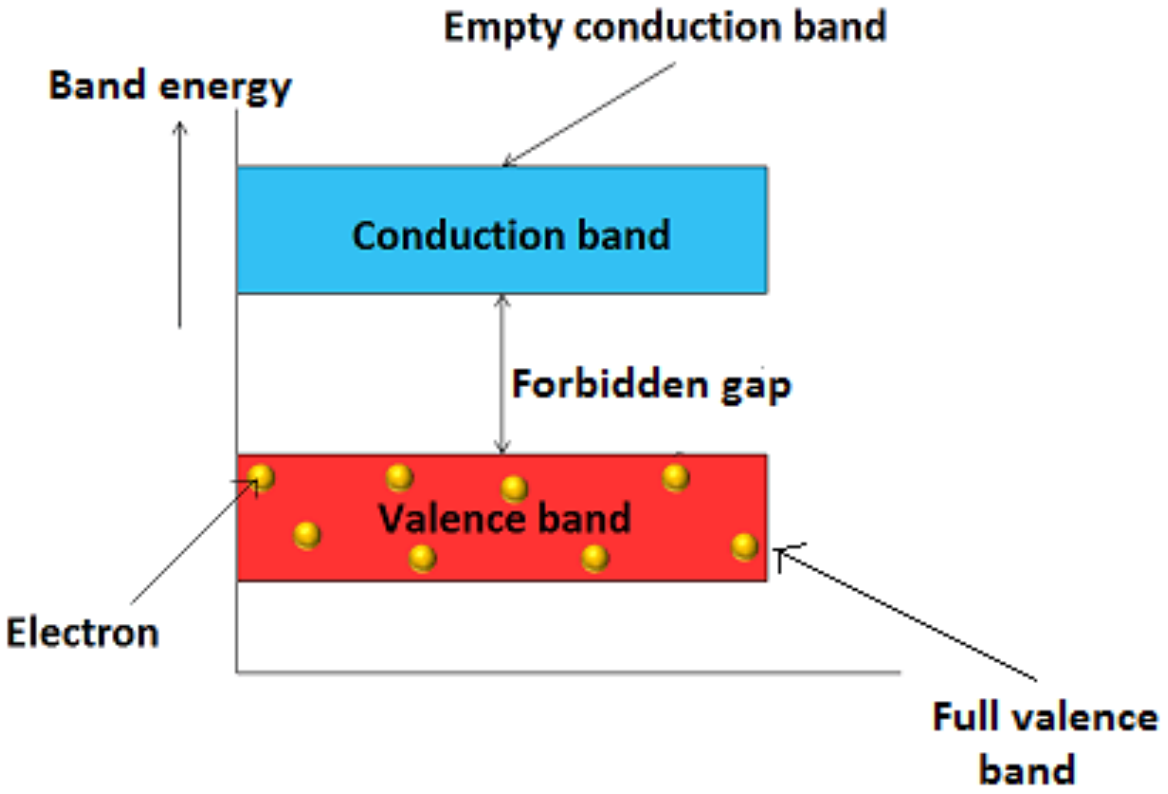
Crystal Structure: Examples

	Crystal	a[Å]	Crystal	a[Å]
Diamond	C	3.567	CuBr	5.691
	Si	5.431	CuCl	5.406
	Ge	5.657	CuI	6.054
	α-Sn	6.491	GaAs	5.654
			GaP	5.451
Zinc-blende	AgI	6.473	GaSb	6.118
	AlAs	5.62	HgS	5.852
	AlP	5.451	HgSe	6.084
	AlSb	6.135	HgTe	6.429
	BAs	4.777	InAs	6.036
	BN	3.615	InP	5.869
	BP	4.538	InSb	6.478
	BeS	4.85	SiC	4.348
	BeSe	5.07	ZnO	4.63
	BeTe	5.54	ZnS	5.409
	CdS	5.818	ZnSe	5.668
	CdSe	6.05	ZnTe	6.089
	CdTe	6.480		

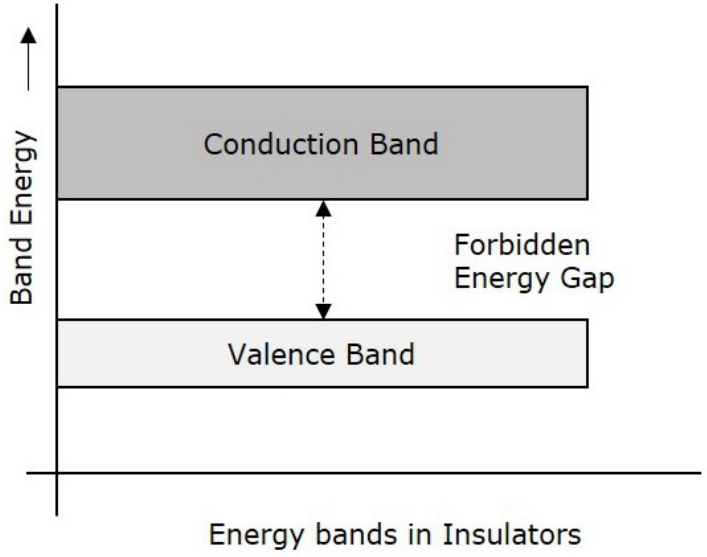
Energy Bands: Isolated atoms and Crystals



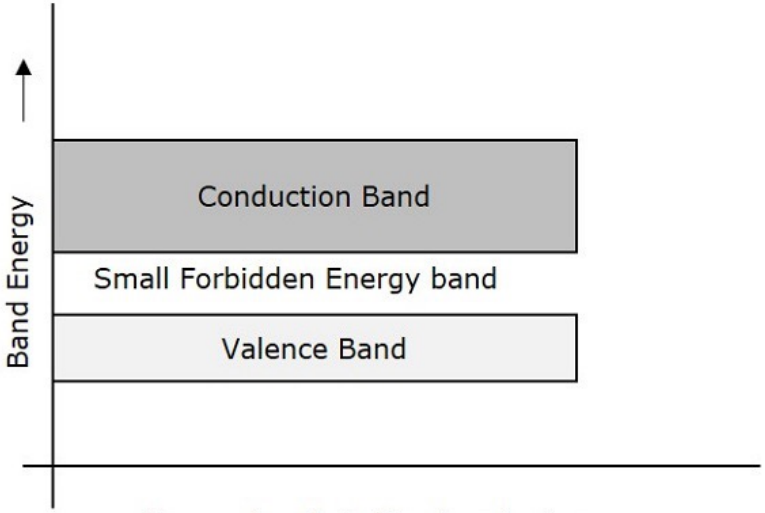
Energy Bands: Isolated atoms and Crystals



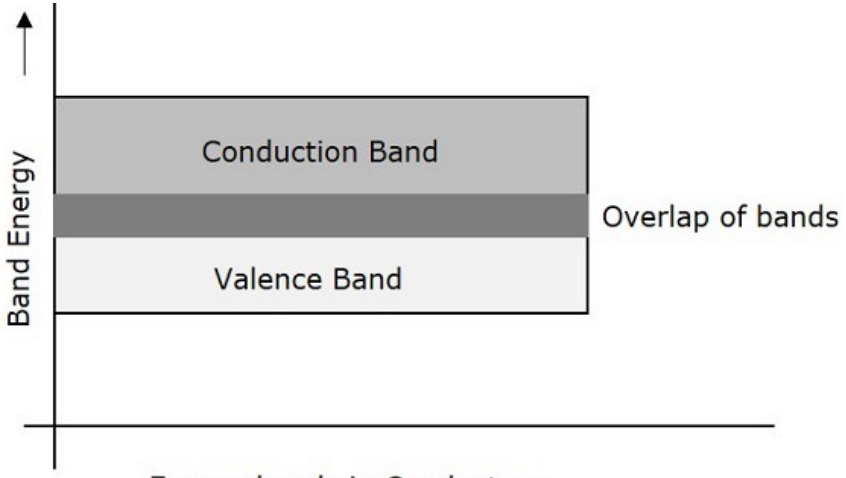
Classification of Materials



Energy bands in Insulators



Energy bands in Semiconductors



Energy bands in Conductors

Examples

Quartz	E_g= 10 eV
Silicon	E_g= 1.12 eV
Copper	E_g= 0 eV

Semiconductors

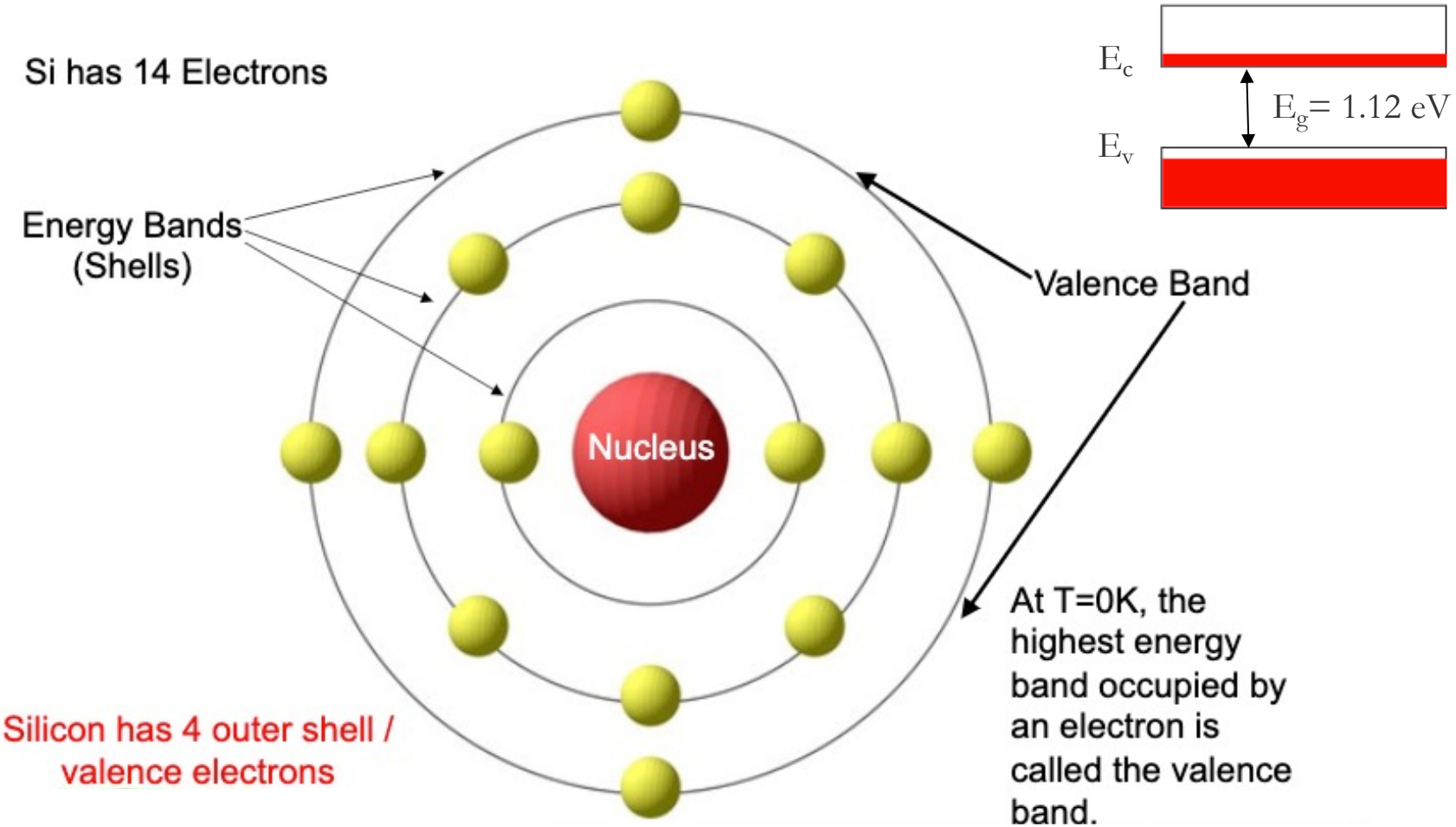
Semiconductor	Bandgap Energy E_G (eV)
Carbon (diamond)	5.47
Silicon	1.12
Germanium	0.66
Tin	0.082
Gallium arsenide	1.42
Gallium nitride	3.49
Indium phosphide	1.35
Boron nitride	7.50
Silicon carbide	3.26
Cadmium selenide	1.70

	IIIA	IVA	VA	VIA
	5 10.811 **B** Boron	6 12.01115 **C** Carbon	7 14.0067 **N** Nitrogen	8 15.9994 **O** Oxygen
	13 26.9815 **Al** Aluminum	14 28.086 **Si** Silicon	15 30.9738 **P** Phosphorus	16 32.064 **S** Sulfur
IIB				
30 65.37 **Zn** Zinc	31 69.72 **Ga** Gallium	32 72.59 **Ge** Germanium	33 74.922 **As** Arsenic	34 78.96 **Se** Selenium
48 112.40 **Cd** Cadmium	49 114.82 **In** Indium	50 118.69 **Sn** Tin	51 121.75 **Sb** Antimony	52 127.60 **Te** Tellurium
80 200.59 **Hg** Mercury	81 204.37 **Tl** Thallium	82 207.19 **Pb** Lead	83 208.980 **Bi** Bismuth	84 (210) **Po** Polonium

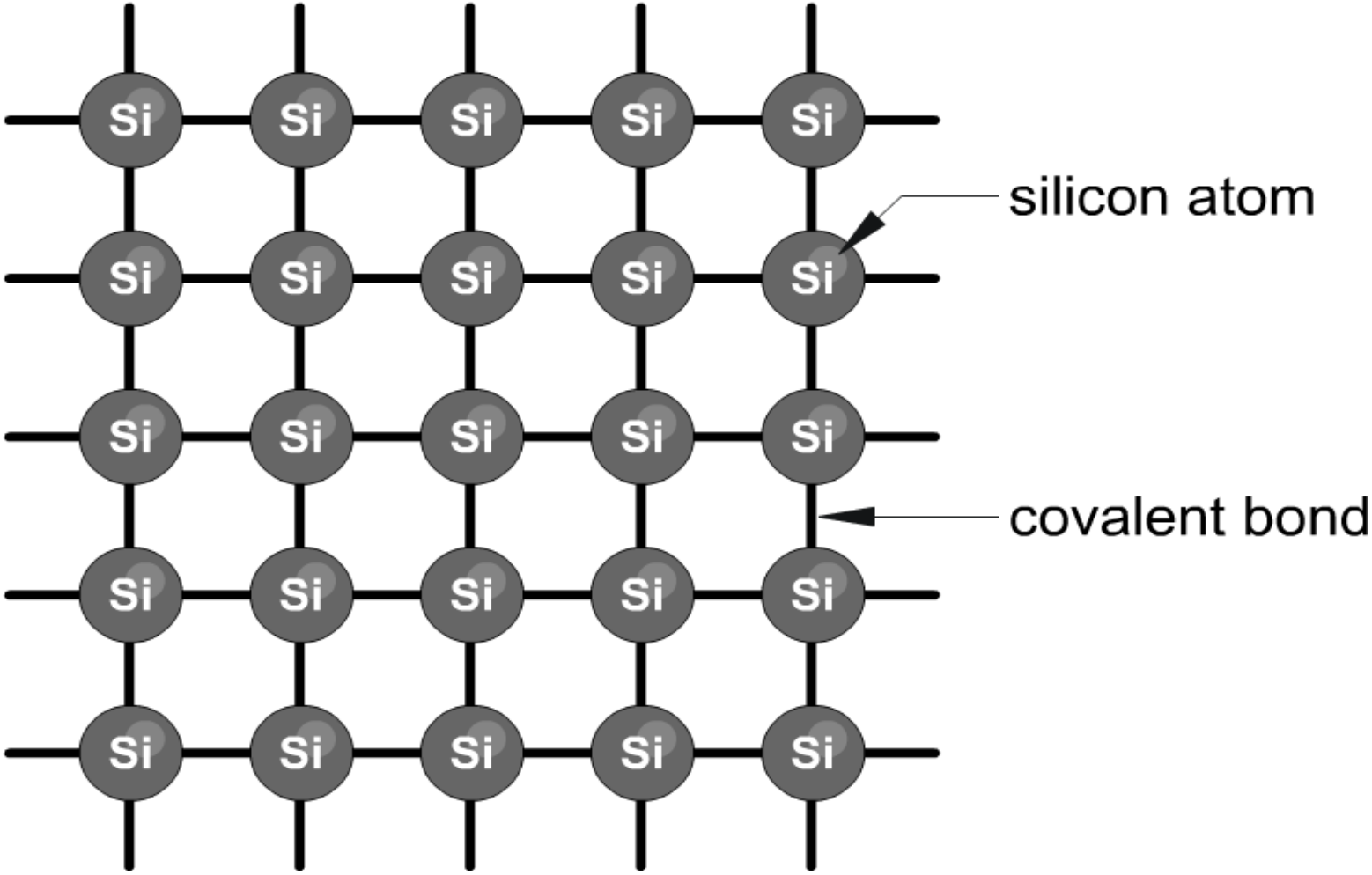
Example: Silicon



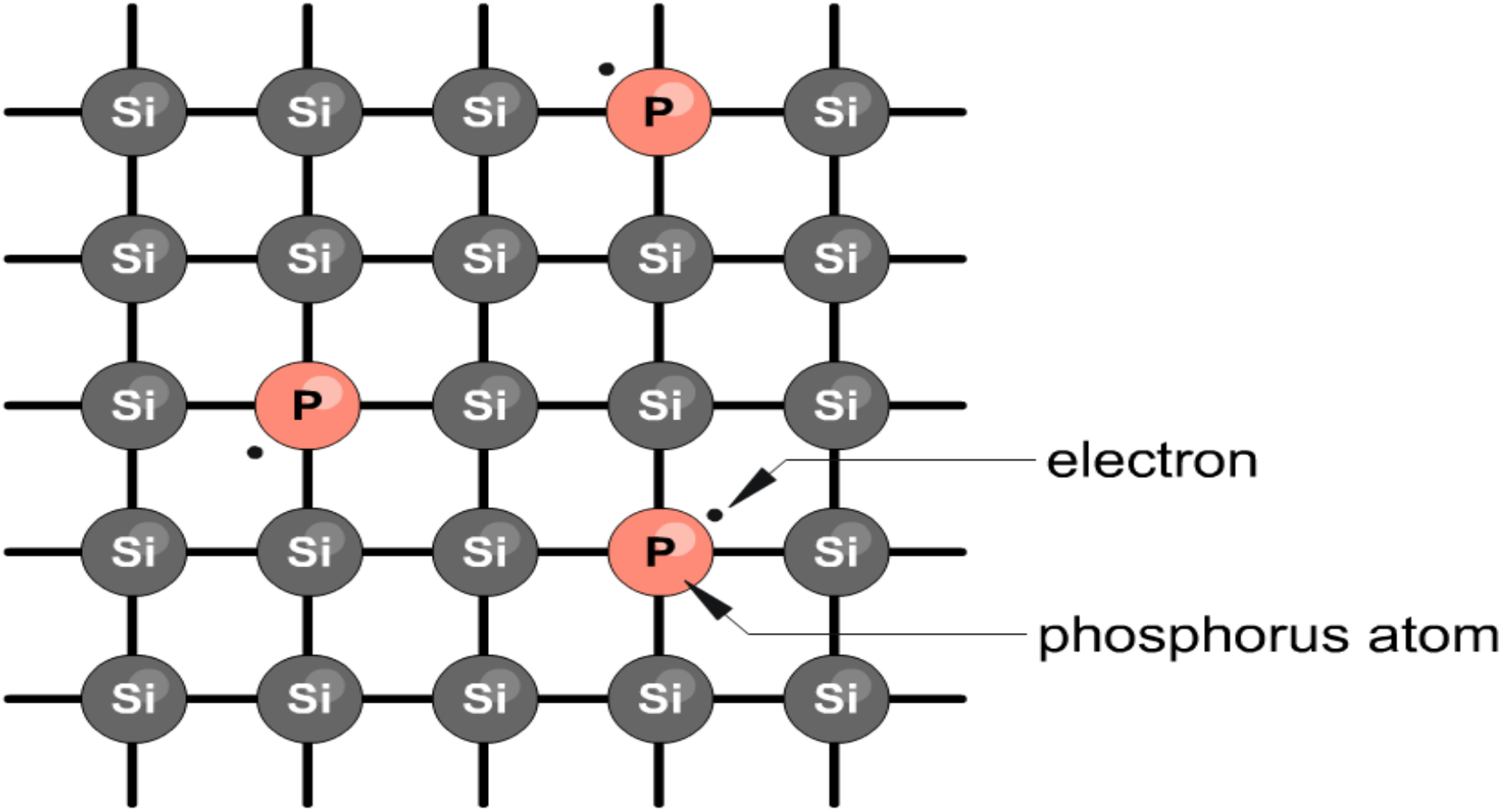
The Silicon Atom



Covalent Bonding in pure Si

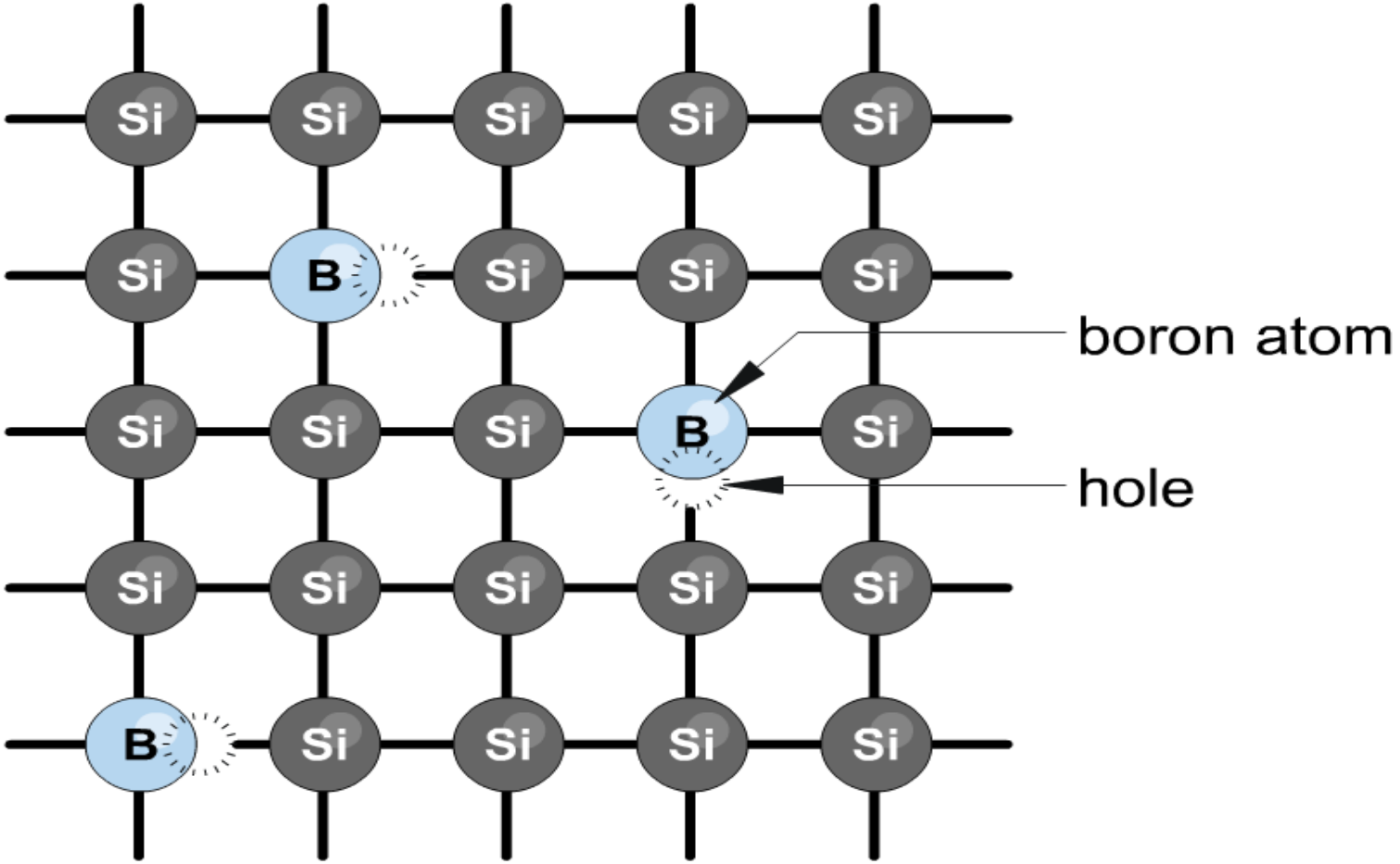


Doping: N-type Si



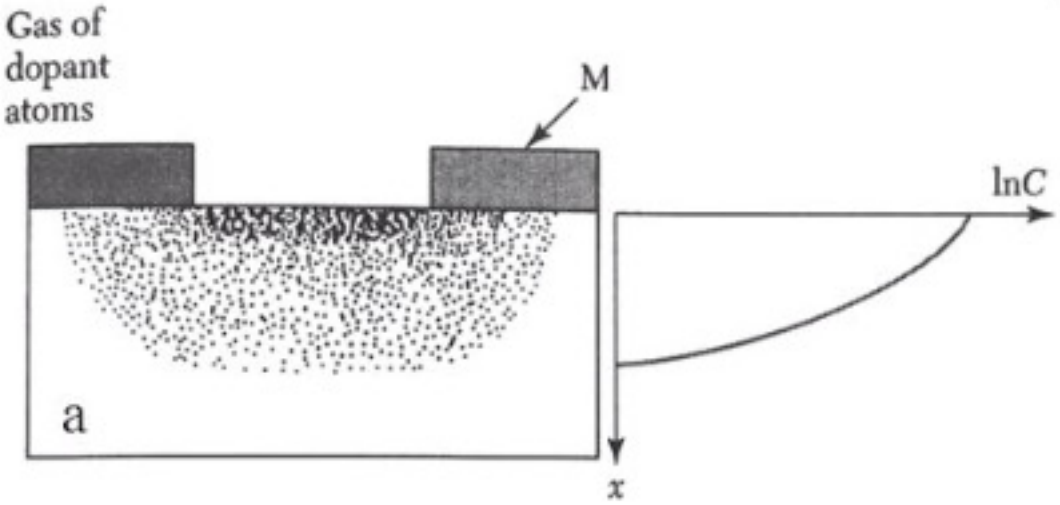
Donor concentration (N_D) 10^{15}-10^{18} atom/cm^3

Doping: P-type Si



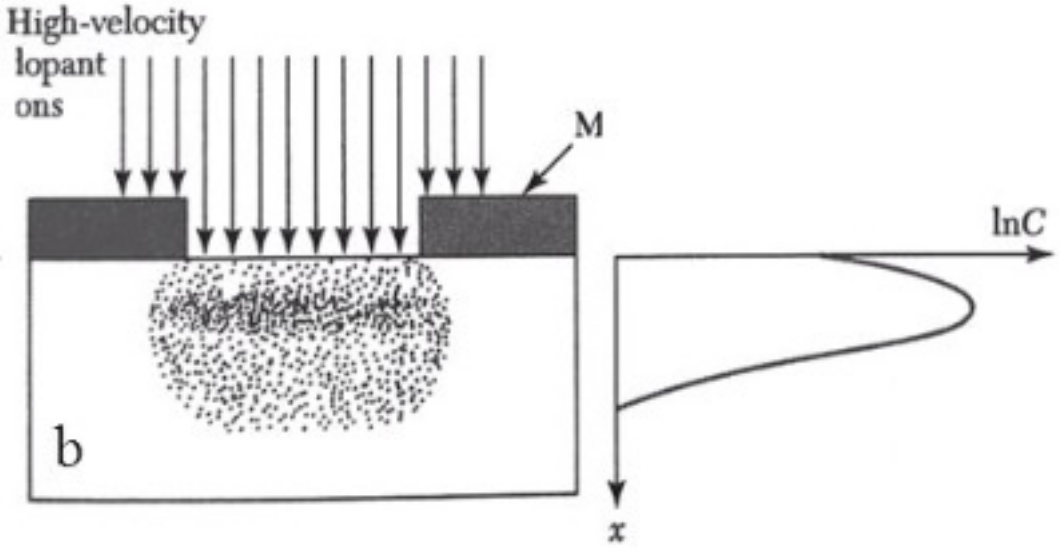
Acceptor concentration (N_A) 10^{13}-10^{16} atom/cm^3

Methods of Doping: Diffusion



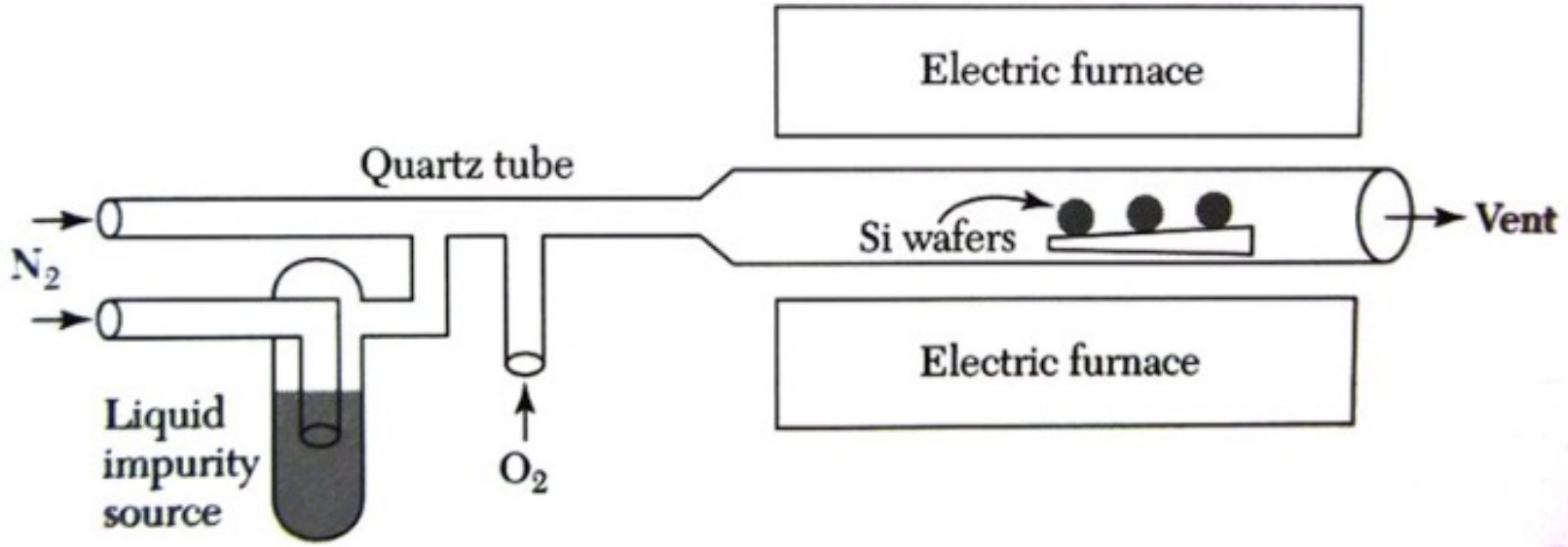
Diffusion is a cheaper and more simplistic method, but can only be performed from the surface of the wafers. Dopants also diffuse unevenly, and interact with each other altering the diffusion rate.

Methods of Doping: Ion implantation



Ion implantation is more expensive and complex. It does not require high temperatures and also allows for greater control of dopant concentration and profile. It is an anisotropic process and therefore does not spread the dopant implant as much as diffusion. This aids in the manufacture of self-aligned structures which greatly improve the performance of MOS transistors.

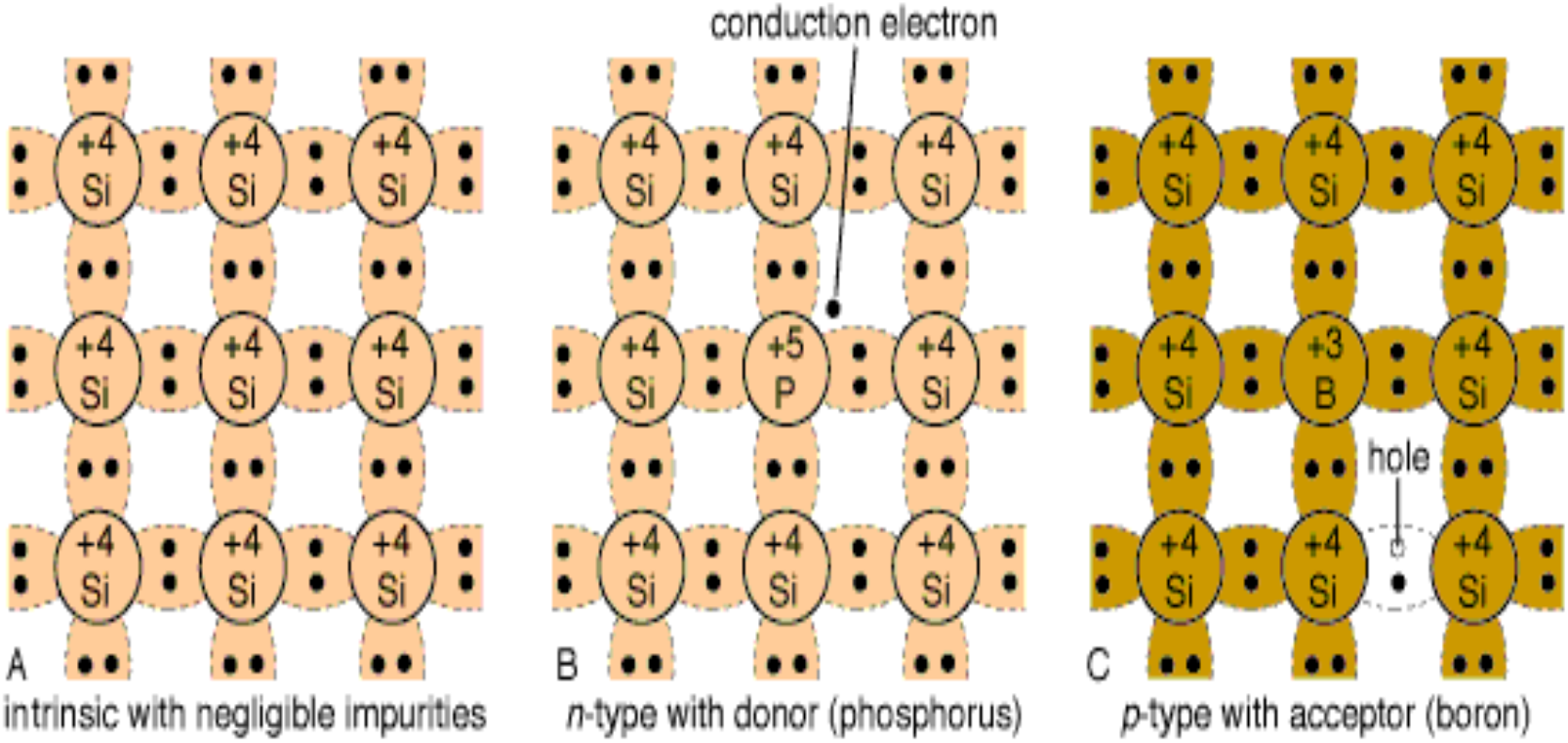
Example: N-type Si (Phosphorus doping)



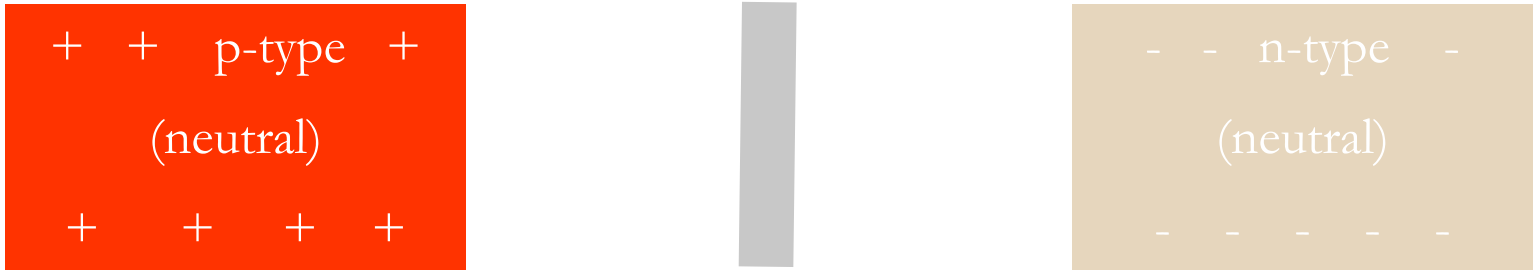
$$4POCl_3 + 3O_2 \rightarrow 2P_2O_5 + 6Cl_2$$

$$2P_2O_5 + 5Si \rightarrow 4P + 5SiO_2$$

Summary


conduction electron
+4 Si
+5 P
+3 B
hole
A
intrinsic with negligible impurities
B
n-type with donor (phosphorus)
C
p-type with acceptor (boron)

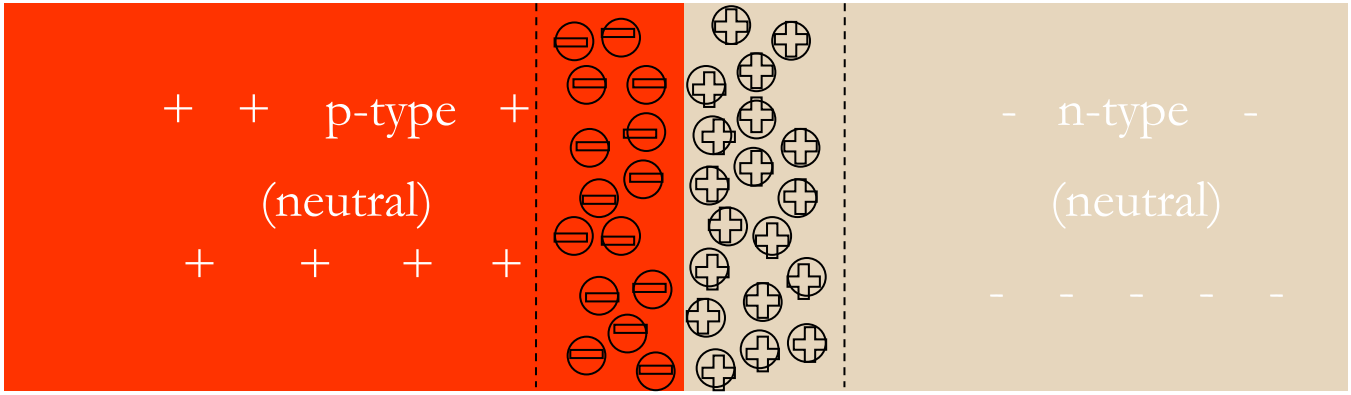
The PN junction



The basic structure of almost all active devices

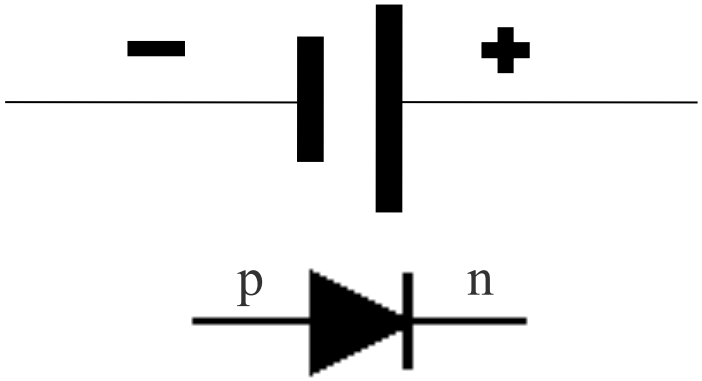
Diodes, transistors, lasers, photocells, solar cells, etc.

The PN junction

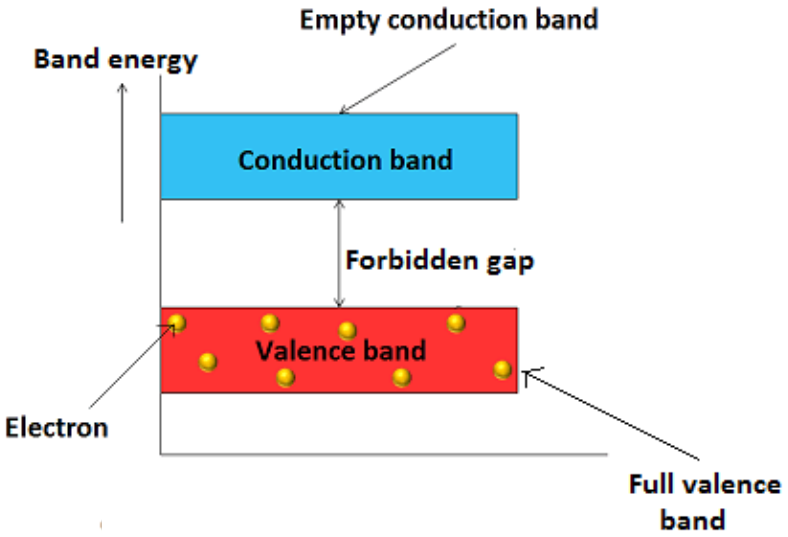


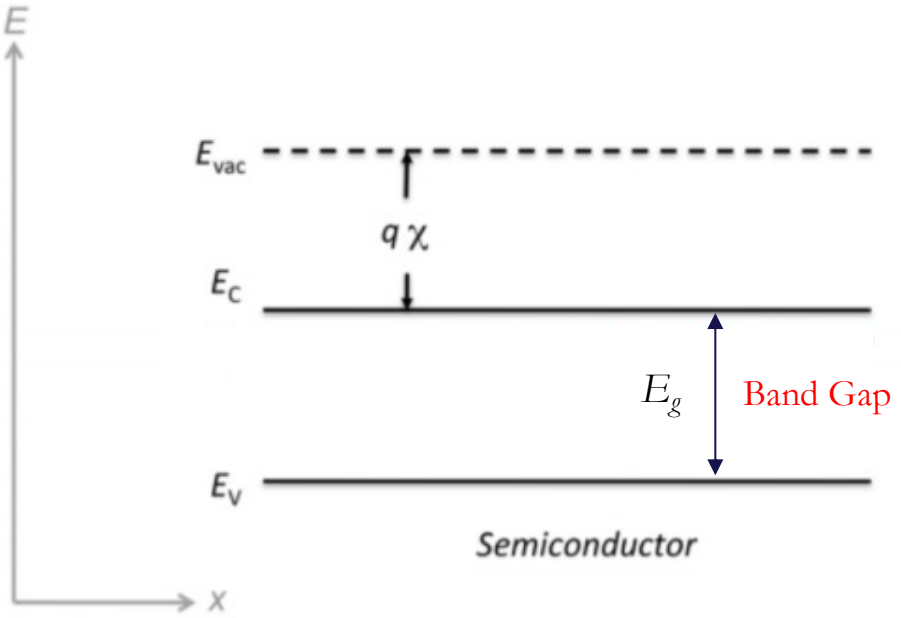
Depletion Region

Built-in Voltage (V_{bi})



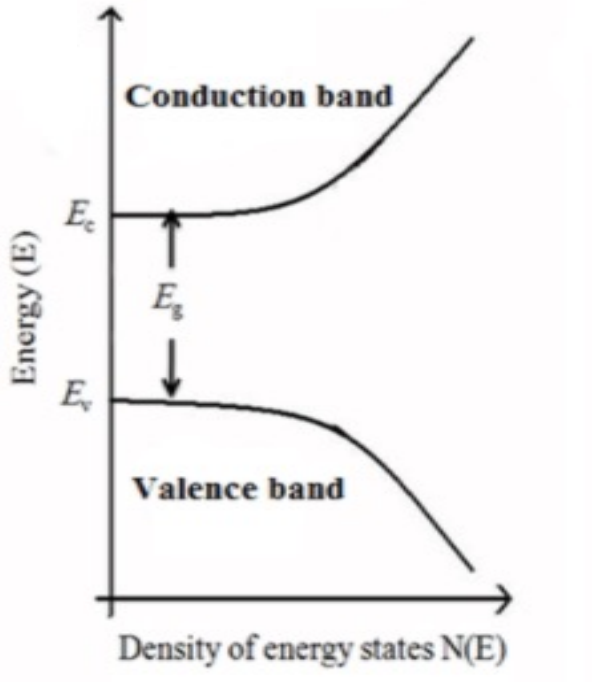
Back to the Energy Bands

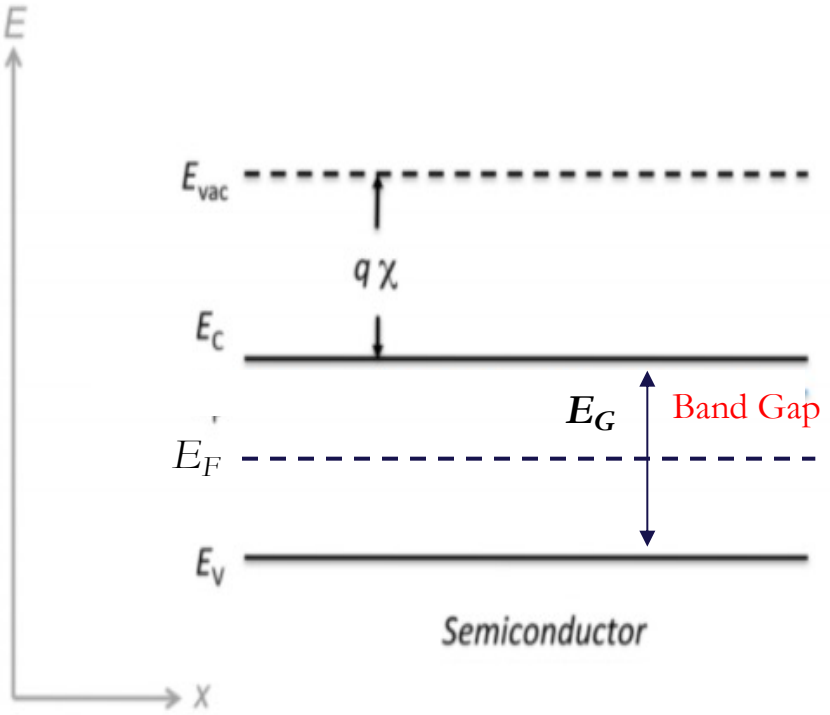




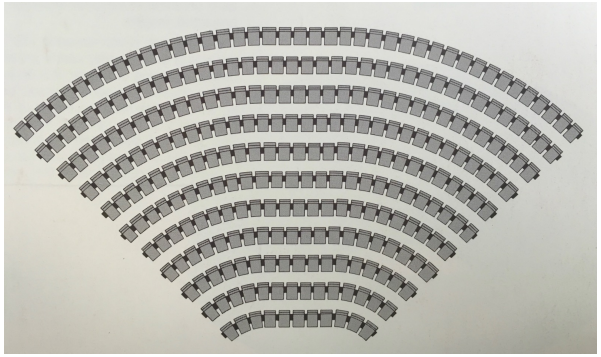
Band Diagram

Back to the Energy Bands



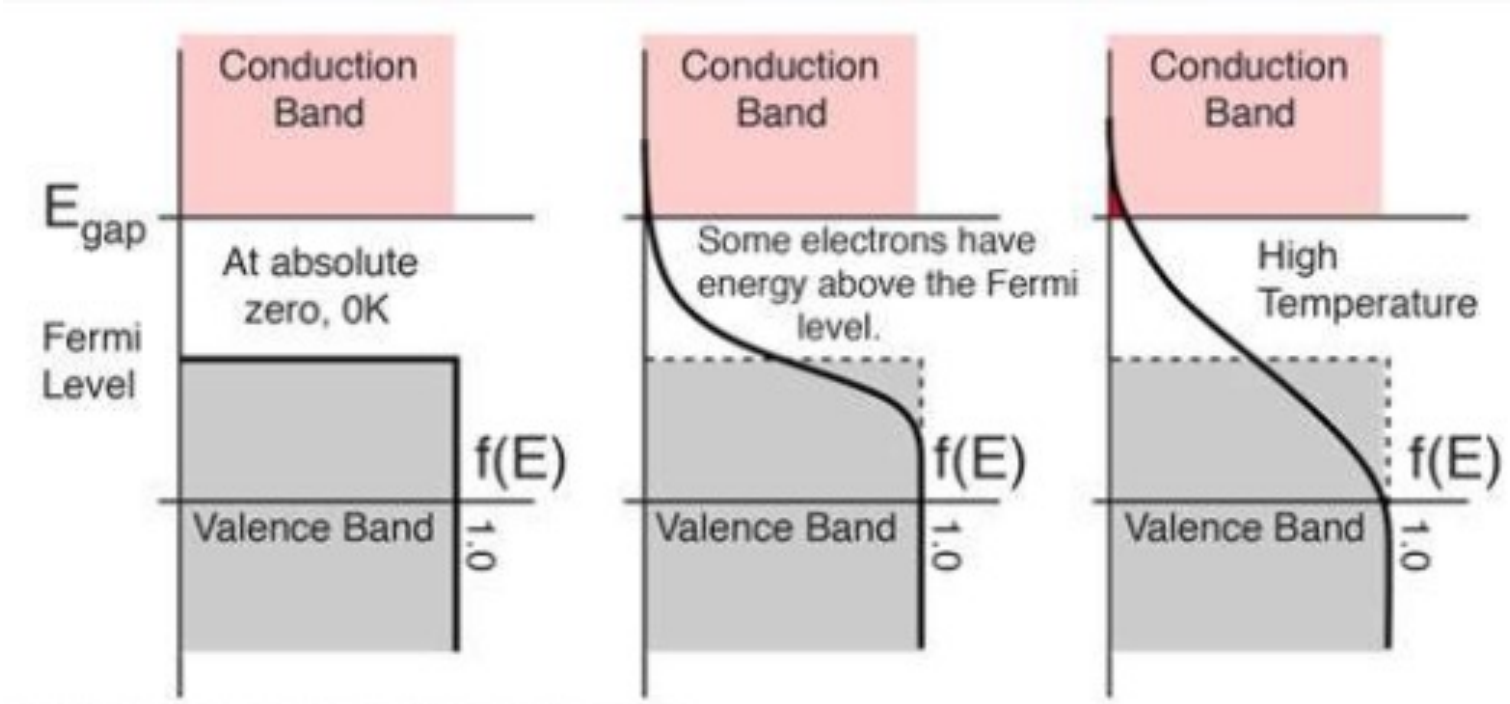


Density of States

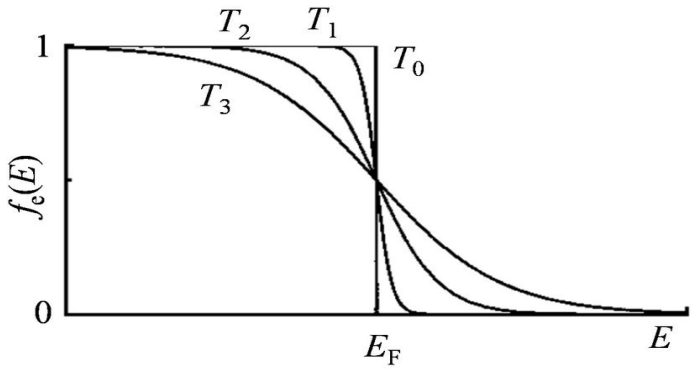


Band Diagram

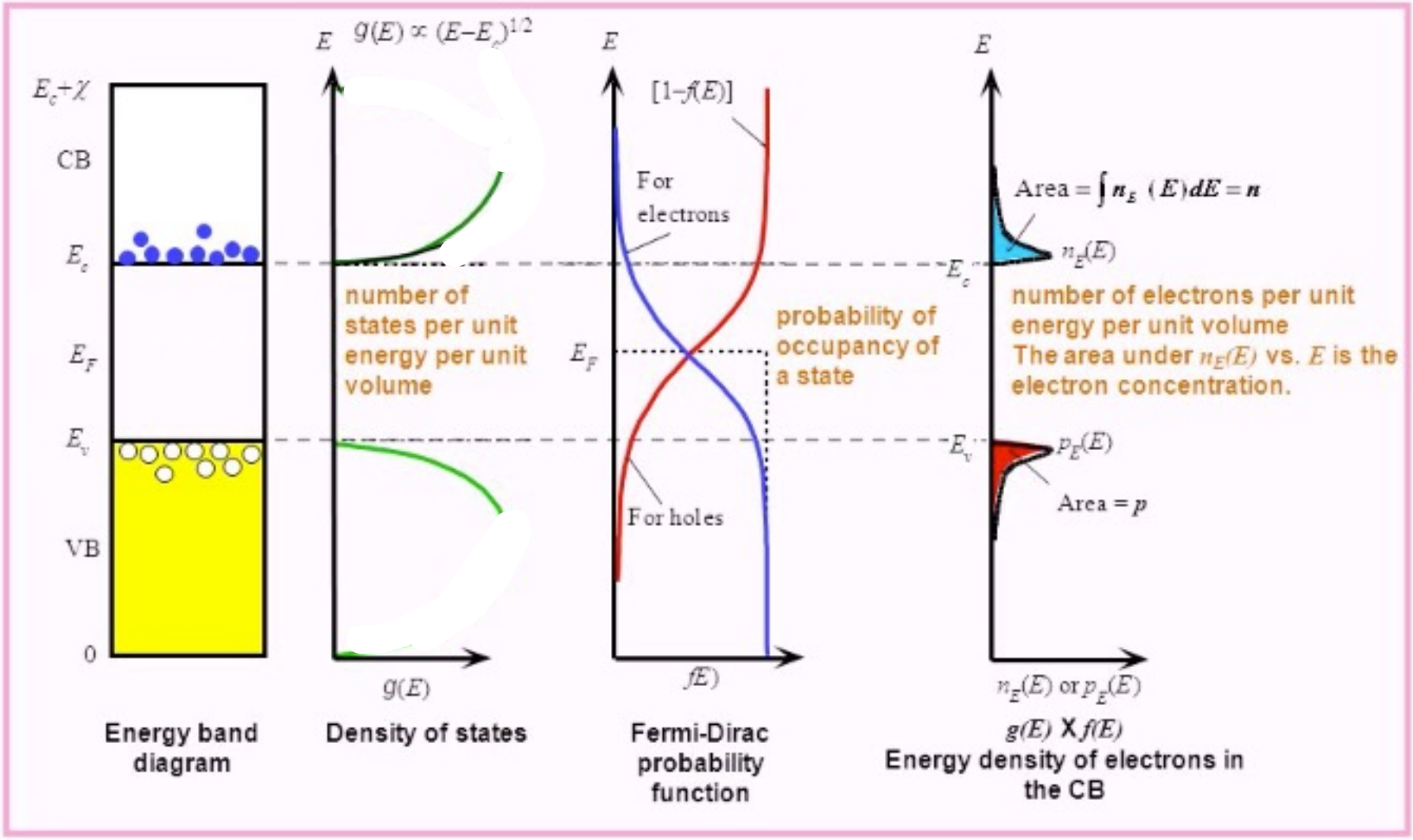
Fermi Distribution



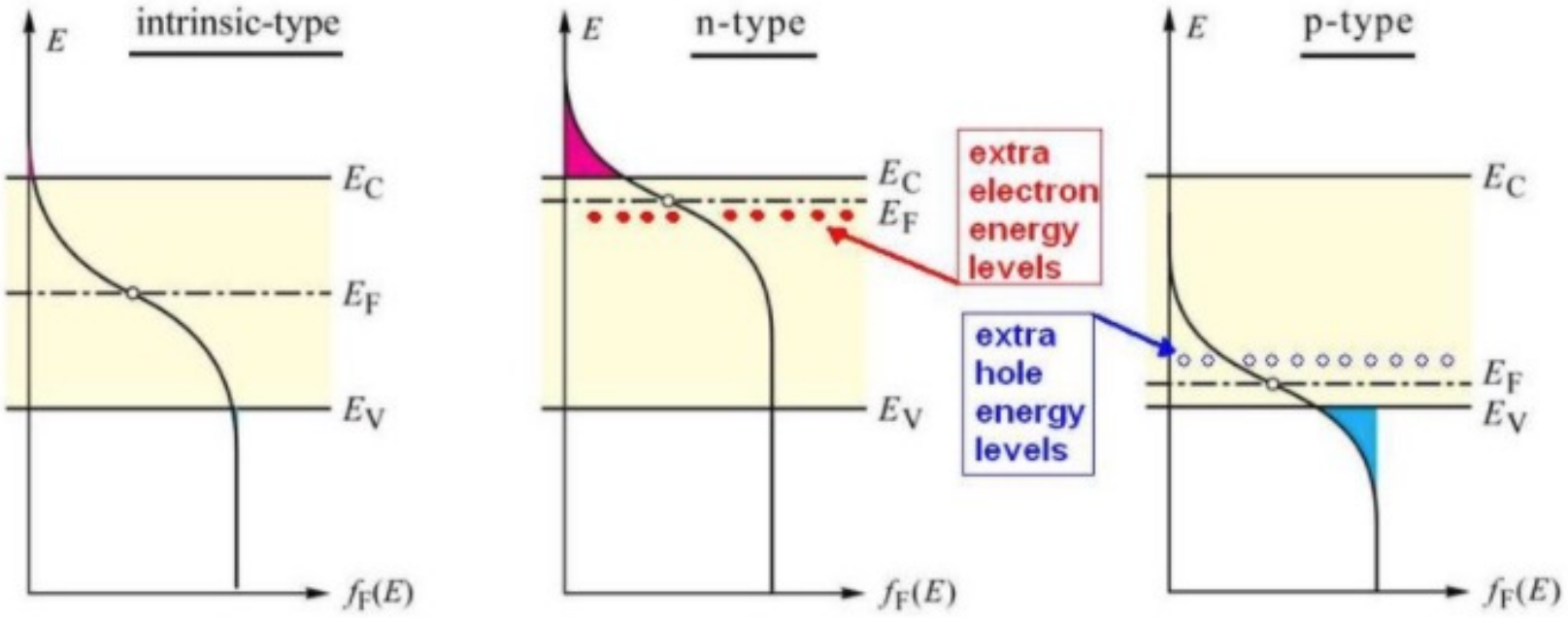
$$f_F(E) = \frac{1}{1 + exp\left(\frac{E - E_F}{kT}\right)}$$



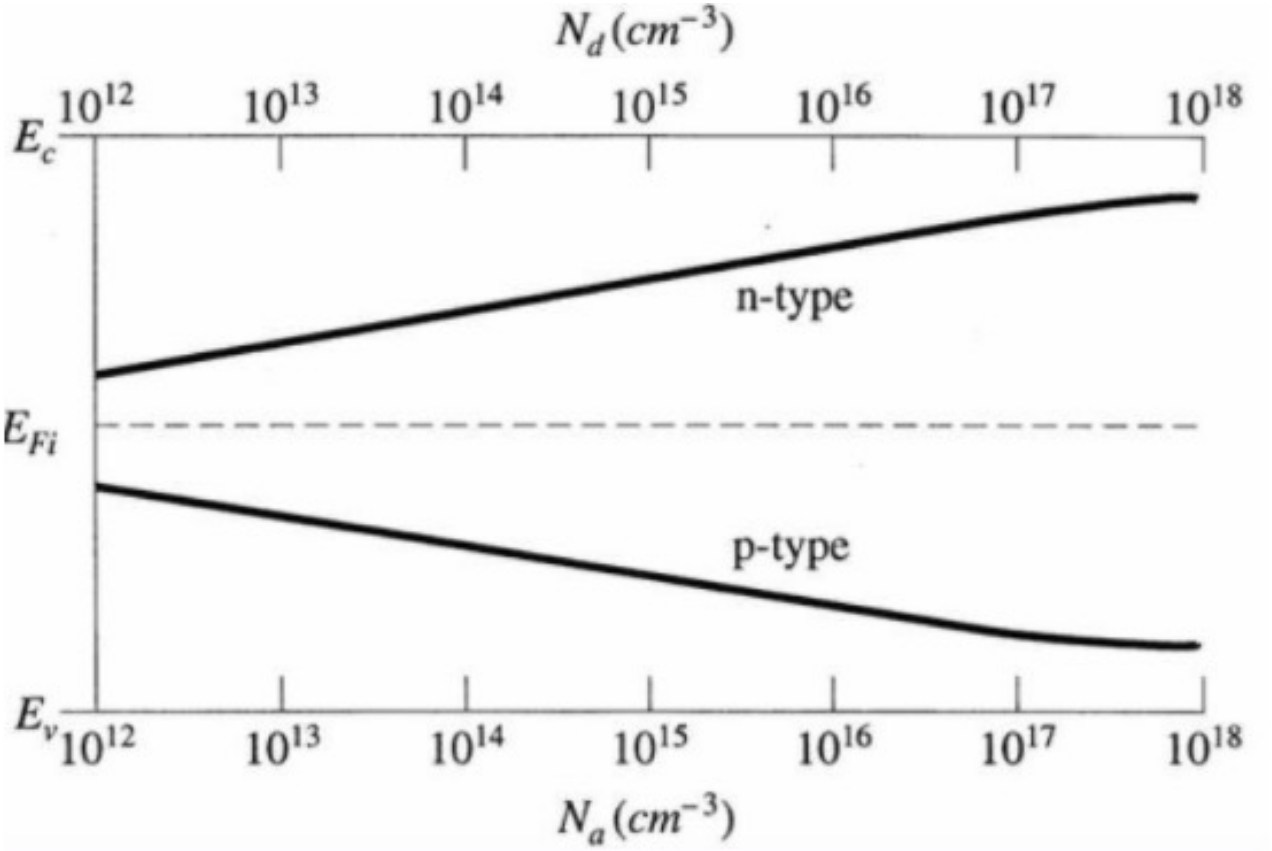
Electron/Hole Density



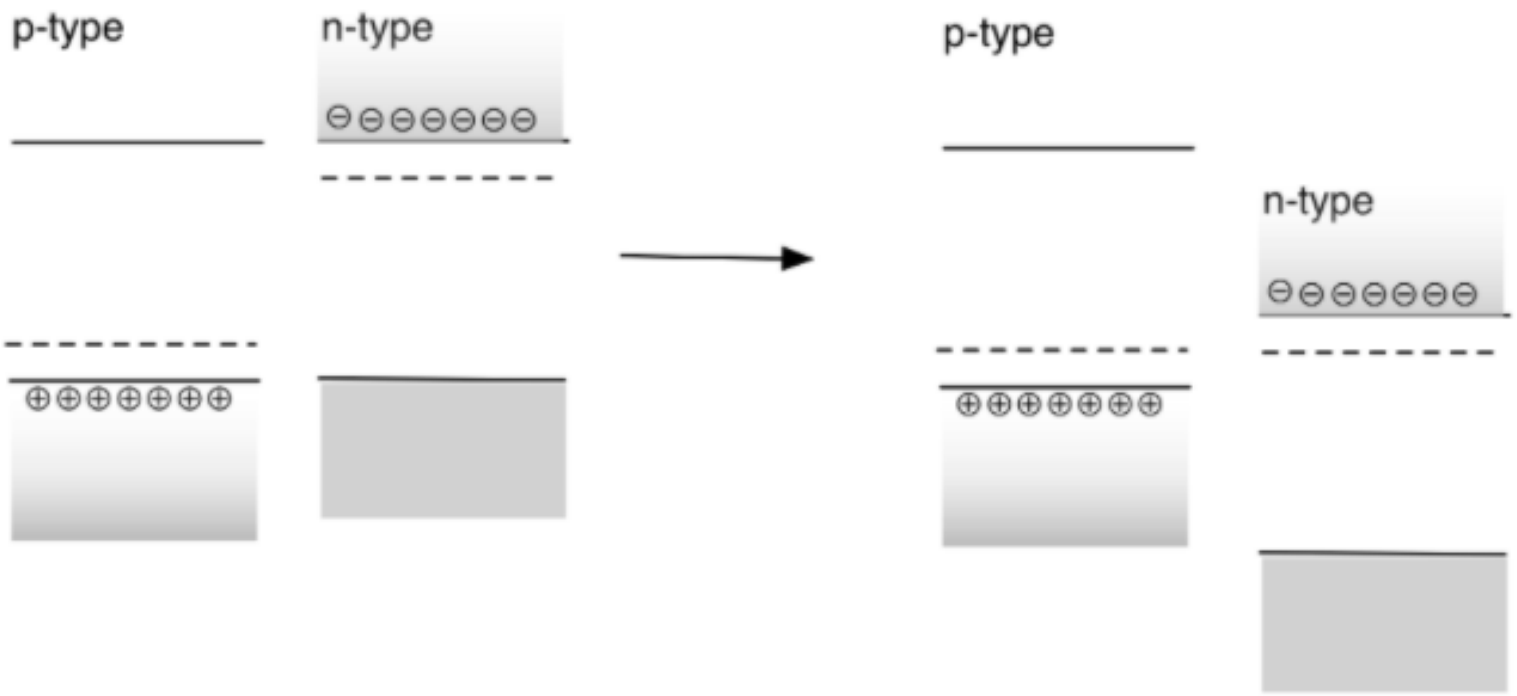
The n-type and p-type Band Diagram



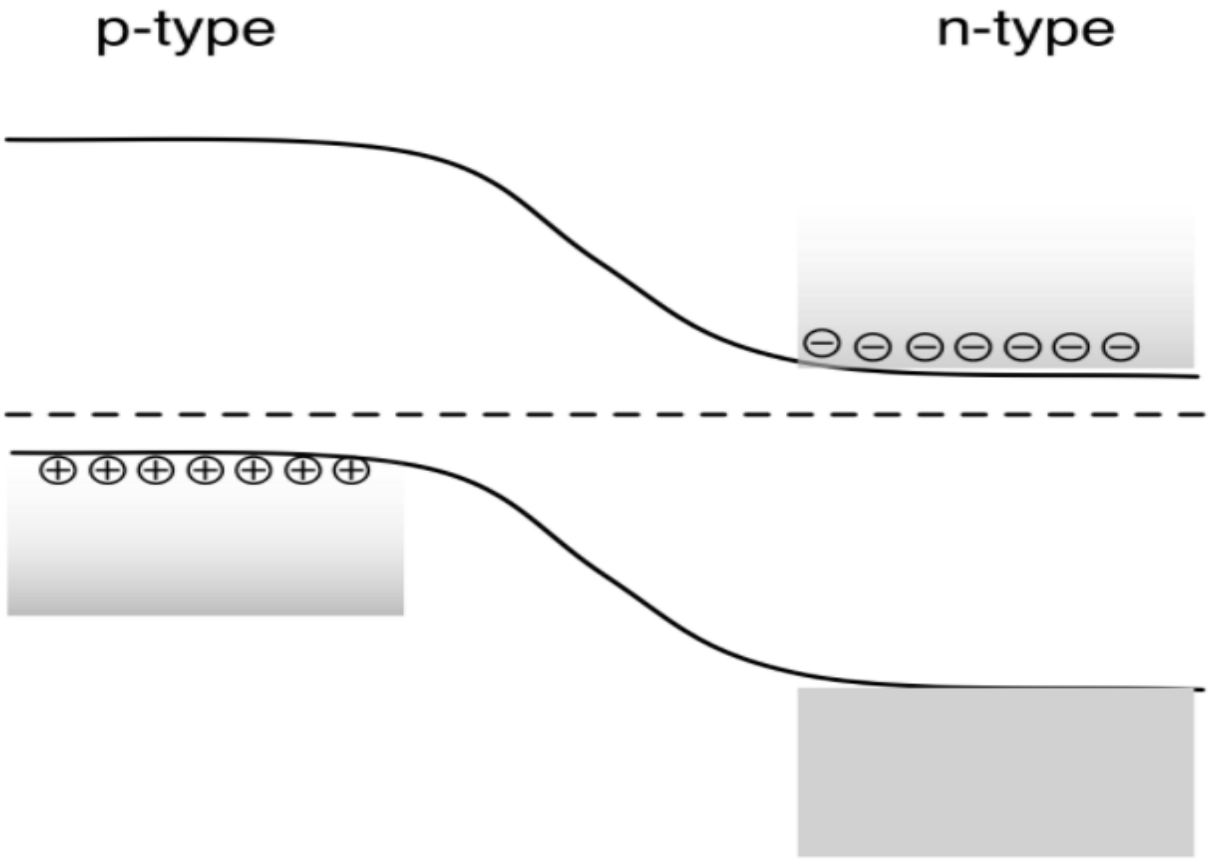
The n-type and p-type Band Diagram



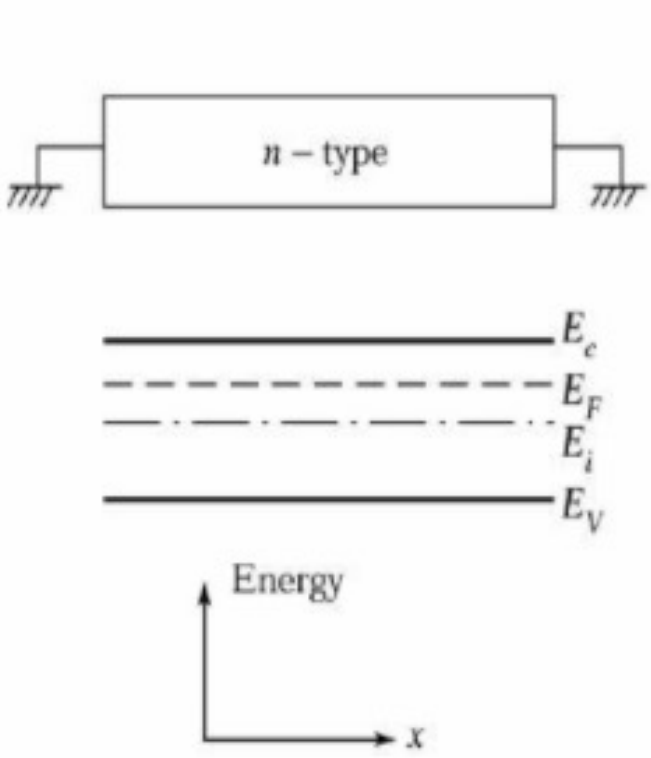
The PN Junction Band Diagram

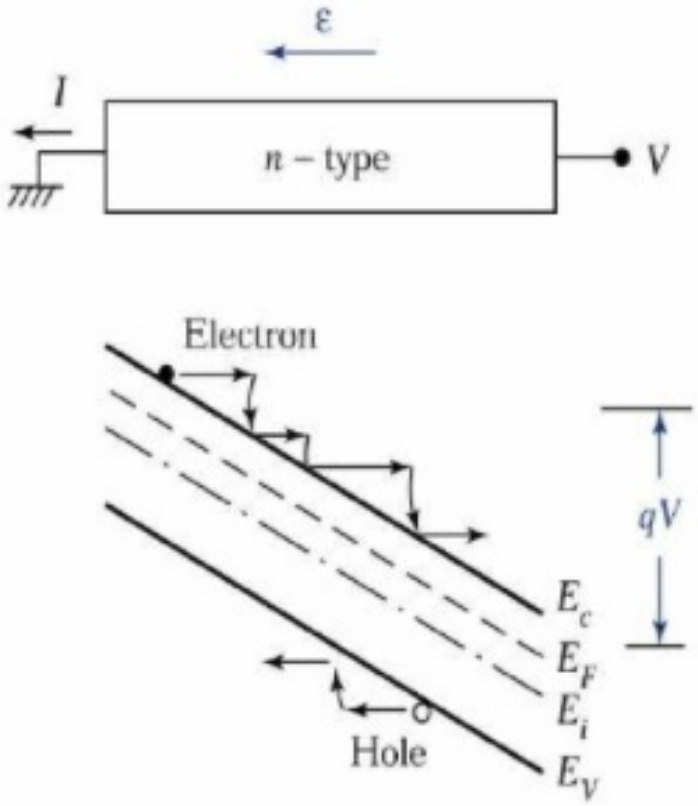


Unbiased PN Junction Band Diagram

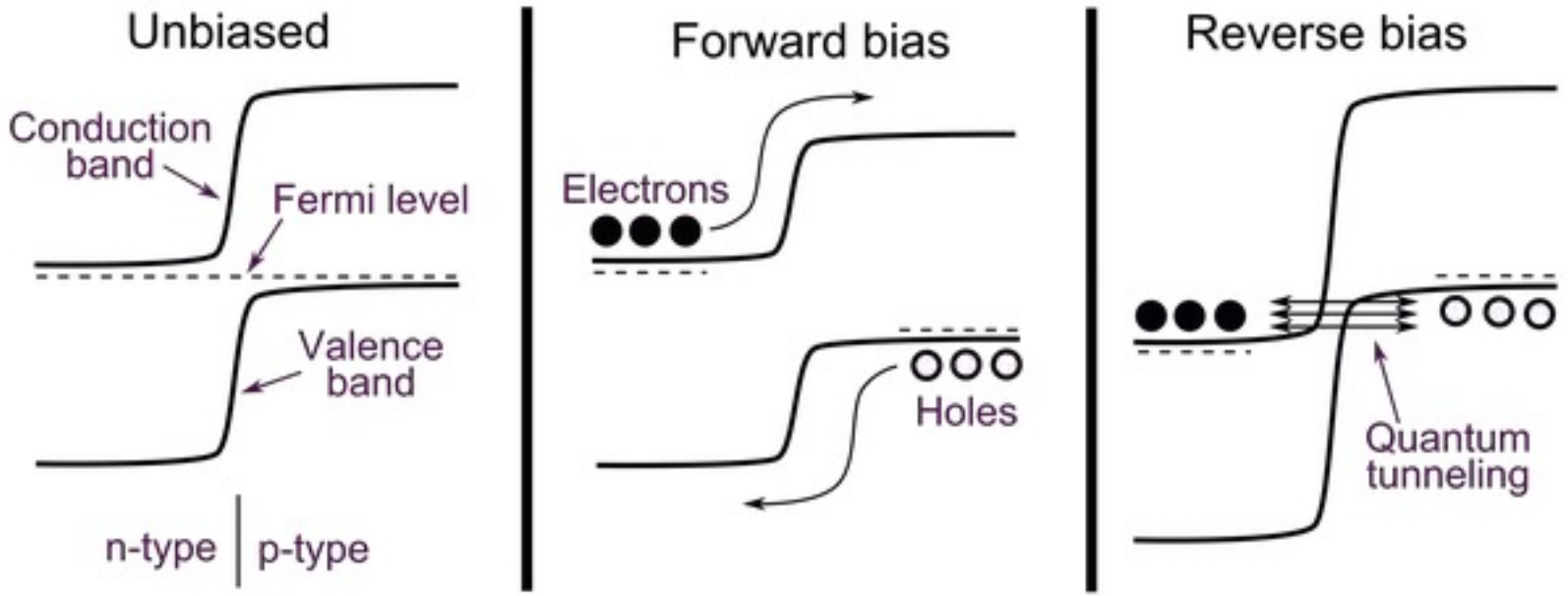


Biased n-type Band Diagram



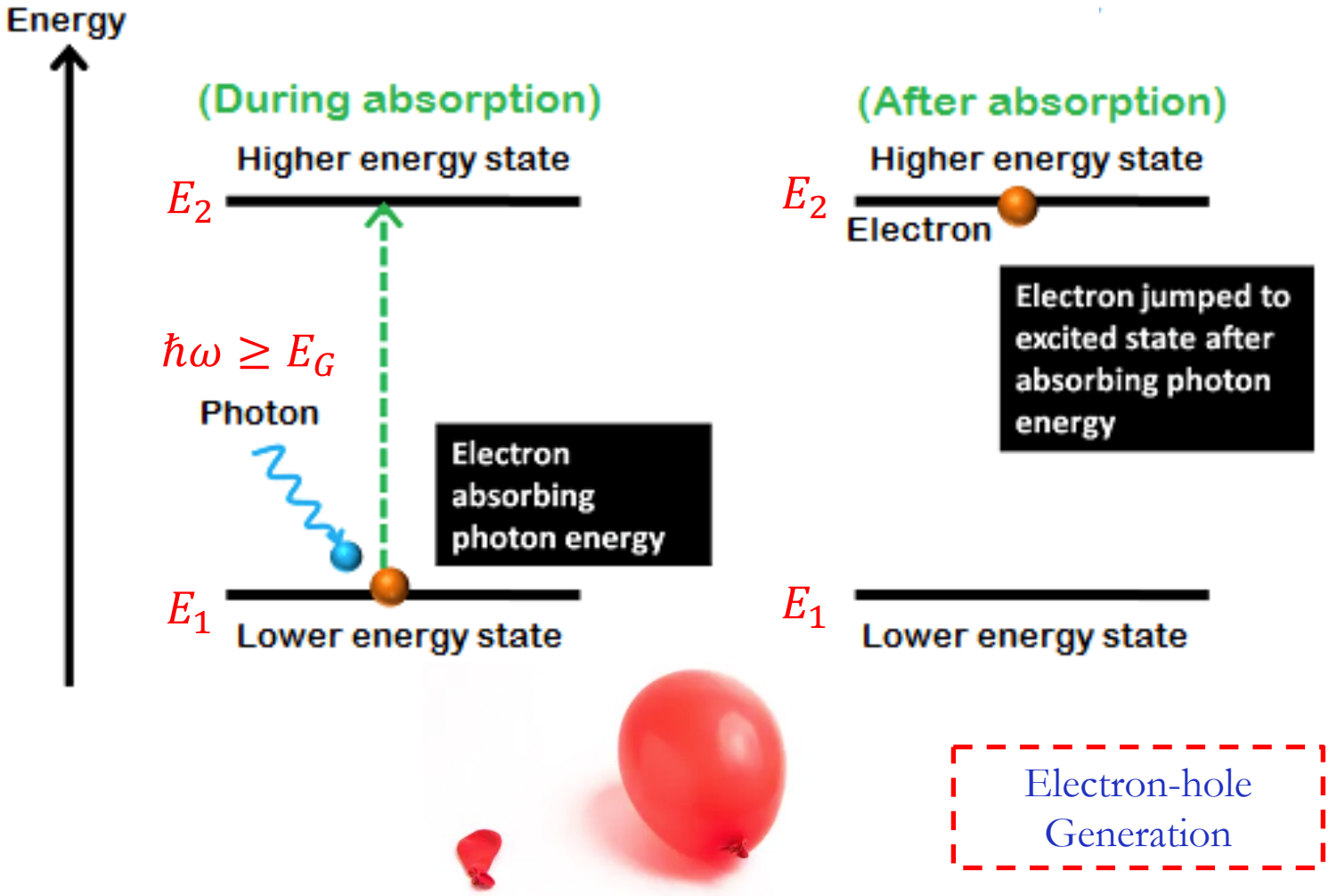


Biased PN Junction Band Diagram



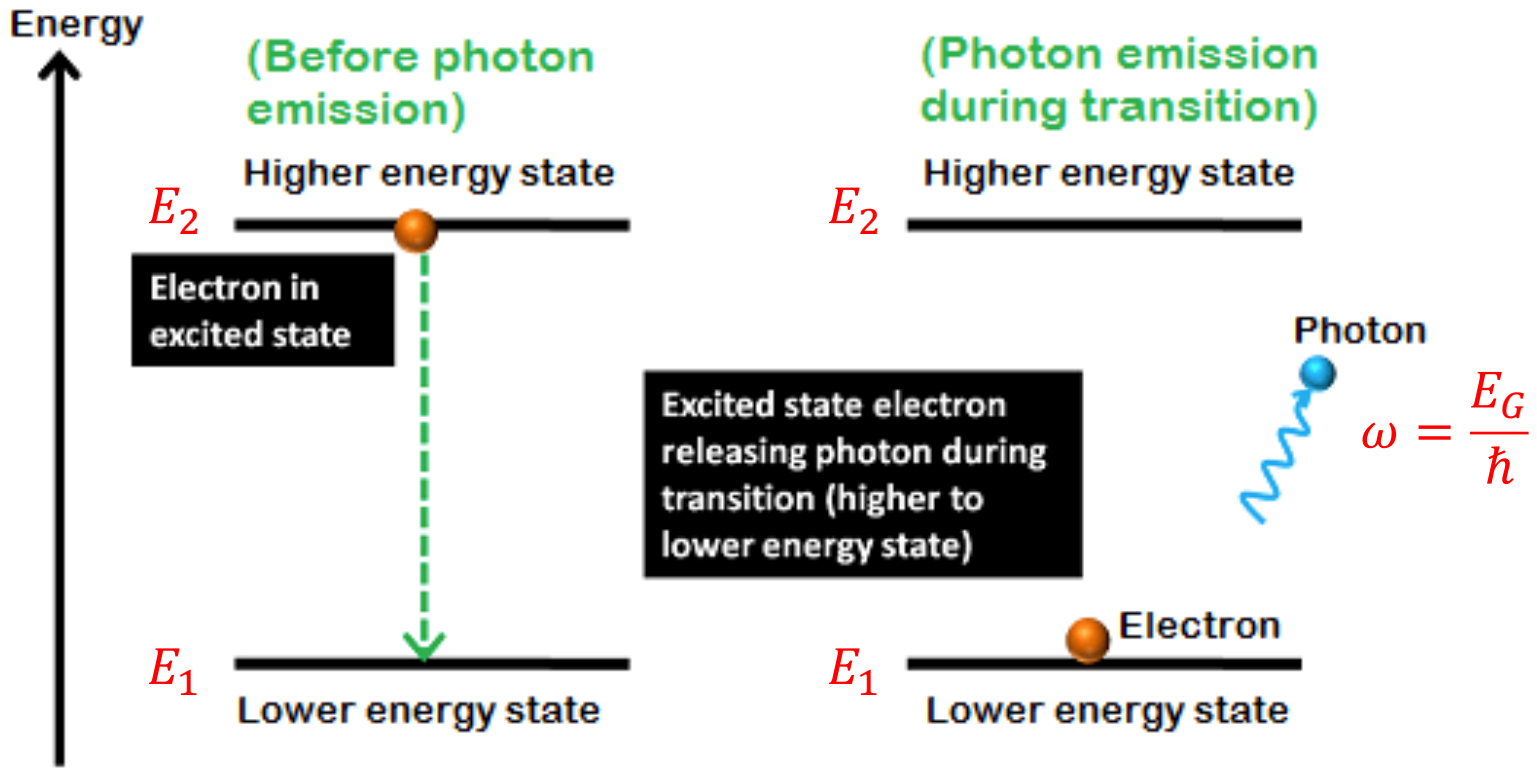
Photon Generation and Absorption

Absorption



Photon Generation and Absorption

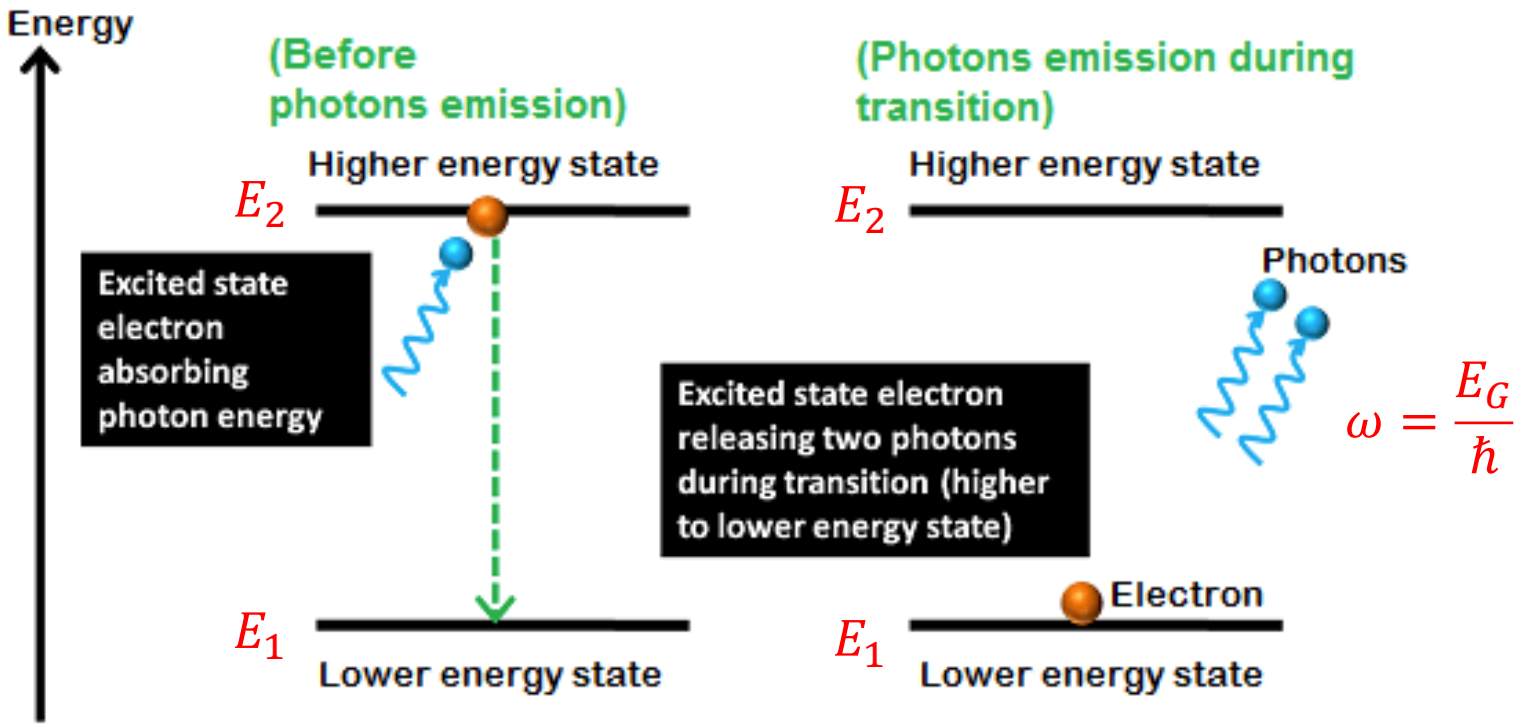
Spontaneous Emission



Electron-hole Re-combination (radiative/nonradiative)

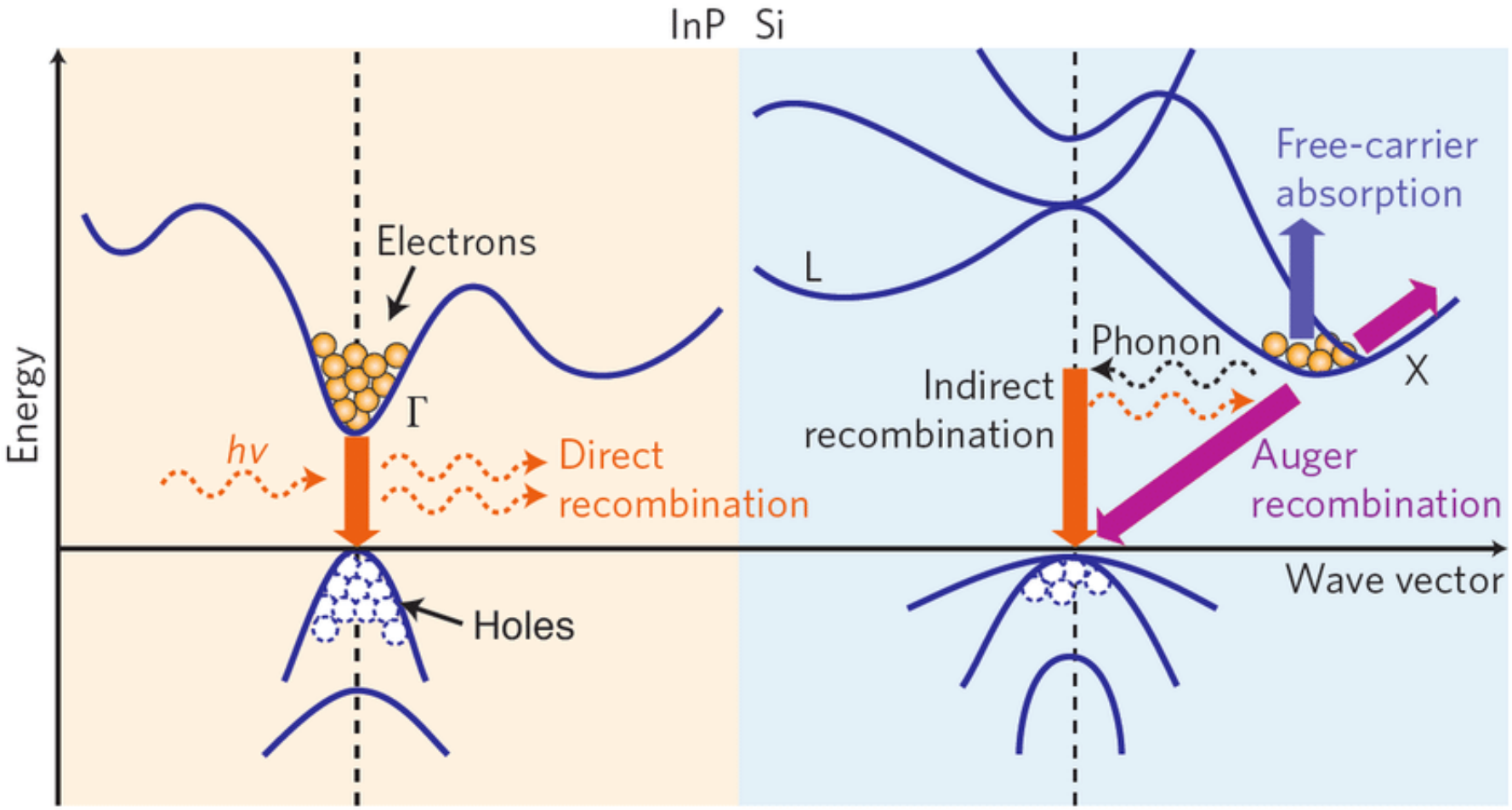
Photon Generation and Absorption

Stimulated Emission



Electron-hole
Re-combination
(radiative/nonradiative)

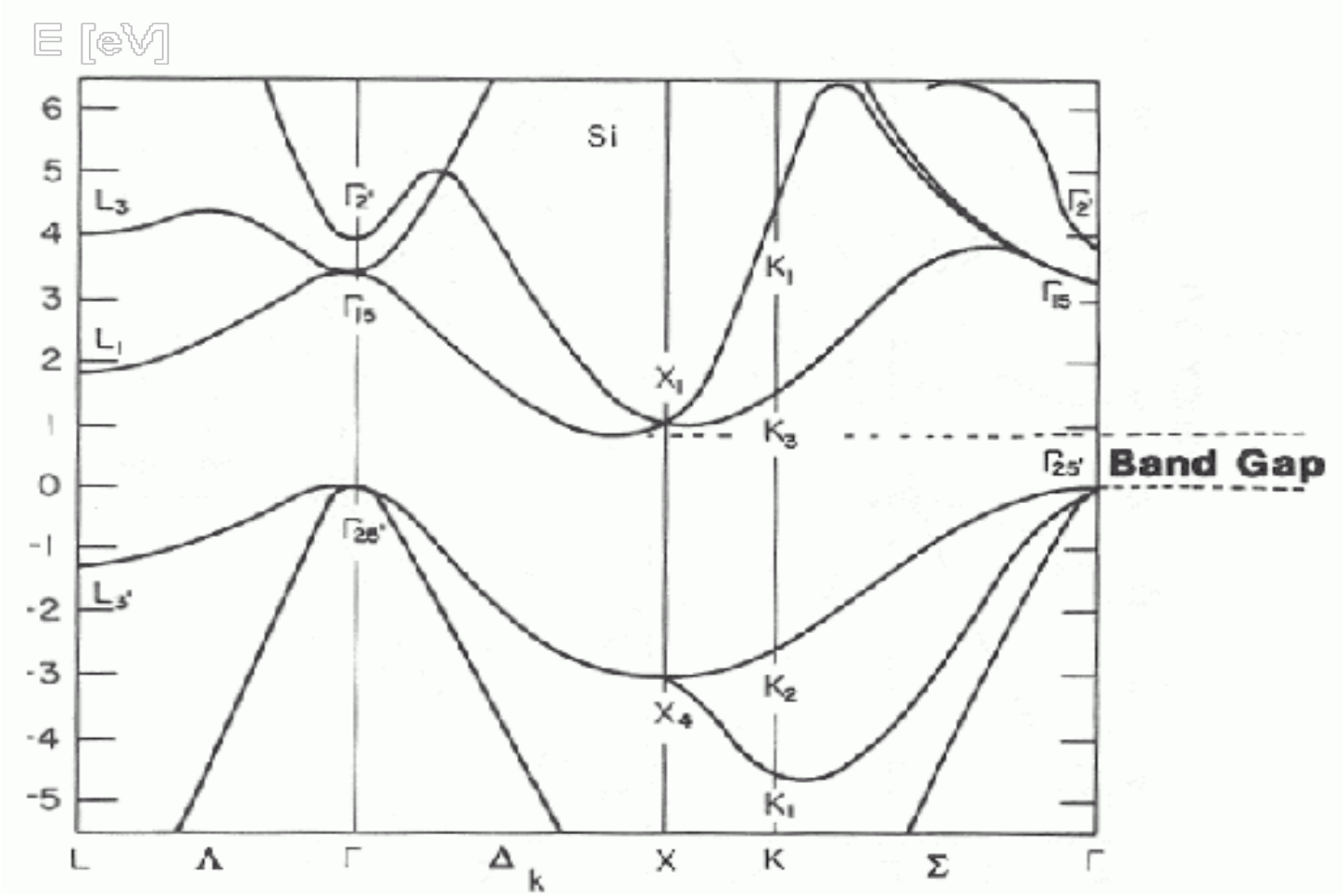
Direct and Indirect Gap Materials



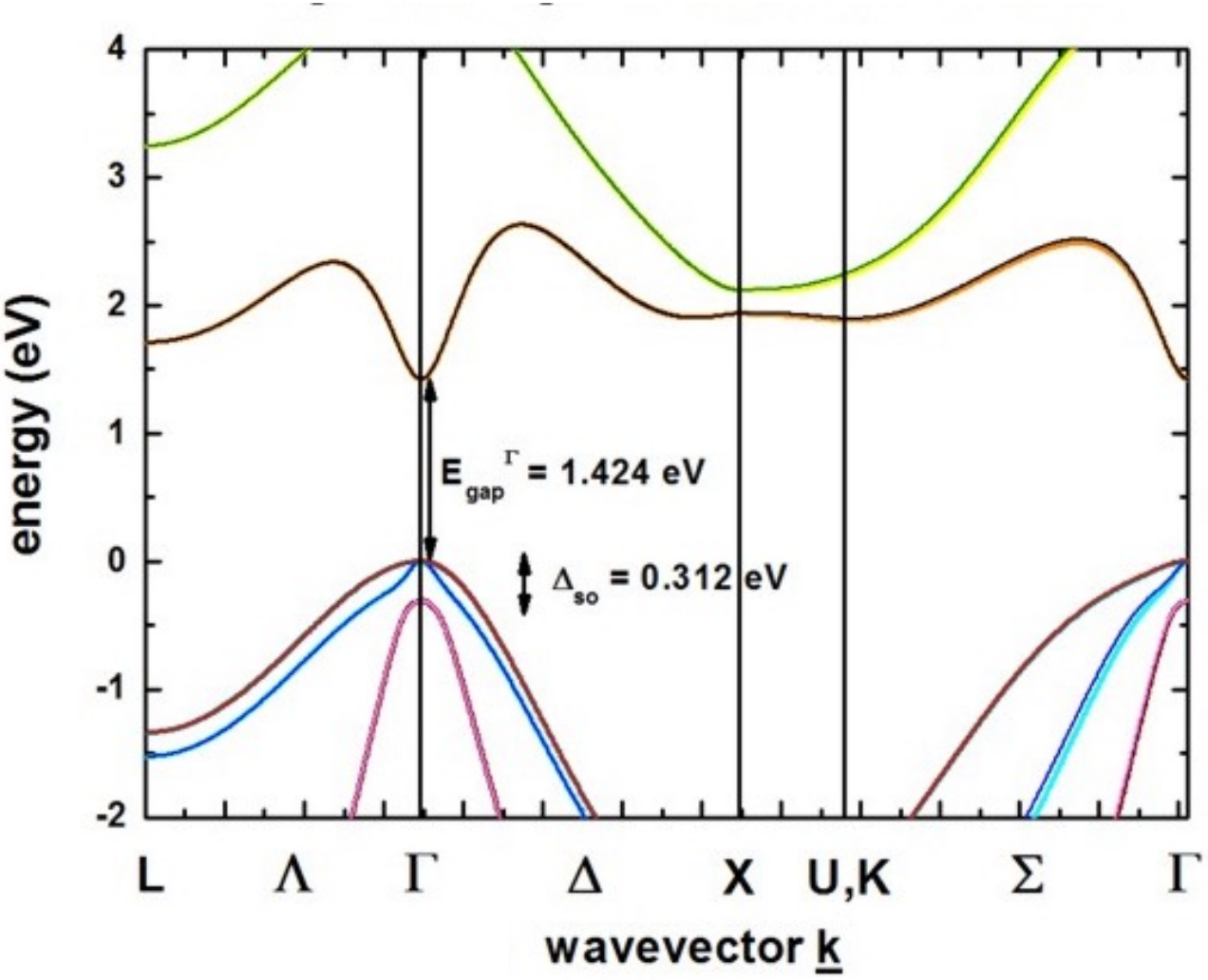
Direct and Indirect Gap Materials

	Material	Direct / IndirectBandgap	Band Gap Energy at 300 K (eV)
Elements	C (di	Indirect	5.47
	Ge	Indirect	0.66
	Si	Indirect	1.12
	Sn	Direct	0.08
Groups III-V compounds	GaAs	Direct	1.42
	InAs	Direct	0.36
	InSb	Direct	0.17
	GaP	Indirect	2.26
	GaN	Direct	3.36
	InN	Direct	0.70
	InP	Direct	1.344
Groups IV-IV compounds	α-SiC	Indirect	2.99
Groups II-VI compounds	ZnO	Direct	3.35
	CdSe	Direct	1.70
	ZnS	Direct	3.68

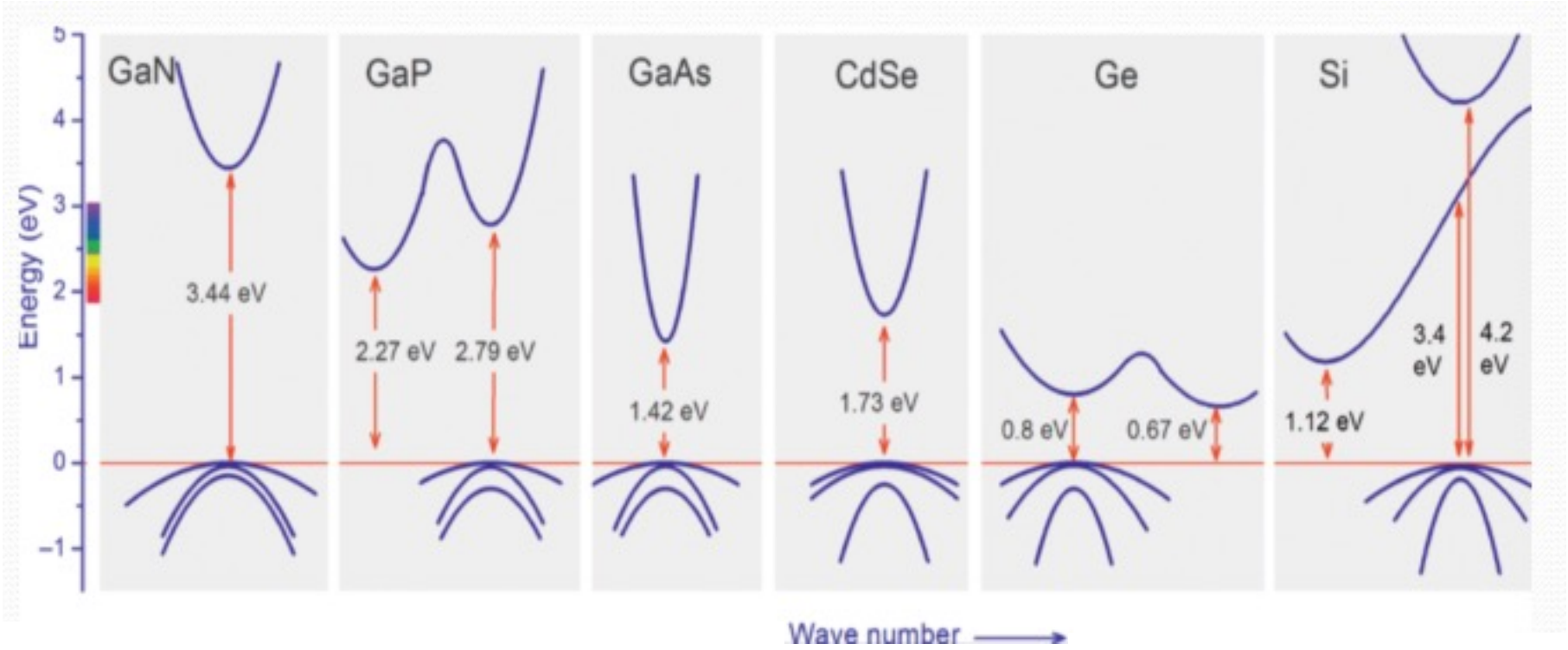
Si: Indirect Gap Material



GaAs: Direct Gap Material



GaAs: Direct Gap Material



Bandgap Engineering

Example: Emission in GaAs, GaN and InN.

$$\omega = \frac{E_G}{\hbar}$$

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega}$$

$$\lambda = \frac{2\pi c\hbar}{E_G}$$

$$\lambda_g = \frac{hc}{E_G}$$

GaAs band gap energy E_G=1.42 eV

$$\lambda_g = \frac{2\pi(3\times10^8)(1.055\times10^{-34})}{(1.42)(1.6\times10^{-19})}$$

$\lambda_g = 875\ nm$ NIR

GaN band gap energy E_G=3.49 eV

$\lambda_g = 356\ nm$ UV

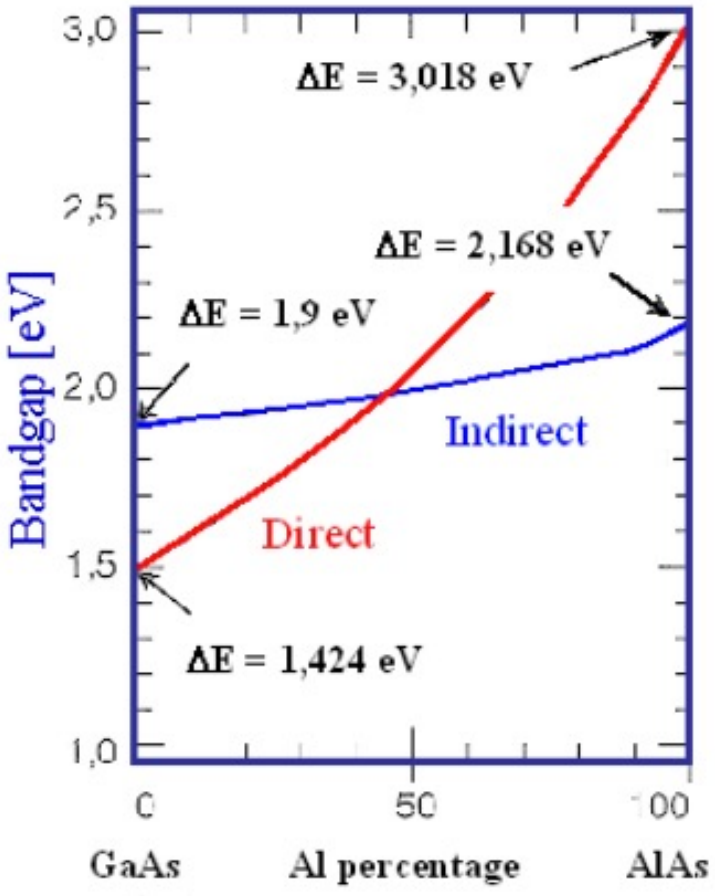
InN band gap energy E_G=0.69 eV

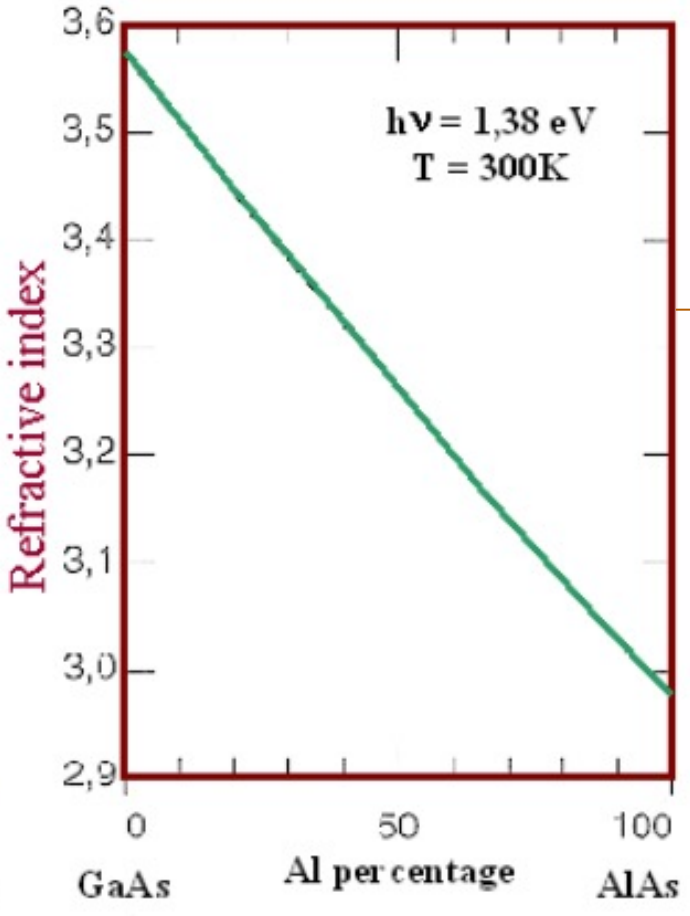
$\lambda_g = 1800\ nm$ IR

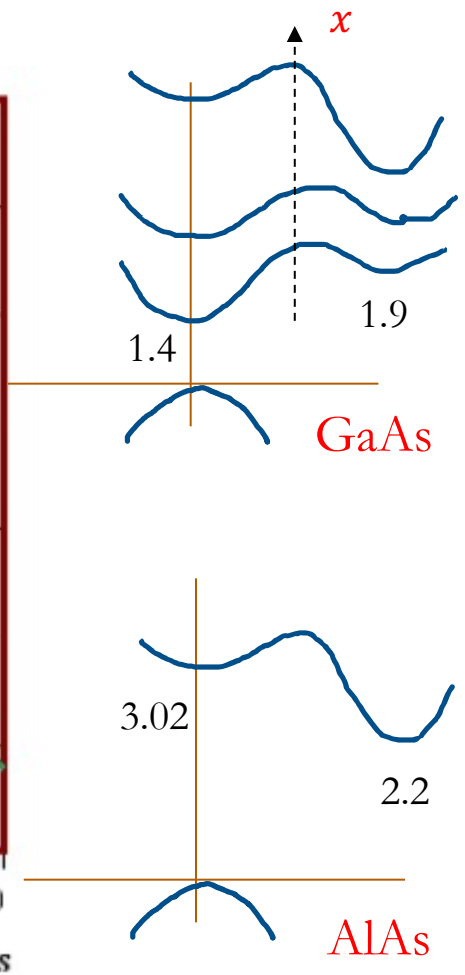
The alloy $In_xGa_{1-x}N$ is very useful for spanning the visible range.

Bandgap Engineering

Example: Emission in $Al_xGa_{1-x}As$.





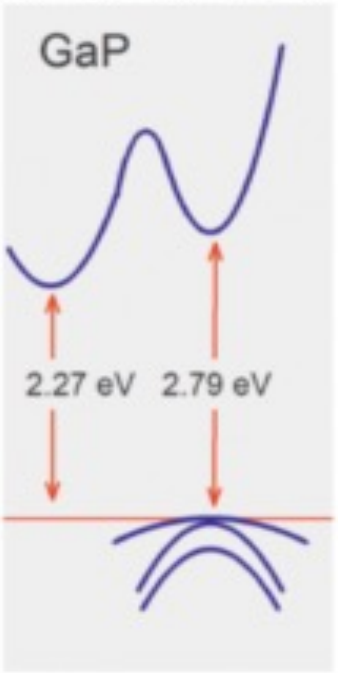


The Green Gap

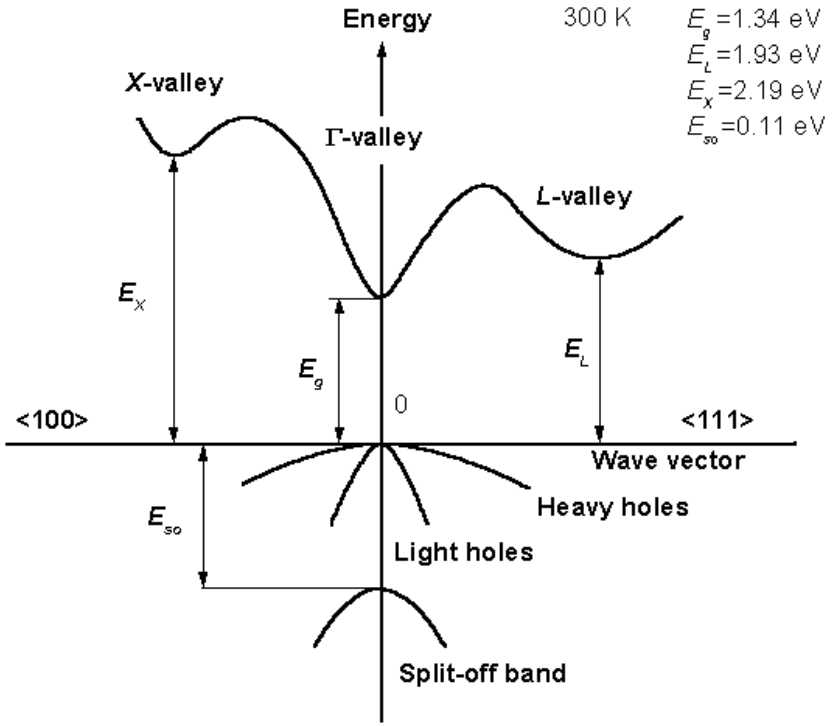
Green Band: 530-570 nm.

Green Band: 2.2-2.3 eV.

Emission in $Ga_xIn_{1-x}P$.



InP

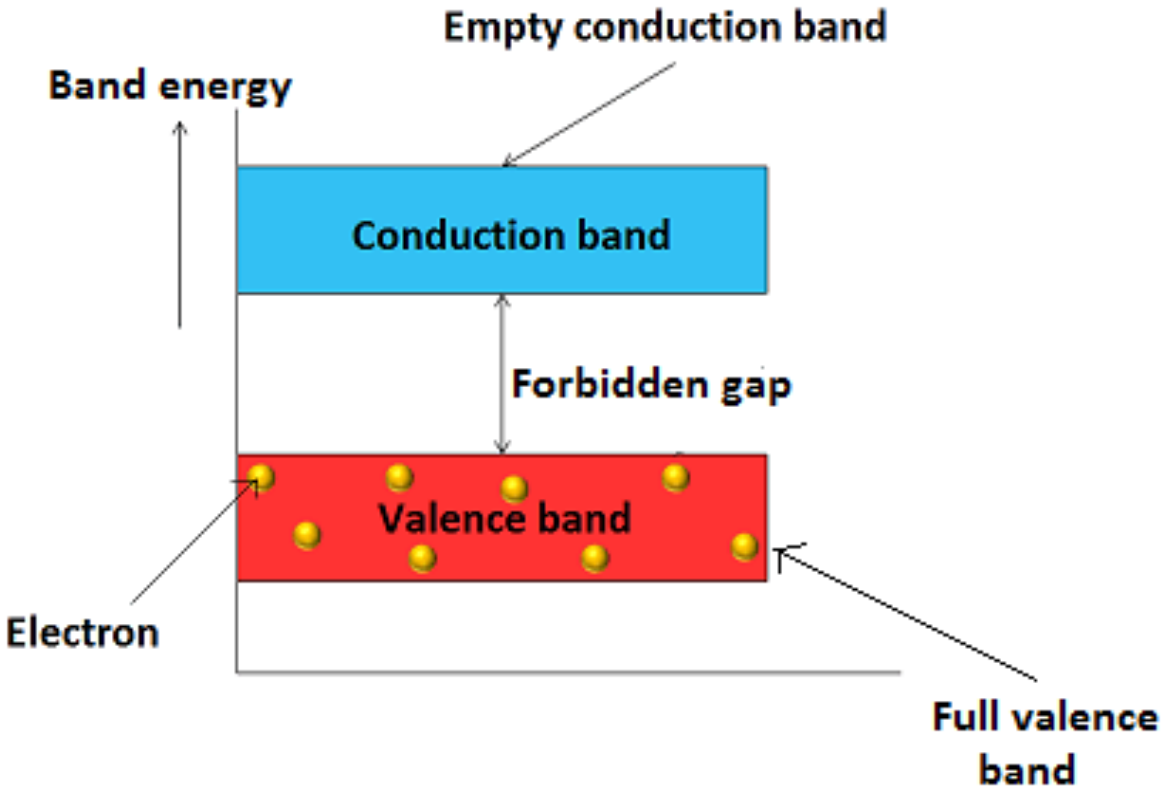


Semiconductors Fundamentals

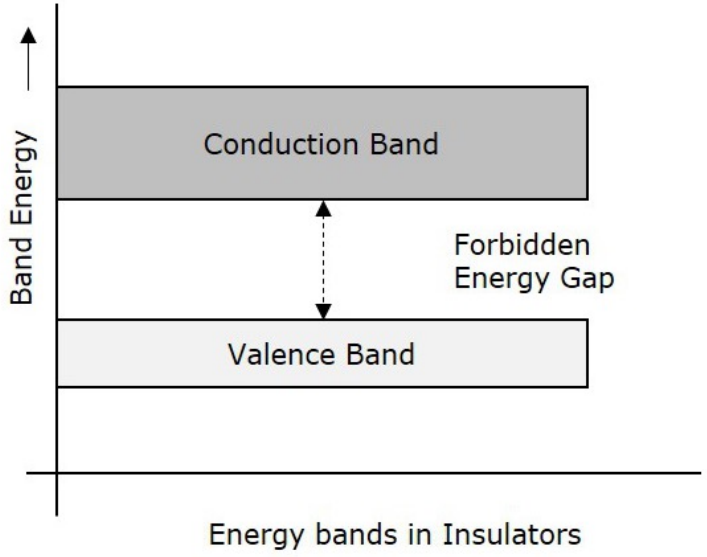
The following subjects will be covered

- Semiconductors in General.
- The PN Junction.
- The Energy Band Diagram.
- Carrier Statistics in Semiconductors.
- PN Junction Band Diagram.
- Photon Generation and Absorption.
- Direct and Indirect Gap Materials.
- Examples.

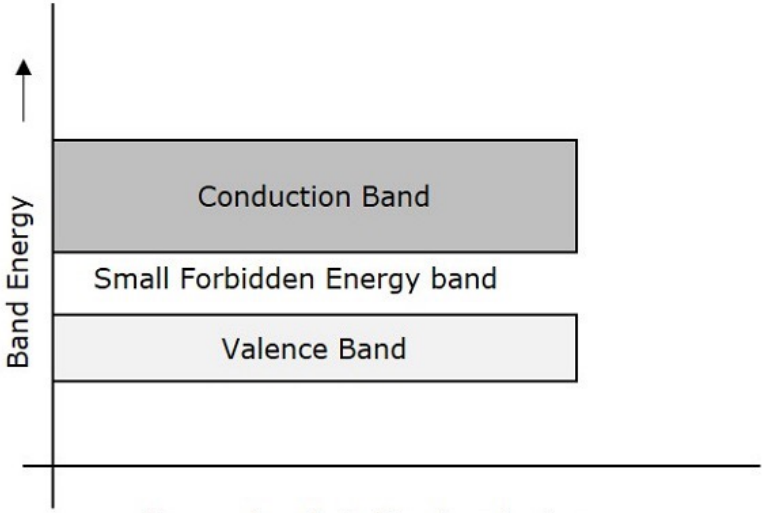
Energy Bands: Isolated atoms and Crystals



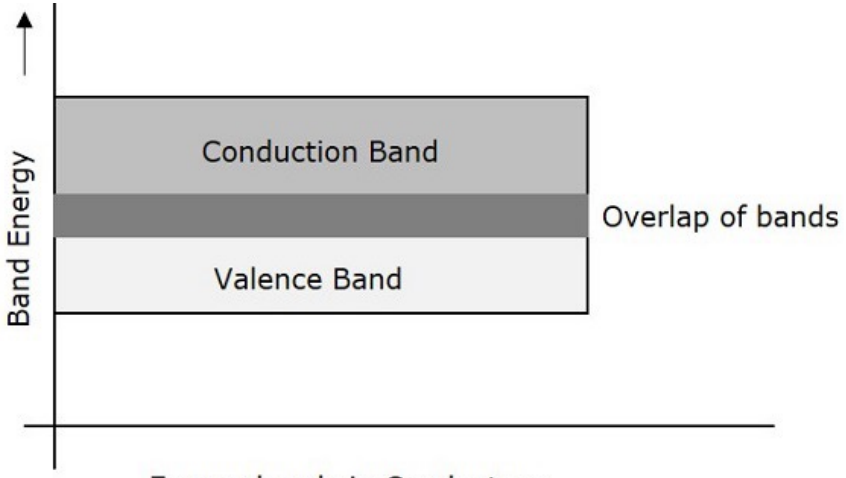
Classification of Materials



Energy bands in Insulators



Energy bands in Semiconductors



Energy bands in Conductors

Examples

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Silicon	E_g= 1.12 eV
Copper	E_g= 0 eV

Semiconductors

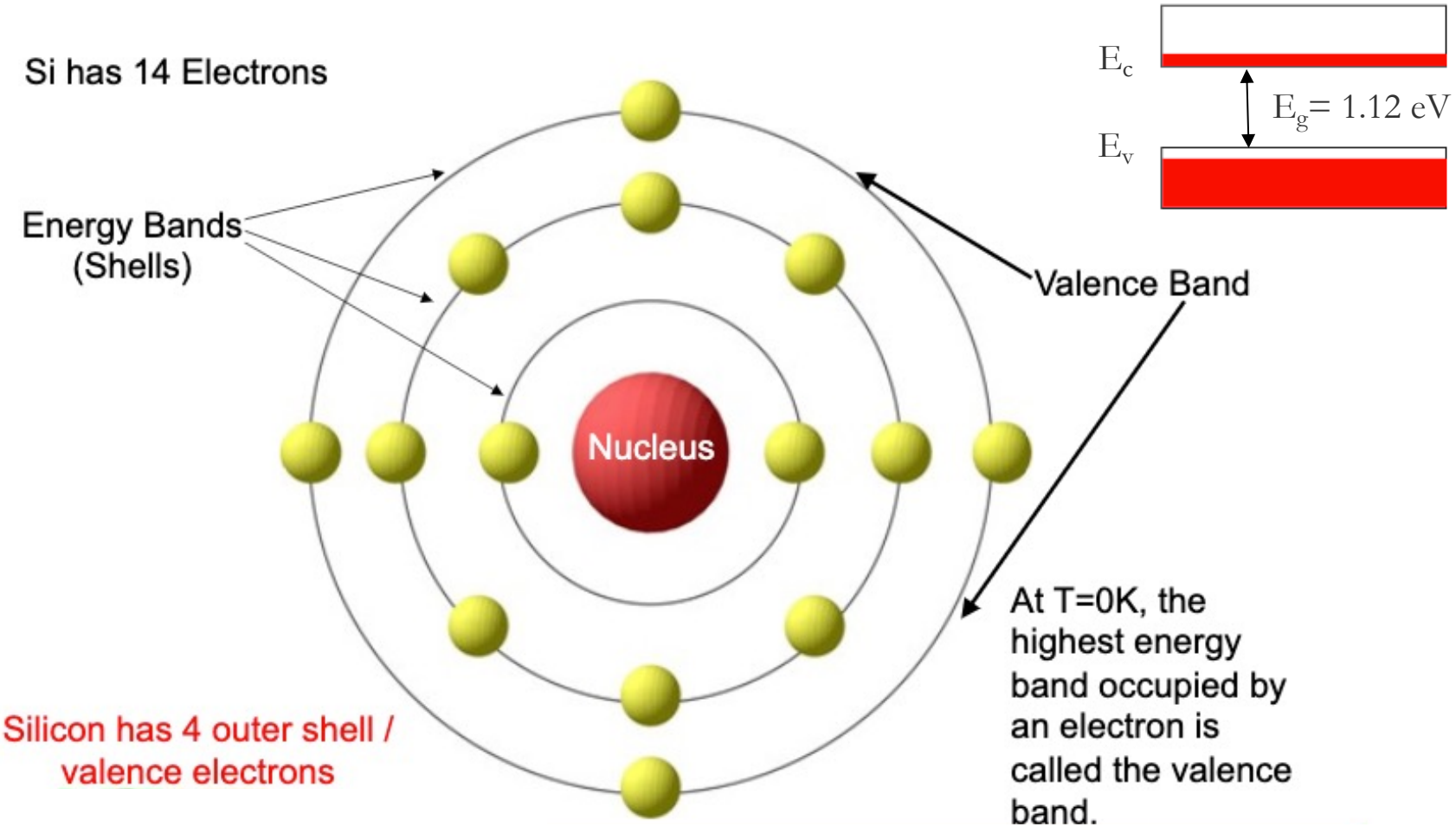
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	IIIA	IVA	VA	VIA
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IIB				
30 65.37 **Zn** Zinc	31 69.72 **Ga** Gallium	32 72.59 **Ge** Germanium	33 74.922 **As** Arsenic	34 78.96 **Se** Selenium
48 112.40 **Cd** Cadmium	49 114.82 **In** Indium	50 118.69 **Sn** Tin	51 121.75 **Sb** Antimony	52 127.60 **Te** Tellurium
80 200.59 **Hg** Mercury	81 204.37 **Tl** Thallium	82 207.19 **Pb** Lead	83 208.980 **Bi** Bismuth	84 (210) **Po** Polonium

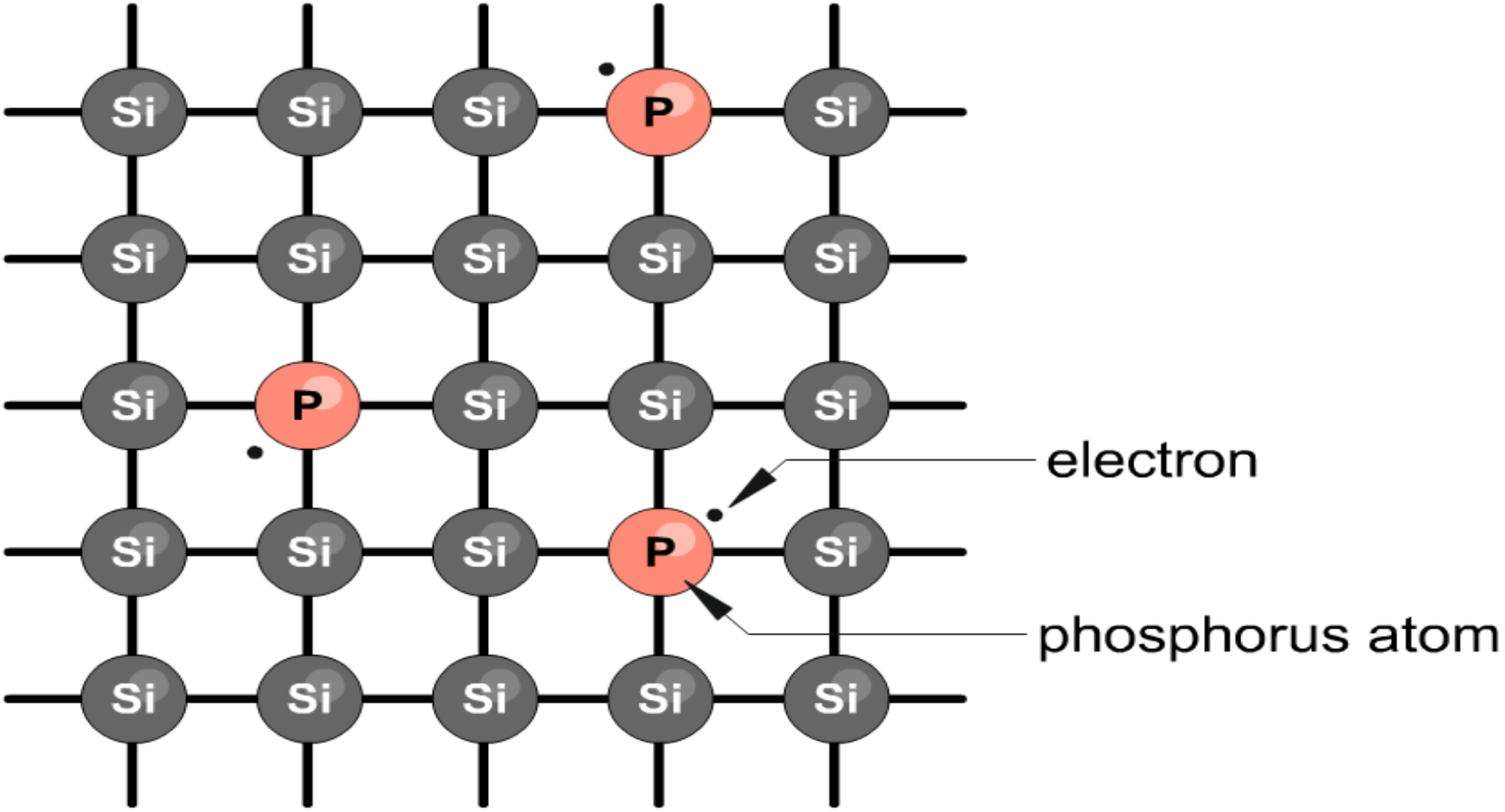
Example: Silicon



The Silicon Atom

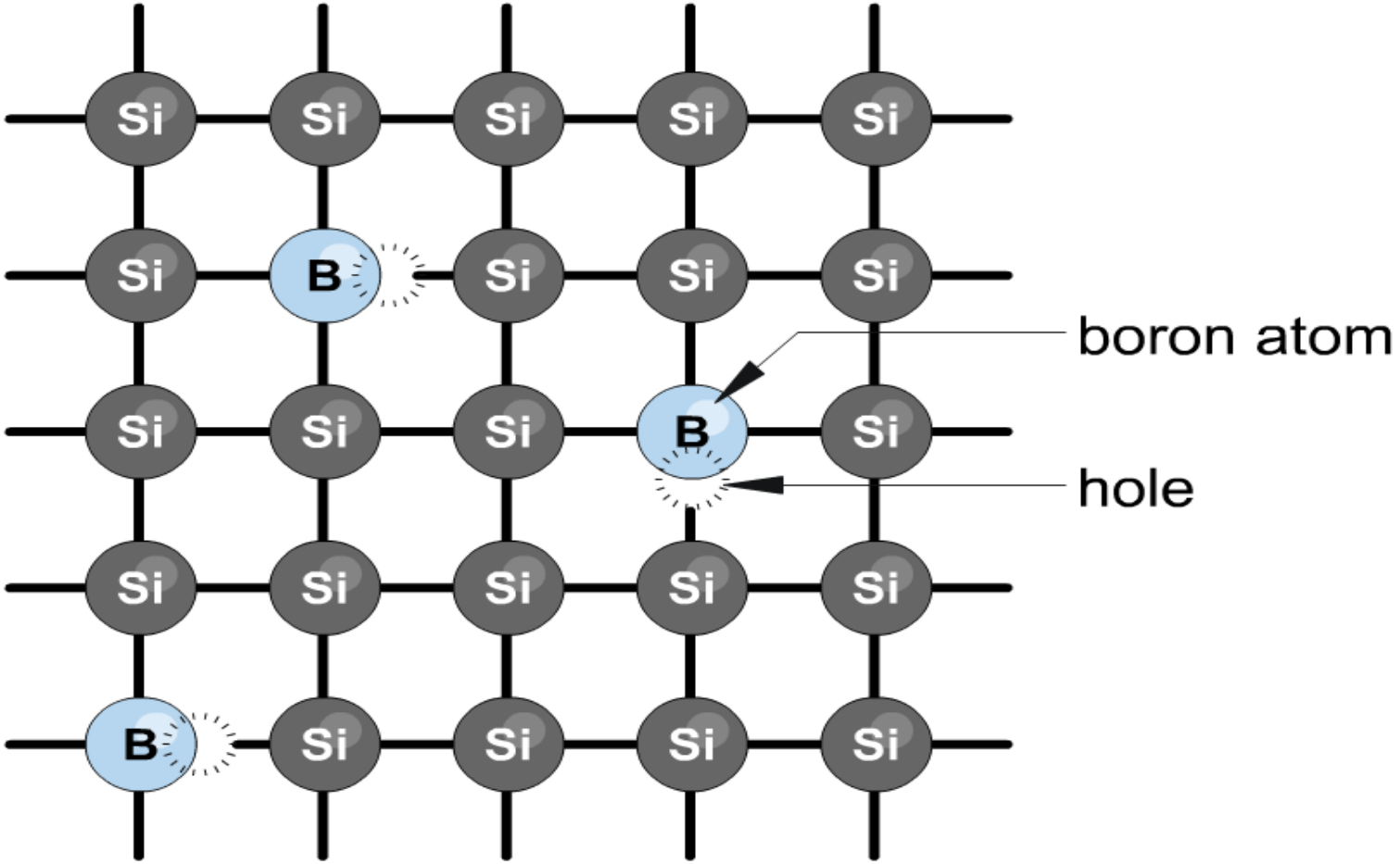


Doping: N-type Si



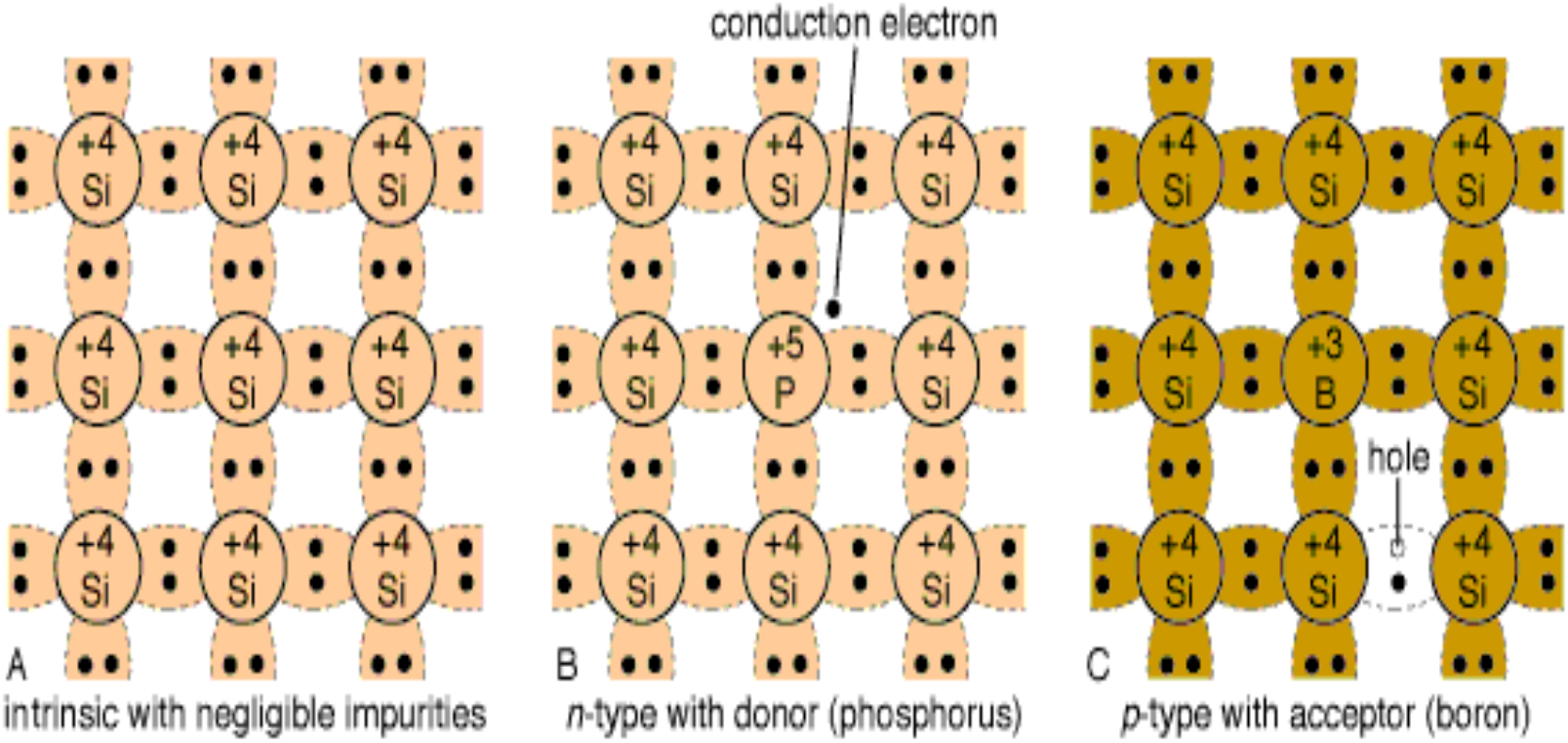
Donor concentration (N_D) 10^{15}-10^{18} atom/cm^3

Doping: P-type Si

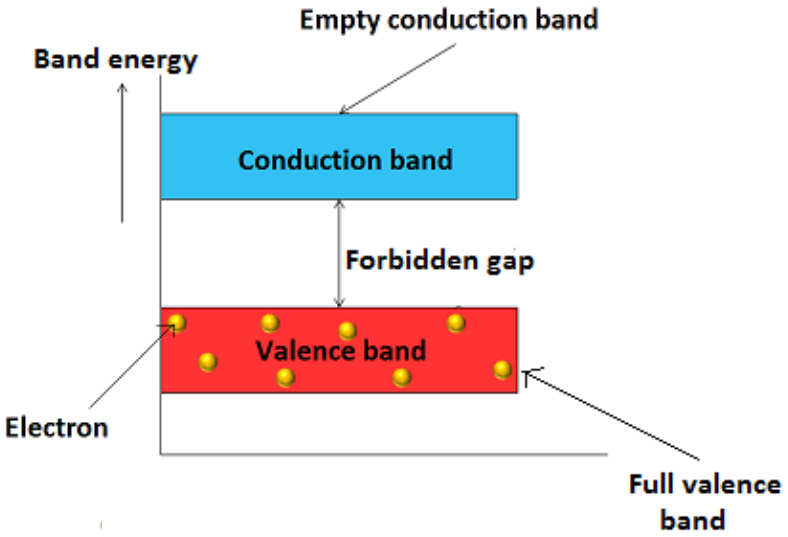


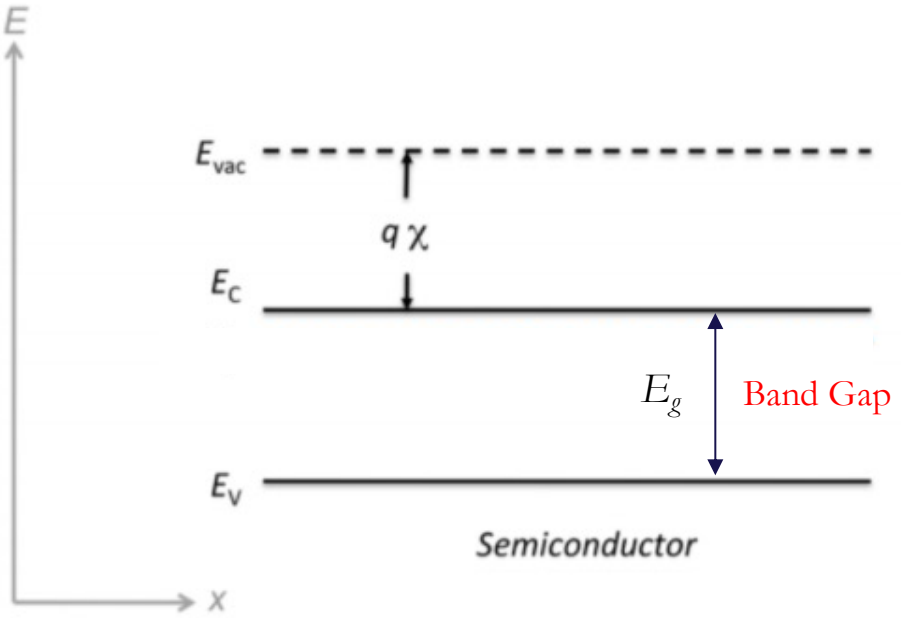
Acceptor concentration (N_A) 10^{13}-10^{16} atom/cm^3

Summary


conduction electron
+4 Si
+5 P
+3 B
hole
A
intrinsic with negligible impurities
B
n-type with donor (phosphorus)
C
p-type with acceptor (boron)

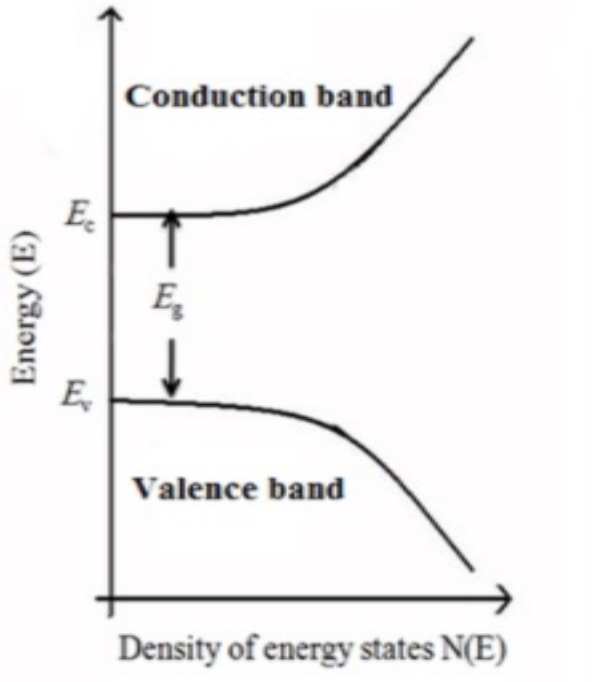
Back to the Energy Bands

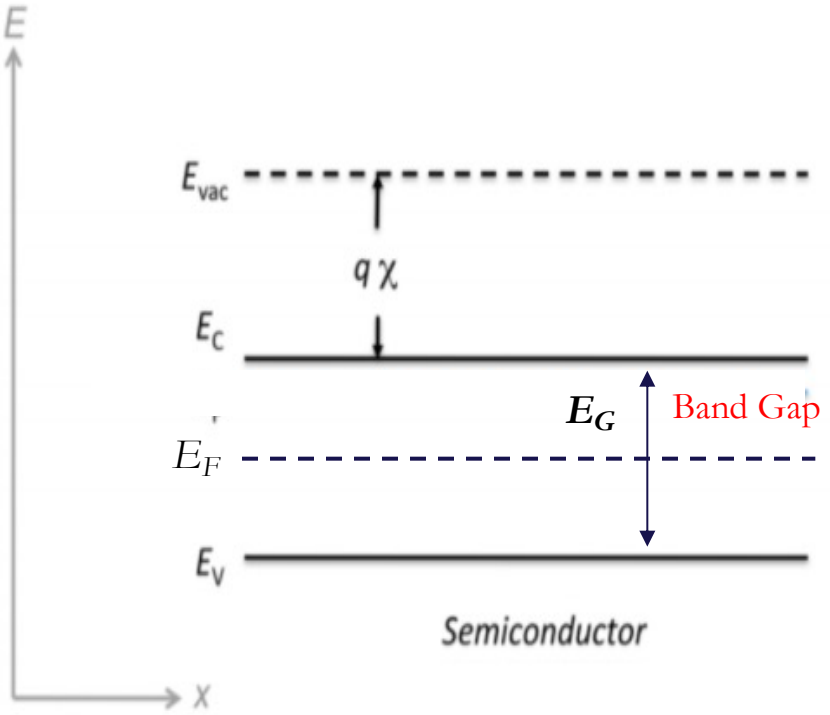




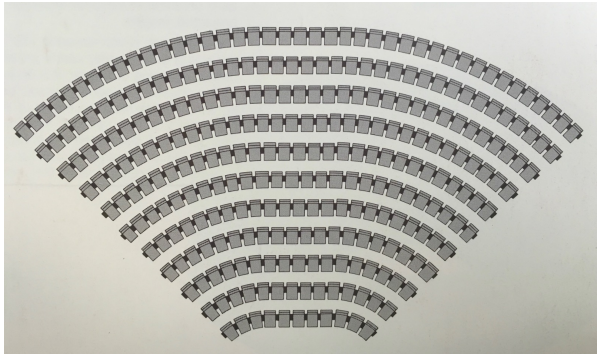
Band Diagram

Back to the Energy Bands



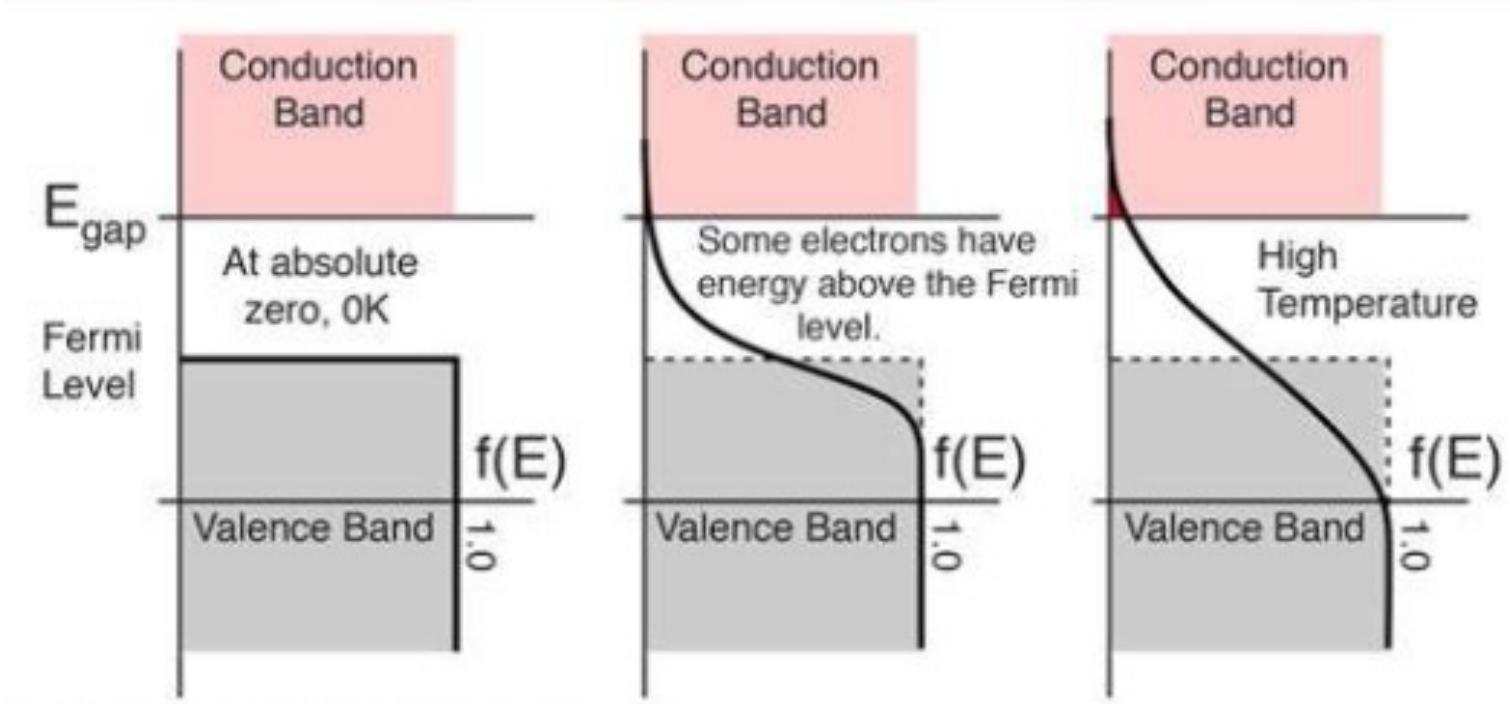


Density of States

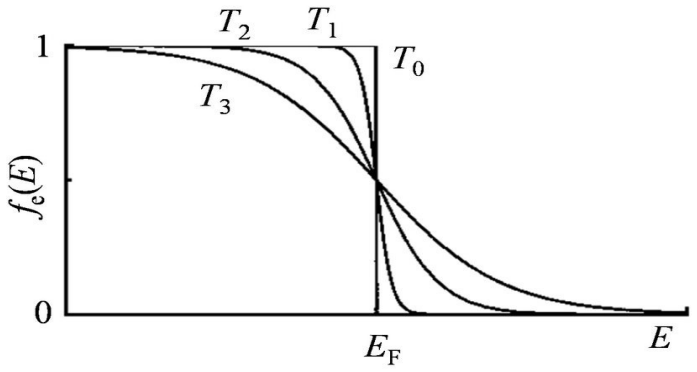


Band Diagram

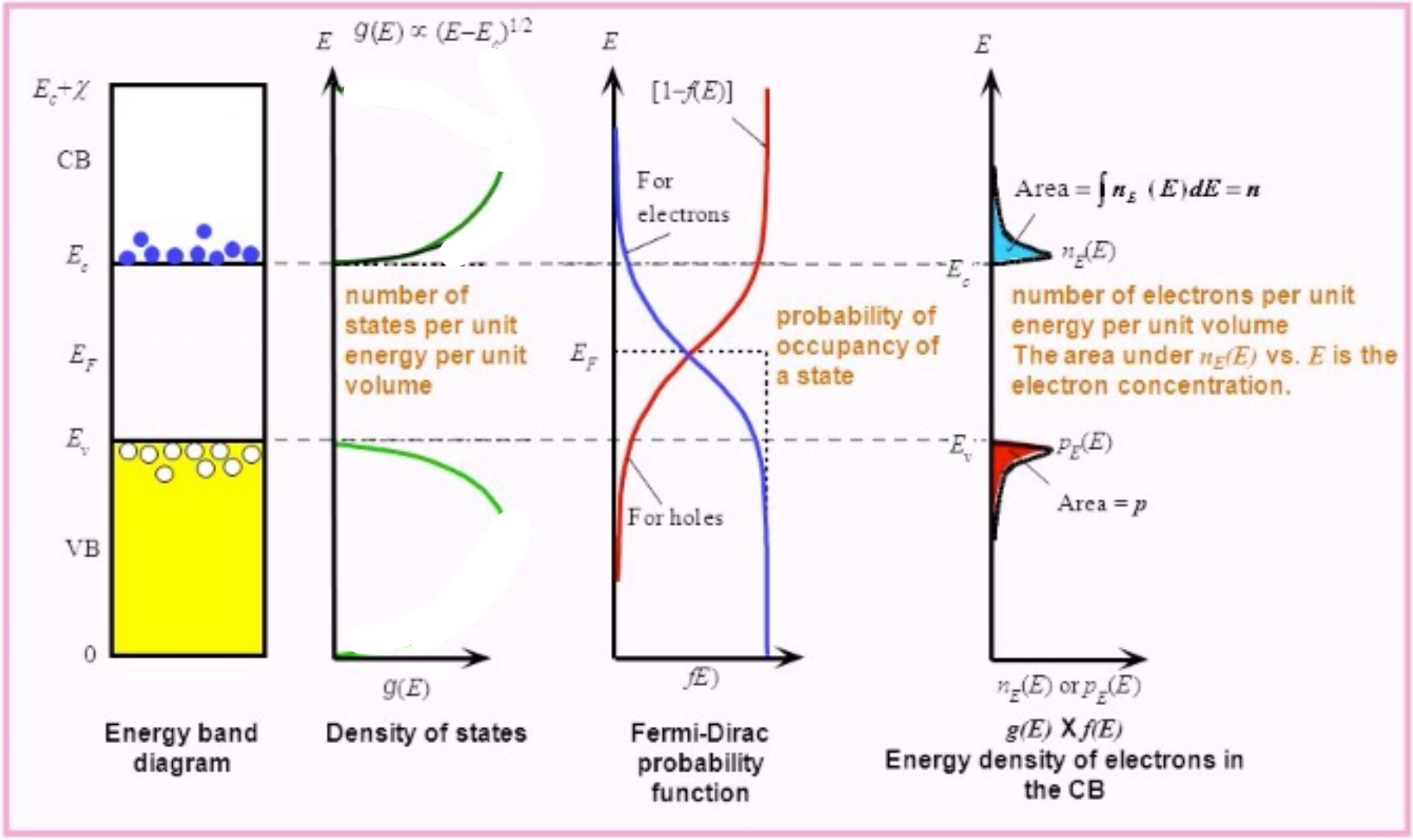
Fermi Distribution



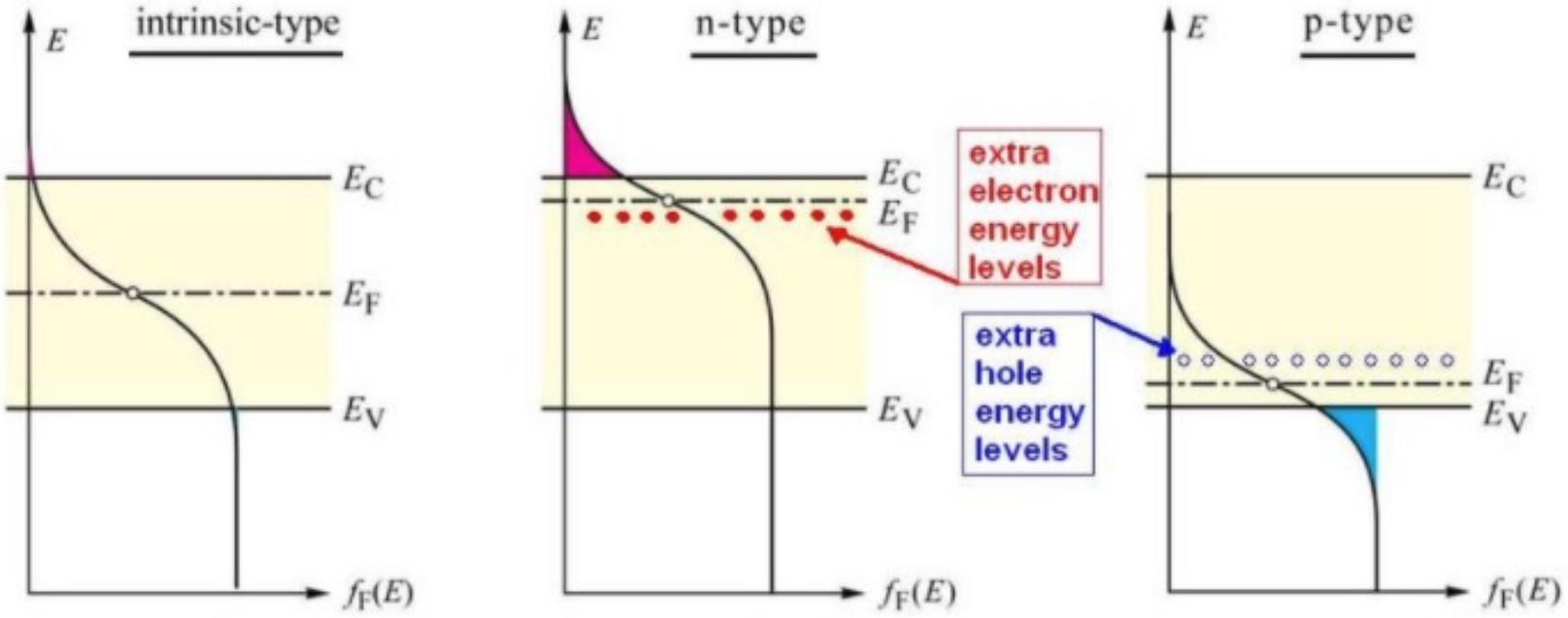
$$f_F(E) = \frac{1}{1 + exp\left(\frac{E - E_F}{kT}\right)}$$



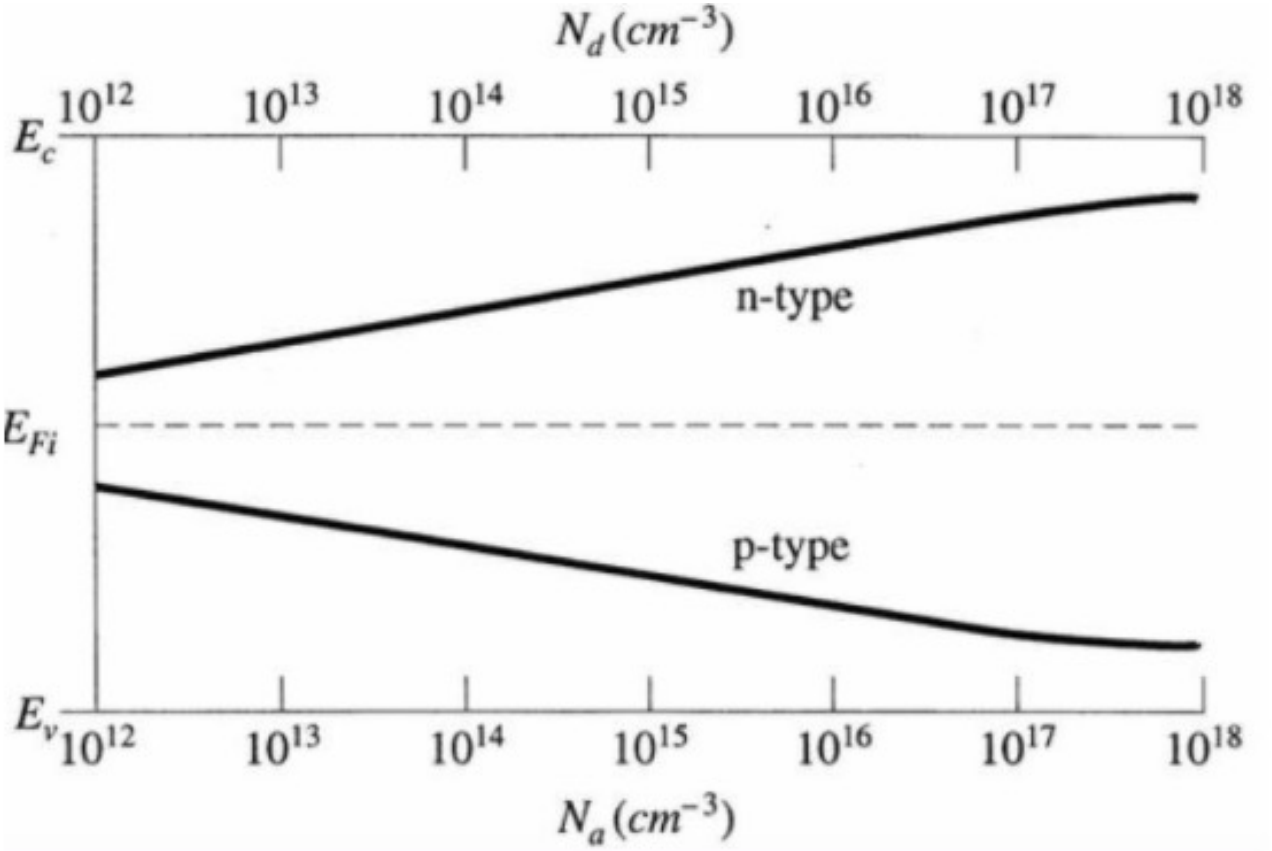
Electron/Hole Density



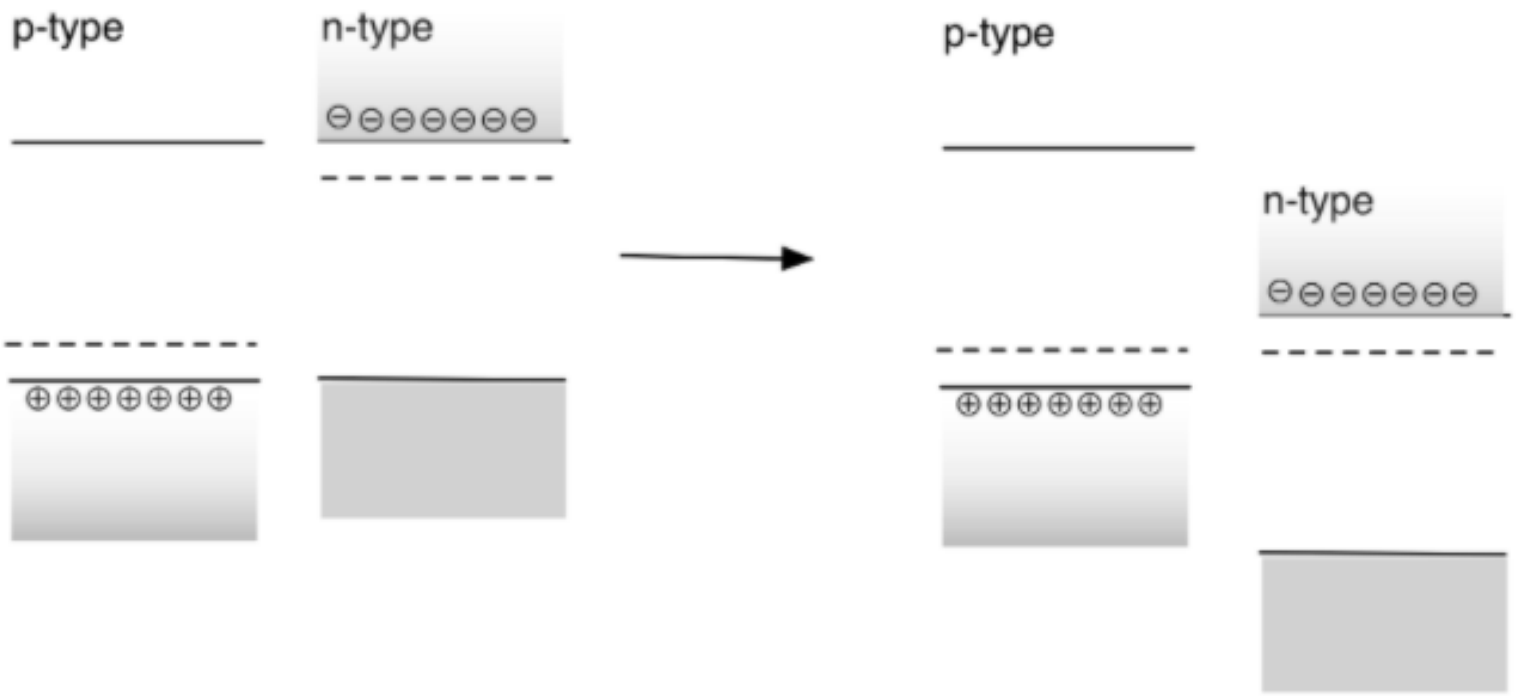
The n-type and p-type Band Diagram



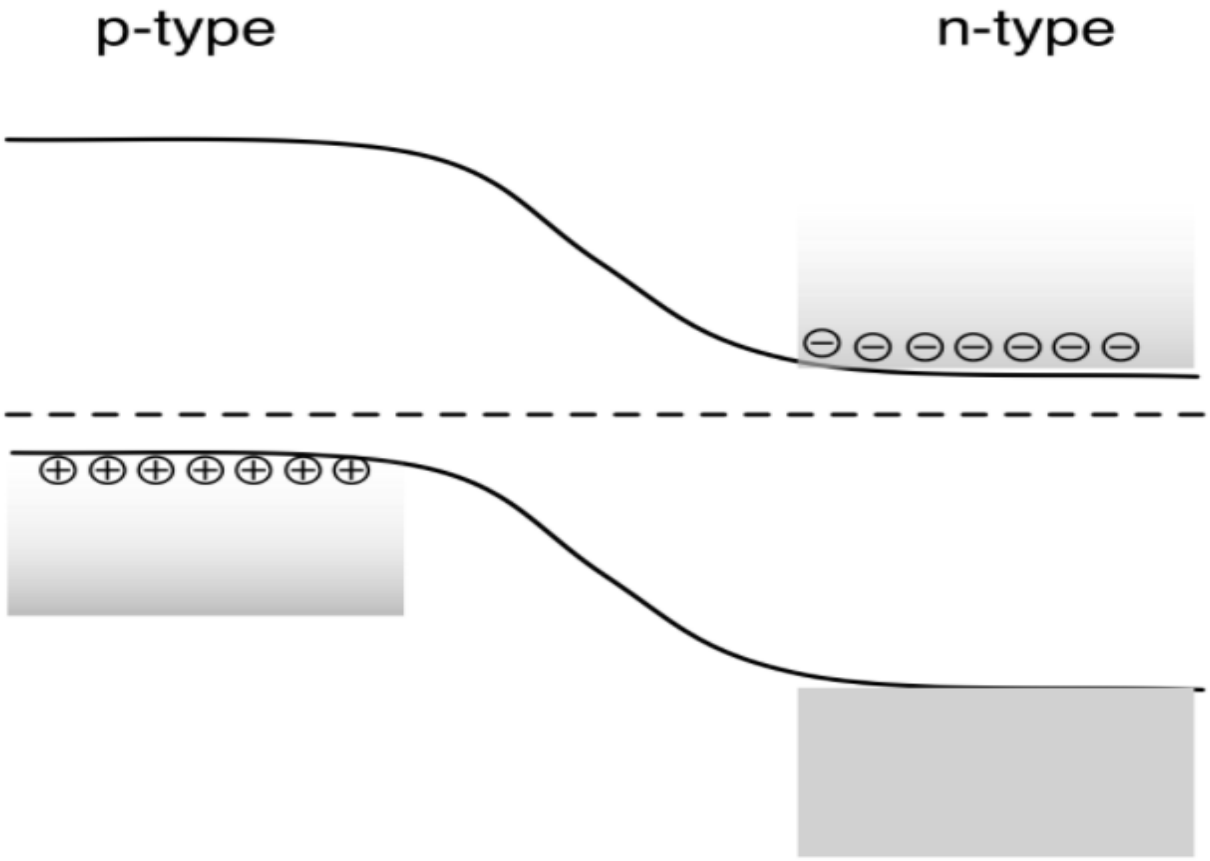
The n-type and p-type Band Diagram



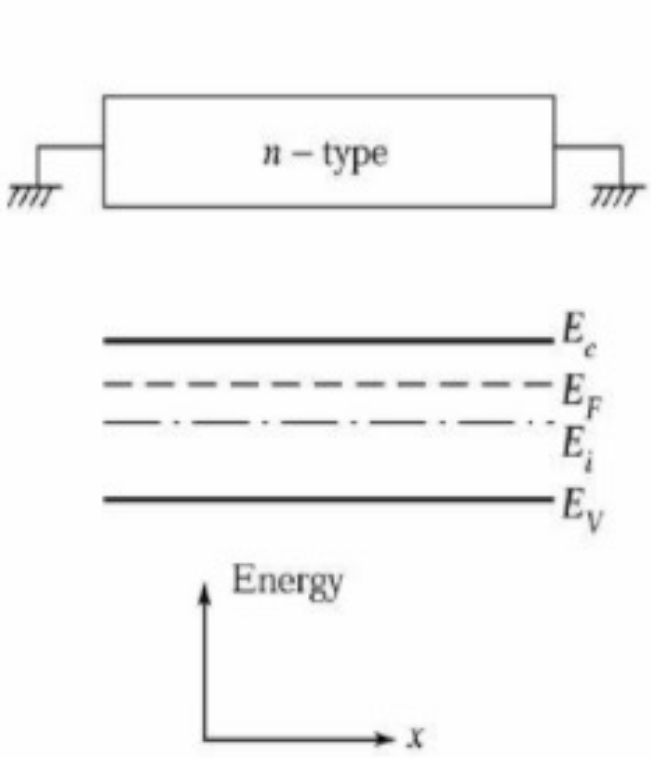
The PN Junction Band Diagram

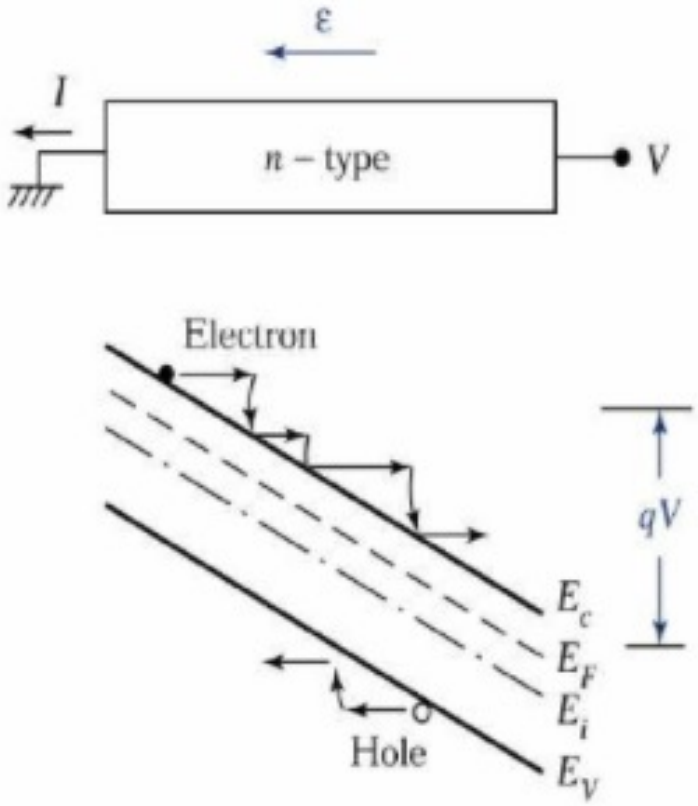


Unbiased PN Junction Band Diagram

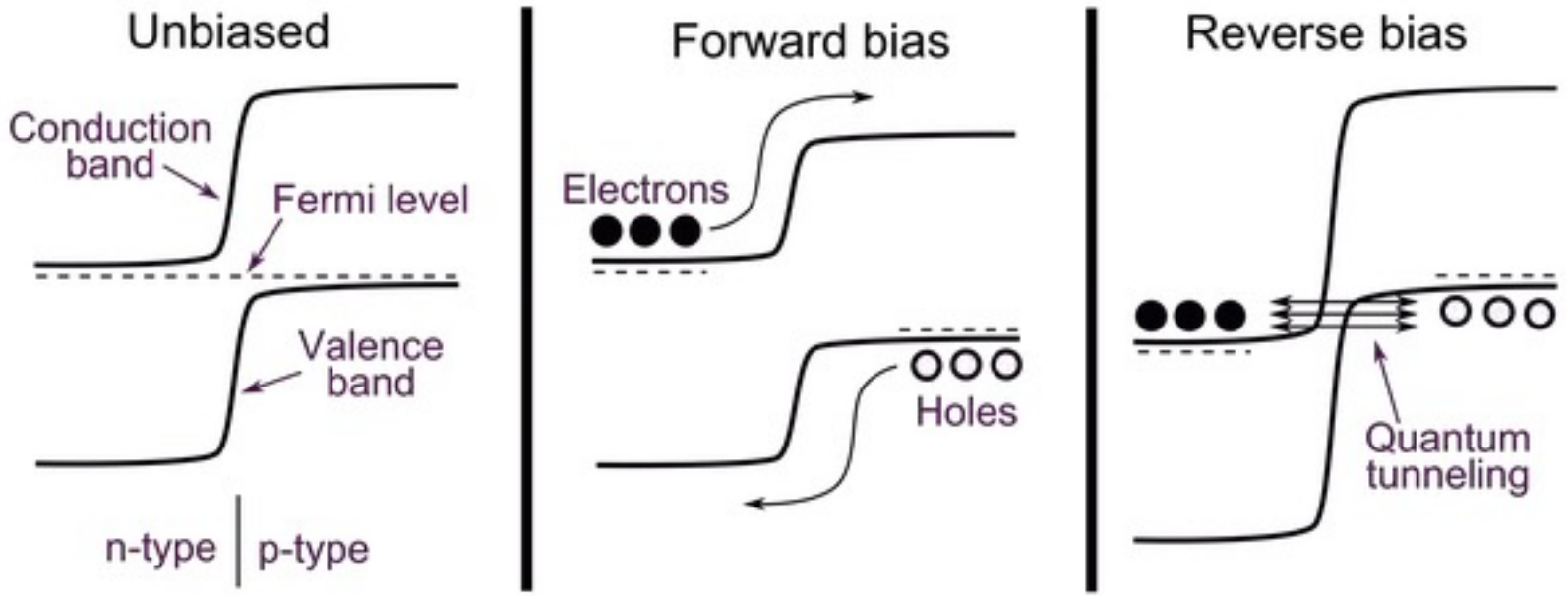


Biased n-type Band Diagram



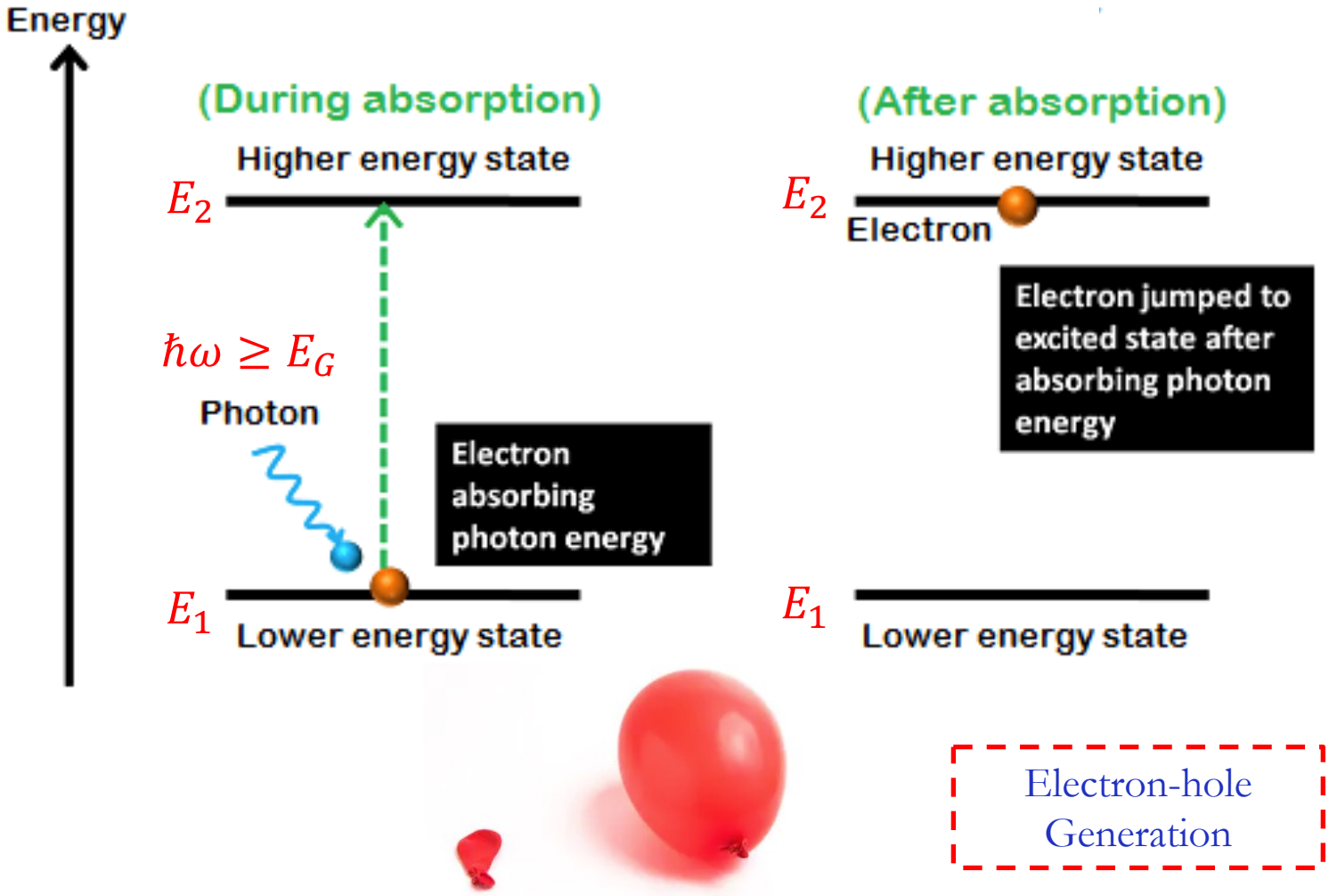


Biased PN Junction Band Diagram



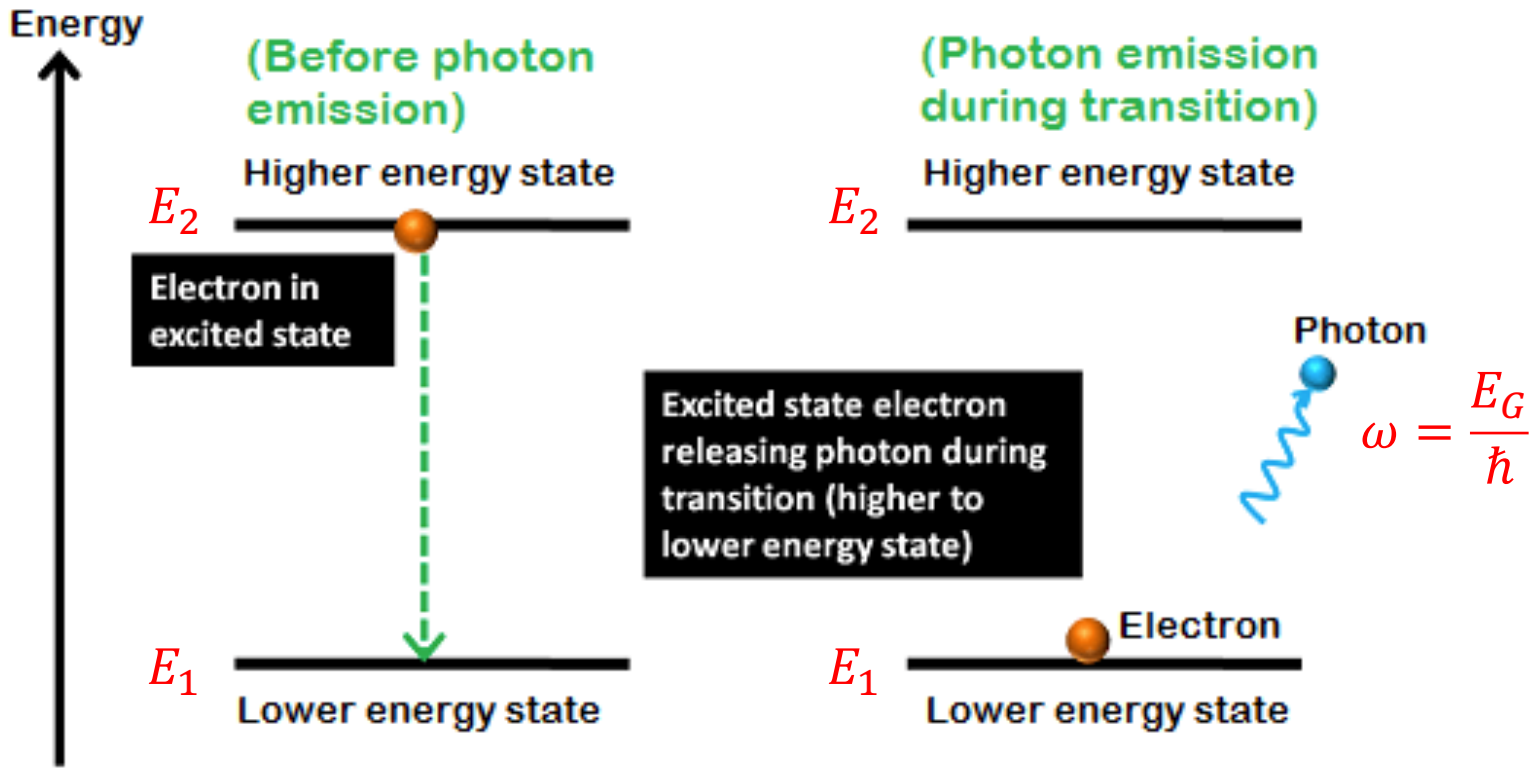
Photon Generation and Absorption

Absorption



Photon Generation and Absorption

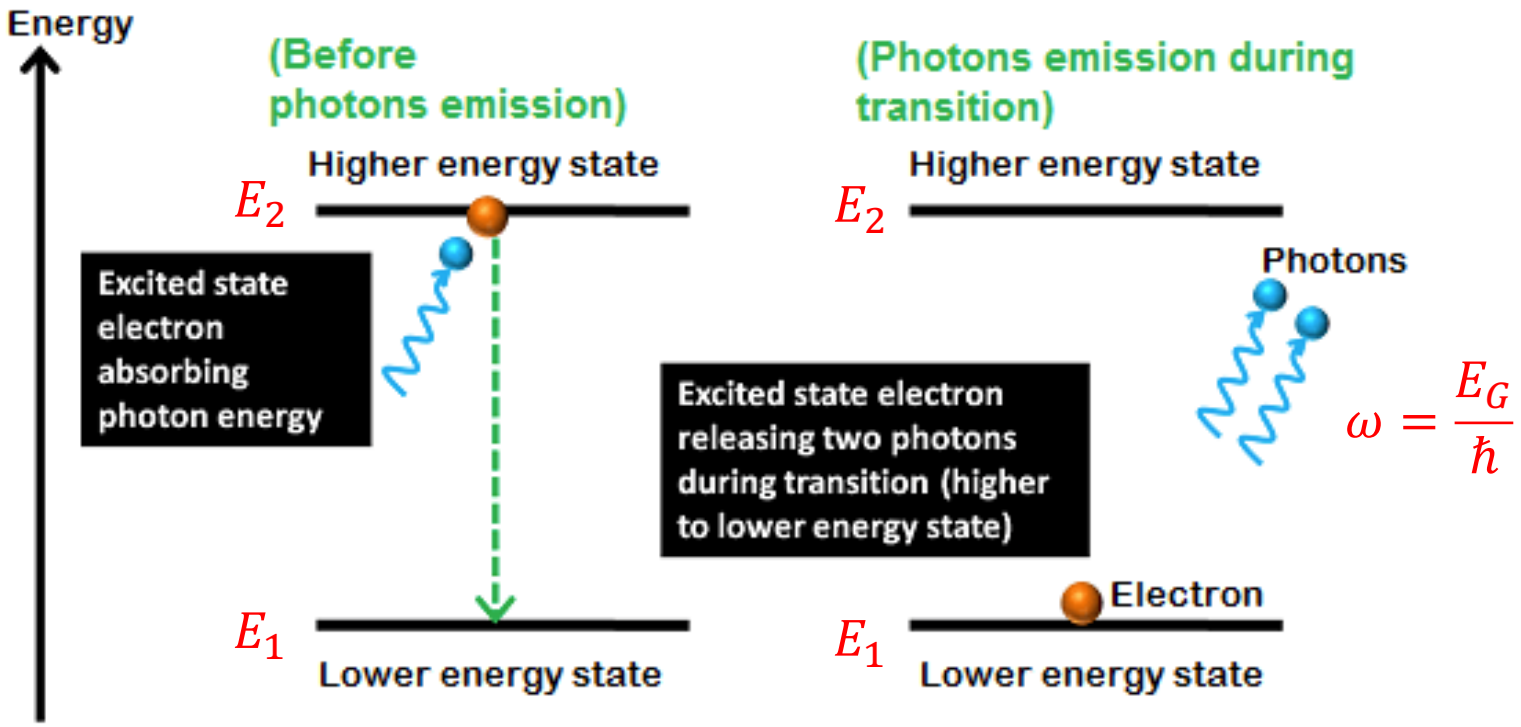
Spontaneous Emission



Electron-hole Re-combination (radiative/nonradiative)

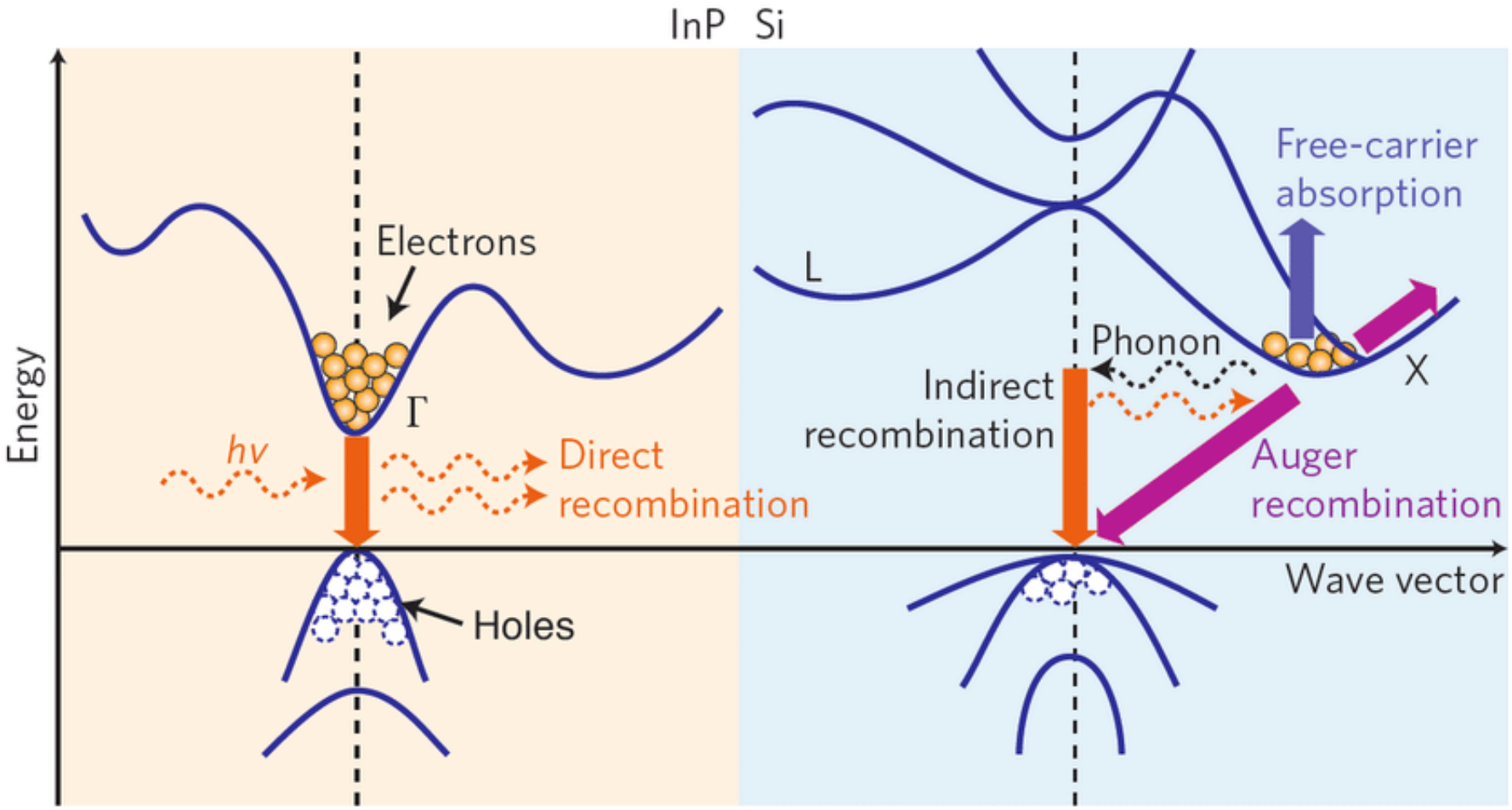
Photon Generation and Absorption

Stimulated Emission



Electron-hole
Re-combination
(radiative/nonradiative)

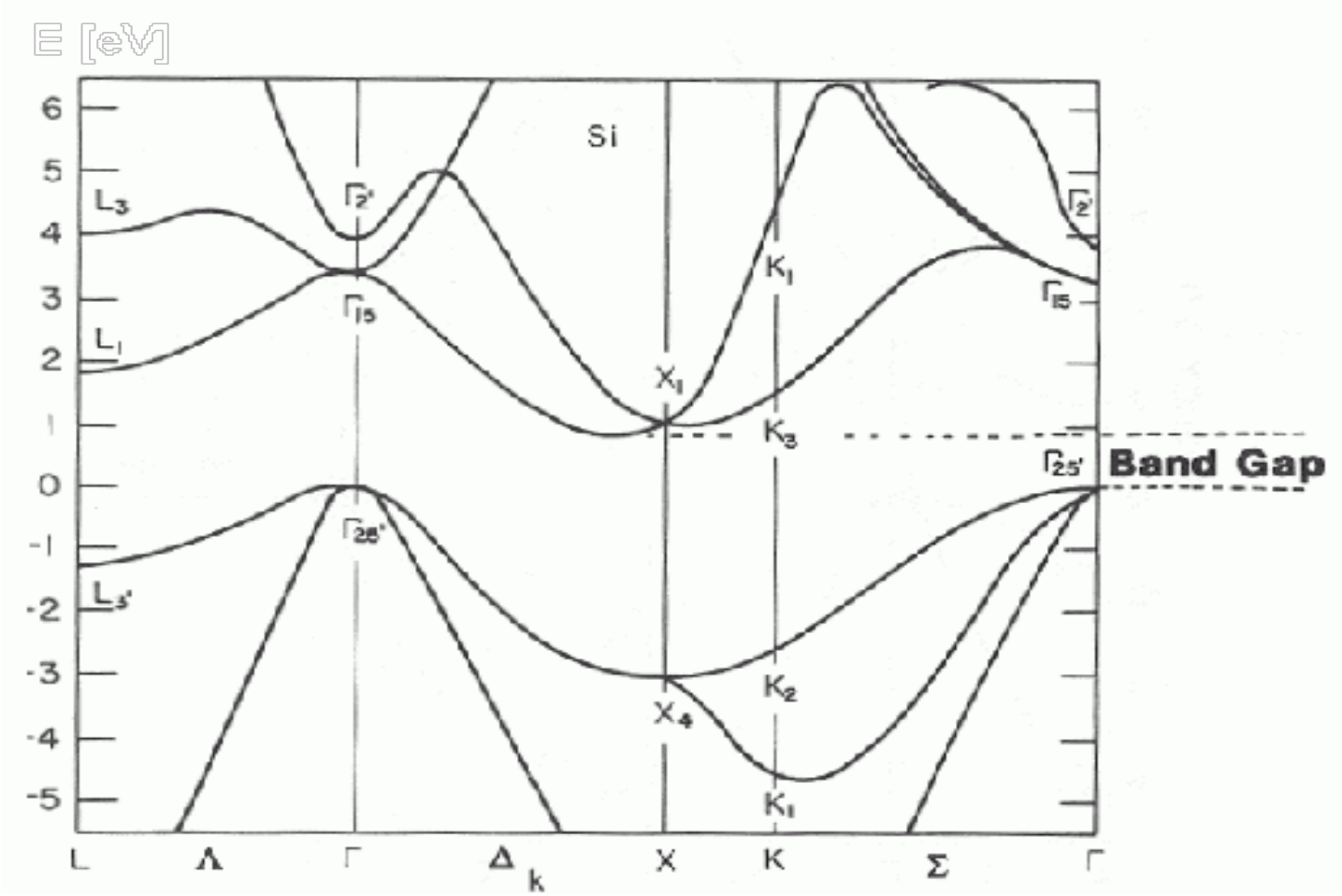
Direct and Indirect Gap Materials



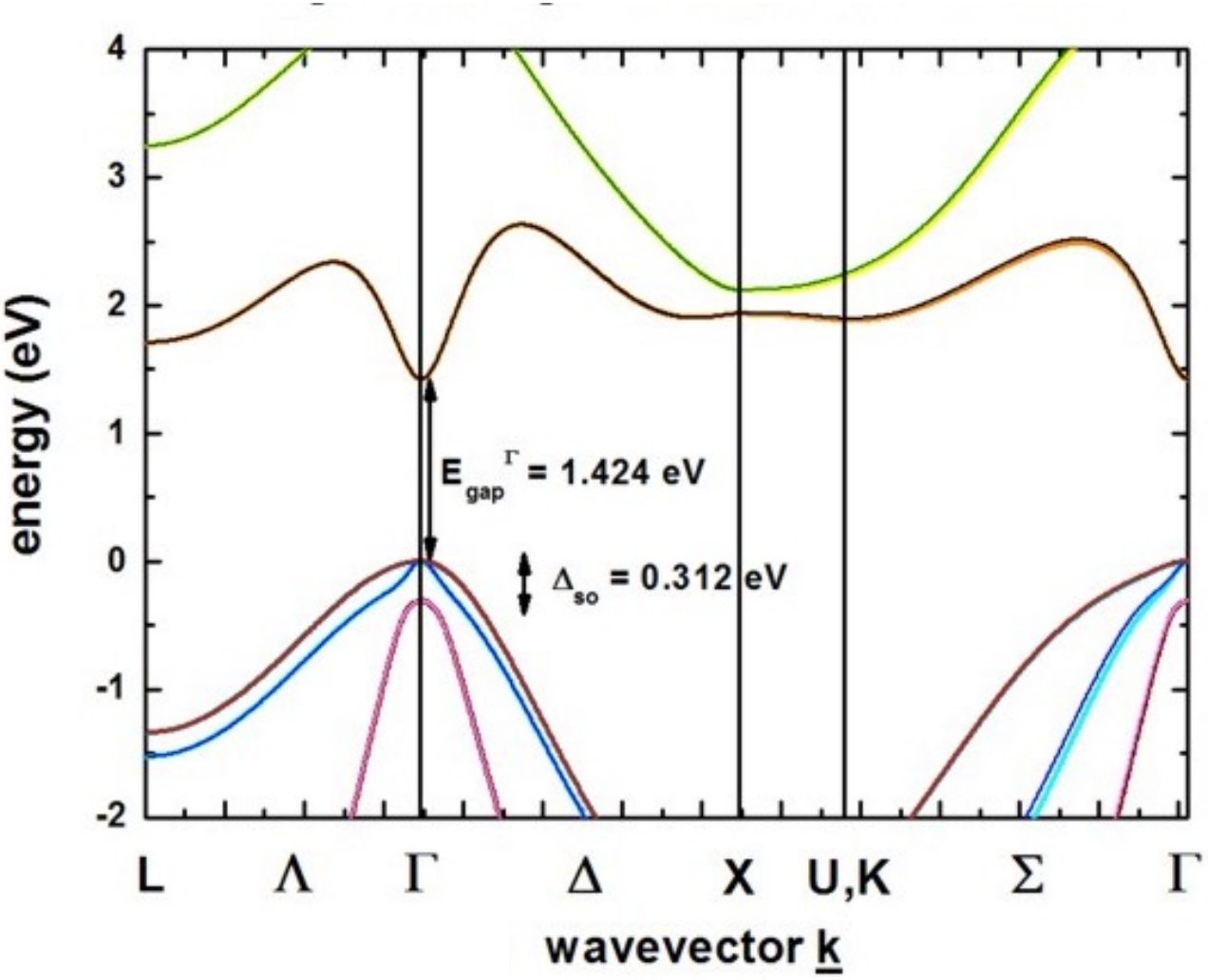
Direct and Indirect Gap Materials

	Material	Direct / IndirectBandgap	Band Gap Energy at 300 K (eV)
Elements	C (di	Indirect	5.47
	Ge	Indirect	0.66
	Si	Indirect	1.12
	Sn	Direct	0.08
Groups III-V compounds	GaAs	Direct	1.42
	InAs	Direct	0.36
	InSb	Direct	0.17
	GaP	Indirect	2.26
	GaN	Direct	3.36
	InN	Direct	0.70
	InP	Direct	1.344
Groups IV-IV compounds	α-SiC	Indirect	2.99
Groups II-VI compounds	ZnO	Direct	3.35
	CdSe	Direct	1.70
	ZnS	Direct	3.68

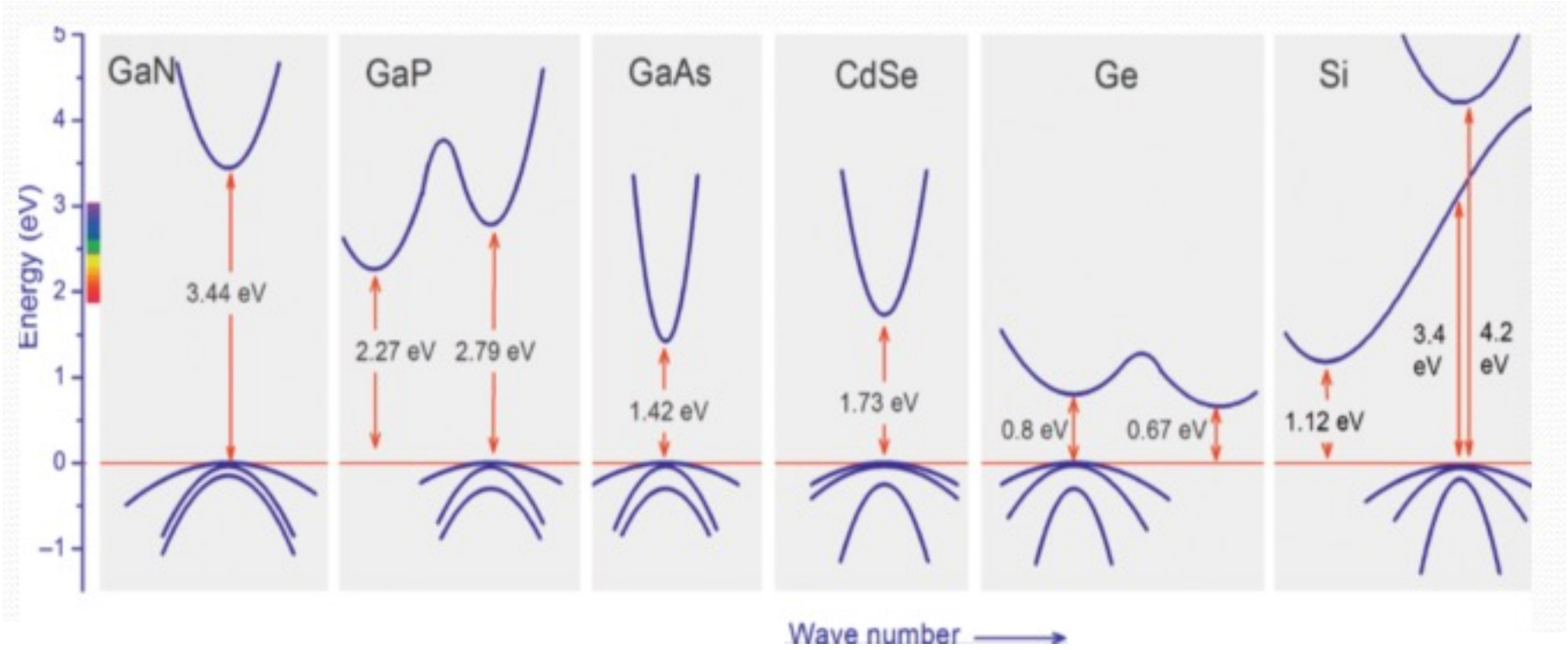
Si: Indirect Gap Material



GaAs: Direct Gap Material



GaAs: Direct Gap Material



Bandgap Engineering

Example: Emission in GaAs, GaN and InN.

$$\omega = \frac{E_G}{\hbar}$$

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega}$$

$$\lambda = \frac{2\pi c\hbar}{E_G}$$

$$\lambda_g = \frac{hc}{E_G}$$

GaAs band gap energy E_G=1.42 eV

$$\lambda_g = \frac{2\pi(3\times10^8)(1.055\times10^{-34})}{(1.42)(1.6\times10^{-19})}$$

$\lambda_g = 875\ nm$ NIR

GaN band gap energy E_G=3.49 eV

$\lambda_g = 356\ nm$ UV

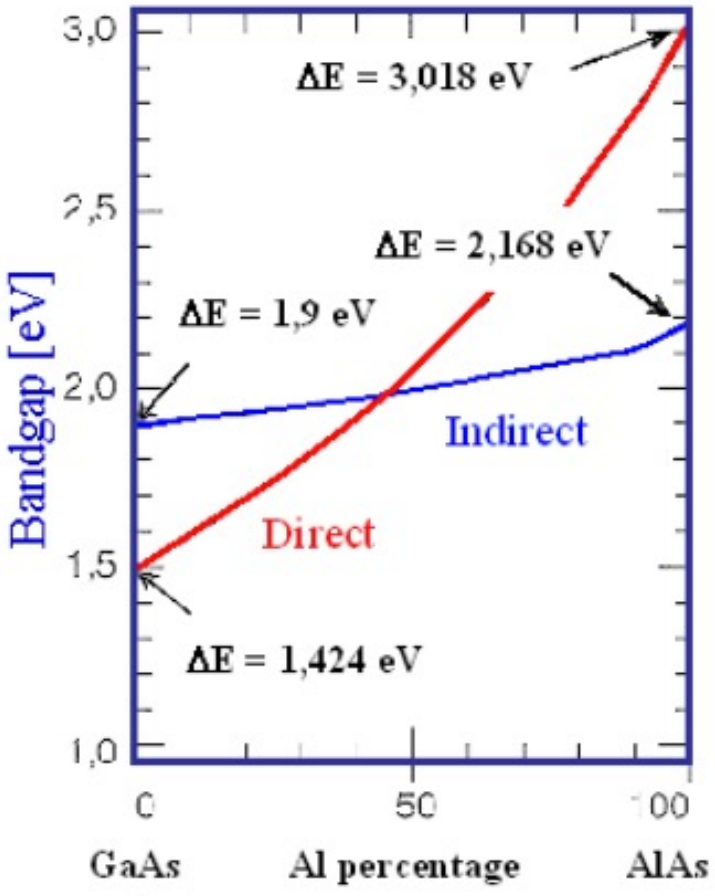
InN band gap energy E_G=0.69 eV

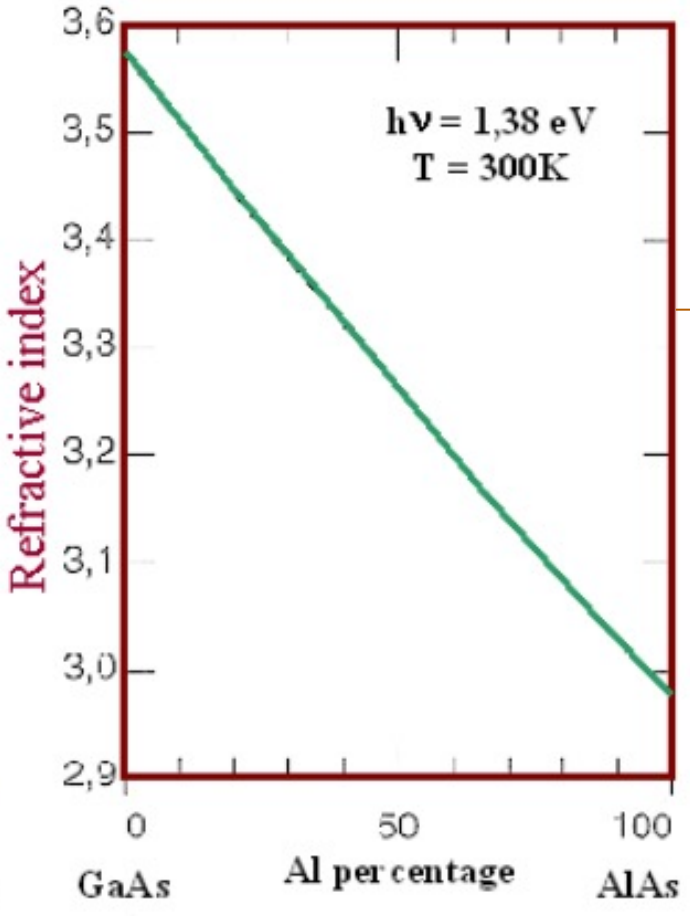
$\lambda_g = 1800\ nm$ IR

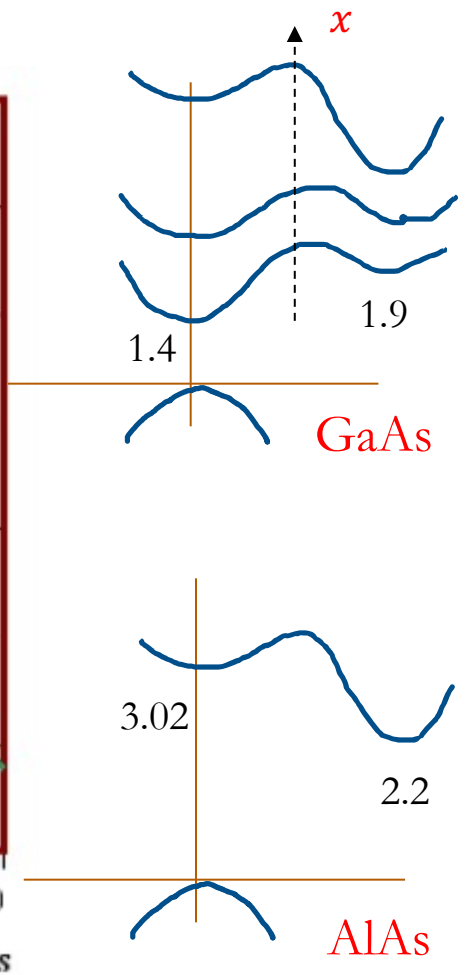
The alloy $In_xGa_{1-x}N$ is very useful for spanning the visible range.

Bandgap Engineering

Example: Emission in $Al_xGa_{1-x}As$.





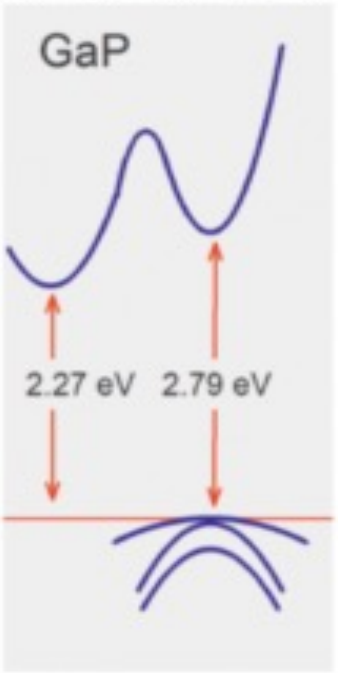


The Green Gap

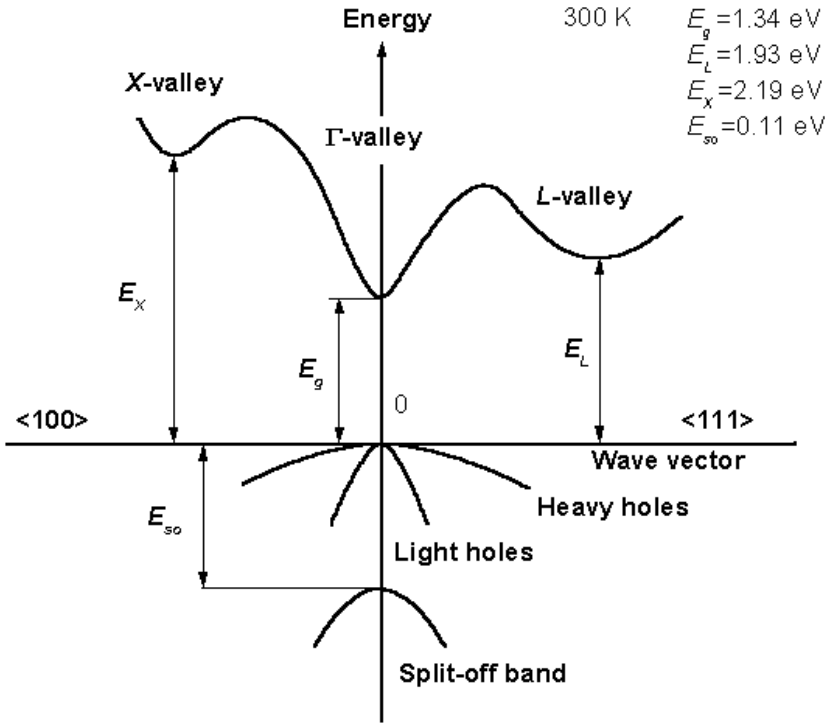
Green Band: 530-570 nm.

Green Band: 2.2-2.3 eV.

Emission in $Ga_xIn_{1-x}P$.



InP



Photodetectors

The following subjects will be covered

- Photodetector Types.
- General Characteristics of Photodiodes.
- Modes of Operation.
- PN, PIN and APD Devices.

Photodetectors

Type (1): Photo-emissive

Light is absorbed, electron-hole pair generated and ejected out of material as current.

Type (2): Photo-conductive

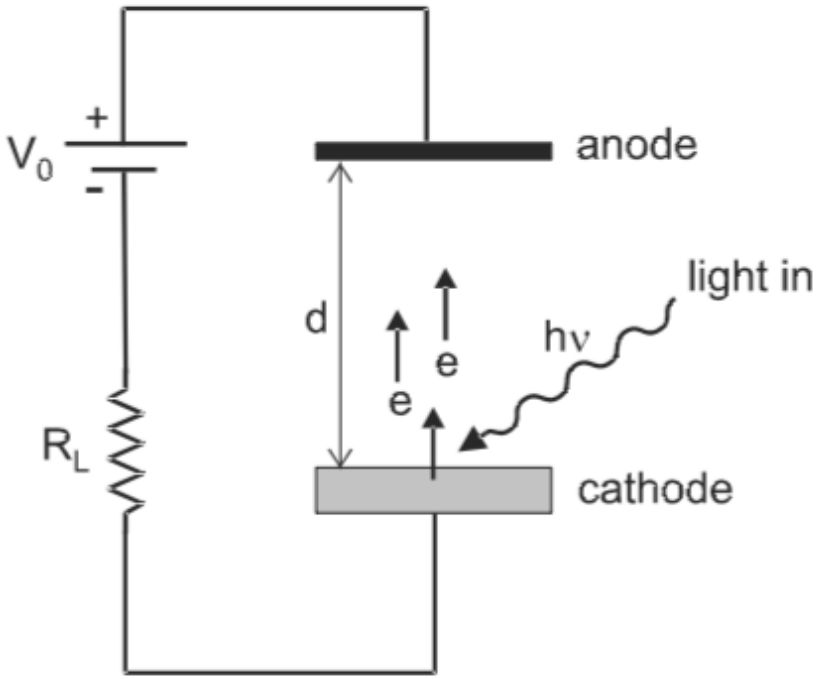
Light is absorbed, electron-hole pair generated and the material conductivity is changed.

Examples:

- Photoconductors: A piece of semiconductor the conductance of which is controlled by light (photoresistors).
- Photodiodes: A pn junction-based device that converts light energy to electron-hole pairs. The ehp are saved and used for operation. (PIN diodes, APDs, Solar cells, etc.)

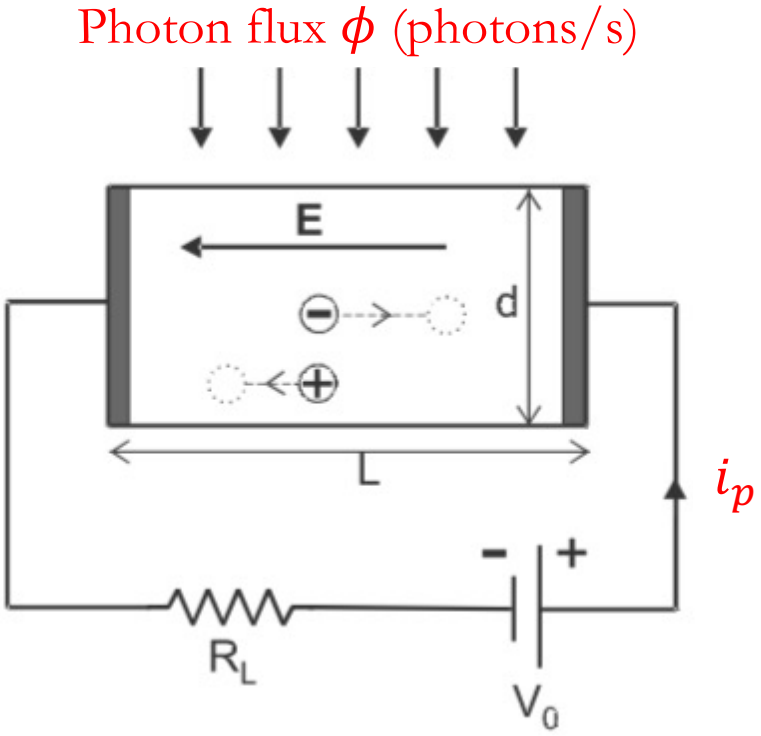
Photo-emissive Devices

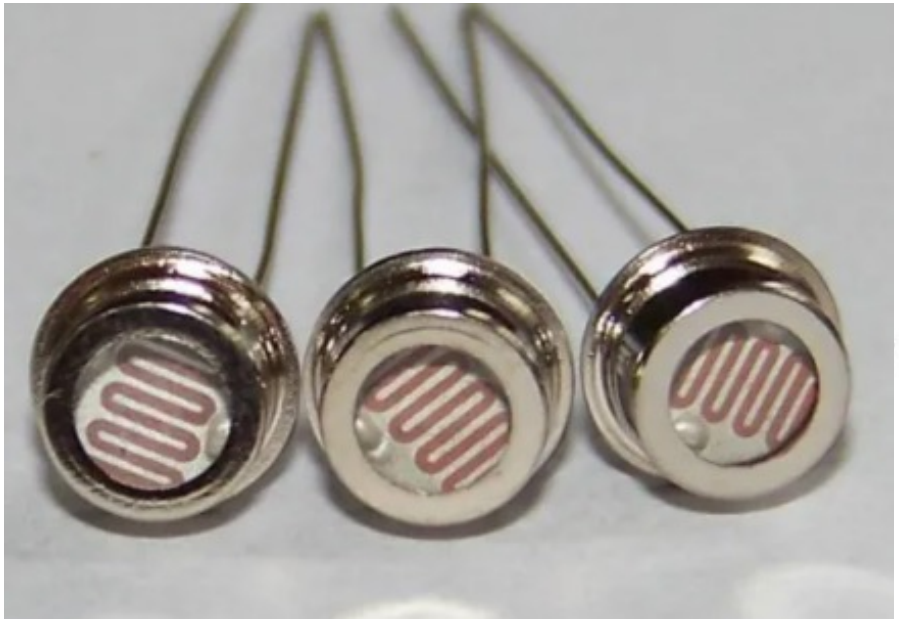
The basic structure of a photo-emissive device is a photoelectric-effect set-up.



Photoconductive Devices

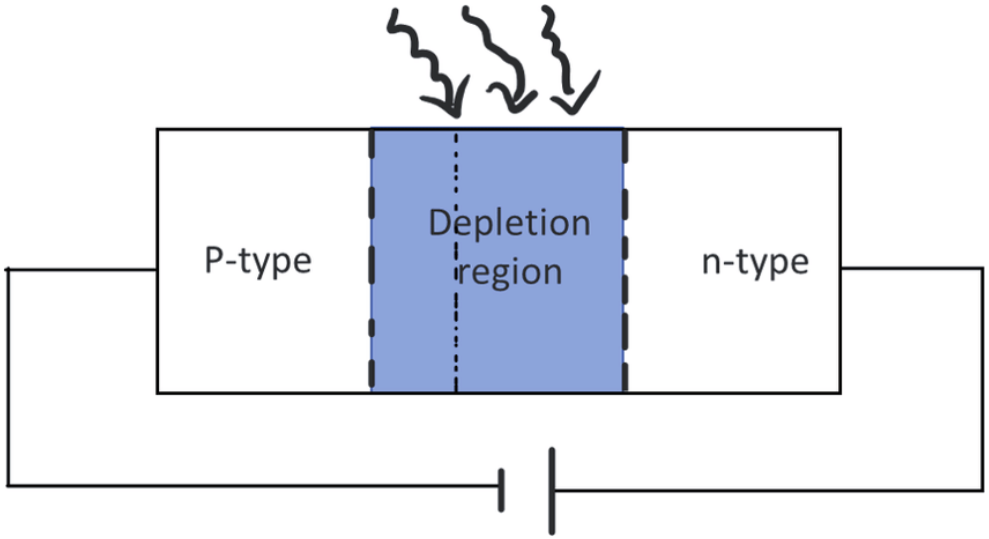
The basic structure of a photoconductive device is a semiconductor material connected across a battery.





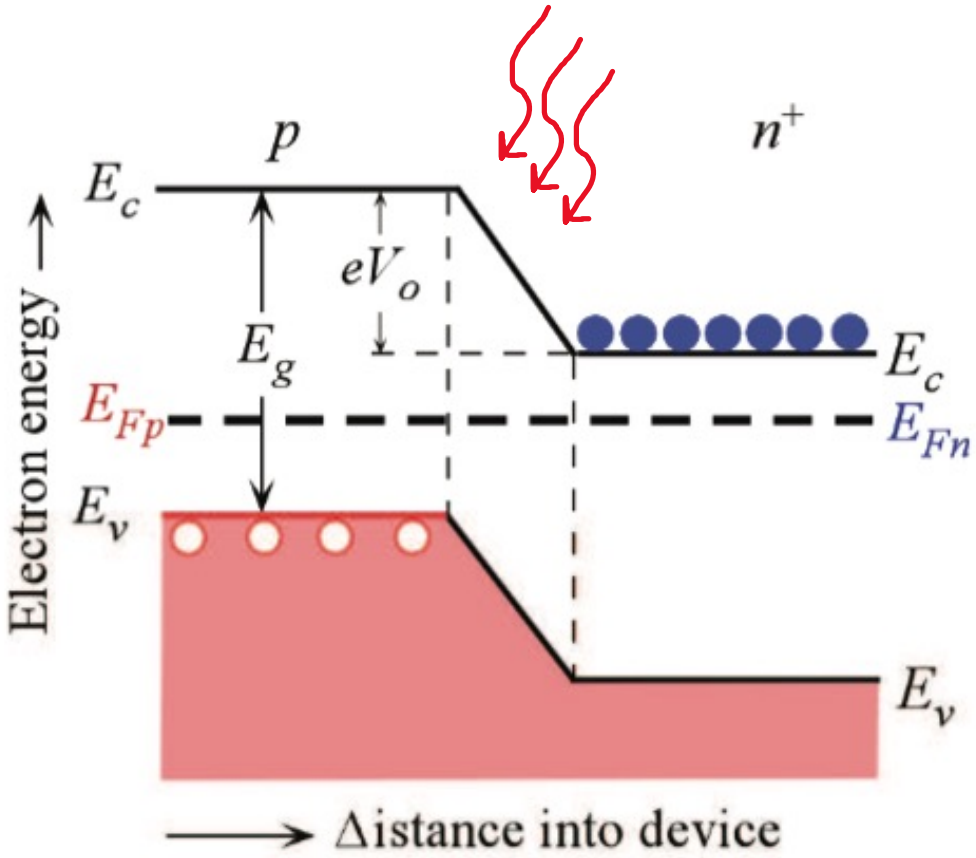
Photodiodes

The basic structure of a photodiode device is a pn junction.



Photodiodes

The basic structure of a photodiode device is a pn junction.



Photodiode Characteristics

Quantum efficiency. The ratio of optical current ehps to incident photon flux.

Responsivity. The ratio of optical current to incident optical power.

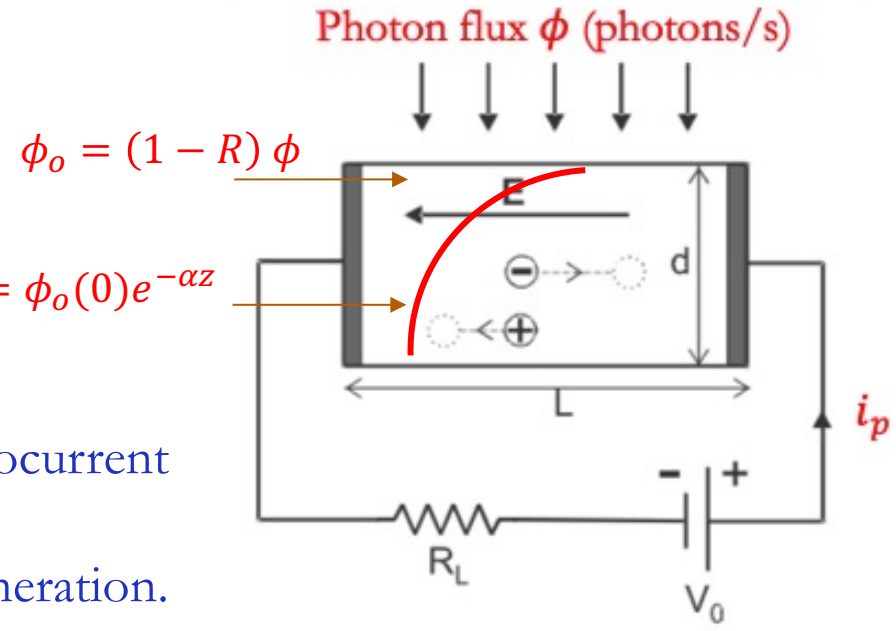
Impulse response. The rise time of the output current or the output bandwidth.

Noise power. The amount of current created by sources other than light.

Photodiode Characteristics: Efficiency

Quantum efficiency. The ratio of created ehp to number of incident photons.

$$\eta = \frac{\text{ehp flux making the optical current}}{\text{Incident photon flux } (\phi)}$$



Absorbed photon flux:

$$(1-R)\left(1-e^{-\alpha d}\right)\phi$$

Not all absorbed photons will make photocurrent due to:
(1) ehp recombination (2) phonon generation.
Introduce the factor $0 \le \xi \le 1$.

Current-making photon flux = $(1-R)\left(1-e^{-\alpha d}\right)\xi\,\phi$

$$\eta = (1-R)\left(1-e^{-\alpha d}\right)\xi$$

The engineering task is to maximize the efficiency.

Photodiode Characteristics: Efficiency

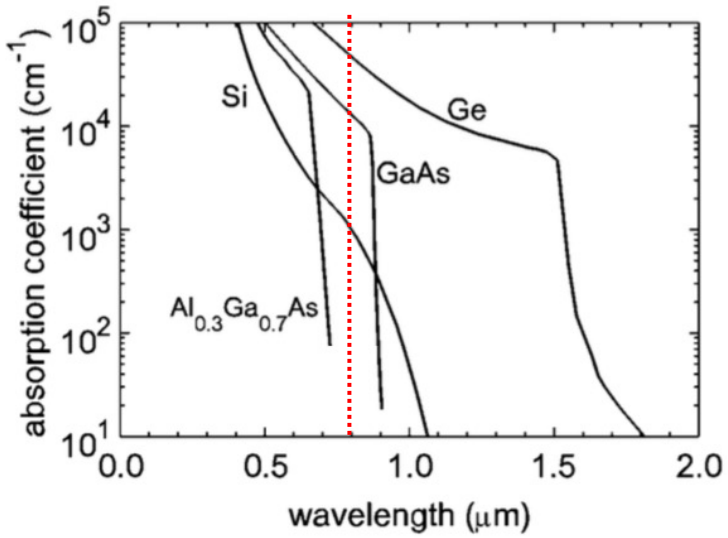
Quantum efficiency. The ratio of created ehp to number of incident photons.

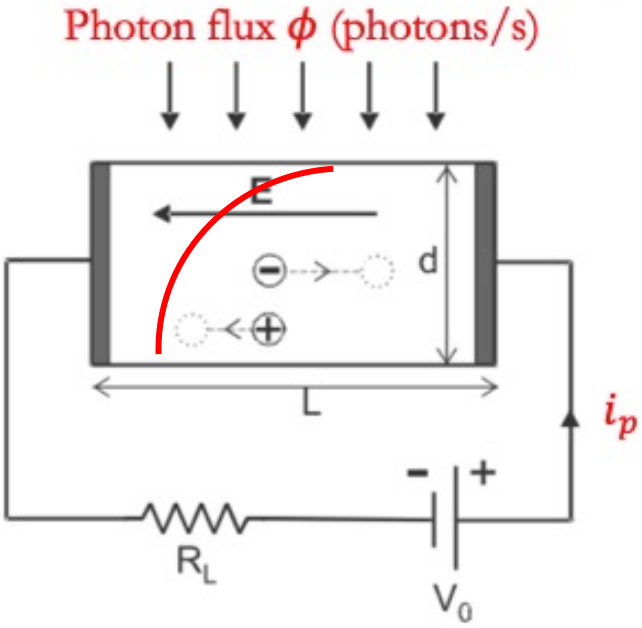
$$\eta = (1 - R)\left(1 - e^{-\alpha d}\right)\xi$$

ξ : Apply bias and clean fabrication process.

R : Use anti-reflection coating (small bandwidth)

αd : Choose best material and thickness.



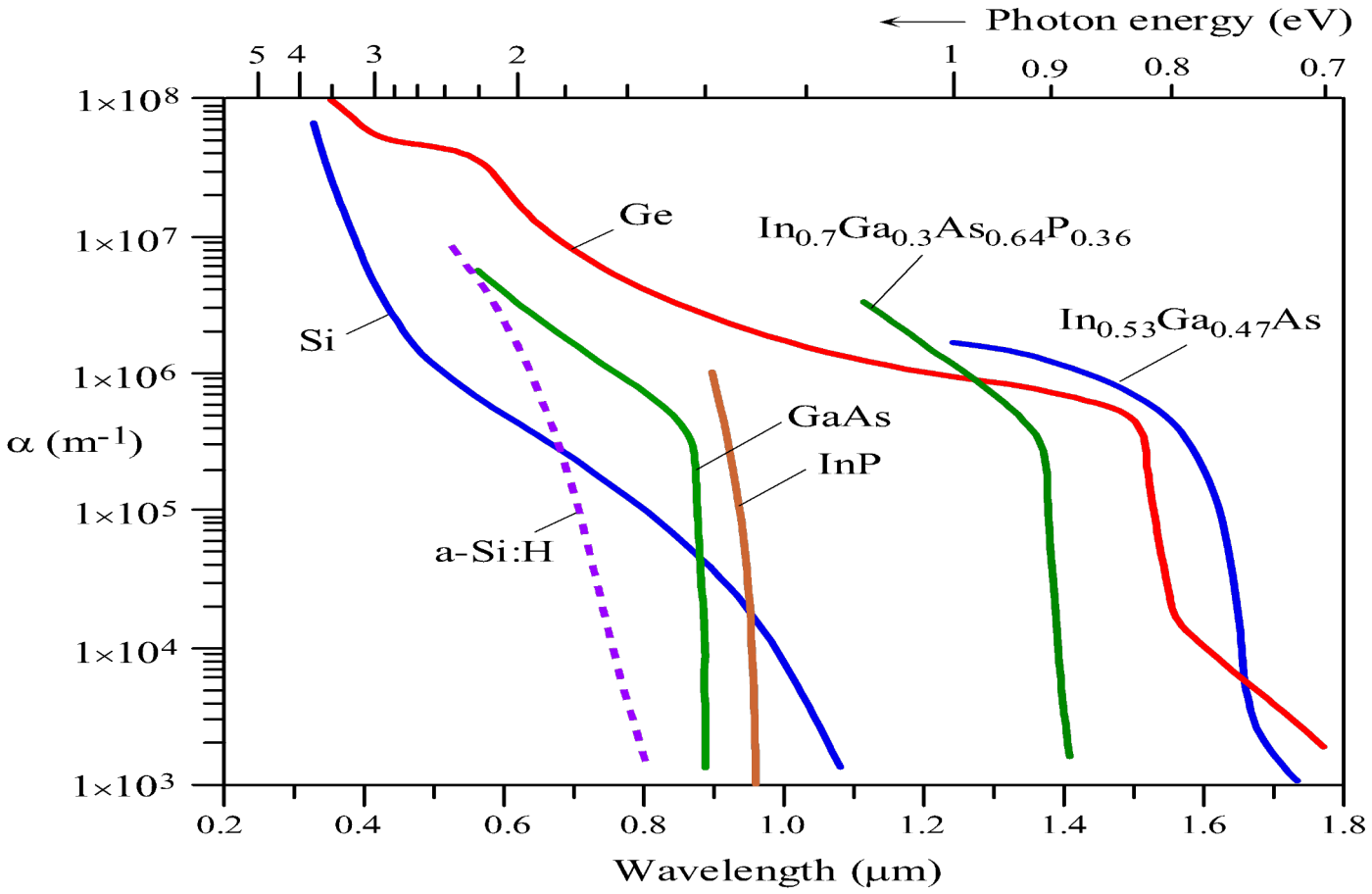


Example: For 95% absorption at 0.8 μm, find material and thickness.

$$1 - e^{-\alpha d} = 0.95$$

Photodiodes Characteristics: Efficiency

Absorption coefficient of common semiconductor materials.

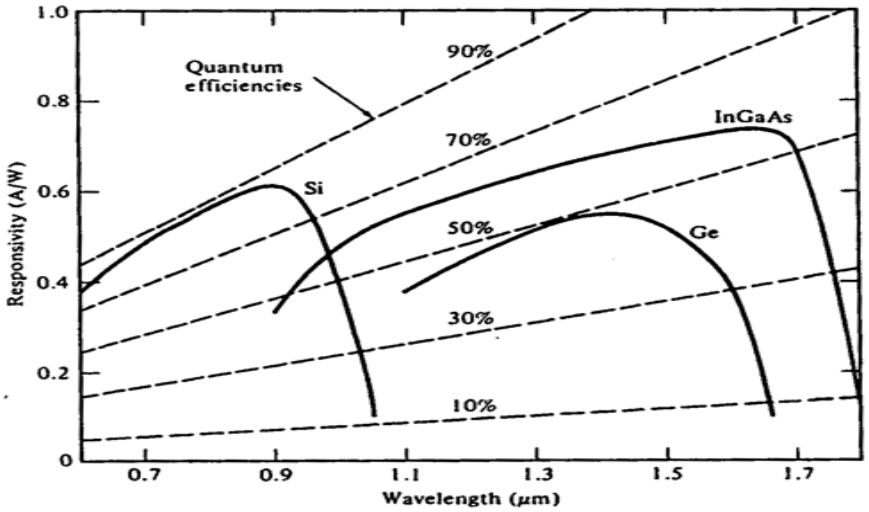


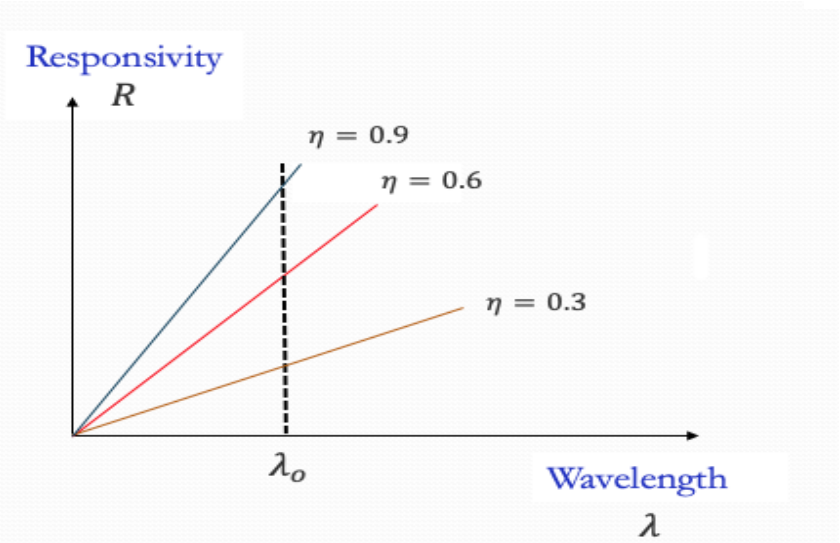
Photodiode Characteristics: Responsivity

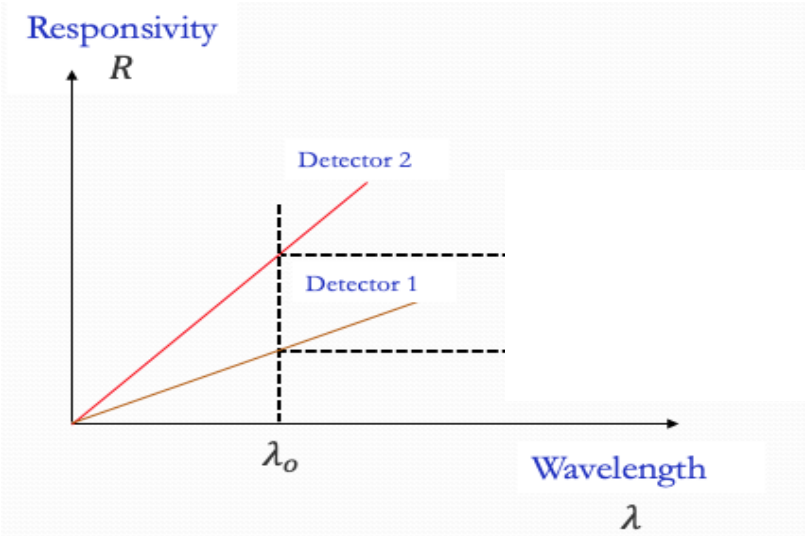
Responsivity. The ratio of optical current to incident optical power.

$$R = \frac{i_p}{P_o} = \eta \frac{e}{\hbar\omega} \qquad \text{(A/W)}$$

$$R = \frac{\eta e}{\hbar 2\pi c}\lambda$$

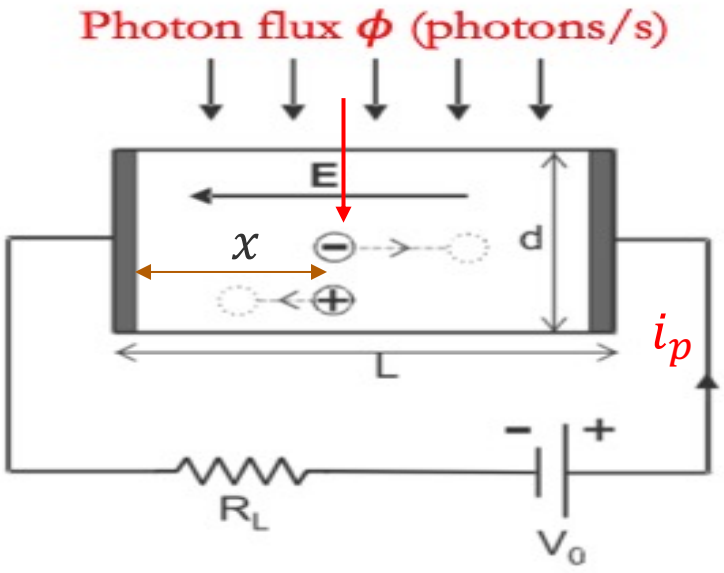


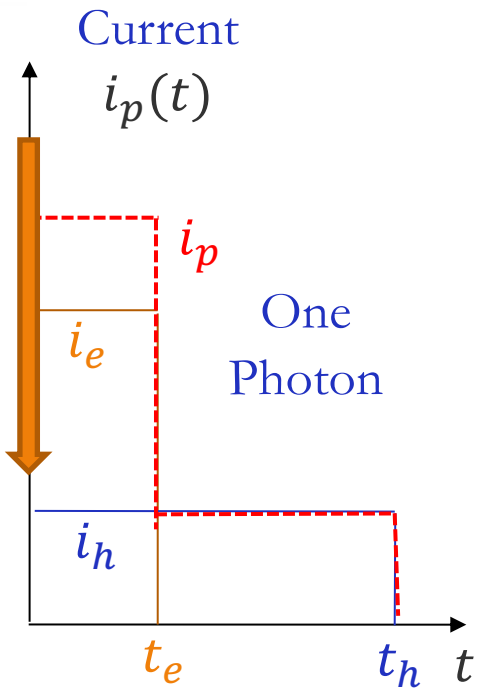


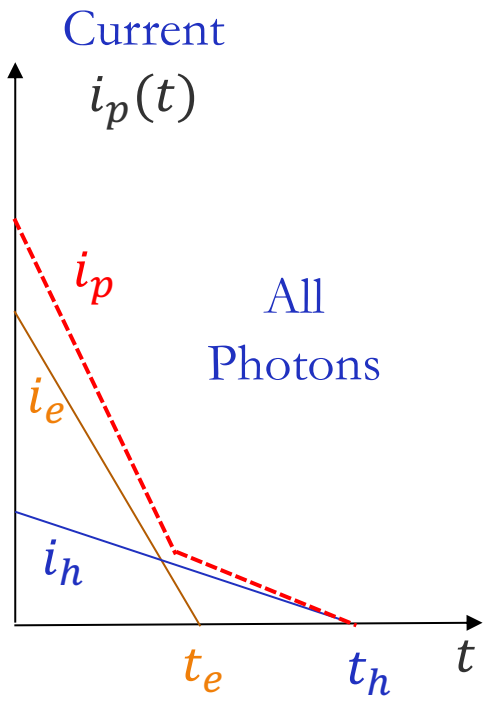


Photodiode Characteristics: Impulse Response

The impulse response is composed of transit time and RC delay.







$$i_e(t) = e\frac{v_e}{l}$$

$$i_h(t) = e\frac{v_h}{l}$$

$$i_p(t) = \frac{e}{l}(v_e + v_h)$$

$$t_e = \frac{L - x}{v_e}$$

$$t_h = \frac{x}{v_h}$$

Example. GaAs material with L= 1 cm, v_e=10^7 cm/s. i_e=1.6 pA.

Photodiode Characteristics: Impulse Response

The impulse response is composed of transit time and RC delay.

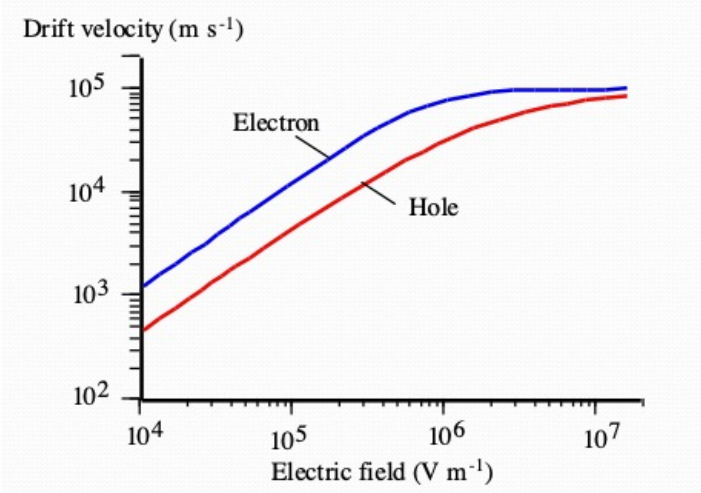
$$i_e(t) = e\frac{v_e}{L}$$

$$i_h(t) = e\frac{v_h}{L}$$

$$i_p(t) = \frac{e}{L}(v_e + v_h)$$

$$v_e = \mu_n E \qquad v_h = \mu_p E$$

cm²/Vs	μ_n	μ_p
Si	1500	450
Ge	3900	1900
InP	4600	150
InGaAs	14000	400



To improve the response:

- High mobility material.
- Small L.
- Drift velocity point.
- Help holes ($t_e = t_h$).

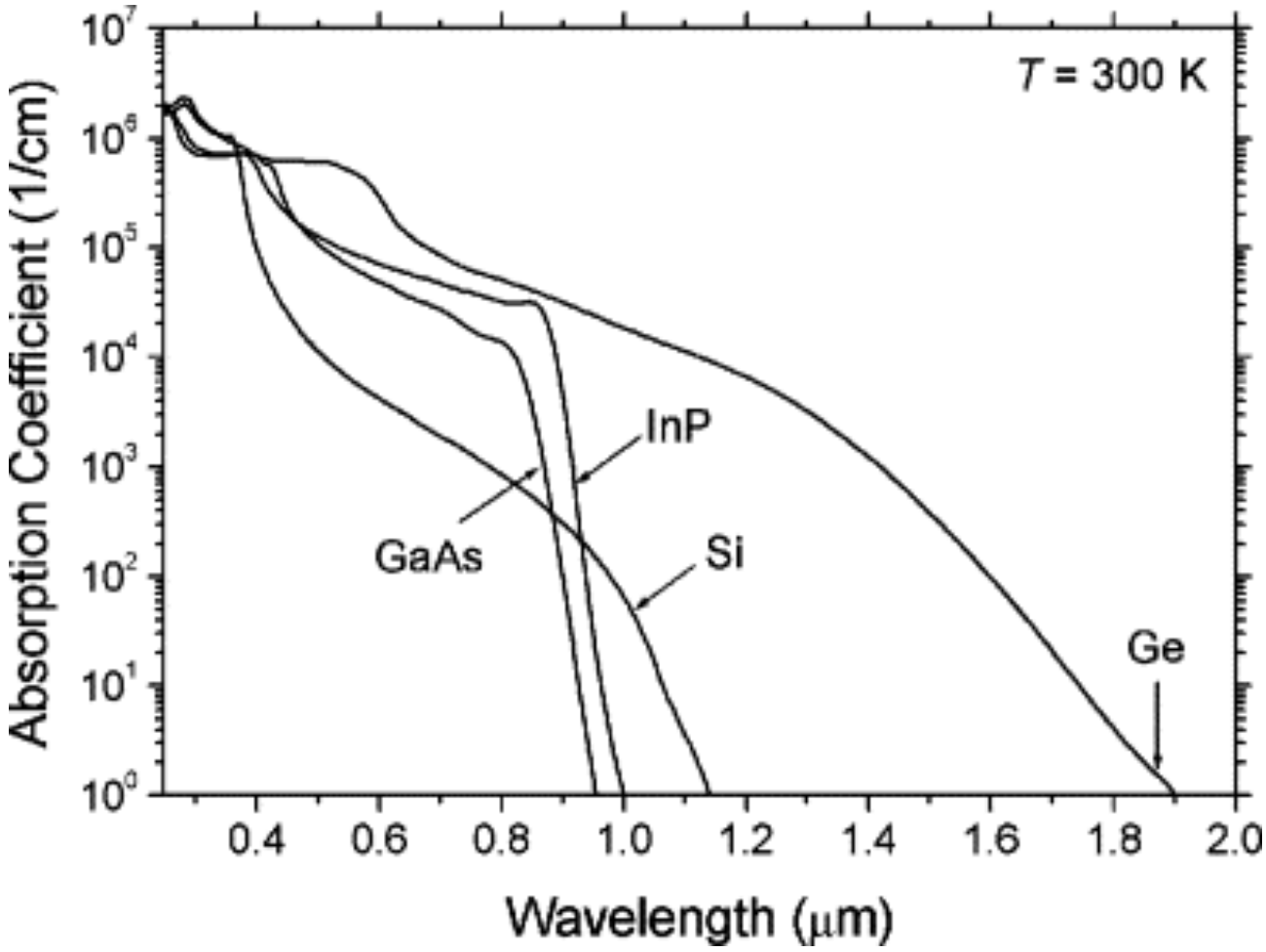
Cut-off Wavelength

$$\hbar\omega \geq E_g$$

$$\lambda_{co} = \frac{hc}{E_g} = \frac{1.24}{E_g}$$

	E_g(eV)	λ_{co}(μm)
Si	1.12	1.11
Ge	0.66	1.87
InP	1.35	0.91
InGaAs	0.75	1.65

Photodiode Characteristics

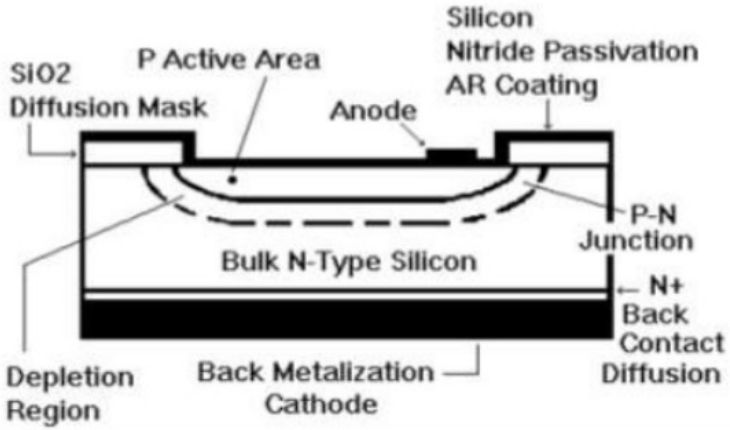


Photodiode Characteristics: Datasheet

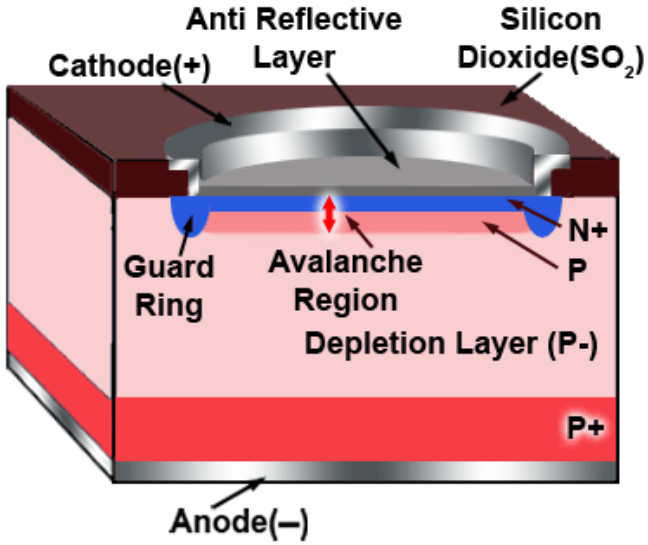
Parameter	Symbol	Unit	Si	Ge	InGaAs
Wavelength range	λ	nm	400-1100	800-1650	1100-1700
Responsivity	R	A/W	0.4-0.6	0.4 –0.5	0.75 – 0.95
Dark current	I_D	nA	1-10	50-500	0.5-2.0
Rise time	τ_r	ns	0.5-1	0.1-0.5	0.05-0.5
Bandwidth	B	GHz	0.3-0.7	0.5-3	1-2
Bias voltage	V_B	V	5	5-10	5

Photodiodes Types and Designs

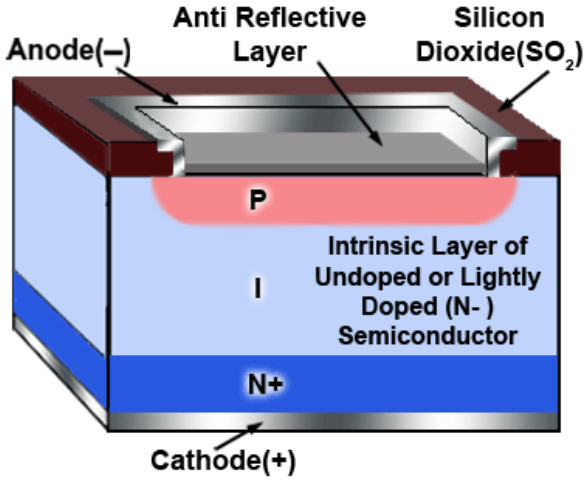
Basic PN photodiode.



Avalanche photodiode (APD).



PIN photodiode.





Examples

(1) A Si PIN with L=20 μm, V_{bias}=100V, λ=900 nm (pulsed). Discuss operation.

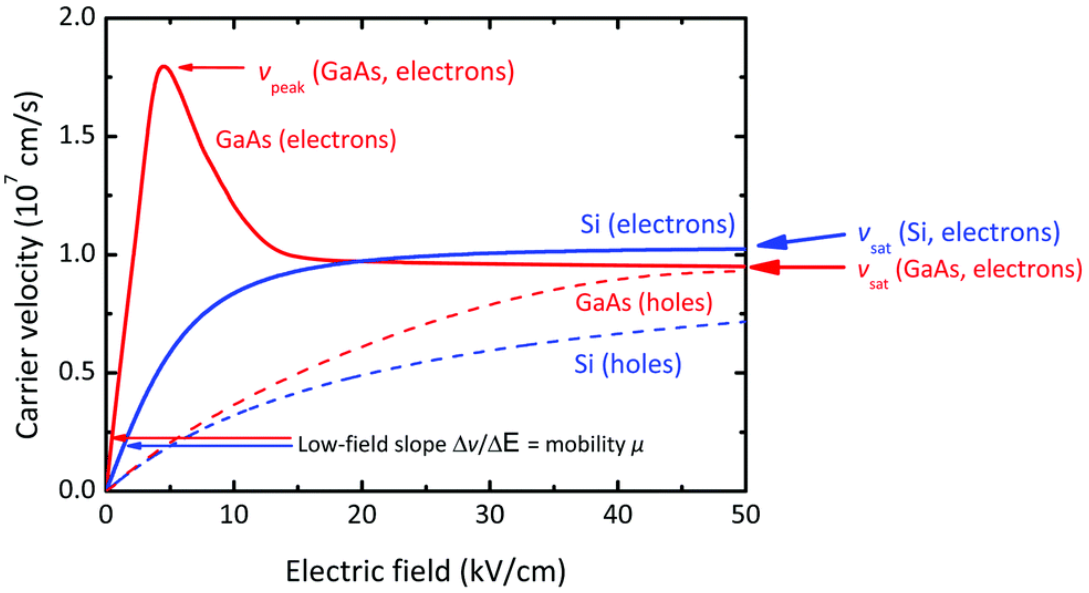
At 900 nm, $\alpha = 3\times10^4\ m^{-1}$

$$E = \frac{V_{bias}}{L} = 5\times10^6\ V/m$$

$$v_e = 10^5\ \ m/s$$

$$v_h = 7\times10^4\ \ m/s$$

$$t_h = \frac{L}{v_h} = 0.286\ ns$$



To improve the speed, decrease L, but that will hurt efficiency and hence responsivity. Find a trade-off.

Examples

(2) A Si PIN with diameter 0.4 mm is illuminated with a 700 nm light of intensity I=0.1 mW/m^2. The optical current produced was measured to be i_p=56.6 nA. Find efficiency and responsivity.

$$P_o = Area \times Intensity = A\,I = \pi(0.2\,mm)^2(0.1\,m)$$

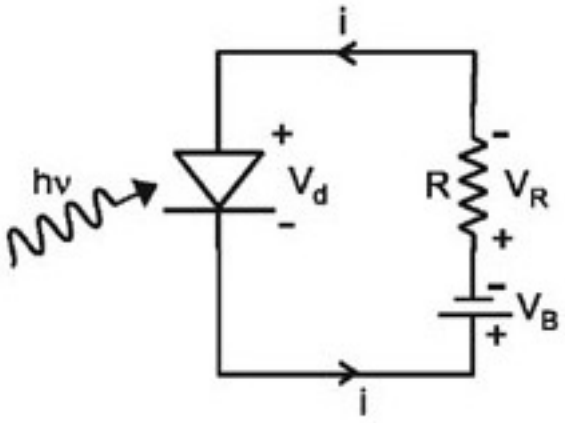
$$P_o = 1.26\times10^{-7}\,W$$

$$R = \frac{i_p}{P_o} = 0.45\,A/W$$

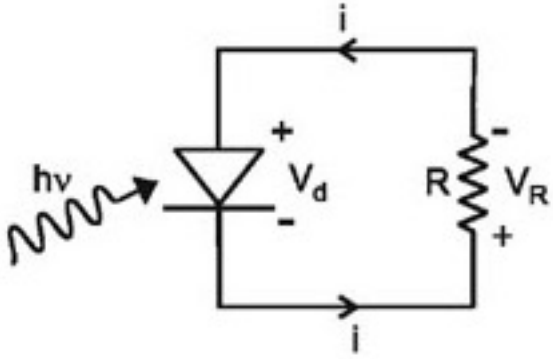
$$\eta = R\frac{hc}{e\lambda} = 80\%$$

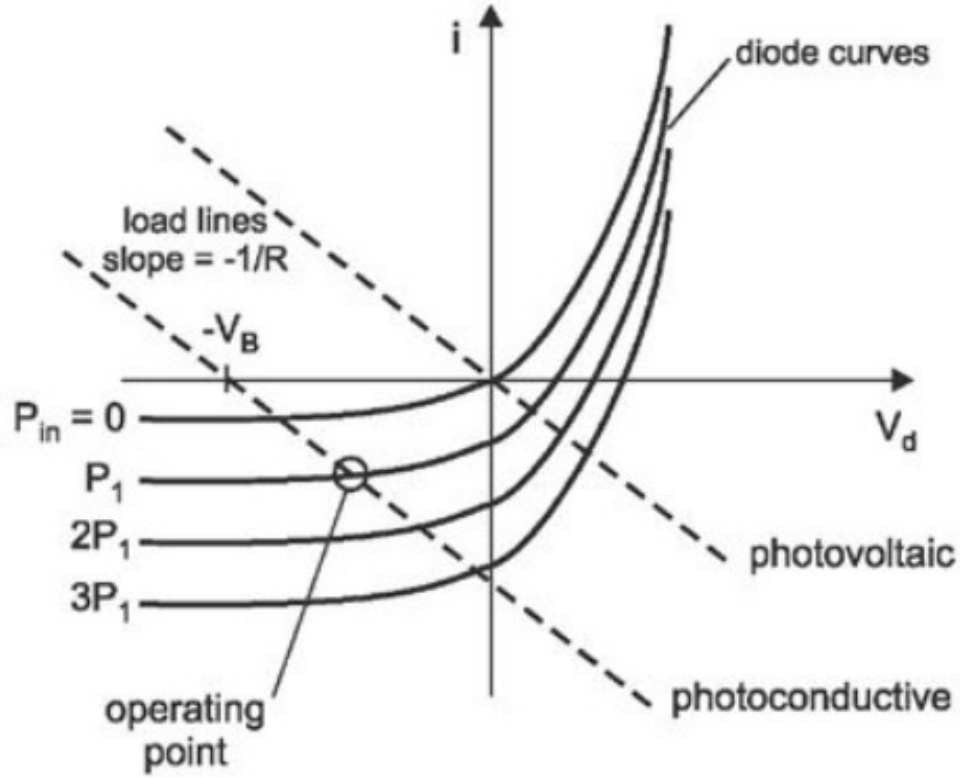
Photodiode Modes of Operation

Photoconductive Mode



Photovoltaic Mode





Photodetectors

The following subjects will be covered

- General Characteristics of Photodiodes.
- Modes of Operation.
- PN, PIN and APD Devices.

Photodiode Characteristics

Quantum efficiency. The ratio of optical current ehps to incident photon flux.

Responsivity. The ratio of optical current to incident optical power.

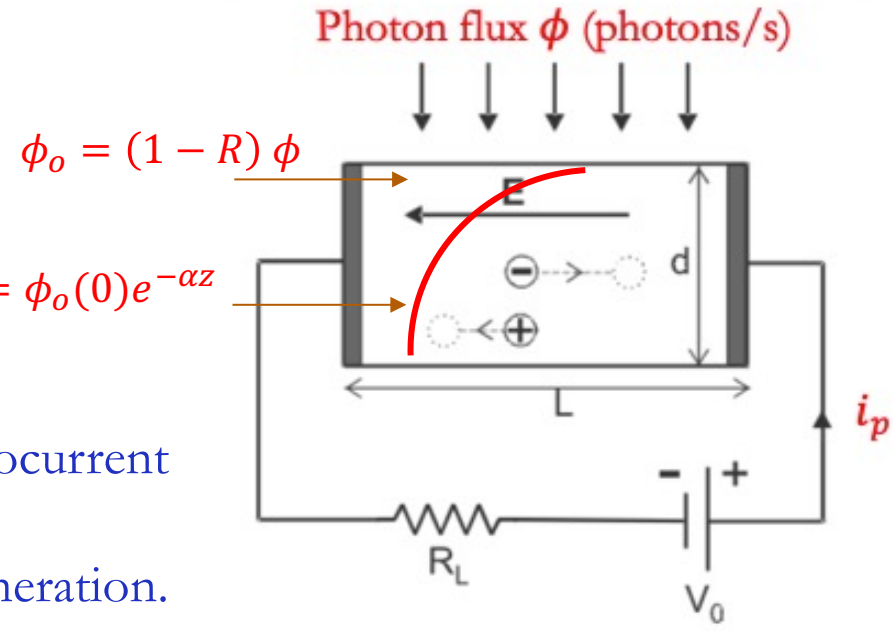
Impulse response. The rise time of the output current or the output bandwidth.

Noise power. The amount of current created by sources other than light.

Photodiode Characteristics: Efficiency

Quantum efficiency. The ratio of created ehp to number of incident photons.

$$\eta = \frac{\text{ehp flux making the optical current}}{\text{Incident photon flux } (\phi)}$$



Absorbed photon flux:

$$(1-R)\left(1-e^{-\alpha d}\right)\phi$$

Not all absorbed photons will make photocurrent due to:
(1) ehp recombination (2) phonon generation.
Introduce the factor $0 \le \xi \le 1$.

Current-making photon flux = $(1-R)\left(1-e^{-\alpha d}\right)\xi\,\phi$

$$\eta = (1-R)\left(1-e^{-\alpha d}\right)\xi$$

The engineering task is to maximize the efficiency.

Photodiode Characteristics: Efficiency

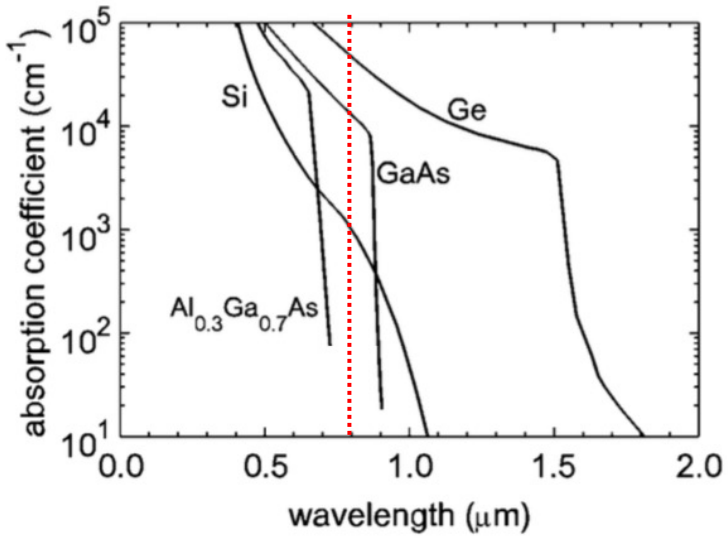
Quantum efficiency. The ratio of created ehp to number of incident photons.

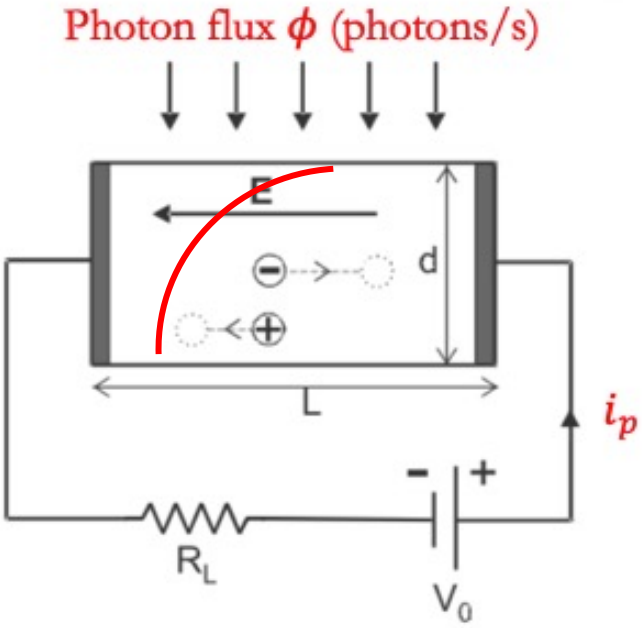
$$\eta = (1 - R)\left(1 - e^{-\alpha d}\right)\xi$$

ξ : Apply bias and clean fabrication process.

R : Use anti-reflection coating (small bandwidth)

αd : Choose best material and thickness.



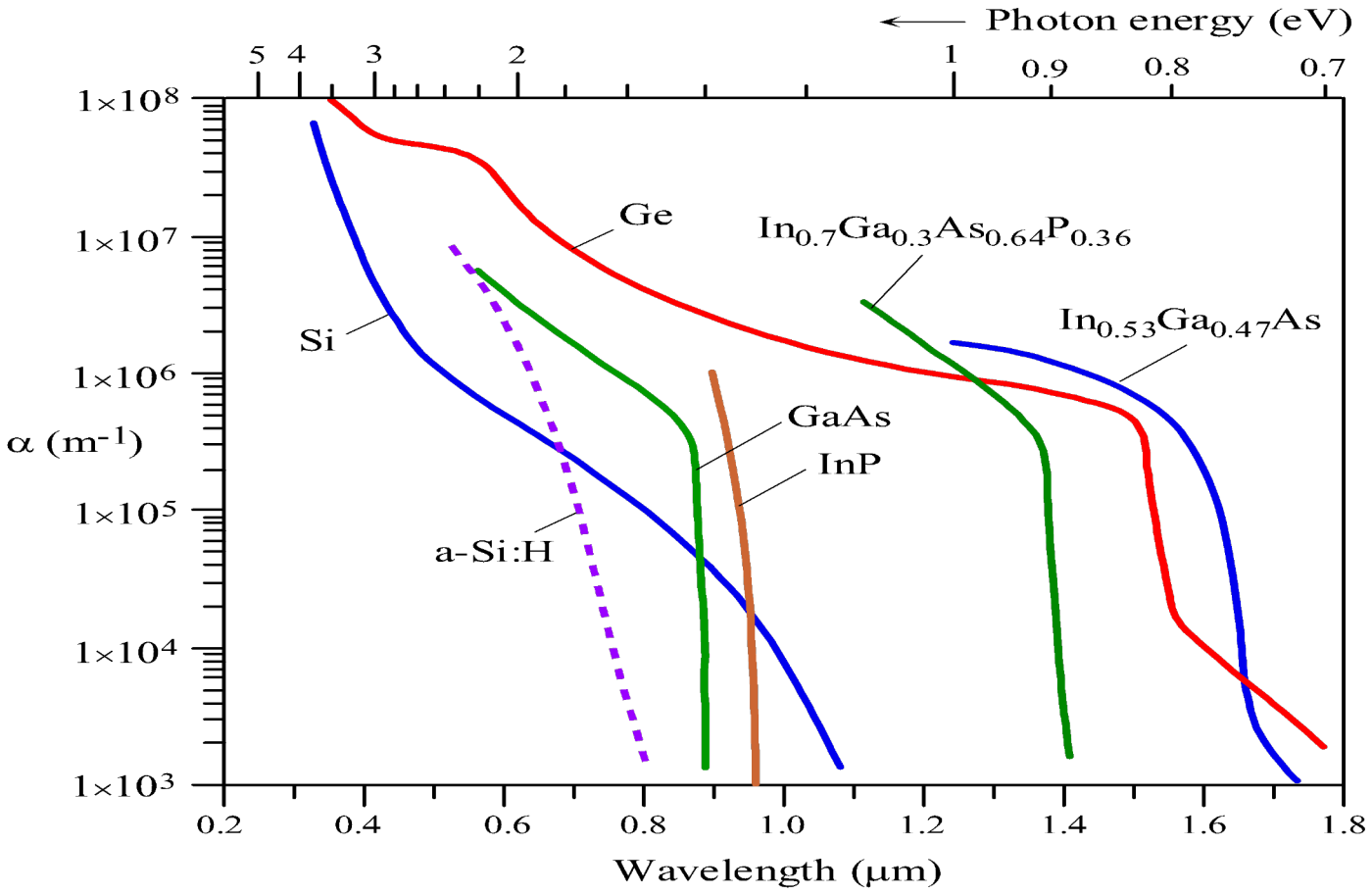


Example: For 95% absorption at 0.8 μm, find material and thickness.

$$1 - e^{-\alpha d} = 0.95$$

Photodiodes Characteristics: Efficiency

Absorption coefficient of common semiconductor materials.

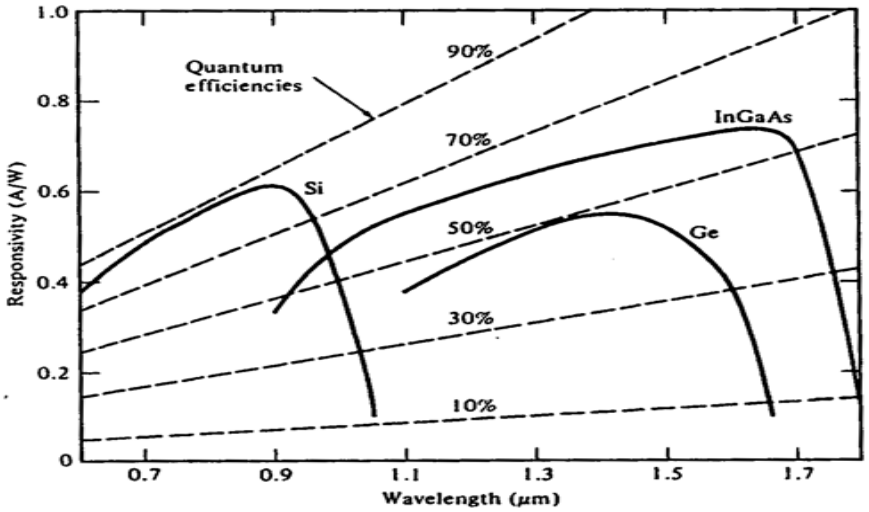


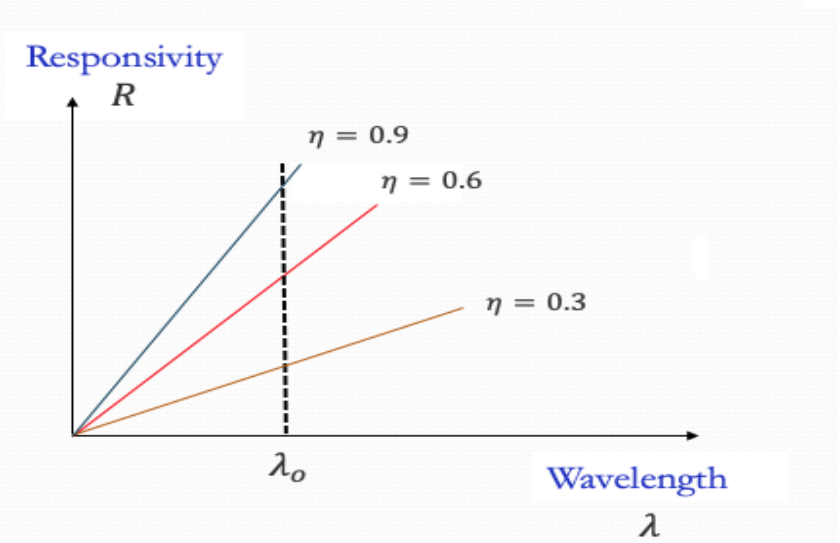
Photodiode Characteristics: Responsivity

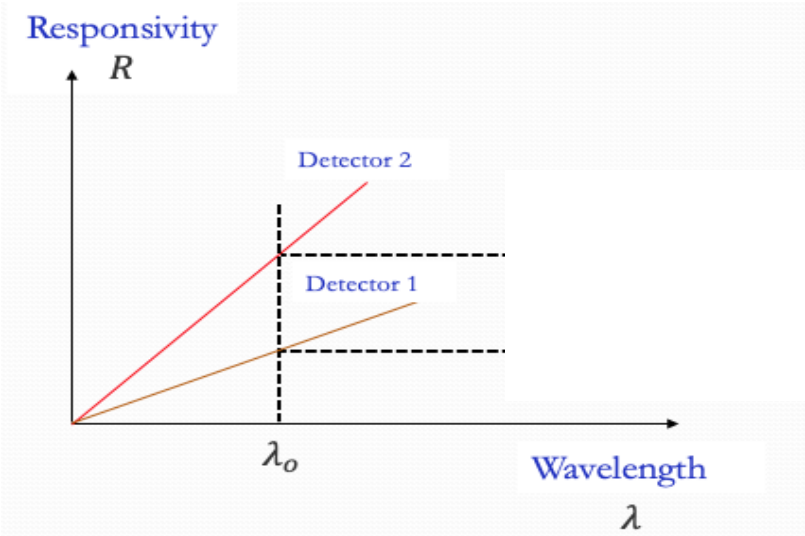
Responsivity. The ratio of optical current to incident optical power.

$$R = \frac{i_p}{P_o} = \eta \frac{e}{\hbar\omega} \qquad \text{(A/W)}$$

$$R = \frac{\eta e}{\hbar 2\pi c}\lambda$$

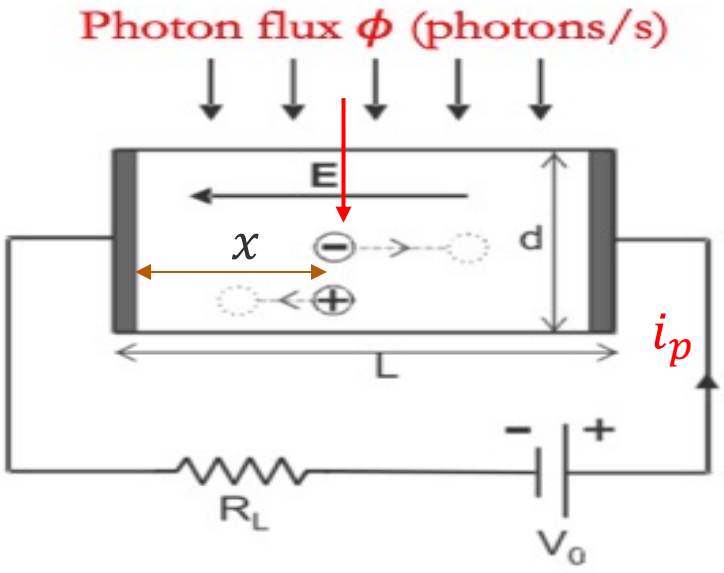


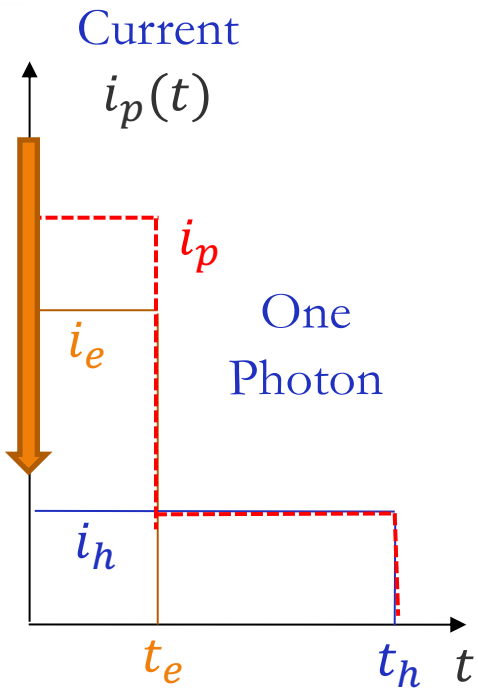


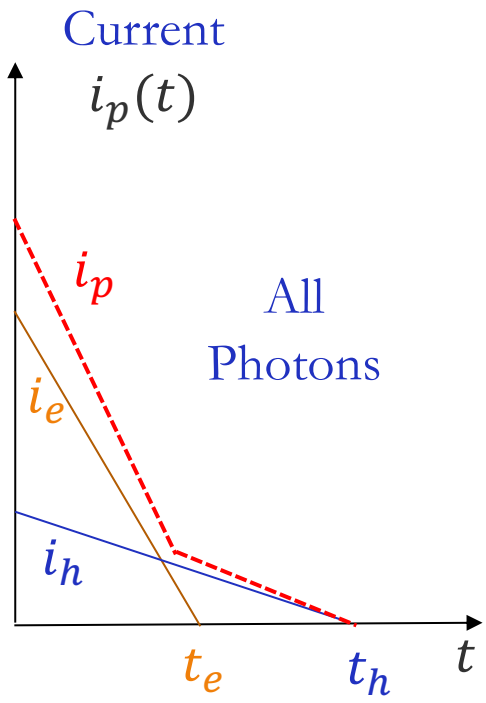


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The impulse response is composed of transit time and RC delay.







$$i_e(t) = e\frac{v_e}{l}$$

$$i_h(t) = e\frac{v_h}{l}$$

$$i_p(t) = \frac{e}{l}(v_e + v_h)$$

$$t_e = \frac{L - x}{v_e}$$

$$t_h = \frac{x}{v_h}$$

Example. GaAs material with L= 1 cm, v_e=10^7 cm/s. i_e=1.6 pA.

Photodiode Characteristics: Impulse Response

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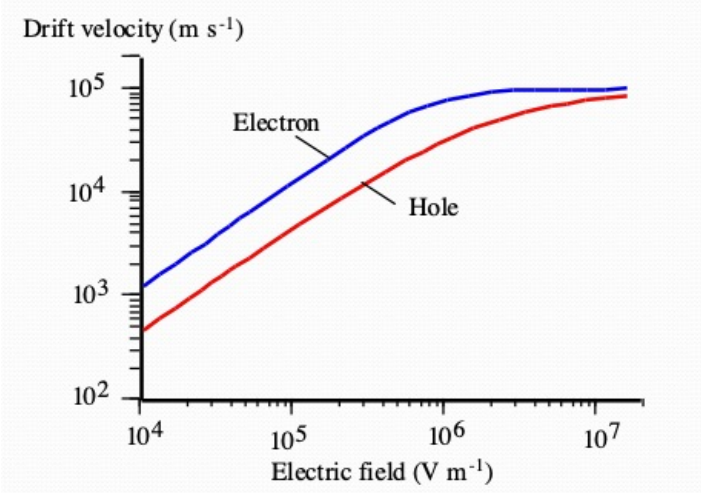
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cm²/Vs	μ_n	μ_p
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To improve the response:

- High mobility material.
- Small L.
- Drift velocity point.
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Cut-off Wavelength

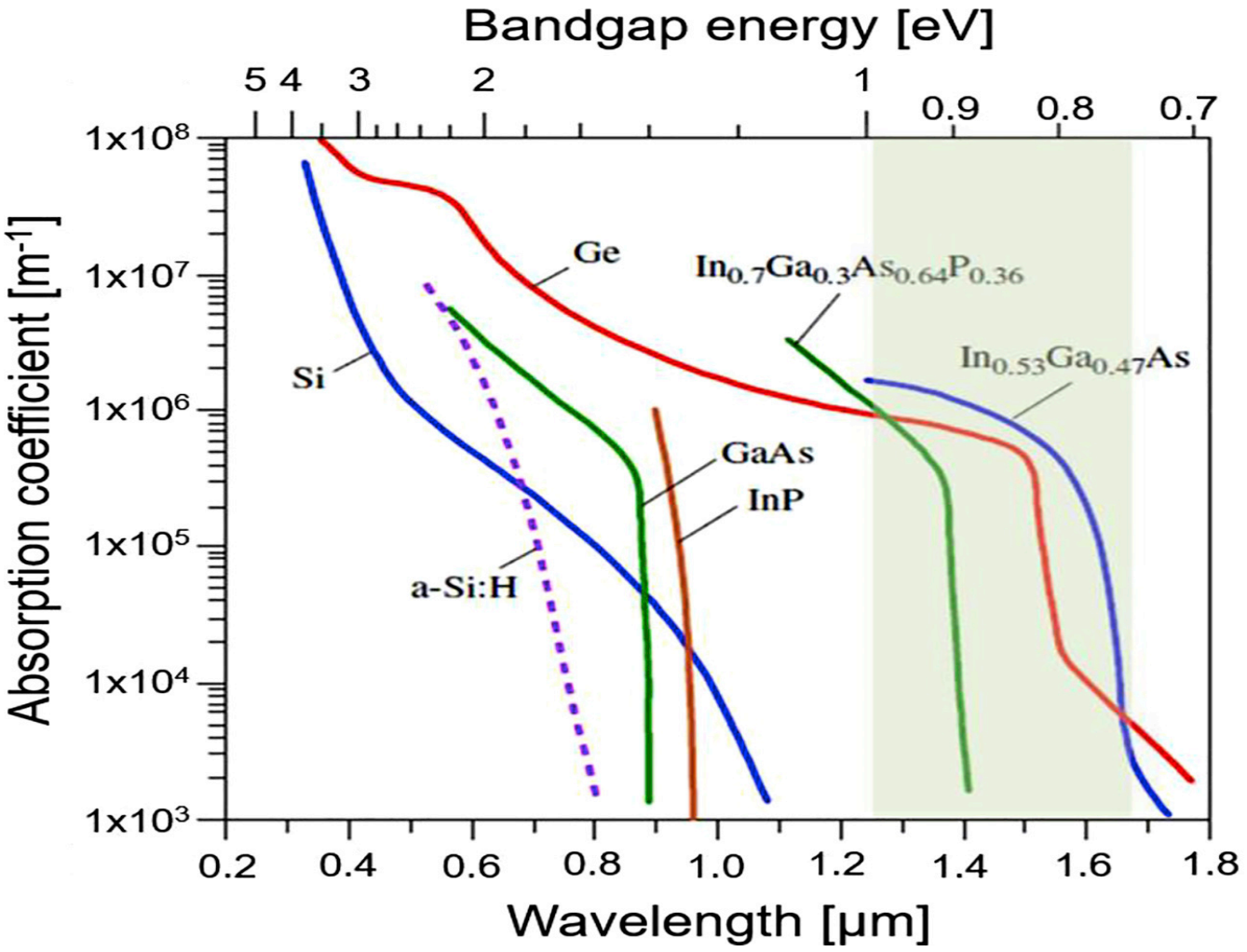
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$$\lambda_{co} = \frac{hc}{E_g} = \frac{1.24}{E_g}$$

	E_g(eV)	λ_{co}(μm)
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Absorption Curves

Absorption coefficient of common semiconductor materials.

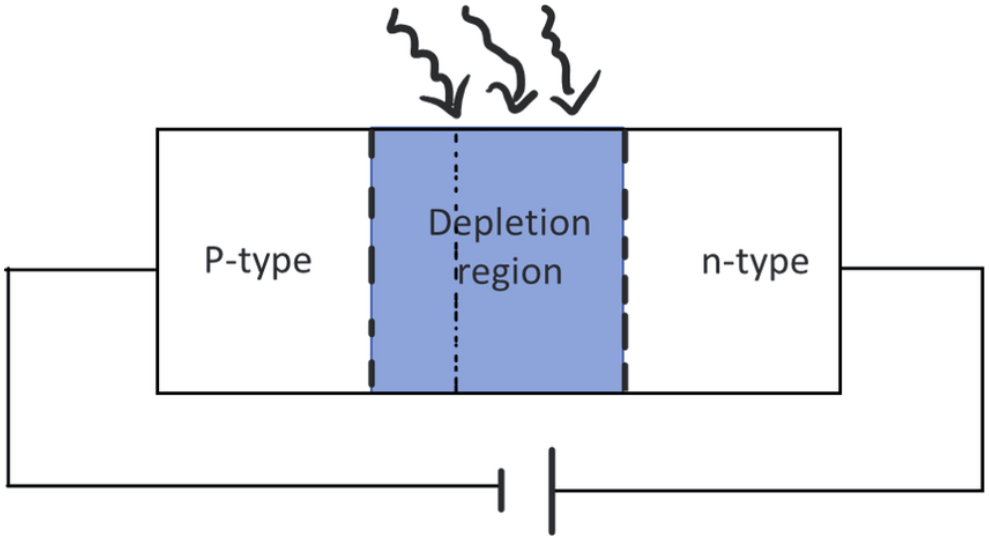


Photodiode Characteristics: Datasheet

Parameter	Symbol	Unit	Si	Ge	InGaAs
Wavelength range	λ	nm	400-1100	800-1650	1100-1700
Responsivity	R	A/W	0.4-0.6	0.4 –0.5	0.75 – 0.95
Dark current	I_D	nA	1-10	50-500	0.5-2.0
Rise time	τ_r	ns	0.5-1	0.1-0.5	0.05-0.5
Bandwidth	B	GHz	0.3-0.7	0.5-3	1-2
Bias voltage	V_B	V	5	5-10	5

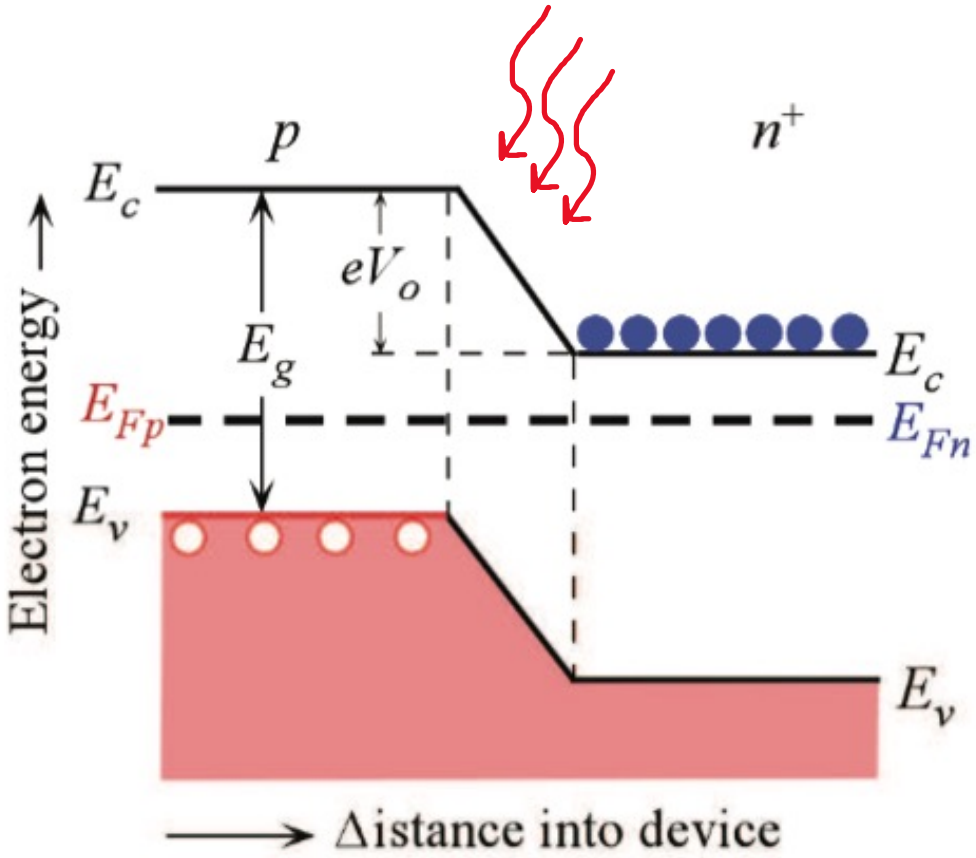
Photodiodes

The basic structure of a photodiode device is a pn junction.



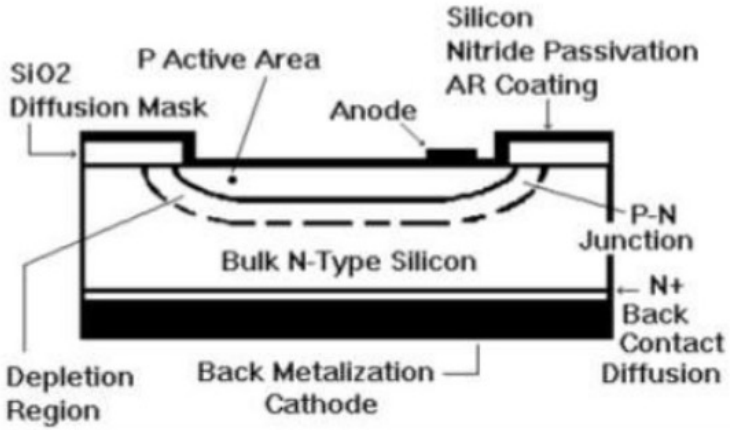
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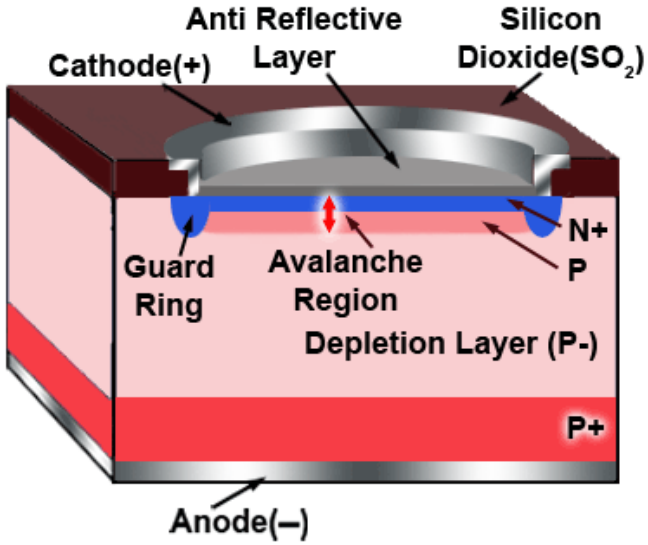


Photodiodes Types and Designs

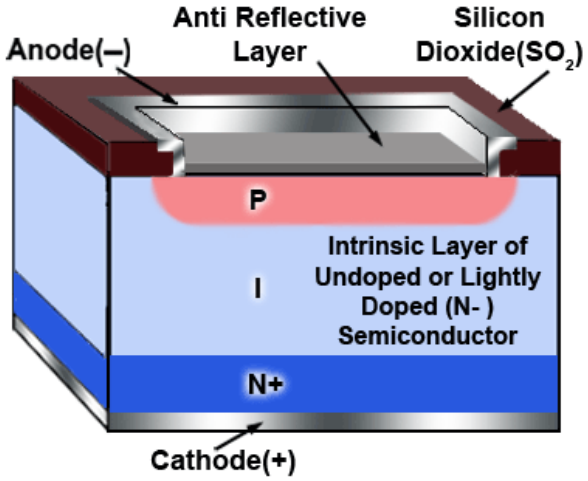
Basic PN photodiode.



Avalanche photodiode (APD).



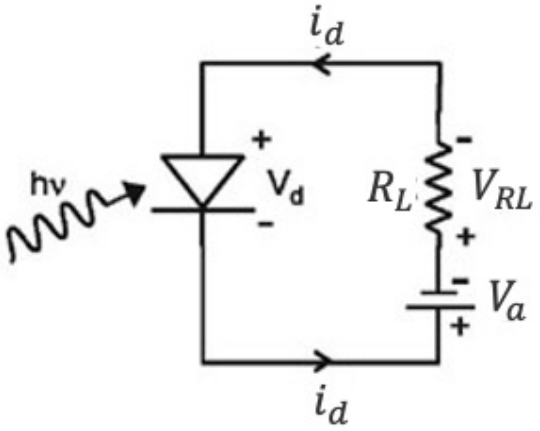
PIN photodiode.



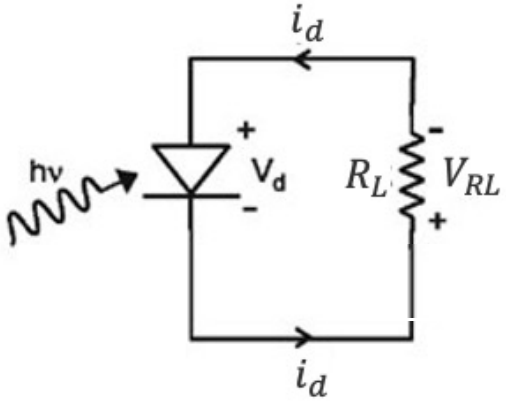


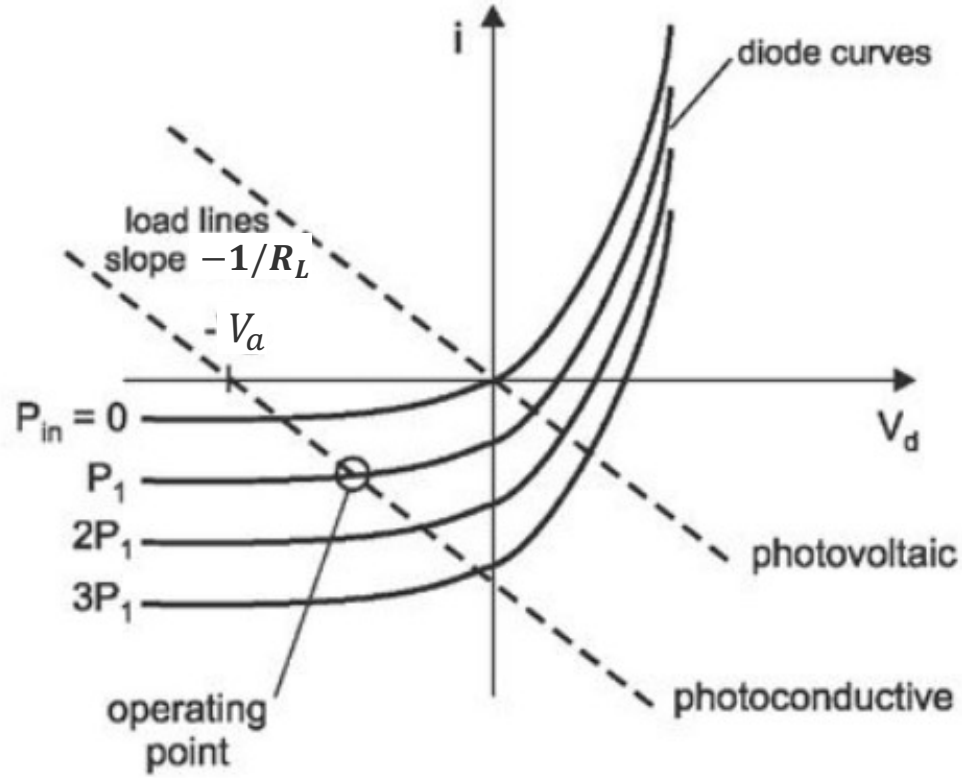
Photodiode Modes of Operation

Photoconductive Mode

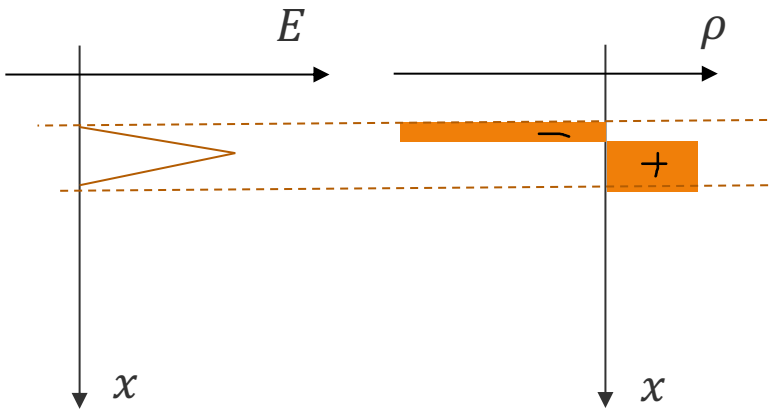


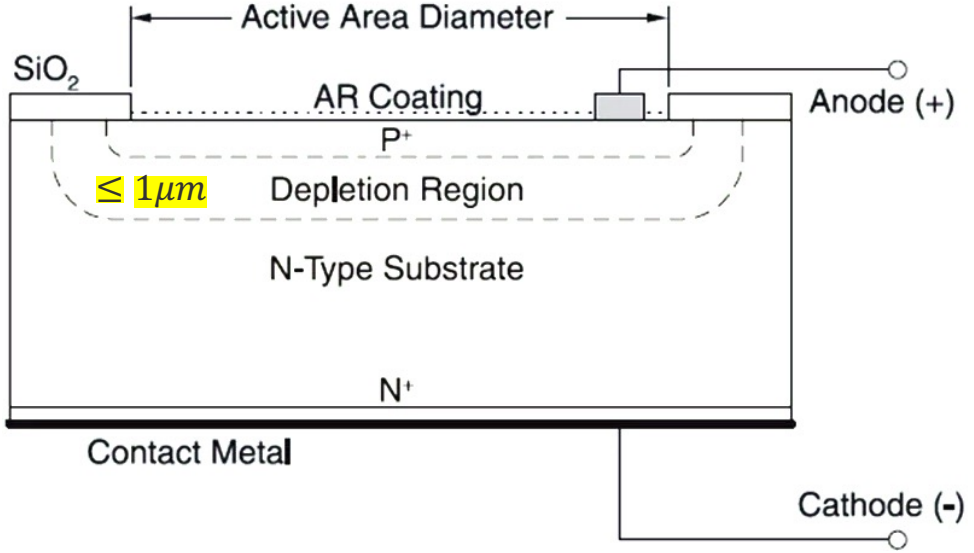
Photovoltaic Mode





Basic PN Photodiode Design

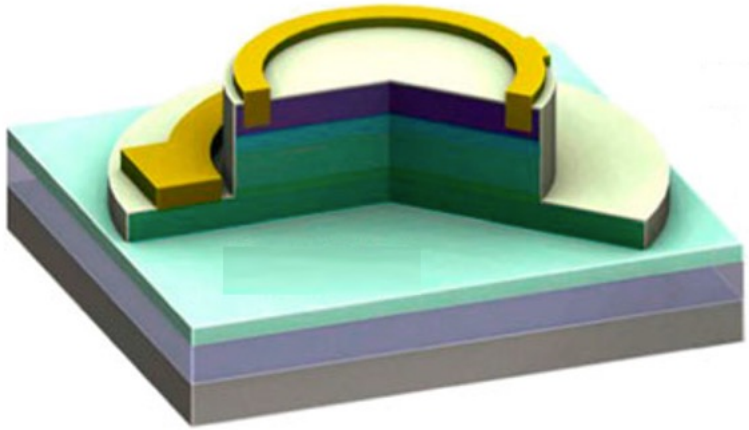




$$C_j = \frac{\varepsilon A}{\sqrt{\frac{2\varepsilon(V_{bi}+V_A)}{q}\frac{(N_A+N_D)}{N_A N_D}}}$$

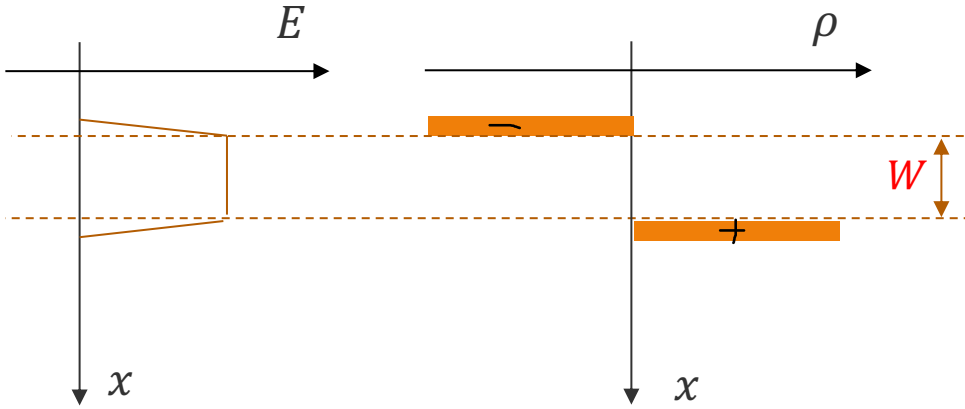
$$C_j \approx A\sqrt{\frac{q\varepsilon N_D}{2(V_{bi}+V_A)}}$$

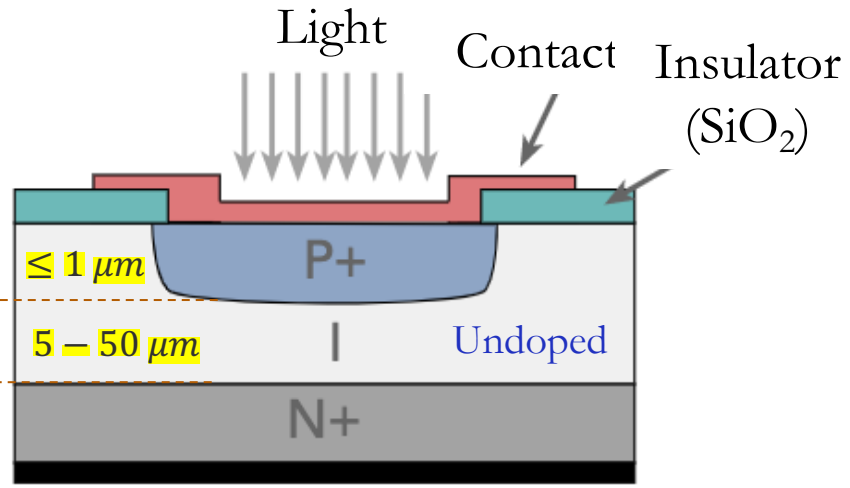
$$N_A \gg N_D$$



The PiN Photodiode

This structure improves absorption, speed and efficiency.





$$E \approx \frac{V_A}{W} \qquad C_j \approx \frac{\varepsilon A}{W} \sim pF$$

Independent of V_A

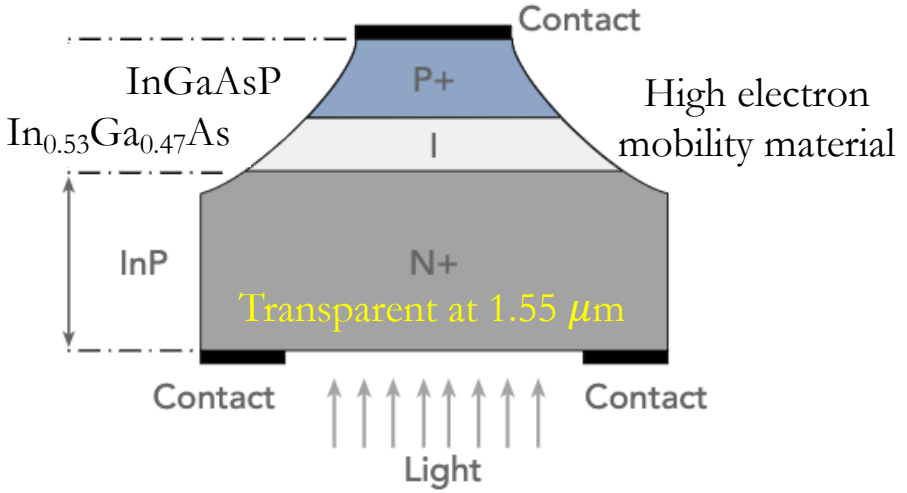
$$RC \sim 50\ pF$$

Transit time can be a problem if W is too large!

$$t = \frac{W}{v_e}$$

Increase V_A and hence v_e.
Typically 1 ns.

An excellent design for optical fiber communications.



Examples

(1) A Si PIN with L=20 μm, V_{bias}=100V, λ=900 nm (pulsed). Discuss operation.

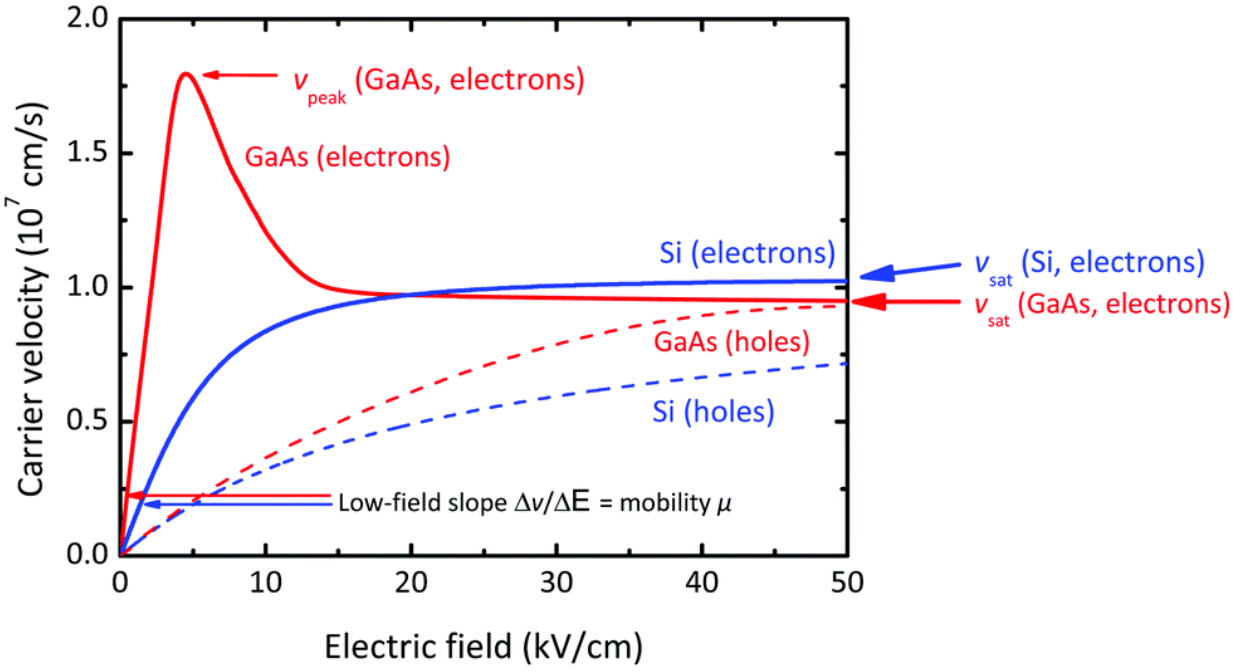
At 900 nm, $\alpha = 3\times10^4\ m^{-1}$

$$E = \frac{V_{bias}}{L} = 5\times10^6\ V/m$$

$$v_e = 10^5\ \ m/s$$

$$v_h = 7\times10^4\ \ m/s$$

$$t_h = \frac{L}{v_h} = 0.286\ ns$$



To improve the speed, decrease L, but that will hurt efficiency and hence responsivity. Find a trade-off.

Examples

(2) A Si PIN with diameter 4 cm is illuminated with a 700 nm light of intensity I=0.1 mW/m^2. The optical current produced was measured to be i_p=56.6 nA. Find efficiency and responsivity.

$$P_o = Area \times Intensity = A\,I = \pi(0.2\,mm)^2(0.1\,m)$$

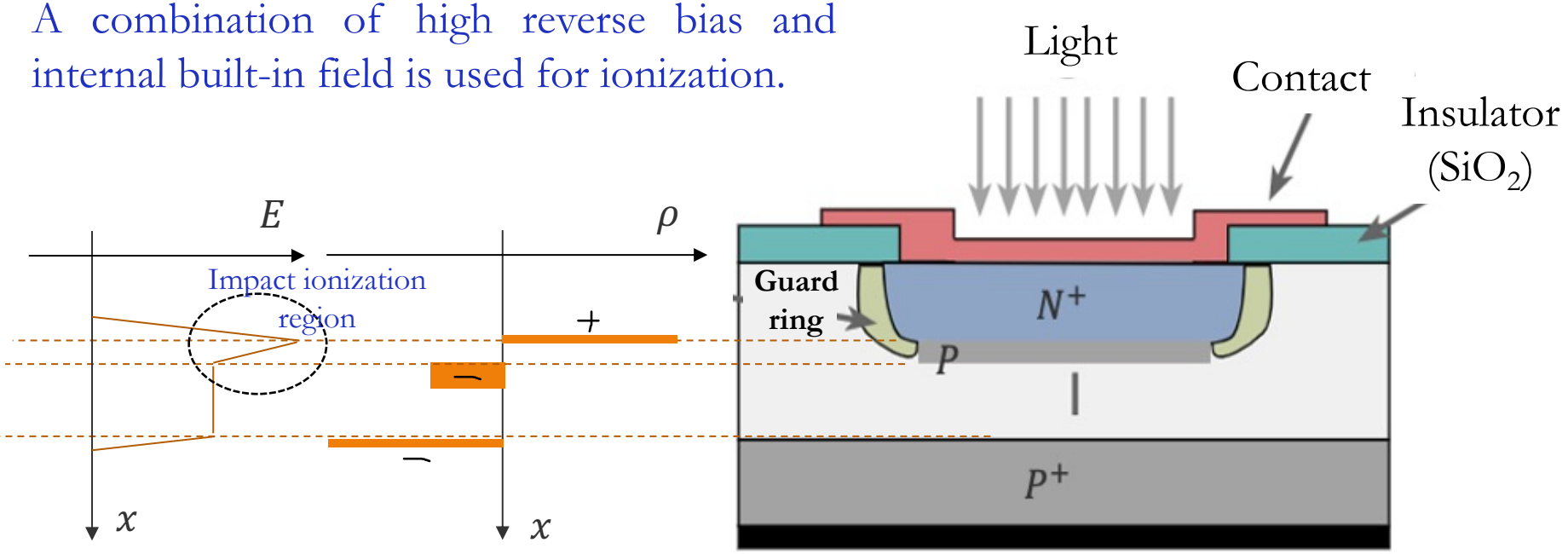
$$P_o = 1.26\times10^{-7}\,W$$

$$R = \frac{i_p}{P_o} = 0.45\,A/W$$

$$\eta = R\frac{hc}{e\lambda} = 80\%$$

The Avalanche Photodiode (APD)

A combination of high reverse bias and internal built-in field is used for ionization.



No hole ionization (lower noise and faster operation).

Ionization (Gain) factor

Si ~ 100

Ge ~ 10

AlGaAs ~ 10

Temperature-dependent

$$G \approx \frac{1}{1-(V_A/V_{br})^n}$$

$$i_p = G i_{po}$$

$$R = G\frac{i_{po}}{P_o}$$

Advantages

Avoid breakdown, high sensitivity, no need for current amplifiers, high speed (0.1 n typical).

Examples

(1) An InGaAs APD has 60% efficiency at 1550 nm with no impact ionization. Find the photocurrent and responsivity of the device if it was biased for G=12, with input power of 20 nW.

$$R_o = \frac{\eta_o q e \lambda}{hc} = 0.75\ A/W$$

$$i_{po} = R_o\ P_o = 15\ nA$$

$$i_p = G\ i_{po} = 0.18\ \mu A$$

$$R = G\ R_o = 9.0\ A/W$$

Examples

(2) A Si APD has 70% efficiency at 830 nm with no impact ionization. The photodiode is then biased for G=100. Find the photocurrent with input power of 10 nW.

$$R_o = \frac{\eta_o q e \lambda}{hc} = 0.47\ A/W$$

$$i_{po} = R_o\ P_o = 4.7\ nA$$

$$i_p = G\ i_{po} = 0.47\ \mu A$$